

ARITHMETICITY OF TWISTED FINITE CORRESPONDENCES FOR LATTICES OVER LOCAL FIELDS

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ABSTRACT. We prove arithmetic exhaustion for bifinite correspondences between arbitrarily scalar-twisted group factors of ICC groups commensurable with property- (T) lattices over characteristic-zero local fields and twisted group factors of arbitrary countable ICC groups. Every such correspondence is a closed summand of a finite direct sum of models obtained from actual finite-index subgroup isomorphisms and finite-dimensional projective representations of the matched cocycle ratio. The proof includes reducible products, mixed real and finite places, and quaternionic and Cayley rank-one factors.

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1. INTRODUCTION

A finite correspondence between two group factors is a finite-dimensional relation in the sense of von Neumann module dimension. For lattice groups, we prove that this analytic relation comes from a finite collection of actual subgroup isomorphisms. Scalar cocycles

remain visible: the finite fibres are projective representations whose multiplier is exactly the ratio of the two cocycles along the subgroup isomorphism.

The source groups include lattices in products of higher-rank real and nonarchimedean groups, together with the quaternionic and Cayley property-(T) rank-one groups. The comparison group is only assumed to be countably infinite and ICC. Thus the geometry used in the proof must be constructed inside the correspondence; no boundary or ambient Lie group is initially available on the comparison side.

1.1. The class, conventions, and main theorem.

Definition 1.1. Let $\mathcal{K}$ be the class of countably infinite ICC groups abstractly commensurable with a lattice in a finite nonempty product

$$G = \prod_{i=1}^r G_i, \quad G_i = \mathbf{H}_i(k_i)^+.$$

Here k_i is $\mathbb{R}$, $\mathbb{C}$, or a finite extension of $\mathbb{Q}_p$; the primes may differ. The algebraic group $\mathbf{H}_i$ is connected, adjoint, and absolutely simple over k_i . The superscript $+$ denotes the subgroup generated by the k_i -points of the unipotent radicals of proper k_i -parabolic subgroups. Each G_i is required to be noncompact and to have Kazhdan's property (T).

A lattice is discrete and has finite Haar covolume. Abstract commensurability means an actual isomorphism between finite-index subgroups. ICC means that every nonidentity conjugacy class is infinite. No irreducibility, cocompactness, product decomposition, or torsion hypothesis is imposed on the lattice.

Property (T) means that every continuous unitary representation with almost invariant unit vectors has a nonzero invariant vector. The real simple factors in Definition 1.1 have higher rank or are locally $\mathrm{Sp}(n, 1)$, $n \geq 2$, or $F_{4(-20)}$. At a finite place their relative rank is at least two. Complex places are treated as real Lie groups when using differential arguments. Standard structure and permanence results are recalled in Section 3; see also [5].

All scalar cocycles take values in $\mathbb{T}$ and are normalized:

$$\mu(g, h)\mu(gh, j) = \mu(h, j)\mu(g, hj), \quad \mu(e, g) = \mu(g, e) = 1.$$

The twisted group factor $L_\mu(\Gamma)$ has canonical unitaries u_g with $u_g u_h = \mu(g, h)u_{gh}$ and trace $\tau(u_g) = \mathbf{1}_{\{e\}}(g)$. For an ICC group it is a factor: the absolute values of Fourier coefficients of a central vector are constant on conjugacy classes, so square summability kills all nonidentity coefficients.

A bifinite M – N correspondence is a Hilbert space with commuting normal unital left M and right N actions and finite von Neumann module dimensions on both sides. The actions need not be mutual commutants, and the correspondence need not be irreducible. Every correspondence in this paper is separable; every factor isomorphism is normal, unital, complex-linear, and $*$ -preserving. For a finite factor Q , write

$$Q^t = p(M_n(\mathbb{C}) \overline{\otimes} Q)p, \quad (\mathrm{Tr}_n \otimes \tau_Q)(p) = t > 0.$$

The matrix trace Tr_n is unnormalized; the corner itself has its normalized trace.

For finite-index subgroups $A \leq \Gamma$ and $B \leq \Lambda$, an isomorphism $\delta : A \rightarrow B$, and a nonzero finite-dimensional projective unitary representation σ satisfying

$$(1.1) \quad \sigma(a)\sigma(b) = \frac{\mu(a, b)}{\omega(\delta(a), \delta(b))} \sigma(ab), \quad \sigma(e) = I,$$

define

$$(1.2) \quad E(A, B, \delta, \sigma) = L^2(M) \overline{\otimes}_{L_{\mu|_A}(A)} (V_\sigma \otimes L^2(N)), \quad M = L_\mu(\Gamma), \quad N = L_\omega(\Lambda).$$

The right N action is standard. The left subgroup action on the second factor is $u_a(\xi \otimes \eta) = \sigma(a)\xi \otimes v_{\delta(a)}\eta$. Equation (1.1) makes this an action with multiplier $\mu|_A$.

A right action is represented by $\rho(h)\xi = \xi v_h^*$, so $\rho(h)\rho(k) = \overline{\omega(h, k)}\rho(hk)$. The left action is $\pi(g)\xi = u_g\xi$. Thus the joint action $U_{(g, h)}\xi = u_g\xi v_h^*$ has multiplier $\alpha((g, h), (g', h')) = \mu(g, g')\overline{\omega(h, h')}$.

Theorem 1.2 (Arithmetic exhaustion). *Let $\Gamma \in \mathcal{K}$, let Λ be any countably infinite ICC group, and let $\mu \in Z^2(\Gamma, \mathbb{T})$ and $\omega \in Z^2(\Lambda, \mathbb{T})$ be normalized scalar cocycles. Every bifinite $L_\mu(\Gamma)$ – $L_\omega(\Lambda)$ correspondence is unitarily isomorphic to a closed bimodule summand of a finite direct sum of the models $E(A, B, \delta, \sigma)$ in Equation (1.2), where $A \leq \Gamma$ and $B \leq \Lambda$ have finite index, $\delta : A \rightarrow B$ is an actual group isomorphism, and σ is a finite-dimensional projective unitary representation satisfying Equation (1.1).*

Conversely, every such model, finite direct sum, and closed bimodule summand is a normal bifinite correspondence. If $n = \dim_{\mathbb{C}} V_\sigma$, the elementary model has dimensions

$$\dim_{L_\mu(\Gamma)} E(A, B, \delta, \sigma) = n[\Lambda : B], \quad \dim E(A, B, \delta, \sigma)_{L_\omega(\Lambda)} = n[\Gamma : A].$$

The theorem retains both finite sums and closed summands. Its displayed dimension formulas concern the elementary models themselves. The finer problem of identifying every summand and every bounded intertwiner is addressed in the companion article [49], which uses Theorem 1.2 as its geometric input. The elementary construction and its normality and dimensions are proved here in Section 2.

Only the matched ratio $\mu/\delta^*\omega$ must admit the finite-dimensional projective representation in Equation (1.1). Neither endpoint cocycle is required to have finite order, finite type, or a virtually trivial restriction, and the image of σ may be infinite. In particular, existence of a nonzero correspondence forces Γ and Λ to be abstractly commensurable. For trivial cocycles the converse follows by taking the trivial one-dimensional fibre. These conclusions apply to nonuniform and reducible lattices as well as to their ICC commensurable groups.

1.2. Rigidity and finite-correspondence precedents. The group-factor reconstruction problem asks which features of a discrete group survive in its von Neumann algebra. Connes proved countability of the fundamental group of a property-(T) group factor [18]; his later problems ask whether nonisomorphic ICC property-(T) groups have nonisomorphic factors and ask for the fundamental groups of those factors [19, Chapter V, Appendix B.ε, Problems 1–2]. Popa’s survey states a stronger recovery question that controls the given isomorphism, up to a character and inner conjugacy, together with its amplification scale [57, Section 3, Equations (3.2) and (3.2’)]. Property (T) alone does not ensure group reconstruction: independent, concurrent constructions of OpenAI and Zhou give nonisomorphic ICC property-(T) groups with isomorphic group factors [48, Theorem 1.2] [66, Theorem A]. The present theorem concerns the lattice class in Definition 1.1. Arithmetic exhaustion is the finite-correspondence input used in the companion article to obtain stable canonical recovery for this class.

Deformation and rigidity methods have produced strong reconstruction theorems in several settings. Ioana’s theorem recovers Bernoulli actions of ICC property-(T) groups from their crossed products [38]. Ioana–Popa–Vaes obtained W^* -superrigid groups among generalized wreath products [39], and Chifan–Ioana–Osin–Sun constructed property-(T) examples using wreath-like products [16]. These results recover groups or actions from operator-algebraic data through constructions different from the lattice geometry developed here.

Finite correspondences retain finite-index relations and their compositions. Jones’ index and module dimensions [40], the finite-index expectation inequality of Pimsner–Popa [55], and Connes’ relative tensor product [19, Chapter V, Appendix B] give the framework. Vaes’ classification for generalized Bernoulli factors already organizes finite-index bimodules by subgroup isomorphisms and finite-dimensional projective representations of cocycle ratios, and computes their fusion by projective induction [65, Theorem 2.2 and Section 6.2]. Donvil–Vaes prove rigidity under arbitrary scalar twists and virtual isomorphism for certain left-right wreath products. Their results include the projective-ratio constraint and reconstruction, up to amplification, of finite-index factor neighbors [27, Theorems A and 6.6]. Chifan–Fernández

Quero–Osin–Tan prove stable canonical recovery for twists of certain property- (T) wreath-like products when the source cocycle is trivial on the abelian base [15, Theorem F]. The exhaustion theorem here treats arbitrary source cocycles in the local-field lattice class.

A geometric line of operator-algebraic rigidity for lattices begins with character and finite-factor representation results. Bekka classified the characters of $\mathrm{SL}_n(\mathbb{Z})$, $n \geq 3$ [3, Theorem 3]; Peterson established a finite-factor representation dichotomy for irreducible higher-rank property- (T) lattices [54, Theorem A]. Boutonnet–Houdayer developed stationary and noncommutative boundary methods that recover character rigidity [11, Theorems A–C]. Their noncommutative factor theorem subsequently classified intermediate von Neumann algebras of specified lattice boundary crossed-product inclusions by parabolic quotients [12, Theorem B and Theorem 3.3]. Related results recover Weyl groups from the distinguished inclusion associated with the action on the quotient by a split-torus centralizer [34, Theorem A and Corollary B], and classify intermediate C^* -algebras for boundary crossed products of irreducible lattices in products of real rank-one groups [35, Corollary B]. These results show how boundary data can retain the ambient geometry. For the finite correspondence considered here, compatible boundary measurements must be constructed in several module copies; their simultaneous law then produces finite-rank subgroup-graph fibres.

The search for a classical equivariant map from an operator-valued coupling is motivated in part by the self-coupling strategy in Furman’s measure-equivalence rigidity theorem [29, Sections 3–4]. The noncommutative classicalization and finite-module reconstruction used here are proved below; they are not consequences of that classical theorem.

1.3. From boundary measurements to subgroup graphs. The proof first passes to an actual lattice and removes the compact projective sectors of the finite commutant. The remaining conjugation action is weakly mixing. In this reduced setting the goal is a fixed orthogonal partition of the correspondence K into finite-dimensional subspaces, permuted by both endpoint groups, with free actions and finitely many orbits. A joint stabilizer of one subspace is then the graph of an actual finite-index subgroup isomorphism, and its action on that subspace is the projective fibre in Equation (1.1).

The geometric construction produces this partition through boundary measurements. Write B for the compact full flag space of the known ambient group G . A projection-valued measure, or PVM, assigns an orthogonal projection to each Borel set, with countable additivity in the strong operator topology; disjoint sets give orthogonal projections and the whole space gives the identity. An angular measurement P_x on B therefore records the possible image of a source flag $x \in B$: $P_x(A)$ is the projection for its image to lie in $A \subset B$. Initially the construction supplies positive operator-valued observations. Proving that they are PVMs, and that observations at different labels commute, are separate parts of the argument.

To compare source copies, use $K_0 \otimes K_1$, with the left Γ -action in each slot and the right Λ -action acting diagonally. The eventual joint angular law is a PVM Θ on the space of nonsingular, essentially invertible maps $B \rightarrow B$. The two left actions transport its source and target labels; diagonal $\rho(\Lambda)$ fixes it. The same construction on $K_0 \otimes K_1 \otimes K_2$ compares slot zero separately with slots one and two. The required conclusion includes

$$[\Theta_{01}(A), \Theta_{02}(C)] = 0 \quad \text{for all Borel sets of boundary maps } A, C.$$

This common-slot commutation has a concrete purpose. Normal slices in the separate branch slots generate an abelian algebra on K_0 . Disintegrating over it gives boundary-map PVMs on the fixed space K_1 . Finite module dimension makes their orbit transversals finite rank; weak mixing makes the resulting projection partition independent of the disintegration parameter. The boundary-map labels themselves need not become deterministic. Section 13 proves this passage and the subsequent graph-stabilizer identification.

Both copies of B in this construction belong to the source lattice. No flag space for the unknown group Λ is assumed. To read the known geometry from Fourier coefficients, omit one left action and put the remaining actions, together with diagonal ρ , in a finite regular column.

Inducing the omitted action to G/Γ gives a finite tracial algebra even when the lattice is reducible. Escape in all ambient factors then creates an isometric boundary link. Escape in a subproduct instead gives a pairing through the fixed algebra's conditional expectation. Ordered Fourier transfer and incidence relations make the positive observations sharp and identify exact residue graphs, while unrelated measurements still retain their original order.

Three local mechanisms control the missing Cartan gaps. At finite places, compact-open fixed algebras and a uniform subgroup logarithm argument recover homogeneous residue maps and upper speed bounds. Real panels inside higher-rank components use optimal finite-dimensional modules: trace inequalities force a nonzero scalar witness in their section spaces, and positivity of its algebraic action excludes excess speed. Isolated quaternionic and Cayley factors use uniform ray estimates and a finite-module spectral gap to obtain positive mass in linear windows on fixed vectors. Pair packing then gives horizontal derivatives, whose closed module core produces an injective positive height with its own-label covariance. The Carnot boundary geometry belongs to the work of Cowling–Dooley–Korányi–Ricci and Pansu [20, 21, 51]; the ray and finite-module estimates used here are proved in Sections 7–8.

Full tuples bring these branches together. Comparing lattice packing with Haar growth calibrates the speeds around the finite type permutation and fixes the exact Jacobians. Ordered imaginary powers then produce a Haar-height isometry. Its common graph core permits locality arguments for the unbounded heights. Locality gives finite jets for the actual spectral control measure, which may be singular; commuting cone multipliers eliminate every positive-order symbol. The principle is related to Peetre's locality theorem [52, 53], but the measured operator-domain argument is proved here. Applying it also to the two shared-slot copies gives the isolated-type commutation needed for the joint law.

For higher-component panels, pairing powers and exact residue graphs supply the corresponding commutation. Simultaneous reflections and panel galleries then glue all local types into the nonsingular, essentially invertible chamber maps described above. After extracting the fixed finite-rank partition and its graph fibres, we restore every compact sector with its exact projective multiplier. Two-sided finite-index induction returns the original correspondence as a closed summand. This explains both the matched cocycle ratio and the summand qualification in Theorem 1.2.

1.4. Organization. Section 2 verifies the elementary arithmetic models. Sections 3–5 construct the finite-module framework and ordered sharp boundary links. The finite-place, isolated real, and higher-component real arguments occupy Sections 6–9; Table 1 records their exact inputs to Section 10. Sections 11–12 prove locality and simultaneous commutation. Section 13 extracts the arithmetic models and completes Theorem 1.2. All geometric and analytic arguments needed for this theorem are given in this article.

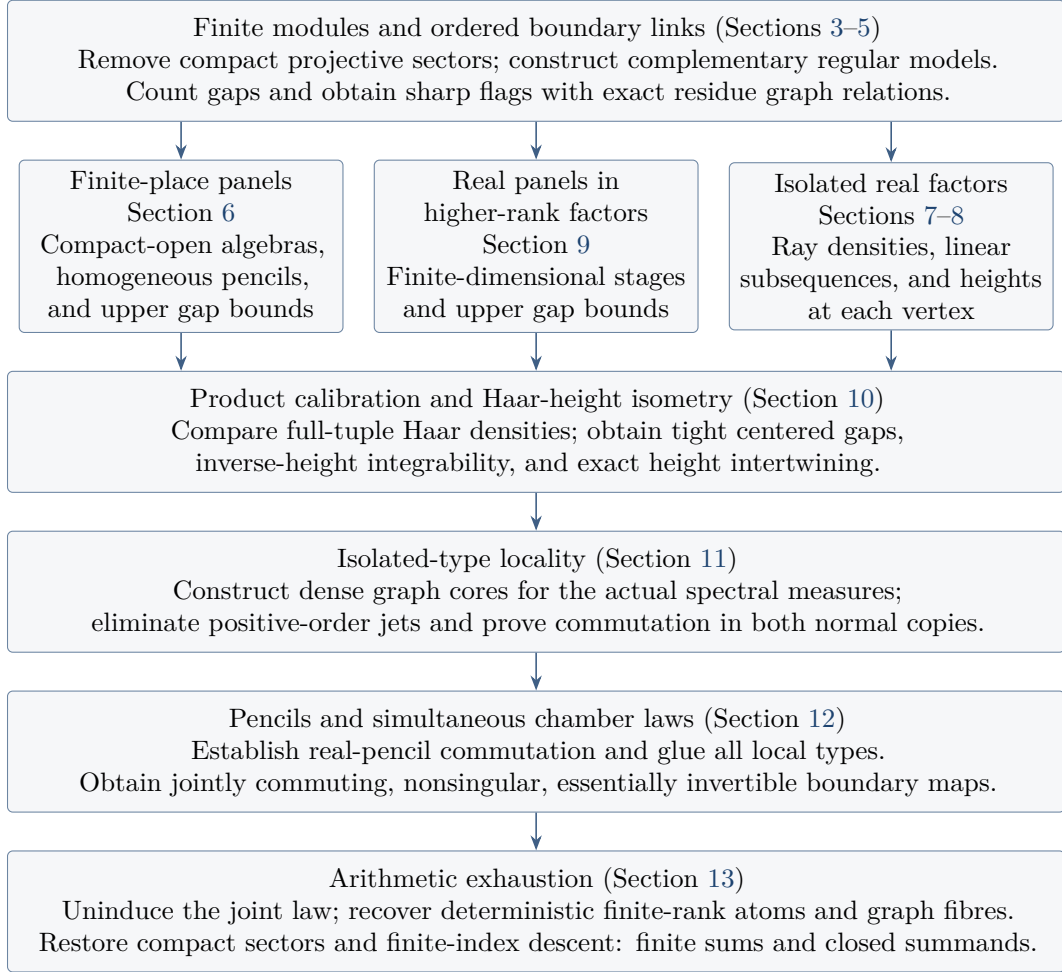


FIGURE 1. The geometric proof of arithmetic exhaustion. Each local branch enters product calibration with its own speed and domain statement; simultaneous commutation is established only afterwards.

2. ELEMENTARY ARITHMETIC CORRESPONDENCES

The models in Theorem 1.2 have an explicit counting-space realization. We give the scalar transport and the normality proof before beginning the geometric argument; neither depends on arithmetic exhaustion. Projective induction is classical [45, Section 4], and these coordinates also belong to the finite-index bimodule calculus of [65, Section 6.2].

Throughout this section Γ, Λ are arbitrary countably infinite ICC groups, $M = L_\mu(\Gamma)$, and $N = L_\omega(\Lambda)$. Put

$$D = \Gamma \times \Lambda, \quad \alpha((g, h), (g', h')) = \mu(g, g') \overline{\omega(h, h')}.$$

For an isomorphism $\delta : A \rightarrow B$ of finite-index subgroups, its graph is $C_\delta = \{(a, \delta(a)) : a \in A\} \leq D$.

2.1. Projective induction. Let $C \leq D$ be a subgroup and let γ be a normalized unitary projective representation on a finite-dimensional space V , with multiplier $\alpha|_{C \times C}$. Write $X = D/C$ for left cosets. Choose representatives $r_x \in D$, with $r_C = e$, and give each $x \in X$ a copy V_x of V . Set $\mathcal{H}(C, \gamma) = \bigoplus_{x \in X} V_x$, with the counting Hilbert norm. For $s \in D$ put

$$(2.1) \quad \begin{aligned} c(s, x) &= r_{sx}^{-1} s r_x, \\ U_s^X(x) &= \frac{\alpha(s, r_x)}{\alpha(r_{sx}, c(s, x))} \gamma(c(s, x)) : V_x \longrightarrow V_{sx}. \end{aligned}$$

Lemma 2.1 (Exact projective induction). *Equation (2.1) defines a unitary projective representation of D with multiplier α . Changing the coset representatives gives a unitarily equivalent bundle by unitary identifications of its individual fibres. If C is the graph of a finite-index subgroup isomorphism $A \rightarrow B$, the bundle has commuting normal M and right N actions, and is bifinite.*

Proof. The correction $c(s, x)$ belongs to C , and

$$c(s, tx)c(t, x) = c(st, x).$$

Using the cocycle identity for α and the projective multiplication of γ in Equation (2.1) gives

$$(2.2) \quad U_s^X(tx)U_t^X(x) = \alpha(s, t)U_{st}^X(x).$$

One can also check all the scalar factors at once in the central extension with multiplication $(z, s)(w, t) = (zw\alpha(s, t), st)$: the representation of its preimage of C is $(z, c) \mapsto z\gamma(c)$, and the equality $sr_x = r_{sx}c(s, x)$ gives exactly the ratio in Equation (2.1). Thus this argument retains, rather than discards, the fixed multiplier.

For explicit independence of representatives, suppose $r'_x = r_x k_x$, where $k_x \in C$. The identification from the new coordinate copy of V_x to the old one is

$$F_x = \alpha(r_x, k_x)^{-1}\gamma(k_x).$$

Indeed identifying a fibre by $U_{r'_x}$ instead of U_{r_x} has precisely this effect. Equation (2.2) then gives $F_{sx}U_s'^X(x) = U_s^X(x)F_x$.

Now assume that C is a graph. Both coordinate groups act freely on X . For example, if $(g, e)r_x C = r_x C$, then $r_x^{-1}(g, e)r_x \in C$ has second coordinate e , so $g = e$. On an orbit of the first coordinate group choose a fibre V_{x_0} . The unitary identification

$$\ell^2(\Gamma) \otimes V_{x_0} \longrightarrow \bigoplus_{g \in \Gamma} V_{(g, e)x_0}, \quad \delta_g \otimes \xi \longmapsto U_{(g, e)}^X \xi$$

intertwines its action with $\lambda_\mu \otimes 1$, since $U_{(a, e)}U_{(g, e)} = \mu(a, g)U_{(ag, e)}$. Hence it extends normally to $L_\mu(\Gamma)$. The same argument for the second coordinate gives copies of $\lambda_{\bar{\omega}}$, which are the standard normal representations of the generators acting by right adjoints $\xi \mapsto \xi v_h^*$. The two actions commute, because the product cocycle has trivial mixed-coordinate factors.

The first coordinate orbits are indexed by Λ/B , and the second coordinate orbits by Γ/A . They are finite in number. Each is a regular module with multiplicity $\dim_{\mathbb{C}} V$, proving bifiniteness as well as the dimension formulas below. $\square$

We next identify this construction with the Connes-product model in Theorem 1.2. This also supplies a direct normality check of the subgroup action used to define that product.

Lemma 2.2 (The graph-bundle unitary). *Let $\delta : A \rightarrow B$ be a finite-index subgroup isomorphism and let $\sigma : A \rightarrow \mathcal{U}(V)$ have multiplier $\beta = \mu|_{A \times A}/\delta^*\omega$. Define $\gamma(a, \delta(a)) = \sigma(a)$. Then*

$$E(A, B, \delta, \sigma) = L^2(M) \bar{\otimes}_{L_{\mu|A}(A)} (V \otimes L^2(N)) \cong \mathcal{H}(C_\delta, \gamma).$$

The unitary sends the vector ξ in the fibre labelled by x to $U_{r_x} j \xi$, where $j \xi = \hat{1} \otimes (\xi \otimes \hat{1})$ in the displayed model.

Proof. On $V \otimes L^2(N)$ the subgroup action is

$$u_a(\xi \otimes \eta) = \sigma(a)\xi \otimes v_{\delta(a)}\eta.$$

Its multiplier is $\beta(a, b)\omega(\delta(a), \delta(b)) = \mu(a, b)$. For normality, choose representatives t for $B \setminus \Lambda$. On the summand spanned by $\xi \otimes v_{\delta(a)}v_t$, the map

$$(2.3) \quad \xi \otimes v_{\delta(a)}v_t \longmapsto \delta_a \otimes \sigma(a)^*\xi \quad \text{in } \ell^2(A) \otimes V$$

is unitary and conjugates the subgroup action to $\lambda_{\mu|A} \otimes 1$. To check the phase, applying u_b before this map produces

$$\omega(\delta(b), \delta(a))\sigma(ba)^*\sigma(b) = \omega(\delta(b), \delta(a))\beta(b, a)\sigma(a)^* = \mu(b, a)\sigma(a)^*.$$

Thus the left subgroup action is normal. Its right N action is the standard action on the second factor and commutes with it.

Choose representatives r for Γ/A . As a right $L_{\mu|A}(A)$ module, $L^2(M)$ has orthonormal free basis u_r ; the assertion follows directly from $E_{L_{\mu|A}(A)}(u_r^* u_{r'}) = \delta_{r,r'} 1$. The relative product therefore identifies with the orthogonal sum

$$(2.4) \quad E(A, B, \delta, \sigma) = \bigoplus_{r \in \Gamma/A} u_r \otimes (V \otimes L^2(N)).$$

At its identity fibre the graph action is exactly

$$U_{(a, \delta(a))} j \xi = u_a \otimes (\xi \otimes v_{\delta(a)}^*) = j \sigma(a) \xi.$$

For $g = ra$, balancing in Equation (2.4) gives the more explicit formula

$$U_{(g, h)} j \xi = \mu(r, a)^{-1} u_r \otimes (\sigma(a) \xi \otimes v_{\delta(a)} v_h^*).$$

The last Fourier coordinate is $\delta(a)h^{-1}$, up to its displayed unit scalar. Thus two translated identity fibres are orthogonal unless their labels in D/C_δ agree: equality of their first cosets and last Fourier coordinates is exactly that coset equality. Conversely these fibres exhaust every summand in Equation (2.4). They are an orthogonal, transitive system with graph action γ at the basepoint. Finally

$$U_s U_{r_x} j = \frac{\alpha(s, r_x)}{\alpha(r_{sx}, c(s, x))} U_{r_{sx}} j \gamma(c(s, x)),$$

which proves that the stated unitary has exactly the transport of Equation (2.1). $\square$

Corollary 2.3 (Elementary normality and dimensions). *The correspondence $E(A, B, \delta, \sigma)$ is normal and bifinite. If $n = \dim_{\mathbb{C}} V_\sigma$, then*

$$\dim_M E(A, B, \delta, \sigma) = n[\Lambda : B], \quad \dim E(A, B, \delta, \sigma)_N = n[\Gamma : A].$$

Proof. Use Lemma 2.2 and the orbit decompositions in Lemma 2.1. Each free coordinate orbit gives n regular copies, of module dimension n with the unnormalized matrix trace convention. The respective orbit counts are $[\Lambda : B]$ and $[\Gamma : A]$. $\square$

Finite direct sums and closed bimodule summands of these models remain normal and bifinite: the two actions restrict normally to a reducing subspace, and module dimension is additive and monotone under orthogonal projections in the commutant [41, Theorem 10.2.1]. The geometric problem is the converse: starting from an arbitrary bifinite correspondence, recover enough subgroup graphs to contain it. We now make the finite-index and compact reductions needed for that construction.

3. FINITE-MODULE REALIZATIONS AND LOCAL-FIELD GEOMETRY

The first analytic goal is to construct a finite regular realization for each choice of an omitted left action on the tensor copies of the correspondence. Each realization includes the other left actions and the diagonal right action. These realizations supply the finite traces used to construct boundary measurements in Section 4. We first remove the compact part of one conjugation action, then form the tensor copies and the induced transports for the omitted source action. Conjugation by these transports gives a genuine action of its ambient group on the finite algebra. The remaining subsections describe the geometric data to be read from the source-lattice Fourier indices.

Fix a nonzero bifinite $L_\mu(\Gamma)$ – $L_\omega(\Lambda)$ correspondence K as in Theorem 1.2. A right action of $L_\omega(\Lambda)$ is written as the projective representation

$$\rho(h) \xi = \xi v_h^*, \quad \rho(h) \rho(k) = \overline{\omega(h, k)} \rho(hk).$$

The left representation is denoted by $\pi(g) \xi = u_g \xi$. Thus π and ρ commute, although their multipliers need not be trivial. Conjugation by either representation is an ordinary group

action. All Hilbert spaces and von Neumann algebras considered below are separable in the senses appropriate to their normal representations.

3.1. Compact reduction and finite-index return. We will prove the geometric reconstruction first for an actual ICC lattice $\Gamma < G$ and under the condition

$$(3.1) \text{ Ad } \rho \text{ on } L^2(\pi(L_\mu(\Gamma))', \text{tr}) \ominus \mathbb{C}1 \quad \text{has no nonzero finite-dimensional subrepresentation.}$$

Here the commutant is a finite factor, equipped with its normalized trace. The following reductions justify this choice. Their final use, after the geometric partition has been constructed, is made in Section 13.

Lemma 3.1 (Twisted regular absorption). *Let α, ν be normalized scalar cocycles on a countable group L , and let v be a ν -projective unitary representation on a Hilbert space V . Then*

$$\lambda_\alpha \otimes v \simeq \lambda_{\alpha\nu} \otimes 1_V.$$

If L is ICC and V is finite-dimensional, tensoring a finite normal $L_\alpha(L)$ -module with v gives a finite normal $L_{\alpha\nu}(L)$ -module. Any additional operators commuting with every v_h are preserved by the absorption identification.

Proof. On $\ell^2(L) \otimes V$ use the unitary

$$T(\delta_g \otimes \xi) = \delta_g \otimes v_g^* \xi.$$

Indeed,

$$\begin{aligned} T(\lambda_\alpha(h) \otimes v_h) T^*(\delta_g \otimes \xi) &= \alpha(h, g) \delta_{hg} \otimes v_{hg}^* v_h v_g \xi \\ &= \alpha(h, g) \nu(h, g) \delta_{hg} \otimes \xi. \end{aligned}$$

A finite normal module over a finite factor embeds in finitely many copies of the regular module [41, Theorems 10.1.1 and 10.2.1]. When L is ICC both twisted group algebras are factors. Applying this unitary to such an embedding proves the second assertion. The formula also proves the last assertion directly. $\square$

Lemma 3.2 (Compact commutant reduction). *Let K be a nonzero bifinite $L_\mu(\Gamma)$ - $L_\omega(\Lambda)$ correspondence, with both groups ICC. There is a finite-index subgroup $\Lambda_0 \leq \Lambda$ such that the restriction of K to the right subgroup factor is a finite orthogonal sum of blocks*

$$K_c = V_c \otimes K'_c, \quad \pi(g)|_{K_c} = 1 \otimes \pi'_c(g), \quad \rho(h)|_{K_c} = v_h^c \otimes \rho'_c(h) \quad (h \in \Lambda_0).$$

The spaces V_c are finite-dimensional. The v_h^c are normalized projective matrix representations, with multipliers ν_c , and

$$(3.2) \quad \rho'_c(h) \rho'_c(k) = \frac{\overline{\omega(h, k)}}{\nu_c(h, k)} \rho'_c(hk).$$

Each K'_c is a finite normal module on both sides and satisfies Equation (3.1) for its reduced representations. There is no assertion that any v^c has finite image.

Proof. Put $M = L_\mu(\Gamma)$, $N = L_\omega(\Lambda)$, and $P = \pi(M)'$. Finite M -module dimension makes P a finite factor. Moreover $N^{\text{op}} \subset P$ has finite index. To see the latter without assuming the endpoint actions are mutual commutants, regard K as the equivalence module for the mutually commuting factors $\pi(M)$ and P . It has finite nonzero P -dimension. Finitely many copies of it therefore contain $L^2(P)$ as a P -module. Their restrictions to N^{op} are finite, by the original right finiteness of K . Consequently $L^2(P)$ has finite N^{op} -dimension, which is the index assertion. We use the standard finite-module and index facts in this argument in the normal, trace-preserving convention; see [40] and [55, Proposition 2.1].

Let D_c be the bounded elements of P whose $\text{Ad } \rho$ -orbits are precompact in $L^2(P)$. The compact vectors of a unitary representation form its closed almost-periodic subspace: they

are the closed span of its finite-dimensional subrepresentations. Bounded compact elements are closed under adjoints and products, since

$$\|xy - x'y'\|_2 \leq \|x\|_\infty \|y - y'\|_2 + \|x - x'\|_2 \|y'\|_\infty$$

on uniformly bounded sets. Their bounded balls are strongly closed, because bounded strong convergence in a finite von Neumann algebra implies L^2 convergence. They therefore form a von Neumann subalgebra. They are dense in the compact Hilbert subspace: the nearest-point projection in $L^2(P)$ onto each closed operator-norm ball is equivariant and nonexpansive, so maps compact orbits to compact orbits, and these projections approximate every L^2 vector.

Twisted conjugation on $L^2(N) \ominus \mathbb{C}1$ has no nonzero finite-dimensional subrepresentation. Indeed, the diagonal of a finite-rank invariant orthogonal projection in the Fourier basis is constant on conjugacy classes: scalar conjugation phases disappear from the diagonal. The diagonal is summable. Every nonidentity conjugacy class is infinite, so all its entries vanish. Only the identity vector remains. The trace-preserving expectation $E_{N^{\text{op}}} : P \rightarrow N^{\text{op}}$ is equivariant and L^2 -contractive. It follows that

$$E_{N^{\text{op}}}(x) = \text{tr}(x)1 \quad (x \in D_c).$$

By the finite-index expectation inequality there is $\varepsilon > 0$ such that $E_{N^{\text{op}}}(x) \geq \varepsilon x$ for $x \geq 0$. Thus a nonzero projection $q \in D_c$ has $\text{tr}(q) \geq \varepsilon$. A finite algebra with such a lower bound has only finitely many orthogonal nonzero projections, and is finite-dimensional.

Restrict Λ to the finite-index subgroup Λ_0 fixing all the central blocks of D_c . A central block z now reduces both endpoint actions. Write

$$zK = V_c \otimes K'_c, \quad zD_c = \mathcal{B}(V_c) \otimes 1.$$

The left action commutes with D_c , hence is $1 \otimes \pi'_c$. Choose normalized matrix implementers v_h^c for $\text{Ad } \rho(h)|_{zD_c}$. Uniqueness of an implementer up to scalar gives $v_h^c v_k^c = \nu_c(h, k) v_{hk}^c$, with ν_c a normalized cocycle. After removing these implementers, $\rho(h)$ belongs to the commutant of $\mathcal{B}(V_c)$ and therefore has the asserted tensor form. Multiplication gives Equation (3.2).

Normality and finiteness of π'_c follow by taking a matrix corner of the finite normal left module. For the right action, first embed the original restricted ρ in a finite multiple of the $\bar{\omega}|_{\Lambda_0}$ -regular representation. This is possible because finite-index restriction preserves finite module dimension. Tensor with $\bar{v}^c$. Lemma 3.1 gives a finite normal module with multiplier $\bar{\nu}_c \bar{\omega}$. In $\bar{V}_c \otimes V_c$ the vector corresponding to the identity matrix is fixed by $\bar{v}_h^c \otimes v_h^c$. Tensoring that line with K'_c exhibits ρ'_c as a summand of the preceding normal finite representation. This proves its normality for exactly the multiplier in Equation (3.2). Equivalently, its right factor has cocycle $\omega|_{\Lambda_0} \nu_c$.

Finally, a compact vector in the reduced commutant embeds as $1_{V_c} \otimes \xi$ on zK and as zero on the other blocks. It is compact for Λ_0 , and hence for Λ : the latter orbit is contained in a finite union of translates of the former compact orbit closure. It therefore belongs to $L^2(D_c)$. Its intersection with the reduced commutant on this block is scalar. This proves Equation (3.1). $\square$

Lemma 3.3 (Finite-index restriction and return). *Let $\Gamma_0 \leq \Gamma$ and $\Lambda_0 \leq \Lambda$ have finite index, and restrict both cocycles literally to these subgroups. Every M – N correspondence is a summand of the two-sided induction of its restriction. The inclusions are given, on the respective sides, by*

$$(3.3) \quad J_L \xi = [\Gamma : \Gamma_0]^{-1/2} \sum_{r\Gamma_0 \in \Gamma/\Gamma_0} u_r \otimes \pi(u_r)^* \xi,$$

$$(3.4) \quad J_R \xi = [\Lambda : \Lambda_0]^{-1/2} \sum_{\Lambda_0 s \in \Lambda_0 \backslash \Lambda} (\xi v_s^*) \otimes v_s.$$

These are bimodular isometries, with all scalar cocycles included. Restriction and induction preserve bifiniteness. An ICC group stays ICC on finite-index restriction.

Proof. The u_r in Equation (3.3) are an orthonormal right basis for $L^2(M)$ over the subgroup factor, because

$$E_{L_{\mu|\Gamma_0}(\Gamma_0)}(u_r^* u_{r'}) = \delta_{r,r'} 1.$$

The analogous left-basis identity for the v_s is $E_{L_{\omega|\Lambda_0}(\Lambda_0)}(v_s v_{s'}^*) = \delta_{s,s'} 1$. The relative tensor-product inner products give the asserted isometries.

For explicit left equivariance, if $gr = r'a$ with $a \in \Gamma_0$, then

$$u_g u_r = \frac{\mu(g, r)}{\mu(r', a)} u_{r'} u_a.$$

Move u_a across the balanced tensor product. The identity $u_r^* u_g = (\mu(g, r)/\mu(r', a)) u_a u_r^*$ shows that the term is exactly the r' -term of $J_L(\pi(u_g)\xi)$. The right calculation is the same calculation in the opposite algebra. Explicitly, when $sh = at$ with $a \in \Lambda_0$, one has

$$v_s v_h = \frac{\omega(s, h)}{\omega(a, t)} v_a v_t, \quad v_h v_t^* = \frac{\omega(s, h)}{\omega(a, t)} v_s^* v_a.$$

Balancing v_a proves right equivariance of Equation (3.4). The other endpoint action commutes with every operation. Thus the two maps compose to a bimodular isometry into two-sided induction, whose range is a closed summand. The finite basis formulas also show preservation of finite module dimensions.

If an element of Γ_0 has finite conjugacy class in Γ_0 , its conjugacy class in Γ is a finite union of translates of that class. ICC of Γ forces the element to be the identity. This proves the last assertion. $\square$

For a source in $\mathcal{K}$, choose the actual isomorphism of finite-index subgroups in Definition 1.1. Its image is a finite-index subgroup of a reference lattice, hence itself a lattice in the same ambient group. Pull the scalar cocycle through that isomorphism, without modifying it by a finite kernel. The restrictions retain property (T) and ICC, as well as bifiniteness; property (T) permanence is used in its usual finite-covolume and finite-index forms [5, Theorem 1.7.1]; see also the original finite-covolume statement in [24, Theorem 3]. Lemmas 3.2 and 3.3 now reduce the geometric work to the actual-lattice, weakly mixing setting. Restoring V_c after a partition of K'_c tensors every atom with V_c and restores the original exact right multiplier. It changes neither the atom permutations nor their freeness. This observation will be used only after that partition is proved.

3.2. Tensor copies and complementary regular realizations. From now until Section 13, let $\Gamma < G$ be an actual ICC lattice and impose Equation (3.1). For a comparison of two source copies we use two copies of K ; for two comparisons sharing their first copy we use three. The right comparison group acts diagonally. We omit one left source action from the regular model so that its induced transports remain available on G/Γ .

Take $m = 1$ or 2 , and put

$$H = K^{\otimes(m+1)}, \quad R = \Gamma_1 \times \cdots \times \Gamma_m \times \Lambda.$$

The slots of H are numbered $0, \dots, m$; π_j denotes the left representation in slot j . Let W be the representation of R given by the π_j , $j \geq 1$, and diagonal ρ . Its split multiplier is

$$(3.5) \quad \alpha_R((g_1, \dots, g_m, h), (g'_1, \dots, g'_m, h')) = \prod_{j=1}^m \mu(g_j, g'_j) \overline{\omega(h, h')}^{m+1}.$$

Lemma 3.4 (Complementary regularity). *The representation W is normal and finite over the twisted factor for R . The same assertion holds with any one of the slots omitted instead of slot zero. The joint commutant of all $\pi_j(\Gamma)$, $0 \leq j \leq m$, and diagonal $\rho(\Lambda)$ is scalar.*

Proof. Embed the omitted slot in a finite regular multiple for its $\bar{\omega}$ -projective right representation. At that regular index absorb the diagonal right actions of the remaining slots by Lemma 3.1. This use of absorption allows an infinite-dimensional auxiliary tensor space.

The absorbing unitaries are formed from the other slots' $\rho(h)$ and therefore commute with all their $\pi_j(g)$. After absorption, the Λ action is regular with multiplier $\bar{\omega}^{m+1}$, and each remaining π_j can independently be embedded in a finite regular multiple for $L_\mu(\Gamma_j)$. The resulting whole representation is a subrepresentation of a finite regular multiple for exactly Equation (3.5). The construction is symmetric in the omitted slot.

Write $P = \pi(M)'$. The final joint commutant is $(P^{\bar{\otimes}(m+1)})^{\text{Ad } \rho^{\otimes(m+1)}}$. Condition (3.1) makes this fixed algebra scalar. For completeness, an invariant tensor with a factor in $L^2(P) \ominus \mathbb{C}1$ gives a Hilbert–Schmidt intertwiner between that factor's representation and another unitary representation. The positive compact operator obtained by composing with its adjoint has a nonzero finite-dimensional invariant spectral space, contrary to weak mixing. Decompose each tensor factor as $\mathbb{C}1 \oplus (L^2(P) \ominus \mathbb{C}1)$ to apply this argument to every nonconstant summand. $\square$

Choose a column realization

$$H = eL^2(B)^k, \quad e \in M_k(B), \quad d = (\text{Tr}_k \otimes \tau_B)(e) > 0.$$

Here B is the opposite of the α_R -twisted factor of R . Using the adjoints of the opposite canonical generators identifies it with the $\bar{\alpha}_R$ -twisted group factor: write these generators as w_r . Thus W_r is right multiplication by w_r^* , which indeed has multiplier α_R .

Set

$$(3.6) \quad Y = G/\Gamma, \quad \mathcal{H} = L^2(Y; H), \quad \mathcal{N} = L^\infty(Y) \bar{\otimes} eM_k(B)e.$$

Give Y its invariant probability measure. The trace on $\mathcal{N}$ is

$$\text{tr}_{\mathcal{N}}(A) = d^{-1} \int_Y (\text{Tr}_k \otimes \tau_B)(A(y)) dy,$$

whereas the Hilbert norm of a column uses the unnormalized entrywise trace sum. All finite-matrix estimates permit rectangular products. Whenever a family of coefficient algebras is varied, coefficient traces are normalized and matrix sizes are fixed, or the corresponding Hilbert-norm factors are retained explicitly. An operator-norm bound alone is not a bound independent of arbitrary trace rescaling.

Choose a Borel section $d_0 : Y \rightarrow G$ and write

$$c(a, y) = d_0(ay)^{-1}ad_0(y), \quad X_a(y) = \pi_0(c(a, y)), \quad (U_af)(ay) = X_a(y)f(y).$$

These transports compose up to a scalar function of modulus one on the base. More explicitly,

$$X_a(by)X_b(y) = \mu(c(a, by), c(b, y))X_{ab}(y).$$

Conjugation on base-decomposable operators is therefore a genuine action, denoted by $\sigma = \text{Ad } U$.

Proposition 3.5 (Induced finite algebras). *The transports U_a are strongly continuous in a , and σ preserves the trace on $\mathcal{N}$. For $1 \leq j \leq m$, let R'_j be R with Γ_j omitted, put $\beta_j = \text{Ad } W_{\Gamma_j}$, and let $\mathcal{A}_j$ be the commutant of the base multipliers and $W|_{R'_j}$. Then*

$$(\mathcal{A}_j)^{\beta_j} = \mathcal{N}, \quad (\mathcal{A}_j)^\sigma \text{ has a faithful finite } \beta_j\text{-invariant trace}, \quad \mathcal{N}^G = \mathbb{C}1.$$

Proof. Translation is continuous in measure on measurable maps on the finite-measure quotient. Applied to the Borel section, this shows that $c(a, y)$ varies continuously in measure as a discrete-valued map. One can verify the assertion first on compact sets where the section is continuous, by Lusin approximation, and then remove sets of arbitrarily small measure. The corresponding projective unitaries are bounded, so their multiplication operators, together with the base translations, vary strongly on $\mathcal{H}$. Conjugation of the corner is inner at each fibre and translation preserves base probability. Thus σ preserves the displayed trace. The first fixed-algebra identity is simply the commutant of all of W in the decomposable algebra.

For the second identity, undo induction. If $A(y)$ is a σ -invariant decomposable field, lift it, for $g = d_0(y)\gamma$, as

$$\tilde{A}(g) = \pi_0(\gamma)^* A(y) \pi_0(\gamma).$$

The cocycle covariance makes this field left G -invariant. It is therefore almost everywhere constant, with value T , and right Γ -covariance says that T commutes with $\pi_0(\Gamma)$. It also commutes with $W|_{R'_j}$. Conversely every such T gives a fixed field. Consequently $(\mathcal{A}_j)^\sigma$ is the commutant of the complementary regular representation obtained by omitting slot j . Lemma 3.4 makes this commutant finite; its normalized trace is β_j -invariant, since the remaining slot acts by unitaries in that commutant. Including all the slots and using the last assertion of that Lemma gives $\mathcal{N}^G = \mathbb{C}1$. $\square$

3.3. The source groups and full versus partial mixing. The induced algebra has only scalar whole-group invariants. To turn that fact into limits along Cartan paths, we need the mixing theorem for the actual root-generated groups. We record the group conventions here; the length and matrix coordinates are introduced afterwards. We use the standard parabolic and root-group structure, Cartan–Iwasawa decompositions, central-cover facts, root-generated simplicity, and highest-weight theory [7, 9, 13, 14, 64, 37].

Convention 3.6 (Factors and panel groups). At a complex place the local group is regarded as a real Lie group. The noncompact property-(T) archimedean factors are the higher-rank ones and the groups locally isomorphic to $Sp(n, 1)$, $n \geq 2$, or $F_{4(-20)}$. At a finite place the factors have relative rank at least two. For higher-rank property (T), see [24, Theorem 6]; the real rank-one classification and its Kostant attribution are recalled in [5, Section 3.3 and Theorem 3.5.4]. Relative rank one at a finite place gives an unbounded action on the Bruhat–Tits tree and is incompatible with property (T) [5, Remark 1.6.3].

In a rank-one semisimple Levi position exactly one almost simple factor is isotropic. Its minimal boundary is the full panel residue. The compact and toric normal factors of the Levi act trivially on this boundary. Source transports use its root-generated group. When an algebraic central cover is needed, use the subgroup generated by its lifted unipotent radicals. The root-group isomorphisms below map this subgroup onto the effective root-generated group with finite central kernel. Every application of Howe–Moore takes invariants for this root-generated group or lift. Projectivities use the effective adjoint group; the adjoint point group is allowed for changes of frame. We never replace a full flag by an oriented or topological covering flag.

To check these conventions, take two opposite parabolics. Their refinement spaces identify with the flags of the common Levi. On the transverse open cell, the radical is removed by projection, and Levi transport acts by its inner algebraic automorphisms. The anisotropic normal factors are compact and have no flag coordinate in the relative building; the central torus acts trivially there as well. Simply connected covers have finite central kernel, contain the root-generated groups in their images on points, and have image of at most finite index in the adjoint point group. They have the same Bruhat flag orbits. At real places any remaining ineffective or component extension is compact or finite.

For the cover assertion, a central covering induces an isomorphism on the corresponding relative root groups, so its image contains the subgroup generated by the unipotent radicals [10, Theorem 2.20(iii) and Proposition 2.24(ii)]. Its kernel on local points is finite. For a central kernel C , the rational-point quotient injects into $H^1(k, C)$, which is finite over the present local fields [8, Propositions 1.12 and 1.17, Theorem 6.1]. This proves the finite-index assertion directly; no stronger identification of all rational points with the root-generated group is needed.

Lemma 3.7 (Product mixing and partial escape). *For the action on $L^2(\mathcal{N})$, escape in every simple factor gives weak operator convergence to the projection onto constants. If a path is the identity outside a subproduct G_0 and escapes in every factor of G_0 , its weak limit is*

the projection onto $L^2(\mathcal{N})^{G_0}$, equivalently the conditional expectation onto $\mathcal{N}^{G_0}$. On the orthogonal complement of a simple factor's invariants, its matrix-coefficient decay remains uniform after composition with arbitrary commuting group unitaries.

Proof. Use the Howe–Moore theorem for the noncompact isotropic almost simple root-generated groups and their root-generated lifts from Convention 3.6 [36]; an independent proof with this root-generated group convention is given in [17, Theorem 1.1 and Definition 4.23]. We first justify the type-I input for these actual groups. For an isotropic simple factor $\mathbf{H}$ over k , choose a nonzero parabolic unipotent radical $\mathbf{U}$. The k -span of the $\mathbf{H}(k)$ -conjugates of $\text{Lie}(\mathbf{U})$ is a nonzero invariant Lie ideal, hence all of $\text{Lie}(\mathbf{H})$. Choose a basis $X_1, \dots, X_d$ from these conjugate subspaces. The map

$$(t_1, \dots, t_d) \mapsto \prod_{j=1}^d \exp(t_j X_j)$$

has invertible differential at zero and takes values in $\mathbf{H}(k)^+$. The local-field inverse function theorem makes this subgroup open; at finite places see [31, Proposition 1.11]. The same proof applies to the root-generated subgroup of an algebraic simply connected cover. At an archimedean place, the generating unipotent groups are connected, so the root-generated group is connected. Being open as well, it is the identity component of the ambient real algebraic group; for a complex group, use restriction of scalars to $\mathbb{R}$. The same applies to its algebraic lift. These groups are therefore type I by [25, Theorem 1]. At a finite place, the ambient reductive group of local points is type I [6], and type I passes to its open root-generated subgroup [4, Proposition 6.E.21(1)].

Restrict a separable unitary representation π to one simple factor S . Its type-I decomposition with multiplicity is

$$\mathcal{H} = \int_{\widehat{S}}^{\oplus} (H_{\xi} \otimes M_{\xi}) d\eta(\xi), \quad \pi(s) = \int_{\widehat{S}}^{\oplus} (\xi(s) \otimes 1) d\eta(\xi).$$

In this central decomposition, every unitary commuting with S acts fibrewise as $1 \otimes V_{\xi}$ [26, Propositions 5.4.7 and 8.6.4, Theorem 8.6.6]; see also [28, Theorem 7.32]. This is the decomposition of the restriction to S , so no invariance of a finer decomposition of the other factors is required. For finite tensor sums in a nontrivial fibre, uniformly over the unitaries V_{ξ} ,

$$\left| \left\langle (\xi(s) \otimes V_{\xi}) \sum_j e_j \otimes u_j, \sum_k f_k \otimes v_k \right\rangle \right| \leq \sum_{j,k} |\langle \xi(s) e_j, f_k \rangle| \|u_j\| \|v_k\|.$$

Howe–Moore makes the right side tend to zero along escape. Approximation by finite tensor sums is uniform because all operators have norm one. For arbitrary fixed fibre vectors the bound $\|a_{\xi}\| \|b_{\xi}\|$ is integrable. Dominated convergence therefore proves the stated uniform decay off the S -invariants.

Let P_i project onto the invariants of the i th simple factor. These projections commute with every group action and with each other. Expand the identity as the finite sum of the mutually orthogonal products obtained from $\prod_i (P_i + (1 - P_i))$. In every sector containing $1 - P_i$, the uniform decay just proved kills its coefficients when that factor escapes, even while all other factor translations vary. Thus for all-factor escape the weak limit is $\prod_i P_i$, which is the projection onto constants by Proposition 3.5. For partial escape, expand only over the factors of G_0 ; the other translations are the identity. The limit is the projection onto $L^2(\mathcal{N})^{G_0}$, namely the trace-preserving conditional expectation. The same argument applies to the simple Levi factors used later. $\square$

3.4. Panel groups, Cartan coordinates, and Haar volumes. The following geometric facts will specify the Fourier-index data used by the boundary measurements. At a panel, only one noncompact rank-one Levi factor acts on the residue; on a Cartan path, exact root coordinates will determine which flag coordinates survive.

An archimedean panel inside a higher-rank component cannot have noncompact Lie algebra $\mathfrak{f}_{4(-20)}$: its complexification would have to occur as a component of a proper absolute Dynkin subdiagram, and the Dynkin list has no such occurrence. At a finite place the effective rank-one Lie algebra is simple also over $\mathbb{Q}_p$. Indeed after restriction of scalars its absolutely simple factors form one Galois orbit, so a $\mathbb{Q}_p$ -ideal is either zero or the whole algebra. An adjoint automorphism centralizing an open subgroup centralizes all sufficiently small inner exponentials, and hence this Lie algebra; it is therefore the identity. These assertions concern the ambient local groups, not a discreteness assumption on a lattice projection.

Let Π be the union of the simple *indivisible* relative roots of the factors. A face of type $I \subset \Pi$ has strictly positive simple coordinates exactly in I ; opposition of types is denoted by a star. Choose a special maximal compact in each factor. Cartan and Iwasawa decompositions can first be made on the simply connected covers and then descended. Write $\mu(g)$ for the Cartan vector only when its argument is a local-group element; this use of μ is distinct from the scalar two-cocycle $\mu(g, h)$. For a positive root α , put

$$m_\alpha = \dim_k \mathfrak{g}_\alpha \quad \text{at a finite place,} \quad 2\rho = \sum_{\alpha > 0} m_\alpha \alpha,$$

and use the analogous real restricted-root multiplicities at archimedean places. Double roots are included in this sum. The fundamental coweight rays e_i are dual to the simple indivisible roots.

Convention 3.8 (Exact finite-place translation coordinates). At a finite place use the module absolute value, with $|\varpi| = q^{-1}$. The Cartan vector is the actual dominant translation vector of a Cartan representative in the split centralizer: its pairing with every centralizer character is the exact logarithm of that character's module. We do not replace this by a coarsely equivalent length in identities involving heights. All displacements and offsets along e_i range on a sufficiently divisible mesh in this logarithmic translation lattice. Cocharacter rationality and common multiples of relative coroot translations and central isogenies ensure that these elements and their semisimple projections to every standard Levi have lifts in the corresponding root-generated covering groups. There are only finitely many standard Levis, so one divisible mesh suffices. At real places the corresponding split exponentials lie in the generated identity lift. No argument at a finite place requires every real parameter to be an actual split displacement.

Bounded parts of the centralizer act equicontinuously, with bounded inverse, on the relevant weight spaces. This permits compact frames while keeping the character normalization exact.

Lemma 3.9 (Haar size of Cartan shells). *For a bounded-thickness shell about a dominant vector L , its Haar measure is at most $C \exp(2\rho(L))$. For one fixed sufficiently large thickness, there is also a lower bound $c \exp(2\rho(L))$. The constants are independent of L on the permissible Cartan mesh. The analogous assertions hold for the finite product, with the sum of the factors' 2ρ .*

Proof. At a real place use the Cartan density, a product of the appropriate powers of $\sinh \alpha(L)$. The upper bound follows on every fixed bounded neighborhood. For the lower bound, even when L lies on a wall, choose inside the shell one fixed bounded offset box whose simple coordinates are bounded away from zero. On this box every positive-root factor is bounded below by a fixed multiple of its exponential. The box has fixed positive Euclidean volume.

At a finite place the Cartan double coset has measure $[K : K \cap aKa^{-1}]$ times the fixed compact measure. This index is comparable to $\exp(2\rho(L))$. One can see the uniform comparison in compact root charts: a sufficiently small analytic compact open subgroup has product coordinates in negative-root, centralizer, and positive-root directions. Under conjugation by the dominant translation, its intersection is obtained by shrinking the expanding root lattices. The index is the product of their module indices, with exponent $\sum_{\alpha > 0} m_\alpha \alpha(L)$. Equivariant exponentials on small root lattices and parabolic product uniqueness give this calculation;

changing the compact frame changes the small subgroups by a uniformly bounded index. Other Cartan representatives are within bounded distance of the chosen split mesh. This proves both bounds there. Multiplication of Haar measures gives the product assertion. $\square$

Fixed left or right multiplication changes vector distances by a bounded amount. We use ordinary compact-rotation probabilities on flags, or equivalently the Haar measure classes in their root coordinates. Proper algebraic incidence degeneracies have measure zero. All incidence and Weyl positions are those of the relative spherical building. A prescribed reduced longest gallery from a chamber parametrizes its open opposition cell by successive panel coordinates, with both measure-null directions preserved; this is the ordered unipotent factorization of the Bruhat cell. The same statement holds at finite places.

3.5. Compactified matrices and admissible scalar tests. The measured geometric index will always lie in a copy Γ_j of the source lattice, inside its known ambient local group. We now give a compact matrix description of the flags and residual projectivities that can survive as that index escapes. This description is algebraic at finite places; only its scalar observables will act on our complex Hilbert spaces.

For each relative vertex type choose a highest-line representation as follows. Take the determinant line of the relevant parabolic nilradical in an exterior power and the cyclic submodule it generates. Over a splitting field, choose a compatible positive system. This line is a highest line, with a weight strictly positive on the omitted absolute nodes and zero on the complementary coroots. Its relative weight is a positive multiple of the relevant fundamental weight. The cyclic highest-weight module is absolutely irreducible and its top split space is one-dimensional. Lowering along an omitted absolute simple root produces a weight separated by a positive multiple of just the corresponding relative simple root. Every other weight drops by a nonnegative combination involving that root. Consequently a normalized Cartan matrix has rank-one limit exactly when that simple gap diverges.

At a finite place these are k -rational modules, used only to form compact projective matrix data. Use compact-invariant lattice norms and compact sup-norm frames, recording both the matrix and its inverse up to projective normalization. At an archimedean target, for Hilbert highest-line amplifications use a real proximal module irreducible on complexification, with a compact-invariant inner product. Taking higher tensor weights is allowed. No Hilbert space is tensored with a vector space over a nonarchimedean field.

Lemma 3.10 (Face blocks and residual projectivities). *Let a sequence have diverging simple gaps of types D , and keep its complementary gaps in a bounded window. Its compactified matrix data determine an image D -flag, a reverse D^* -flag, and an invertible residual Levi transport between their transverse refinement spaces. These data are independent of compact frame choices. On a deeper stratum the already determined image vertices persist. On compact transverse windows the residual rank-one panel lengths differ by bounded amounts from the actual omitted Cartan gaps.*

Proof. Pass to convergent compact Cartan frames. In each highest-weight module, precisely the weights with maximal value on the open face survive projective normalization. They are the submodule generated from the highest line by the complementary Levi. Bounded complementary gaps keep the surviving block and its inverse bounded in those frames. Thus the surviving block acts by an invertible Levi transformation. The same analysis of the inverse matrix identifies the reverse face.

Equivalently, write an ordinary flag in the open Bruhat cell transverse to the reverse face. Conjugating its radical coordinates by the Cartan translation contracts the coordinates belonging to the diverging gaps; the complementary Levi coordinates remain. First project to the reverse refinement space, then apply the surviving Levi transformation. This is the intrinsic residual projectivity. A change of compact frame conjugates these coordinates by the corresponding compact Levi projectivity and leaves the intrinsic map unchanged. Passing to a deeper stratum preserves all preceding image vertices. In a face omitting one type in a

higher-rank component, the surviving rank-one split displacement is exactly that omitted coordinate; bounded frames change its length by a uniformly bounded amount. The argument works simultaneously for one omitted type in several components.

In an archimedean module the surviving top block contains the highest lines of all incident flags, and those lines span it by the Levi action. At a finite place this is an algebraic statement about the k -module only; the later operator-algebra argument instead uses scalar clopen ranges. Exhausting buffered transverse patches gives the assertions on the full stratum. $\square$

Absolute growth on a highest line is measured with the compact-equivariant norm. Relative to the exact Cartan exponent, its correction is the test on the input flag direction. On a transverse cell the normalized ratio converges to the corresponding absolute pairing. At a finite place the character valuations of the nonsplit centralizer give the same statement with Convention 3.8; an arbitrary bounded-error replacement of the character exponent is not made.

Convention 3.11 (Smooth and clopen symbols). On real coordinate windows, a smooth symbol has the specified uniformly bounded finite number of derivatives, increased whenever a separated expansion is needed. On finite-place coordinate boxes, its scalar-test counterpart is locally constant at a uniformly fixed clopen level. The underlying coordinate maps themselves remain analytic. Uniformity means that normalized denominators are bounded away from zero on a buffered larger window, so that all the chosen clopen levels pull back to a common level. In the bounded-width case of Equation (5.1), a test of one argument may instead be an arbitrary bounded function of that argument.

Here is the elementary expansion fact implicit in this convention. On a fixed product of buffered real coordinate boxes, extend the symbol smoothly to a product of tori. Fourier coefficients, after more differentiations than the total coordinate dimension, are bounded by a fixed summable sequence. On a finite-place box there are only finitely many cells at the prescribed level. Combining Fourier characters with their cell indicators expresses a joint symbol as an absolutely summable sum of products of unary symbols. For a uniformly bounded family of smooth seminorms the coefficient bound is the stronger uniform bound

$$\sum_n \sup_{\theta} |c_n(\theta)| < \infty,$$

where θ denotes any additional parameter. This is the bound needed for the Gram-sequence estimates later.

On a buffered transverse matrix-product window, projective normalization has a uniformly nonzero denominator. Matrix multiplication and its inverse-coordinate version therefore have uniform smooth bounds at real places and uniform clopen levels at finite places. At a finite place the logarithm of a weight-matrix product is the sum of its two entry logarithms and a bounded locally constant correction of uniform level. Additional logarithmic phases are kept as unary factors on the two entries; only the correction is expanded in the compact coordinates. Growing radial cutoffs are used in the unshifted regimes specified in Section 10. In particular this convention makes no claim that an arbitrarily translated radial cutoff confines a rescaled logarithm to a fixed compact box.

Common panel equalities will be used as exact substitutions by finite clopen partitions in the finite-place slots, followed by regular smooth division in the real slots. Countably many determining window and angular identities are established before multiplying by arbitrary Borel output projections. One need not assume a uniform transverse estimate for arbitrary discontinuous joint masks.

Lemma 3.12 (Interpolation in an affine building). *Fix an endpoint x in one of the affine buildings above. In an apartment containing x, y , multiply the dominant simple coordinates of $y - x$ by independent numbers $t_i \in [0, 1]$. The resulting point is independent of the apartment and depends on y with a uniform Lipschitz constant. The same uniform assertion holds*

for variation of x , after opposition and replacement of the fractions by their complements. Constants for a finite product depend only on its root systems.

Proof. Write $L_i = \alpha_i(y - x) \geq 0$ and let T_t denote the linear map with simple coordinates $t_i L_i$. For every positive root $\alpha = \sum_i n_i \alpha_i$, where $n_i \geq 0$,

$$0 \leq \alpha(T_t(y - x)) = \sum_i n_i t_i L_i \leq \sum_i n_i L_i = \alpha(y - x).$$

Thus the new point belongs to every affine root half-apartment containing both endpoints: it lies in their root enclosure. Apartment intersections are enclosed. The apartment transition maps agree on this enclosure, and the Weyl stabilizer of a wall displacement also fixes its transformed displacement, since its zero simple coordinates stay zero. This proves independence.

On any apartment and Weyl sector, T_t has a uniformly bounded Euclidean operator norm, independent of the fractions. Subdivide a segment of varying endpoints into cells and Weyl sectors relative to x , and choose apartments containing x and those cells. Local finiteness makes this a finite subdivision on a compact segment. Sum the Euclidean bounds and use apartment independence at the interfaces. Reversing the endpoints applies opposition and the complementary fractions, proving the other Lipschitz bound. $\square$

We use successive limits along sequences, realized by free ultrafilters when convenient. A faster time is always sent to infinity with all slower parameters fixed. Contracted-radical estimates are invoked only in that order. Noncompact isotropic Levi factors contain discrete nonabelian free subgroups: choose two rank-one translations with separated attracting and repelling data and sufficiently large powers. The usual ping-pong domains, in the rank-one symmetric space or the locally finite tree, give a discrete Schottky subgroup. The full chamber action is a topologically amenable homogeneous-space action because its minimal parabolic is amenable; the anisotropic part is compact modulo the central torus. Restriction to a closed subgroup preserves this amenability [1, Propositions 2.2 and 2.5, Example 2.7(5)]. These facts are used for the source local group and its lattice, not for an action of the unknown comparison group on a building.

Convention 3.13 (Operators and Fourier coordinates). An invariance under U is always a conjugation invariance of decomposable operators, so base scalar phases do not change it. An “other group” in a commutant includes Λ unless Λ is explicitly the measured group. A target Fourier arrow in the geometric arguments records $w_{(h,\cdot)}$ and the Cartan or flag data of $h \in \Gamma$ in a measured source-lattice copy; it does not record geometric data of the Λ index. A bounded column is an operator-bounded tracial matrix column, essentially uniformly bounded when it depends on y . The closed range of a positive effect means its support space when a range inclusion is asserted.

4. BOUNDARY OBSTRUCTIONS, LINKS, AND STRICT GAP GROWTH

An escaping source translation acts on a finite regular array. We will record the limiting flags of its measured Fourier index and compare the measurements seen from the two ends of the translation. The first task is to construct these measurements and their input forms. The second is to prove that a vertex translation produces exactly one divergent target gap, and that successively completing such a vertex-origin path adds exactly one target gap at each stage.

We retain the regular realizations and induced actions of Section 3. In particular, $H = eL^2(B)^k$, $\mathcal{H} = L^2(Y; H)$, and $d = (\text{Tr} \otimes \tau_B)(e)$. For an integrable corner field write

$$\mathsf{T}(C) = \int_Y (\text{Tr} \otimes \tau_B)(C(y)) dy, \quad \tau_{\mathcal{N}}(C) = d^{-1} \mathsf{T}(C).$$

Thus $\tau_{\mathcal{N}}$ is a probability trace, whereas the Hilbert norm of a column satisfies $\|f\|_2^2 = \mathsf{T}(ff^*)$. All matrix sizes in this section are fixed and finite. An operator-bounded column means an

essentially bounded field of bounded tracial columns. Such columns form a dense subspace of $\mathcal{H}$. The constant corner frame will be denoted by

$$b_r = e\varepsilon_r, \quad 1 \leq r \leq k, \quad \sum_{r=1}^k b_r b_r^* = e.$$

Here and below a constant column can still have entries in B .

4.1. Compact matrix data and the surviving flags. Write Π for the indivisible relative simple roots, over all components. A face of type $I \subset \Pi$ has strictly positive simple coordinates exactly in I ; opposition of types is denoted by a star. The Cartan projection is written $\mu(h)$ in this geometric discussion, and is distinct from a scalar cocycle. Partial flags are transverse when they are in the open relative position of subfaces of opposite chambers. Incidence means containment in one chamber.

Lemma 4.1 (Matrix compactification and residual projectivities). *There is a finite family of projectively normalized matrix coordinates, including coordinates of the inverse, with the following properties.*

- (i) *The set $D \subset \Pi$ of diverging simple gaps is detected by rank one in the fundamental highest-line coordinates. A stratum with gap set D carries an image D -flag, a reverse D^* -flag, and a residual projectivity between their Levi refinement spaces. Its closure has only gap sets containing D .*
- (ii) *On a bounded complementary-gap window, the residual blocks are invertible in compact frames. They send a full flag transverse to the reverse face, by transverse parabolic projection followed by Levi transport, to a full flag incident with the image face.*
- (iii) *Fixed left and right group multiplications preserve gap sets. Right multiplication leaves the image flag unchanged; left multiplication transports it by the usual target action. Reverse data obey the opposite rule, and residual projectivities change by the corresponding compositions.*
- (iv) *In an archimedean highest-line module, each surviving top face space is spanned by the highest lines of flags incident with the observed face. A rank-one Levi pencil has the effective rank-one projectivity group and its omitted simple gap, up to the bounded changes made by compact endpoint frames.*

The matrix coordinates are real smooth coordinates at archimedean places and projective local-field coordinates at finite places.

Proof. Use the highest-line modules, their normalized matrices and their inverses from Section 3. At archimedean places normalize by the Hilbert–Schmidt norm; at finite places use projective ratio charts with a maximal coordinate. Lemma 3.10 identifies the surviving Levi blocks when the gap set is D , proves their invertibility on a bounded complementary-gap window, and describes the intrinsic transverse projection followed by residual transport. In each fundamental highest-line coordinate, rank one detects divergence of the corresponding simple gap. These finitely many coordinates therefore record the gap set, its image and reverse flags, and the residual maps. If an additional gap diverges, the previous image vertices persist by that lemma. This proves the stratum and closure assertions in (i), and the residual-map assertion in (ii).

We verify the fixed-multiplication rule that will be used on Fourier indices. For a fixed matrix a , both a and a^{-1} have finite norm. Thus projectively normalized multiplication by a has a denominator bounded away from zero. Applying this to every chosen module and its inverse shows that fixed multiplication changes each simple Cartan gap by a bounded amount. Left multiplication carries the image top space by a , whereas right multiplication leaves it unchanged. Applying the same calculation to inverse matrices gives the reverse-flag rule. The residual map transforms by the corresponding compositions, since its description by transverse projection and surviving Levi transport is intrinsic. This proves (iii).

The last assertion is the spanning and length statement of Lemma 3.10: the Levi orbit of the highest line spans its surviving top space, and on a panel the only noncompact isotropic factor supplies the omitted simple gap. Compact changes of endpoint frame change that length by a bounded amount. At a finite place the modules remain vector spaces over the local field; no Hilbert-space amplification by them is involved. $\square$

Compact windows in a stratum are exhausted by bounds on its complementary gaps and by compact transverse projective charts. Their tests may be chosen with slack: a slightly larger window retains the same large-gap separation and the same lower bounds for the invertible residual blocks. At real places these tests are smooth functions of matrix coordinates. At finite places they are locally constant on compact boxes at a fixed level. Increasing a highest weight by tensor powers in later polynomial tests does not change the scalar projective multiplier of the original Hilbert representation.

4.2. Fourier transport before compression. The Fourier tests in this section act on the full regular arrays. They are not followed by a corner projection before the next operation. This distinction allows us to use a diagonal spectral calculus without claiming that an angular Fourier cutoff preserves the operator norm of a column.

Choose a path a with a^{-1} split toward a source face F of type I , with its coordinates in I tending to infinity. Conjugate split frames are allowed. A *lexicographic face path* starts with a vertex translation and adds one new simple coordinate at each successively slower scale. Its fastest parameter is taken to its limit first, while all slower parameters are fixed; the next limit is then taken, and so on. A general face path may instead have several coordinates tending to infinity simultaneously. The general gap upper bound below applies to these paths as well; its exact successive count is asserted for the specified vertex-origin chains. Finite-place parameters range on the divisible cocharacter meshes fixed in Section 3.

The measured label is the compact matrix data of h^{-1} , where h is the Fourier index in the chosen copy Γ_j . The other group indices remain coefficient indices. On the full regular arrays, a scalar test of this label is a diagonal Hilbert-space multiplier. These tests are not followed by corner compression before another array operation.

There are two input maps. The one-column map is

$$(V_a f)(y) = X_a(y) f(y).$$

It is an isometry with the integrated tracial Hilbert norm. Its Hilbert ultralimit V is therefore an isometry. On the minimal spectral span generated by $V\mathcal{H}$, the continuous diagonal tests give a projection-valued measure $E^{(1)}$. Its compression

$$\Phi(A) = V^* E^{(1)}(A) V$$

is a positive effect on the original input. When A records only a gap stratum and its image flag, this is the image measurement. It need not be a projection.

The two-column map, initially on operator-bounded columns, is

$$(4.1) \quad J(\bar{b}, f) = \sqrt{d} (b(ay)^* X_a(y) f(y))^{\lim}.$$

Its output has its own diagonal spectral measure, denoted by E . The link identities below determine the norm of this map and the Hilbert inputs to which it extends. Thus the symbol J at this point specifies a map on bounded column pairs, not a tensor-product isometry.

In both constructions, iterated ultraproducts realize the specified successive limits. Restrict to the minimal closed spectral span generated by the input ranges and a countable determining algebra of continuous tests; these spaces are separable. A Borel stratum always denotes a projection of the limiting spectral measure, rather than a discontinuous cutoff applied pointwise before taking the limit.

Lemma 4.2 (Fourier–Gram transport). *On the limiting arrays, a fixed left or right Fourier shift transports continuous compact matrix tests by its limiting geometric map, and hence transports the associated Borel spectral projections. For a stratum and its image flag, a left*

Fourier matrix multiplier can be removed from a quadratic form through its uncut Gram factor. This remains true for bounded base-dependent multipliers and for compact families of fixed source translations reindexed at the end base. All projective defects cancel in these sesquilinear identities.

Proof. For a fixed shift, normalized matrix multiplication and the inverse formula have denominators bounded away from zero by Lemma 4.1. Their action on the compact label space is continuous. A diagonal continuous test therefore intertwines the shift in the ultralimit. Equality of the two representations on continuous functions implies equality of their spectral measures, so the assertion extends to Borel projections of the limiting variables. Base multiplication and the reindexing of the base commute with diagonal Fourier tests.

If a Fourier index h is changed to rh by a left multiplier, the recorded inverse is changed to $h^{-1}r^{-1}$. Thus its gap set and image flag are unchanged in the limit. Let D be a bounded test of those data and T a bounded left Fourier matrix multiplier. Choose uniformly bounded finite Fourier approximants $T_l \rightarrow T$ in trace L^2 . In a form tested on uncut operator-bounded columns, replacing either occurrence of T by T_l before moving D has error bounded by

$$C\|D\|\|T - T_l\|_2,$$

where C depends only on the fixed column bounds and matrix sizes. For each finite Fourier sum, move D through its individual shifts by the already proved limiting identity. The remaining product is the uncut Gram factor $T_l^*T_l$. Replacing this by T^*T costs another quantity tending to zero, since

$$\|T_l^*T_l - T^*T\|_2 \leq (\|T_l\|_\infty + \|T\|_\infty)\|T_l - T\|_2.$$

All these errors are estimated before applying the cutoff to the columns. In particular the argument applies to a partial unitary with $T^*T = e$ and does not assign an operator bound to Df .

For a field reindexed at ay , invariant probability on Y gives $\int \|T(ay) - T_l(ay)\|_2^2 dy = \|T - T_l\|_2^2$. The same estimates therefore apply after path transport. For a compact family of fixed translations, the cocycle formula and continuity of translations in measure make their matrix fields a compact family in trace L^2 . To verify this assertion directly, approximate the Borel section by continuous data on sets of large measure; along a convergent sequence of translations the discrete cocycle labels then agree eventually off sets of small measure. Unitarity bounds the remaining error. A finite net gives uniform finite-Fourier approximation of the compact family. No common finite-Fourier approximation of the escaping paths themselves is used.

Finally, the group-valued section cocycle satisfies its exact composition identity. Applying the projective source representation gives only a scalar function of the base in each composition formula. Pulling a fixed source translation from one end of an array to the other thus multiplies the *whole* array by one scalar of modulus one. It commutes with diagonal Fourier tests and cancels between the two entries of every sesquilinear form. The same observation applies to a quotient-path calculation: after base reindexing, $X_t X_a^*$ is the transporter for ta^{-1} times a common scalar, so its localized Fourier norm is unchanged. Products in these identities are the actual twisted matrix products; their coefficient phases cannot be replaced by absolute values of untwisted convolutions. $\square$

When there is one measured coordinate, the other group indices, finite matrix factors, and base algebra may be left in the coefficient algebra. The multiplier splits across these group factors, so the proof just given applies without an additional commutation assumption on coefficient entries.

4.3. Link forms, scalar masses, and the finite frame. We first compute the input norm and the covariances of the measurements. For a stratum of positive scalar mass m_D , divide its image measurement by m_D to obtain a unital measurement. Scalarity is part of the next proposition, rather than an assumption on the construction.

Proposition 4.3 (Link identities). *The one-column measurements and the forms (4.1) have the following properties.*

- (i) *The image measurement in one measured copy lies in $\mathcal{A}_j$, is β_j -equivariant, and is invariant under the source face parabolic. Its gap-stratum masses belong to $\mathcal{N}$ and are scalar. Conjugating the source split frame transports the measurement by σ .*
- (ii) *If the path escapes in every simple component, the link is an isometry $\overline{\mathcal{H}} \otimes \mathcal{H} \rightarrow \mathcal{K}$. Its output forms are decomposable at the two bases, covariant under the two independent target shifts, invariant under the independently contracted source radicals, and invariant under simultaneous source centralizers. An image-and-stratum effect depends only on the start input and is $1 \otimes \Phi(A)$; the reverse rule is symmetric.*
- (iii) *If the path is active precisely in a subproduct G_0 , put $N_0 = \mathcal{N}^{G_0}$, and let $E_{N_0} : \mathcal{N} \rightarrow N_0$ be the trace-preserving conditional expectation. On bounded columns the link instead has the form*

$$(4.2) \quad \|J(\overline{b}, f)\|^2 = d^2 \tau_{\mathcal{N}}(E_{N_0}(f f^*) E_{N_0}(b b^*)).$$

Its polarized extension on finite sums is a positive semidefinite form. The same Gram covariances hold on this form space; they do not assert a tensor-product isometry. The precise Hilbert-input extensions are given in Lemma 4.4.

- (iv) *For any Borel event A determined by a gap stratum and its image flag, the finite-frame identity is*

$$(4.3) \quad \langle f, \Phi(A)f \rangle = \frac{1}{d} \sum_{r=1}^k \|E(A)J(\overline{b}_r, f)\|^2.$$

It holds on mixed and nonmixed paths. More generally it holds for events in additional limiting coordinates which are unchanged by the ignored left Fourier shifts. When a proof starts with windows that also constrain the reverse flag or residual projectivity, those windows must first be exhausted to such an invariant event.

- (v) *The image measurements are independent of the finite regular embedding. Adding another original tensor slot gives their normal copies, tensored with the identity on that slot. This assertion concerns these compressed measurements and operators determined by their stated one-ended covariance and uniqueness.*

All statements use the specified order of successive limits. A contracted translate must tend to the identity in that order.

Proof. (i) One-ended covariance. Base multipliers and the companion actions commute with the relevant diagonal tests and intertwine the input column map. The compressed image effects therefore lie in $\mathcal{A}_j$. A measured target shift changes the inverse Fourier label by the corresponding left geometric action. Lemma 4.2 consequently gives β_j -equivariance, with the action on functions given by composition with the inverse target element.

Let n be in the radical contracted by the path. Sliding U_n through U_a puts $U_{ana^{-1}}$ at the far end, up to a common base phase. This translation tends strongly to the identity. In the section model its discrete cocycle is the identity off a set of measure tending to zero, after its pure base shift has been undone. On bounded inputs both descriptions give a vanishing Hilbert error, even with a bounded diagonal test inserted. This proves radical invariance of the full limiting data.

For a fixed centralizing source element s , the corresponding identity slides U_s to the far end with the same s . Its base reindexing has no effect on an integrated form, and its left corner multiplier can be removed for image-and-stratum tests by Lemma 4.2. Its uncut Gram factor is the corner identity. Thus the image measurement is centralizer invariant. The radical and this Levi centralizer generate the face parabolic. The same calculation for a conjugated split path removes the end conjugation and leaves precisely source transport of its image marginal.

The mass of a gap stratum is target invariant, hence belongs to $\mathcal{N}$. It is invariant under the entire source face parabolic. Every inactive simple factor is in the centralizer. In an active factor the contracted radical contains an escaping sequence. Apply Howe–Moore to the representation of that factor on $L^2(\mathcal{N})$: a radical-invariant vector orthogonal to the factor-invariant subspace would have a matrix coefficient both constant and tending to zero. It must therefore be factor invariant. The mass is invariant under all simple factors and so is scalar, because $\mathcal{N}^G = \mathbb{C}$. This argument applies to a pure vertex path as well as to a mixed path.

(ii) *The mixed norm and its polarization.* For bounded b, f , tracial cyclicity and the definition of the induced action give

$$\|J(\bar{b}, f)\|^2 = d \lim_a \mathrm{T}(f f^* \sigma_{a^{-1}}(bb^*)).$$

On a mixed path, trace mixing makes the last expression

$$d^2 \tau_{\mathcal{N}}(f f^*) \tau_{\mathcal{N}}(bb^*) = \|f\|_2^2 \|b\|_2^2.$$

Polarization, or the same calculation on finite sums, gives

$$\left\| \sum_i J(\bar{b}_i, f_i) \right\|^2 = d^2 \sum_{i,l} \tau_{\mathcal{N}}(f_i f_i^*) \tau_{\mathcal{N}}(b_i b_l^*).$$

This is exactly the norm on $\overline{\mathcal{H}} \otimes \mathcal{H}$. The factor $\sqrt{d}$ in (4.1) is essential for this normalization.

Multiplication of an input by a start base function is represented on arrays by that function at y ; an end base function is represented by its value at ay . The diagonal tests commute with both. Their limiting forms are therefore decomposable at the two bases. Moving an independent target shift in either input uses the left or right Fourier transport formula and proves the two covariances. A source radical in either input is slid to its contracted translate in the other input, as above. The fixed translated input converges strongly, its column bound is preserved, and the common phase cancels. Simultaneous centralizing translations use the same exact precomposition and finite-Fourier Gram argument.

For an image-and-stratum event, the ignored end Fourier shifts put each end slice of the compressed effect in $\mathcal{N}$. Independent reverse-radical invariance makes this slice scalar by the factorwise Howe–Moore argument, since every factor is active. Thus the compressed effect has the form $1 \otimes C$. The frame computation below identifies $C = \Phi(A)$. Reversing the two inputs proves the reverse statement.

(iii) *The nonmixed form.* For an escape in G_0 , the weak limit of the trace-preserving action on $L^2(\mathcal{N})$ is the projection onto $L^2(N_0)$. Repeating the norm computation gives (4.2). For a finite sum the formula is

$$(4.4) \quad \left\| \sum_i J(\bar{b}_i, f_i) \right\|^2 = d^2 \sum_{i,l} \tau_{\mathcal{N}}(E_{N_0}(f_i f_i^*) E_{N_0}(b_i b_l^*)).$$

It is positive because it is the limit of squared norms. Quotient by its null space and complete when a Hilbert form space is needed. Every covariance proved before the mixed norm calculation remains an equality of these Gram forms. In particular it can be used on the relative-boundedness domains below. The mass calculation in the first part of the proof is separate from this formula and does not replace N_0 by the scalars.

(iv) *Frame recovery.* For a continuous test of the image and stratum, approximate each left multiplier $b_r(ay)^*$ by bounded finite Fourier matrices. The inverse-index convention makes the test invariant under their left Fourier shifts. Lemma 4.2 therefore moves the test past these multipliers. Summing their uncut Gram factors gives $\sum_r b_r(ay) b_r(ay)^* = e$. The two-column normalization contributes exactly d , proving (4.3). Equality of the limiting spectral measures extends it to invariant Borel events. The proof used neither factorization of the end variable nor mixed trace mixing.

For a window involving residual or reverse data the left shifts may change the window, so that argument does not apply to the individual window. Suppose instead that all end vectors have been annihilated on each member of an exhaustion of the desired image-and-stratum event. Monotone spectral convergence first gives annihilation on that event, and only then may the finite frame be inserted. This proves the stated order of operations. The same proof covers extra limiting coordinates whenever they too are unchanged by the ignored shifts.

(v) *Embedding independence and normal copies.* An intertwiner between two regular embeddings is a bounded Fourier matrix on the relevant finite modules. For the image-and-stratum tests, its uncut Gram factor is the appropriate input corner identity. Lemma 4.2 consequently identifies the two compressed measurements.

For the normal-copy assertion, begin with a regular-multiple embedding for the first two slots and enlarge it by the entire extra slot. At the regular Λ -index h , conjugate the extra slot by the projective inverse of its simultaneous $\rho(h)$ action. Projective absorption transfers its multiplier to the regular Λ multiplier. This conjugation commutes with the extra Γ -action, which can now be embedded in its own finite regular multiple. Both operations commute with scalar Fourier cutoffs on the original measured Γ_j -coordinate. In the original input coordinates, the one-ended path acts only in the zeroth slot. Its compressed one-ended tests are therefore the original tests tensored with identity on the added slot. The embedding-independence just proved identifies this realization with any subsequent one. This proves the assertion for the measurements, and hence for the normal algebras and uniquely determined operators constructed from them. It makes no assertion that all uncompressed spectral arrays are embedding independent. $\square$

The two norm formulas give different input spaces. On an all-factor path, J extends to the ordinary Hilbert tensor product. On a partial path its norm still contains $E_{N_0}(ff^*)$, which can be unbounded for a Hilbert column. The useful replacement is to bound this conditional expectation for one input. Such a bound lets the other input vary through its whole Hilbert space and survives the image effects used later.

Lemma 4.4 (Relative boundedness and Hilbert inputs). *For a path supported in G_0 , call a column f relatively bounded if the positive affiliated operator $E_{N_0}(ff^*)$ is bounded. Then:*

- (i) *for fixed relatively bounded f , the link extends continuously to every Hilbert end column b , and conversely with the inputs reversed;*
- (ii) *the image measurements of paths in factors contained in G_0 commute with N_0 ; their generated algebra, matrix effects, and support projections preserve relative boundedness;*
- (iii) *a closed N_0 -invariant subspace has a dense relatively bounded part, and a relatively bounded column has operator-bounded approximants with the same bound on its conditional expectation;*
- (iv) *the link covariances hold on relatively bounded pairs and when one input is operator-bounded and the other is a Hilbert column. A one-column gap inclusion valid for all inputs also holds for bounded pairs in (4.1).*

Proof. Conditional expectation extends to positive L^1 elements. Put $A = E_{N_0}(ff^*)$. If A is bounded, (4.2) gives

$$(4.5) \quad \|J(\bar{b}, f)\|^2 \leq d\|A\|_\infty\|b\|_2^2.$$

This proves the first continuous extension from bounded b . The construction always takes the link limit before this extension; it does not define a finite-time product of two arbitrary Hilbert columns.

Let $c \in N_0$. Along a path in G_0 , c is a fixed source-invariant left multiplier. It can be moved from the input to the end and removed from an image test by the same finite Fourier–Gram argument as a centralizer multiplier. Polarization shows that every such image effect commutes with c . If a bounded operator Q commutes with N_0 , then, for positive

$c \in N_0$,

$$\langle Qf, cQf \rangle \leq \|Q\|^2 \langle f, cf \rangle.$$

The characterization of conditional expectation by these tests implies

$$E_{N_0}((Qf)(Qf)^*) \leq \|Q\|^2 E_{N_0}(ff^*).$$

This applies to the generated algebra and its support projections, and also after any fixed finite matrix amplification.

If a closed subspace is invariant under N_0 , the spectral projections $p_n = 1_{[0,n]}(E_{N_0}(ff^*))$ preserve it. The columns $p_n f$ are relatively bounded and converge to f , because

$$\|(1 - p_n)f\|_2^2 = d\tau_N((1 - p_n)E_{N_0}(ff^*)) \rightarrow 0.$$

For a relatively bounded f , ordinary left spectral truncation $f_n = 1_{[0,n]}(ff^*)f$ gives operator-bounded columns converging in Hilbert norm, with $E_{N_0}(f_n f_n^*) \leq E_{N_0}(ff^*)$.

These approximants converge in the link form against a fixed Hilbert b . Indeed first replace b by a bounded b_l . For that bounded input, (4.2) bounds the error by $d\|b_l\|_\infty^2 \|f_n - f\|_2^2$. The tails $b - b_l$ are uniformly controlled by (4.5), since

$$E_{N_0}((f_n - f)(f_n - f)^*) \leq 4E_{N_0}(ff^*).$$

Take $n \rightarrow \infty$ first and then $l \rightarrow \infty$. This argument, and its version with the two inputs reversed, extends all the bounded-input Gram identities to the stated domains.

Finally, suppose a gap event annihilates every one-column input. For a bounded end column in the two-column link, first approximate its left multiplier by a finite Fourier sum. Each fixed shift preserves the gap event in the limit, so each resulting term vanishes. The approximation error is estimated on the uncut bounded columns by Lemma 4.2. Taking the approximation limit proves the same inclusion for bounded pairs, with no use of a mixed tensor decomposition. $\square$

Lemma 4.5 (Additional invariant limiting variables). *Adjoin to the compact spectral data any bounded limiting tests which are asymptotically unchanged by fixed left and right Fourier shifts. The corresponding limiting effects retain base decomposability and the measured equivariance established above. If a one-column effect is also source face-parabolic invariant, including its inactive simple factors, and its event or test is invariant under the measured target shifts, that effect is scalar. For a mixed two-ended effect, independent target-shift invariance and independent contracted-radical invariance make its mass scalar on both inputs. These assertions apply to invariant Borel events of the limiting variables.*

Proof. For a fixed Fourier shift, the assumed asymptotic identity of bounded tests gives precisely the limiting intertwining identity used in Lemma 4.2. Its finite-Fourier approximation proof is unchanged. Base decomposability and commutation with companions follow before taking limits. An invariant one-column test therefore gives an element of $\mathcal{N}$. The source face-parabolic invariance, followed by the active-factor radical argument and the inactive-factor centralizer argument in Proposition 4.3, puts this element in $\mathcal{N}^G = \mathbb{C}$.

For mixed links, independent target invariance puts the respective slices in $\overline{\mathcal{N}}$ and $\mathcal{N}$. Independent contracted-radical invariance makes each slice scalar; polarization and normal slice separation give the scalar joint effect. This reasoning works for a submeasurement with additional spectral slots when its indicated event has these invariances.

For example, fixed left and right multiplications change Cartan coordinates by bounded amounts. If a radial variable is divided by a scale tending to infinity, bounded continuous tests in its chosen compactification are asymptotically unchanged whenever this bounded displacement tends uniformly to zero in that compactified coordinate. For the measurements obtained by adjoining such rescaled radial coordinates to the original face data, the needed source face-parabolic invariance also follows from the construction. A contracted radical preserves the entire enlarged data law: after sliding it to the far end, the near-identity transporter changes the uncut bounded input by a Hilbert-small error, uniformly against all

bounded diagonal tests. A fixed centralizer element, including an element of an inactive simple factor, can be slid to that same end. Its base reindexing is removed by invariant probability, and its left multiplier is removed by the finite-Fourier Gram argument. The additional radial test causes no extra term in this argument, because each fixed Fourier shift changes its limiting coordinate by zero. Thus image-and-stratum tests together with these radial coordinates are automatically face-parabolic invariant. In particular their target-invariant masses are scalar even on a pure vertex ray. This conclusion does not require that the nonmixed link norm factor through scalars.

The argument supplies the required invariant limiting tests; no derivative estimate for an arbitrary finite-scale discontinuous mask is asserted. Establish the continuous determining identities first. Their spectral measures then transport identically, giving the assertion for invariant Borel events of the limiting variables. $\square$

4.4. Ordering and genuine source actions. There are two separate kinds of spectral operation. A single limiting compact variable has its usual Borel functional calculus. For several possibly noncommuting compressed effects, an ordered integral initially means a product of single-slot continuous tests. Smooth joint functions on compact real charts can be inserted by absolutely summable separated expansions, and fixed-level finite-place functions by finite separated partitions. The coefficient bounds make these operations converge in norm. Support of an ordered product on a closed relation means that it vanishes on open rectangles disjoint from that relation; support alone does not supply a general ideal of discontinuous joint functions.

On a buffered transverse projective box, normalized matrix multiplication has denominators bounded below. Over a finite local field choose ratio charts with a maximal coordinate. Matrix operations on these compact charts are equicontinuous, so a clopen test at a fixed level pulls back to tests at one uniformly bounded level. Units and choices of a projective representative do not alter this conclusion. At real places the corresponding maps have bounded derivatives of each fixed order. The geometric estimates for the variable shifts that occur in transverse products are proved in Section 5; bounded shifts there are counted in a compact subset of the *whole* ambient product, whose intersection with the lattice is finite.

The order of factors also matters in incidence arguments. Homogeneous local sections let one shrink a violation neighborhood so that its nearby flags admit the required incident values in fixed test patches. Off a closed position relation, it is enough first to prove vanishing on such rectangles. For a graph substitution in totally disconnected slots, refine to finite clopen partitions and insert the projection of the shared panel at the two consecutive slots. Their locally constant factors then agree exactly. At real slots, the corresponding substitution uses the regular smooth graph equations established in the transfer argument. This prescription retains the stated product order throughout.

Lemma 4.6 (Cancelling source multipliers). *After establishing the decomposable link forms, one may enlarge their source actions to genuine strongly continuous group representations by an opposite-multiplier regular factor at each used input base. The compressed effects become their pointwise normal ampliations. This enlargement imposes no new mixing or scalar-commutant assertion.*

Proof. Write the projective source multiplier as η . The regular representation with multiplier $\bar{\eta}$, induced with the same section, has the conjugate of the scalar defect of $X_a(y) = \pi_0(c(a, y))$. The product of these two fibre transporters consequently satisfies the exact group composition law. The section continuity-in-measure argument gives strong continuity of the resulting induced representation.

The added coordinate is unmeasured. At one base, its two transported vectors have exactly their original inner product, so a base-decomposable compressed form is tensored pointwise with the identity. For a two-ended form the two base algebras commute with its effect, and the same construction is performed independently at the two inputs. The generator identification

of minimal spectral dilations gives the corresponding enlarged dilation. In particular a face spectral measure already invariant under the projective transport now commutes with a genuine source representation and can be disintegrated with that representation. All mixing and scalarity statements continue to refer to the original finite algebra $\mathcal{N}$; none is transferred to the enlarged commutant. $\square$

4.5. Escape from bounded target coordinates.

Lemma 4.7 (No all-bounded stratum). *On every escaping source face path, the all-bounded gap stratum has zero mass in a single measured lattice copy. This holds for the one-column measurement and for bounded pairs in the link.*

Proof. We first record the finite-module Bessel estimate used here. For a bounded column $b \in eL^2(B)^k$ and a Hilbert column ξ , Parseval in the regular coordinates gives

$$\sum_{h \in R} |\langle bw_h^*, \xi \rangle|^2 = \|b^* \xi\|_2^2 \leq \|b\|_\infty^2 \|\xi\|_2^2.$$

The harmless projective phases do not change the squared absolute values. The same bound holds for a subset of translates, in particular the R'_j -translates. In finite matrix versions one sums this identity over the fixed row and column indices.

For bounded end and start columns, the squared norm of a coefficient at a fixed measured index $g \in \Gamma_j$ in $b(ay)^* X_a(y) f(y)$ is the sum over R'_j of squared frame coefficients of the pair. Equivalently these are inner products of $X_a(y) f(y)$ with the corresponding partial translates of $b(ay)$, with g absorbed into one of the columns. This is just Parseval in the remaining group coordinates, and remains true with their twisted coefficient phases.

Use now the complementary regular realization for $\Gamma_0 \times R'_j$. Project each column to a finite Γ_0 -coordinate window. These projections commute with the partial R'_j -translates, so the Bessel bounds are preserved. Source transport is unitary and commutes with those partial translates, and the fixed measured shift likewise preserves their bounds. Both projected columns converge in integrated Hilbert norm to the original columns as the finite windows increase. The Bessel estimate on the other column bounds the sum of squared coefficient errors by that Hilbert error. Interchanging the two columns gives the estimate for the second error. Thus the finite-window approximation is uniform in the path parameter. Reindexing the end column at ay does not change its averaged Hilbert error.

For fixed finite windows, the translated supports are disjoint unless the source cocycle $c(a, y)$ belongs to a fixed finite subset of Γ_0 . That exceptional event has measure tending to zero. Indeed, for any $\delta > 0$, choose a compact set containing the section values $d_0(y)$ outside a set of measure δ . Invariance of quotient probability gives the same assertion for $d_0(ay)$. On the intersection of the two good sets, if $c(a, y)$ belongs to a fixed finite set L , then

$$a = d_0(ay) c(a, y) d_0(y)^{-1}$$

belongs to one fixed compact subset of G . This is impossible eventually on an escaping path. The exceptional measure is at most 2δ , and δ is arbitrary.

Off that exceptional set, every partial-translate coefficient of the finite-window pair is zero. On it, their summed squares are uniformly bounded by the Bessel estimate and the column bounds. Integration gives convergence to zero. Remove the finite-window approximation afterward. We have proved vanishing of the norm at every fixed measured Γ_j -index.

A bounded Cartan set is compact in the whole ambient product, so contains only finitely many elements of the discrete lattice. Summing the preceding conclusion gives zero link mass on each bounded Cartan window. Increasing those windows removes the all-bounded stratum. This is a count in the whole lattice and requires no discreteness of a projection to a proper subproduct. Finally apply (4.3), with its finite corner frame, to obtain the one-column assertion. $\square$

4.6. An amenable-boundary obstruction. The measured target boundary in the following argument is the full flag space of the copy Γ_j . The arbitrary comparison group Λ is among the commuting companion actions.

Lemma 4.8 (Full-flag obstruction). *Let S be the root-generated group of a noncompact isotropic almost simple component of a semisimple source Levi subgroup, using its root-generated lift from Convention 3.6 when the ambient action requires it. There is no nonzero positive map from the continuous functions on the full target flag space to $\mathcal{A}_j$ which is target-equivariant and fixed by the source action $\sigma|_S$.*

Proof. Write $Z = G/P$ for the full target flag space. Suppose that $\Phi : C(Z) \rightarrow \mathcal{A}_j$ is such a map. Its mass $a = \Phi(1)$ belongs to $\mathcal{N}^S$: target equivariance makes it commute with the measured target action, and membership in $\mathcal{A}_j$ gives the remaining commutations. Choose $\epsilon > 0$ such that $p = 1_{[\epsilon, \infty)}(a) \neq 0$. On the reducing space $p\mathcal{H}$, the formula

$$\Psi(u) = pa^{-1/2}\Phi(u)a^{-1/2}p$$

defines a unital completely positive map with the same covariances. A positive map from the commutative algebra $C(Z)$ is completely positive. The displayed normalization does not require a to commute with $\Phi(u)$.

Choose a discrete free subgroup $F \cong \mathbb{F}_2$ in S . One obtains it from two sufficiently strong rank-one contractions with disjoint attracting and repelling neighborhoods; at a finite place the same construction uses translations in the rank-one tree. Passing through a finite central cover leaves a free subgroup after choosing the generators in that cover. Let $\mathcal{D}$ be the concrete C^* -algebra on $p\mathcal{H}$ generated by the cut base multipliers, the companion actions $W|_{R'_j}$, and U_f , $f \in F$. These operators commute with both the measured target action and every value of Ψ .

We first obtain a completely contractive minimum-tensor-norm estimate. We use the minimal Stinespring construction and its commutant lifting [61] and [2, Theorems 1.1.1 and 1.3.1]. In the minimal construction of Ψ , the algebraic vectors are $u \otimes \xi$, with form

$$\left\langle \sum_i u_i \otimes \xi_i, \sum_l v_l \otimes \eta_l \right\rangle = \sum_{i,l} \langle \xi_i, \Psi(\overline{u_i} v_l) \eta_l \rangle.$$

The lift of $D \in \mathcal{D}$ is $\widehat{D}(u \otimes \xi) = u \otimes D\xi$. Commutation with the matrix of values of Ψ gives $\|\widehat{D}\| \leq \|D\|$, with the concrete input norm on the right. Thus the companion representation is controlled before dilation; no larger norm on a newly presented companion algebra is introduced. If $W_g W_h = \zeta(g, h) W_{gh}$ in the measured copy, the covariant lift

$$T_g(u \otimes \xi) = (g \cdot u) \otimes W_g \xi$$

has the same multiplier ζ .

The action of Γ_j on Z is topologically amenable. By [1, Proposition 2.5], choose continuous fields $\eta_i : Z \rightarrow \ell^2(\Gamma_j)$ with finite coordinate support, whose norms tend to one and whose equivariance errors tend to zero uniformly on Z for each fixed $g \in \Gamma_j$. Compactness of Z makes these norms uniformly positive eventually; normalizing pointwise gives continuous unit fields $\xi_i(z) = \eta_i(z)/\|\eta_i(z)\|$ with the same uniform asymptotic equivariance. If π_Z is the spectral representation of $C(Z)$ in the dilation, set

$$Q_i \eta = \sum_{t \in \Gamma_j} \delta_t \otimes \pi_Z(\xi_i(\cdot)_t) \eta.$$

The maps Q_i are isometries. They intertwine the companion algebra exactly, and $Q_i T_g - (\lambda_g \otimes T_g) Q_i$ tends to zero in operator norm for each fixed g . Projective absorption is explicit: the unitary

$$A(\delta_t \otimes \eta) = \delta_t \otimes T_t^* \eta$$

conjugates $\lambda_g \otimes T_g$ to $\lambda_\zeta(g) \otimes 1$. It leaves the companion representation on the second factor unchanged. Taking limits and restricting to the original input proves, at every finite matrix level,

$$(4.6) \quad \left\| \sum_g W_g D_g \right\| \leq \left\| \sum_g \lambda_\zeta(g) \otimes D_g \right\|_{C_{r,\zeta}^*(\Gamma_j) \otimes_{\min} \mathcal{D}}.$$

This also fixes the multiplier convention: the multiplier in this formula is that of W , and is conjugated if one instead writes the right action with the adjoint generators of B .

Complementary regularity makes H a subrepresentation of a finite regular multiple for $\Gamma_0 \times R'_j$, with split multiplier. On inducing the Γ_0 coordinate, the identification

$$(y, h) \mapsto d_0(y)h$$

turns it into the G coordinate. In these coordinates U_f acts as left translation on G , times a scalar function of modulus one; the projective regular multiplication only contributes that scalar. The companions $W|_{R'_j}$ use a separate twisted regular coordinate. Tensoring with the separate measured regular representation in (4.6), a simple-base finite Fourier matrix field from the full split group algebra of R acts fibrewise with its original matrix norm.

Consequently, for contraction fields C_f , scalar phases θ_f , and a finitely supported probability ν on F , the operator in this spatial model satisfies

$$\left\| \sum_{f \in F} \nu(f) C_f \theta_f L_f \right\| \leq \|\lambda_F(\nu)\|.$$

Indeed, pointwise Hilbert norms are bounded by $\sum_f \nu(f) \|\eta(f^{-1}g)\|$. Disintegrating Haar measure along the left F -orbits in G gives the ordinary positive regular convolution bound on each orbit. There is no commutation assumption between C_f and L_f . For the concrete cut representation this reasoning is applied to the indicated generator polynomials and then restricted to their invariant input subspace; it does not require an extension of every abstract element of the cut companion algebra to the enlarged regular space.

We contradict this estimate on one matrix vector. Regard the matrix $E(y) = p(y)$ as k columns, and put, with $x = f^{-1}y$,

$$T_f(y) = p(x) X_f(x)^* p(y).$$

This is a contraction. Since p is σ_f -invariant, $X_f(x) p(x) X_f(x)^* = p(y)$, and right multiplication after the shift gives

$$(R_{T_f} U_f^{(k)} E)(y) = X_f(x) p(x) p(x) X_f(x)^* p(y) = p(y).$$

Right matrix multiplication has the opposite convention $(R_T)_{\beta\alpha} = R_{T_{\alpha\beta}}$; the transpose of matrix indices implements $M_k(B)^{\text{op}} \cong M_k(B^{\text{op}})$. In particular it preserves the contraction norm required by the spatial estimate.

For the finitely many f in the support of ν , choose contraction fields $T_{f,n}$, simple in the base and finite in Fourier support, with $T_{f,n} \rightarrow T_f$ in integrated tracial L^2 . To see the contractive approximation directly, first approximate in the unit ball by simple-base fields, use Kaplansky density of the reduced group algebra in its group von Neumann algebra, then approximate each reduced-algebra entry in operator norm by a Fourier polynomial and rescale by its at most $1 + o(1)$ norm. On the bounded matrix vector just used,

$$\|(R_{T_{f,n}} - R_{T_f}) U_f^{(k)} E\|_2 \leq \|T_{f,n} - T_f\|_2.$$

Thus $S_n = \sum_f \nu(f) R_{T_{f,n}} U_f^{(k)}$ satisfies $S_n E \rightarrow E \neq 0$. On the other hand, (4.6) and the spatial estimate give $\|S_n\| \leq \|\lambda_F(\nu)\|$. For the uniform walk on two free generators and their inverses this norm is at most $\sqrt{3}/2 < 1$, the classical free-group estimate of [43, Theorem 3]. For the needed upper bound, the weight $3^{-|h|/2}$ on the four-regular tree gives

this bound by the weighted Schur estimate. The two conclusions are incompatible because $0 < \|E\|_2^2 = \mathsf{T}(p) < \infty$. $\square$

4.7. Strict growth under a slower Levi translation. The next argument follows one positive stratum through an outer limit. It uses invariance of its image marginal, but does not assume invariance of its reverse data or residual projectivity.

Lemma 4.9 (A slower Levi translation removes every compact window). *Let an inner source face path have a positive target gap stratum Ω_D . After the inner limit, append a slower commuting split translation whose projection escapes in a noncompact simple component S of the remaining source Levi. Additional split parts commuting with S are allowed. Every fixed compact window in Ω_D has mass tending to zero in the outer limit. The entire positive contribution of Ω_D then passes to strata $\Omega_{D'}$ with $D' \supsetneq D$.*

Proof. Its scalar mass is some $m_D > 0$, by Proposition 4.3. Divide the stratum measurement by m_D , and denote the resulting unital full-data measurement by Ψ . If $D = \Pi$, its image measurement is a nonzero full-flag map fixed by the remaining source component S , contradicting Lemma 4.8. Thus in the situation under consideration $D \neq \Pi$.

The marginal inclusion. Let $\Psi(F) = V_\Omega^* M_F V_\Omega$ be a minimal full-data dilation, and let $x : \Omega_D \rightarrow X_D$ be its image-flag map. Write $V : \mathcal{H} \rightarrow \mathcal{K}_X$ for the minimal image dilation. The generator assignment

$$M_u V f \mapsto M_{u \circ x} V_\Omega f$$

preserves inner products. It extends to an isometric inclusion $i : \mathcal{K}_X \rightarrow \mathcal{K}_\Omega$, with $iV = V_\Omega$. This inclusion intertwines the image spectral measure, base multiplication, all companions, and the canonical target lifts. Disintegrating over the image variable gives $\mathcal{K}_X = \int^\oplus K_x d\vartheta(x)$ and isometric fibre inclusions i_x . The base variables and their commuting decompositions are retained in this notation.

Positive density forms. Let $\mathcal{F}_x$ be the compact full-refinement space of the image face x , with compact-rotation probability η_x . The residual projectivity of a full datum $\omega \in \Omega_D$ transports the compact-rotation probability on its reverse refinement space to a probability on $\mathcal{F}_x$. Write this probability as

$$k_\omega(l) d\eta_x(l), \quad k_\omega(l) > 0, \quad \int_{\mathcal{F}_x} k_\omega(l) d\eta_x(l) = 1.$$

On a compact data window $C \subset \Omega_D$, the density has bounds $0 < c_C \leq k_\omega(l) \leq C_C < \infty$, uniformly in $\omega \in C$ and $l \in \mathcal{F}_x$. Indeed the residual transformations and their inverses range in a compact set on such a window. Their Jacobians against the compact-rotation measures are positive and uniformly bounded. At a finite place the same conclusion uses the analytic Haar densities in finitely many compact-open charts and bounded absolute Jacobians. No real differential operator is used there.

For fixed (x, l) , define bounded increasing operators

$$D_{n,x}(l) = i_x^* M_{\min(k_\omega(l), n)} i_x.$$

Their limiting quadratic form is

$$q_{x,l}(v) = \|M_{\sqrt{k_\omega(l)}} i_x v\|^2.$$

This form is closed: multiplication by $\sqrt{k_\omega(l)}$ is closed and its composition with the isometric inclusion i_x is closed. Its domain is dense for almost every (x, l) . To see this, choose countably many input vectors f_r such that the sections $V f_r(x)$ are total in K_x almost everywhere. Indeed, the measurable projection onto their fibrewise orthogonal complement kills every angular multiple of $V\mathcal{H}$; minimality of the image dilation makes that projection zero. Tonelli and the normalization of k give

$$\int_{X_D} \int_{\mathcal{F}_x} q_{x,l}(V f_r(x)) d\eta_x(l) d\vartheta(x) = \|f_r\|^2.$$

Thus their span lies in the form domain almost everywhere, on one common conull set. Let $D_x(l)$ be the associated positive self-adjoint operator. The bounded measurable approximating forms above, equivalently their resolvents, give its measurable dependence. For every $v \in K_x$, integration of the nonnegative form yields

$$\int_{\mathcal{F}_x} q_{x,l}(v) d\eta_x(l) = \|v\|^2.$$

Consequently

$$(4.7) \quad (Tf)(x, l) = D_x(l)^{1/2}(Vf)(x)$$

is a bounded isometry from $\mathcal{H}$ to

$$\mathcal{R} = \int_{X_D}^{\oplus} L^2(\mathcal{F}_x, \eta_x; K_x) d\vartheta(x).$$

Equivariance and source lifts. For a target element γ , set

$$r_{\gamma,x}(l') = \frac{d(\gamma_*\eta_x)}{d\eta_{\gamma x}}(l').$$

Transport of the residual probability gives $k_{\gamma\omega}(\gamma l) = r_{\gamma,x}(\gamma l)k_{\omega}(l)$. Since i intertwines the canonical target lifts, the compressed density forms have the same covariance. If $u_{\gamma,x} : K_x \rightarrow K_{\gamma x}$ is the fibre-unitary part of the image lift, their operator form is

$$D_{\gamma x}(\gamma l) = r_{\gamma,x}(\gamma l) u_{\gamma,x} D_x(l) u_{\gamma,x}^*.$$

The target lift on $\mathcal{R}$ includes the half-density $r_{\gamma,x}^{1/2}$, in addition to any factor already present in the image direct integral. Taking square roots proves that T intertwines the target action. Base and companion operators commute with full-data multiplication and intertwine i , so T intertwines them as well. Thus T intertwines base multiplication and all of W_R .

For a source centralizer element s , invariance of the image marginal gives a canonical lift on its minimal dilation, defined on generators by

$$U_s^X(M_u V f) = M_u V U_s f.$$

It fixes the image coordinate and is strongly continuous, first on these generators and then by density. Extend it to $\tilde{U}_s$ on $\mathcal{R}$ by its action on the K_x fibre, with the induced base reindexing, and the identity on l . It commutes with multiplication by the full refinement flag. These lifts use only the image marginal; no lift preserving the full-data law is being assumed.

The finite-trace intertwiner space. Let $\mathcal{E}_0$ be the bounded intertwiners from $\mathcal{H}$ to $\mathcal{R}$ for base multiplication and all target actions W_R . For $B, C \in \mathcal{E}_0$, $B^*C \in \mathcal{N}$, so define

$$\langle B, C \rangle_{\mathcal{E}} = \tau_{\mathcal{N}}(B^*C)$$

and take the Hilbert completion $\mathcal{E}$. There are two useful exact finite-trace estimates:

$$(4.8) \quad \|B\|_{\mathcal{E}}^2 = \frac{1}{d} \sum_{r=1}^k \|Bb_r\|^2, \quad \|Bf\|^2 \leq d\|f\|_{\infty}^2 \|B\|_{\mathcal{E}}^2 \quad (f \text{ operator-bounded}).$$

Both follow by testing B^*B against the corner frame, or against the bounded positive matrix ff^* , respectively.

Source conjugation $\mathcal{U}_s B = \tilde{U}_s B U_s^*$ is isometric for this norm, because σ_s preserves $\tau_{\mathcal{N}}$. Its base-valued projective defects cancel since B intertwines the base algebra. Hence it is a genuine unitary representation on $\mathcal{E}$. Strong continuity follows from the frame equality in (4.8), strong continuity of the two source lifts, and boundedness of B , then extends by density to the completion.

Here S is the root-generated Levi group, or its root-generated lift from Convention 3.6; invariant spaces are taken for that group. This representation has no S -invariant vector. If it did, choose a bounded intertwiner B whose orthogonal projection onto the invariant subspace is nonzero. The closed convex hull of its orbit contains that projection. Finite convex

combinations B_n therefore converge to it in $\mathcal{E}$, and $\|B_n\|_{\text{op}} \leq \|B\|_{\text{op}}$. The bounded-column estimate in (4.8) gives strong convergence on bounded columns. Their density and the common operator bound give a bounded limiting operator B_∞ . It still intertwines the base and target actions, represents the nonzero invariant vector, and is S -fixed. The map

$$u \mapsto B_\infty^* M_{u(l)} B_\infty$$

is then a nonzero positive full-flag map into $\mathcal{A}_j$, target-equivariant and S -invariant. This contradicts Lemma 4.8.

Howe–Moore now gives weak decay of translates of T along escape in S , also when multiplied by the additional commuting translations. For precision, disintegrate the representation into type-I irreducibles of S with multiplicity. There is no trivial summand. Commuting unitaries act on the multiplicity spaces. Approximation by finite tensor sums reduces the assertion to finitely many matrix coefficients of the irreducible S -representations, which tend to zero uniformly over the multiplicity unitaries. Dominated convergence finishes the argument.

The positive compact-window comparison. For a compact window $C \subset \Omega_D$, let $L_C = i^* M_{1_C} i$. This is a decomposable positive contraction on $\mathcal{K}_X$, and $V^* L_C V = \Psi(C)$. Its density lower bound gives $D_x(l) \geq c_C L_C(x)$ in form order. Operator monotonicity of the square root, followed by $L_C^{1/2} \geq L_C$, yields

$$(4.9) \quad D_x(l)^{1/2} \geq \sqrt{c_C} L_C(x).$$

One may obtain this inequality directly from the bounded $D_{n,x}(l)$ for $n \geq c_C$ and then pass to their closed-form limit.

Write the extended path as $a_t b_s$, with the inner t -limit first. The exact cocycle formula and the change of base variable $z = b_s y$ show that the inner full-data law is precomposed by U_{b_s} :

$$\Psi_s(F) = U_{b_s}^* \Psi(F) U_{b_s}.$$

Its image marginal is unchanged. The appropriate translated intertwiner is therefore $T^{(s)} = \tilde{U}_{b_s}^* T U_{b_s}$. Because the marginal lift fixes x, l , its fibre expression is the positive operator $(U_{b_s}^X)^* D_x(l)^{1/2} U_{b_s}^X$ applied to Vf , with base reindexing understood. Inequality (4.9) remains valid after this conjugation and bounds the correspondingly translated L_C .

Put $(J_0 f)(x, l) = V f(x)$, the constant-refinement section. For bounded f , the functional $B \mapsto \langle Bf, J_0 f \rangle$ is continuous on $\mathcal{E}$, by (4.8). It is unnecessary that J_0 itself intertwine the target action. Weak decay and the positive inequality give

$$0 \leq \sqrt{c_C} \langle f, \Psi_s(C) f \rangle \leq \langle T^{(s)} f, J_0 f \rangle \longrightarrow 0.$$

The right side is real and nonnegative by its positive fibre expression. Density of bounded columns and $0 \leq \Psi_s(C) \leq 1$ extend the vanishing to every input. This proves disappearance of the actual translated window mass, rather than merely weak convergence of an unrelated auxiliary operator.

The entire stratum contribution. Each precomposed law Ψ_s has mass one and takes values on Ω_D . Take its outer limit on $\overline{\Omega_D}$. Every continuous test compactly supported in Ω_D has zero limiting effect by the window estimate. A countable compact-window exhaustion therefore shows that the outer law gives no mass to Ω_D . Lemma 4.1 places its remaining support in

$$\overline{\Omega_D} \setminus \Omega_D \subseteq \bigcup_{D' \supsetneq D} \Omega_{D'}.$$

Restoring the scalar factor m_D proves the assertion for the entire original positive contribution. There are only finitely many gap sets, so the same outer limit tracks all of their contributions. Further subpieces are dominated by these positive total measurements. At the next stage the covariance and scalar-mass statements apply anew to each positive total stratum of the enlarged source face; invariance of the original full-data law was never needed. $\square$

Proposition 4.10 (Gap count and vertex quotients). *In one measured copy, every positive stratum of a source path toward a face of type I has at most $|I|$ infinite gaps. A vertex path has exactly one infinite gap. In a successive single-gap lex chain starting at a vertex, the stage with k source gaps has exactly k target gaps; in particular each such extension adds exactly one.*

For a quotient of a faster and a slower vertex translation in a common apartment, there is exactly one target gap if the two source vertices are coaxial, and at least two if they are not. Here coaxial means equal or opposite vertices on the same translation axis.

Proof. Put $r = |\Pi|$, and start with a positive target stratum D on a source face of type I . Append the $r - |I|$ missing source coordinates at successively slower scales. At each stage the fundamental split vector for a missing root has a nonzero projection to the remaining-Levi component containing that root; its other split parts commute with that component. Thus Lemma 4.9 applies at each stage. If a full target stratum were reached before the source face were complete, the obstruction in that lemma would already give a contradiction. Otherwise the tracked positive contribution must pass through a strictly increasing chain of gap sets of length $r - |I|$. Since a target has only r simple types,

$$|D| + r - |I| \leq r,$$

which proves $|D| \leq |I|$. If $I = \Pi$, this is just the automatic ceiling $|D| \leq r$, and no remaining-Levi obstruction is invoked.

For a vertex, Lemma 4.7 excludes zero gaps and the upper bound excludes two or more. The same strict-growth argument, applied successively from that vertex, gives at least k gaps at stage k , whereas the upper bound gives at most k . This proves the exact lex counts. More generally, an extension from a stage already attaining its upper bound has an exact increment of one. No exact-increment claim from an arbitrary unsaturated simultaneous multi-gap initial path is needed.

Finally choose the common apartment for the two vertex translations and make the faster vertex a dominant fundamental ray v , of type j . If the slower vertex is coaxial, their quotient is again a vertex translation at the faster scale, possibly with the opposed orientation, so the vertex conclusion applies. If its vector is w and is not coaxial, then it has a nonzero projection to a simple factor of the remaining Levi. Indeed the common kernel of the remaining simple roots $\Pi \setminus \{j\}$ is precisely $\mathbb{R}v$. Thus a zero projection would force w to lie on that line, which is the coaxial case. After choosing a dominant chamber inside the remaining Levi, the slower quotient component therefore escapes in a noncompact simple factor; the other split components commute with it. Strict growth from the one-gap faster limit forces at least one additional target gap. All these statements retain the faster-first order of limits. $\square$

5. TRANSVERSE FOURIER PRODUCTS AND GEOMETRIC TRANSFER

We continue with the induced finite-module realization of Section 4. All target indices in this section are indices of a measured copy of the source lattice Γ . The comparison group Λ is retained in the coefficient algebra. In particular, none of the flag spaces used below is assigned an action of Λ .

Our objective is a sharp flag measurement at each source flag, together with an exact equation for the residual projectivity on a missing panel. The Fourier product estimate first compares vertex paths and identifies their target types. Isolated types then require a separation argument for angular forms; in higher-rank components, incidence range containments propagate sharpness through the building. The resulting ordered gallery identities supply the residue equation used by the local panel arguments.

5.1. A transverse product estimate. We first state the estimate with its trace dependence visible. A finite coefficient algebra has a fixed faithful finite trace. Matrix sizes are fixed during every limit. We may normalize the coefficient trace for the proof; returning to a trace of mass t multiplies an operator-norm-only L^2 bound by $\sqrt{t}$. More generally, the homogeneous estimate

below uses the actual Hilbert norms and does not require a normalization. Rectangular matrices are included by embedding them in one fixed square matrix algebra and using the corresponding corners.

For a scalar function f of the measured Fourier index, let D_f denote diagonal multiplication of the Fourier array. It is an L^2 operator, and we do not assume that it preserves operator-bounded columns. Consider

$$(5.1) \quad (D_g B_0)(D_f A_0), \quad \text{with index product } qp.$$

Here A_0, B_0 are operator bounded, p is an index of A_0 , and q is an index of B_0 . Write their Fourier expansions as $A_0 = \sum_p a_p w_p$ and $B_0 = \sum_q b_q w_q$, with operator-valued coefficients in the unmeasured algebra.

Proposition 5.1 (Transverse products). *Fix face types, a thickness bound, and sufficiently small compact transverse face patches at the basepoint o . Suppose the supports of f and g lie within that thickness of the respective positive cones, with all specified positive gaps sufficiently large, and that the ideal directions of po and $q^{-1}o$ belong to the fixed transverse patches. Then*

$$\mu(qp) = \mu(q) + \mu(p) + O(1),$$

where the error is uniform on the supports. The product in (5.1) has a uniform tracial L^2 bound in either of the following situations:

- (i) *For each coordinate contributed by both cones, one of its two supports is restricted to a moving interval of a fixed bounded width. The separate cutoff factors may be arbitrary uniformly bounded functions.*
- (ii) *One of the face types is a single vertex. The cutoff factors are smooth with uniformly bounded sufficiently high seminorms in fixed compact normalized matrix charts; at finite places this means locally constant with a uniformly bounded level.*

To specify the retained cutoffs in the estimate, write $f = h_A f_1$ and $g = h_B g_1$. In case (i), these four factors may be arbitrary uniformly bounded scalar functions. In case (ii), each factor has the smooth seminorm bounds or fixed clopen level just specified. Then

$$(5.2) \quad \|(D_g B_0)(D_f A_0)\|_2^2 \leq C \|A_0\|_\infty \|B_0\|_\infty \|D_{h_A} A_0\|_2 \|D_{h_B} B_0\|_2.$$

The constant includes the stated bounds on the scalar factors. The same estimate holds for a joint symbol, meaning the bilinear Fourier sum

$$\sum_{q,p} h_B(q) h_A(p) m(q, p) b_q a_p w_q w_p,$$

where $w_q w_p$ is the actual twisted monomial product and m is a smooth joint function on the fixed compact normalized charts, with sufficiently many uniformly bounded derivatives. At finite places use a fixed clopen level. The support and transverse restrictions remain in force. Consequently an additional smooth cutoff factor whose one-column Hilbert norm tends to zero makes the product tend to zero, provided the other bounds and the other cut Hilbert norm remain bounded. In particular, for a fixed finite trace of mass t ,

$$\|(D_g B_0)(D_f A_0)\|_2 \leq C \sqrt{t} \|A_0\|_\infty \|B_0\|_\infty.$$

Proof. We separate the geometric and Fourier parts of the argument.

Common apartments and addition. On compact transverse patches, the two ideal faces admit common apartment or common flat sections whose origins remain in a bounded set. After erasing the bounded thickness, write the two endpoints in such coordinates as

$$po = z + A + O(1), \quad q^{-1}o = z - B + O(1),$$

with A, B in the specified positive faces and z uniformly bounded. The vector distance between these endpoints is $A + B + O(1)$. The vector-distance inequality under bounded endpoint perturbations therefore proves the Cartan addition assertion. This argument is

made in the whole ambient product; it uses no discreteness of a proper projection of the lattice.

We need the following stability of interpolation. From a fixed endpoint, multiply the simple coordinates L_i of a dominant vector by arbitrary fractions $t_i \in [0, 1]$. The resulting interpolation is uniformly Lipschitz in both endpoints. In an archimedean polar chart, the angular differential in a positive-root direction is multiplied by

$$\frac{\sinh(\alpha(t \cdot L))}{\sinh(\alpha(L))} \leq 1,$$

and the radial differential is the uniformly bounded linear map $L \mapsto t \cdot L$. A wall stabilizer fixes the interpolated vector because zero simple coordinates remain zero. Thus the map extends continuously across the walls; approximation of paths by regular paths gives the same Lipschitz bound there. Reversing the endpoints uses opposition and the complementary fractions.

For a finite-place building, the corresponding apartment-independent interpolation and its uniform Lipschitz bounds in both endpoints are Lemma 3.12. The product of these real and finite-place bounds gives the assertion for the whole ambient space.

The common shift. Suppose $qp = q'p'$ and set $k = p(p')^{-1}$. Then k sends the primed endpoint pair to the unprimed pair. Write the primed coordinates as $z' + A', z' - B'$, with the same bounded errors. Cartan addition gives

$$A + B = A' + B' + O(1).$$

Interpolate from the negative endpoint using the fractions

$$t_i = \frac{B'_i}{A'_i + B'_i},$$

with any choice in $[0, 1]$ when the denominator vanishes. These fractions return z' in the primed model. The Lipschitz bound and equivariance of interpolation yield

$$(5.3) \quad ko = z + B' - B + O(1).$$

There is no small-denominator loss: the fractions are bounded and are held fixed when applying the Lipschitz estimate.

In case (i), every overlapping coordinate has bounded variation on at least one side. The relation between $A + B$ and $A' + B'$ then bounds the corresponding coordinate of $B' - B$. Outside the overlap, one side is bounded by the thickness, so those coordinates are bounded as well. Thus k belongs to a fixed compact subset of the whole product G . Its intersection with Γ is finite, uniformly in the locations of the moving width intervals.

In case (ii), an unbounded difference can occur only in the single overlap coordinate i ; if there is no overlap we are already in the bounded-shift case. Put $\delta = B'_i - B_i$. All other coordinates of $B' - B$ are bounded, and $A_i - A'_i = \delta + O(1)$. The collision identity gives

$$p = kp', \quad q' = qk, \quad (q')^{-1} = k^{-1}q^{-1}.$$

Apply Equation (5.3) also to the reversed pair. In the unprimed and primed apartments, respectively, it gives

$$ko = z + \delta e_i + O(1), \quad k^{-1}o = z' - \delta e_i + O(1).$$

The origins and both errors are uniformly bounded. Let U be the patch of type- i vertices incident to the positive endpoint patch, and V the patch of type- i^* vertices incident to the negative endpoint patch. Choose buffered larger patches U^+, V^+ whose every pair is opposite. For sufficiently large positive δ , the forward vertex of k lies in U^+ and its reverse vertex, the forward vertex of k^{-1} , lies in V^+ . These conclusions follow uniformly from the bounded-distance ray formulas. For large negative δ the memberships reverse. The larger patches can be chosen disjoint in the visual boundary, so a fixed sufficiently large k cannot

have both orientations. Bounded δ gives bounded k , already treated above. The negative ray has the opposed type i^* ; the two types need not coincide.

In the positive orientation the shorter arguments are p' and q^{-1} , and the length-increasing substitutions are

$$p' \mapsto kp', \quad q^{-1} \mapsto k^{-1}q^{-1}.$$

The reverse vertex of the first multiplier is in V^+ , transverse to the entire buffered positive input patch. The reverse vertex of the second is in U^+ , transverse to the entire negative input patch. Thus the witness for k gives uniform transversality on both whole input patches. The negative orientation interchanges the primed and unprimed indices and replaces k by k^{-1} .

For clarity, let R be one of the fixed finite-dimensional representations and put $S(g) = R(g)/\|R(g)\|$. On either substitution $r \mapsto ur$, the transverse Cartan-addition estimate applies to every r in the buffered initial patch. Highest-weight norm comparison gives

$$\|S(u)S(r)\| \geq c > 0, \quad \|S(r^{-1})S(u^{-1})\| \geq c > 0,$$

uniformly in the supported u, r . The second bound applies opposition to Cartan addition and uses $(ur)^{-1} = r^{-1}u^{-1}$. The inverse coordinates are the separately normalized matrices of g^{-1} , so no inverse of a possibly singular limiting matrix is being taken. There are finitely many chosen modules, and the buffer preserves a common lower bound on fixed larger coordinate boxes. Normalized multiplication consequently has uniformly bounded smooth derivatives on those boxes. At a finite place, maximal-coordinate ratio charts have uniformly nonzero denominators; the analytic substitutions pull a fixed clopen test back to a common bounded level.

Gram coefficients and twists. Expand the squared norm in (5.1) and group the pairs of indices with $qp = q'p'$ by $k = p(p')^{-1}$. After trace cyclicity, the summand is the pairing of two Gram coefficients, one made from $a_p a_p^*$ and the other from $b_q^* b_{q'}$, with adjoints changed if the opposite orientation is used. Here the lower-case letters are the operator-valued Fourier coefficients. The cocycle factors are exactly the factors in these two actual adjoint products: this follows by associativity of the twisted Fourier monomials, or directly by the scalar cocycle identity. In particular, we do not replace the operator-valued convolution by an absolute-value convolution.

For the finitely many bounded shifts, row and column Cauchy–Schwarz bound each weighted Gram coefficient in operator norm by the appropriate uncut column norm squared. Its trace norm is bounded by the relevant cut Hilbert norm squared. Combining the two bounds symmetrically and then applying tracial Cauchy–Schwarz gives (5.2), with a factor equal to the uniformly bounded number of shifts. Arbitrary separate bounded cutoffs cause no problem in this part.

For the unbounded shifts, use the orientation just described. When $p = kp'$ is the length-increased side, the first Gram weight is

$$f(kp')\overline{f(p')} = \overline{h_A(p')} f(kp')\overline{f_1(p')}.$$

Retain $\overline{h_A(p')}$ and expand the remaining factor in the shorter argument p' . On the other side, retain $h_B(q)$ in

$$g(q)\overline{g(qk)} = h_B(q) g_1(q)\overline{g(qk)};$$

its shorter geometric argument is q^{-1} . Conjugating a Gram coefficient may conjugate these formulas, with no effect on the cut Hilbert norm. In the opposite orientation interchange primes, retaining $h_A(p)$ and $h_B(q')$. The longer-argument occurrence of each retained factor remains inside the expanded expression; its uniform substitution bounds are therefore still required in case (ii).

Extend each remaining expression using an initial-patch cutoff with slack. For the first Gram sequence, choose conjugated basis functions so that its full weight has the form

$$(5.4) \quad w_k(r) = \sum_n c_n(k) \overline{h(r)} u_n(r), \quad \|u_n\|_\infty \leq 1, \quad \sum_n \sup_k |c_n(k)| < \infty.$$

The last, stronger, order of sum and supremum is essential. It follows from Fourier expansion on a slightly larger real coordinate box and uniform bounds on sufficiently many derivatives. At finite places there is instead a finite basis of fixed-level clopen masks. Products of these constructions cover mixed charts. The other Gram sequence has the same form after the indicated adjunction.

For example, when $p = kp'$ is the length-increased side, $h = h_A$ and the first Gram sequence is a sum of coefficients of $A_0(D_{hu_n}A_0)^*$ with scalar weights $c_n(k)$. Minkowski, Plancherel, and the Hilbert ideal inequality give

$$\left\| \left(\sum_n c_n(k) [A_0(D_{hu_n}A_0)^*]_k \right)_k \right\|_{\ell^2(L^2)} \leq \left(\sum_n \sup_k |c_n(k)| \right) \|A_0\|_\infty \|D_h A_0\|_2.$$

On the other side, use the shorter of q^{-1} , $(q')^{-1}$ to obtain the analogous bound for $B_0^*B_0$. If $B' - B$ is large and positive, the initial arguments are p' and q^{-1} ; the negative orientation is symmetric. Cauchy–Schwarz in the shift variable proves (5.2). The estimate also proves the small-factor assertion; a square-root rate for the product norm is sufficient.

Smooth joint symbols are handled by summably separated expansions, retaining the thickness and transverse restrictions with slack in every term. Finally approximate general bounded columns by uniformly bounded Fourier polynomials. Unary products converge in L^1 , and lower semicontinuity gives the asserted L^2 bound. No step requires operator-norm control of an angularly cut column. $\square$

Remark 5.2 (Uniformity used below). In case (i), the constant is insensitive to the locations of the bounded-width intervals and to arbitrary bounded separate masks. In case (ii), all smooth seminorms and all finite-place clopen levels must be held fixed during a use of the estimate. A gap threshold may be fixed first and exhausted only after the relevant inner limit. For the fixed matrix amplifications below, the cut Hilbert norms in (5.2) are uniformly bounded. Thus no estimate independent of an unbounded unnormalized matrix trace is needed.

5.2. Fullness and the permutation of types. For an archimedean target vertex module V , let $n(v)$ be a unit representative of its highest line; the rank-one projector $n(v)n(v)^*$ is independent of the choice of phase. We also allow the trivial one-dimensional module. At a finite target place only this trivial-module version is used in the next lemma.

Lemma 5.3 (Fullness of open patches). *Let Φ_z be a nonzero family of image measurements constructed in Proposition 4.3, transported over its source label space and normalized by its positive scalar stratum mass to be unital. For every nonempty open target patch O , the closed sum, over conull source labels z , of the ranges of*

$$(5.5) \quad \int_O n(v)n(v)^* \otimes \Phi_z(dv)$$

is the full amplified input space. For the trivial module this is the ordinary patch-fullness assertion. The assertion passes to the normal copies of the measurements in the three-slot system.

Proof. First omit O . Source covariance makes the uncut range join a projection over the finite algebra $(\mathcal{A}_1)^\sigma$. Target covariance transports its range by the finite-dimensional linear target matrices; positive changes in the normalization of $n(v)$ do not change that range. Induce the target lattice action, with its finite invariant trace, to the ambient group. The induced range field then intertwines a trace-isometric ambient action with the inverse linear transport on V .

We spell out the finite-module fact used here. Let a positive matrix A have weight spaces $V_1, \dots, V_s$ with strictly increasing eigenvalues, and let P be a projection over a finite algebra. Filter $\text{ran } P$ by its intersections with $V_1 \oplus \dots \oplus V_j$. Take the closures of the leading projections of these intersections onto V_j . Polar decomposition shows that their dimensions are the successive differences of the filtration dimensions, so their orthogonal sum Q has the same

trace as P . If a vector belongs to the j th filtered intersection, its image under A^n , divided by the j th eigenvalue to the power n , converges to its leading projection. Consequently vectors in $\text{ran } Q$ are strongly approximated by vectors in $A^n \text{ran } P$. Since the latter range projections all have the same finite trace, they converge to Q in trace L^2 , equivalently in trace norm for projections.

Apply this pointwise to the induced field and integrate, using the finite trace. A convergent orbit of a trace-isometric action is fixed: the distance between two consecutive orbit points is constant and also tends to zero. The field is therefore invariant under every positive split matrix and its conjugates. These generate the relevant connected simple group. Irreducibility of the complexified real module forces the range to split off the whole matrix factor. The remaining projection descends to the scalar joint fixed algebra, so the uncut join is full. For an amplified real module in a product, use the matrix representation only in that archimedean component and the trivial representation in all others; the same argument applies.

Now restore O . The finite trace dimension of the patch join is constant along target lattice translates, because the transporting linear matrices are invertible. Given any finite scalar test measure on the target flags, choose a regular contracting direction whose attractor is in O and whose repeller is transverse to almost every point of that measure. Such a repeller exists by the nullity of the proper incidence loci. All-factor mixing on the lattice quotient supplies lattice elements whose two compact frames approximate the chosen frames along increasingly long regular translations: two fixed open frame neighborhoods have positive translated intersection in the quotient, and the neighborhoods may subsequently be shrunk. Contraction away from the repeller implies that suitable translates of O exhaust the test measure.

Apply this to a faithful scalarization of a countable total family of label-averaged effects. The complements of the translated patch joins then have trace tending to zero. One way to see the last implication is to take a faithful positive element generated by that total family and truncate it spectrally away from zero. The constant trace dimension of the translated joins must therefore be the full trace. This proves patch fullness. The normal-copy assertion follows from the normal ampliation statement in Proposition 4.3. $\square$

A normalized image branch of a pure vertex path will be called a *candidate*. Its label z is a source vertex, whereas its values are target vertices. Different candidate choices initially include different paths, limits, and positive strata. Each such choice at a fixed source vertex gives a transported family by Proposition 4.3. Face-parabolic invariance makes this transport well defined on the vertex space. Strong continuity of U and local sections of the homogeneous quotient then make its effects strongly continuous in the source label for each fixed target test. This continuity is used only for these constructed families.

Proposition 5.4 (Type comparison). *There is a unique permutation p of the full relative type list Π such that every candidate at source type i takes values in target type $p(i)$. It preserves opposition: $p(i^*) = p(i)^*$. For noncoaxial source vertices of dual types, all ordered products of candidates vanish on transverse target patches. The permutation preserves the collection of isolated types.*

Proof. Choose paths t, a whose inverses tend to the two source vertices in a common apartment, using parabolic invariance to choose the split frames. Take successive limits, with t faster. Pairing their ordered cut effects between bounded columns and using trace cyclicity gives a pairing of

$$(D_A X_t f)(D_B X_a f')^*$$

with the quotient-path matrix $X_t X_a^*$, after the appropriate base reindexing. The latter is the transporter of ta^{-1} up to a common scalar base phase. Its index product is pq^{-1} , and the intermediate ideal directions are $p^{-1}o, q^{-1}o$, in accordance with the inverse Fourier convention.

If the source vertices are coaxial, the quotient path has exactly one target gap by Proposition 4.10. Nondual transverse target types would force two distinct large gaps in the cut product. The bounded-overlap version of Proposition 5.1 bounds its norm uniformly at fixed angular patches and thickness, independently of the large-gap thresholds. The localized norm of the uncut quotient matrix tends to zero on the forbidden gap strata; this is checked on the finite frame columns. Cauchy–Schwarz therefore makes the pairing zero.

If the source vertices are noncoaxial, the quotient has at least two gaps. Dual transverse target vertex cones put the cut product in a fixed thickened single-vertex cone. The smooth version of Proposition 5.1, with fixed smooth cuts, again compares it with a forbidden localized quotient norm and gives zero. Fix the smooth thickness and angular cuts first, then exhaust the large-gap and thickness windows. This proves the stated ordered vanishing without a claim about uncut operator norms.

Fullness supplies a nonzero ordered cut product for generic source pairs: successively apply Lemma 5.3 to the two families. Generic dual source vertices are opposite and hence coaxial, whereas generic nondual source types are noncoaxial. Thus any candidate choices at dual source types have dual target types, and choices at nondual source types have nondual target types. If T_i is the set of possible target types at source type i , every $j \in T_i$ and $k \in T_{i^*}$ satisfy $k = j^*$. Since the sets are nonempty, each T_i is a singleton. Preservation of both duality and nonduality makes the resulting map injective, and the finite common type list makes it a permutation.

At equality of higher-rank self-opposite source vertices, the nonopposition exclusion extends by continuity. Indeed such a pair is a limit of distinct nonopposite pairs, obtained by varying a chamber across an adjacent panel in the connected diagram. If the image type were isolated, nonopposition would mean equality. Diagonal support of two unital effects would then make them sharp and equal. The same argument applies along the adjacent chamber changes, and gallery connectivity would make the candidate family constant. It would lie in the finite source-fixed algebra and be target equivariant. Taking its invariant trace would give an invariant target flag probability, contradicting flag contraction as in Lemma 5.3. Thus a higher-rank self-opposite type cannot map to an isolated type. A non-self-opposite type cannot do so either, by opposition preservation. Since p is a permutation, it permutes the isolated types. $\square$

5.3. Equality at isolated types. All isolated types under consideration are archimedean. Fix one such source type and its matched target boundary Z . The sharpness argument will produce a scalar functional bounded by the angular norms at two independently moved source labels. We first show that such a nonzero functional cannot exist. The proof encodes equivariant positive maps by frame functionals, uses double ergodicity to separate them, and returns from dominated maps to angular forms. Neither candidate family is assumed sharp.

Lemma 5.5 (Disjoint angular forms). *For almost every independent pair of source labels, possibly taken from two different candidate families at the fixed isolated type, there is no common nonzero target-equivariant positive map dominated by the two candidate maps, where domination is complete-positive domination. After induction, there is no common nonzero linear or conjugate-linear functional on $C^\infty(Z) \odot \mathcal{H}$ bounded by both angular form norms*

$$(5.6) \quad b \mapsto \left(\int_Z \langle b(v), \Phi_x(dv)b(v) \rangle \right)^{1/2}.$$

The generic conclusion holds throughout the transverse source orbit by source covariance.

Proof. Undo source induction. Simultaneous covariance of the base and source label yields measurable unital completely positive maps

$$F_x : C(Z) \longrightarrow B(H), \quad F_{\gamma x} = \pi_0(\gamma) F_x \pi_0(\gamma)^*.$$

They are equivariant for R , where the designated measured Gamma factor acts on Z and the other factors act trivially. The assertion follows either from the lattice cocycle formula or from

Haar disintegration on the source homogeneous space. Scalar projective phases disappear in conjugation. The source lattice action on pairs of this boundary is ergodic: the open-pair stabilizer contains a split subgroup escaping in the active factor and all complementary ambient factors, so all-factor mixing on the finite-volume quotient applies.

Let $\mathfrak{C}$ be the separable matrix-amplified full twisted crossed product associated with $C(Z)$ and R . Encode F_x by its minimal covariant Stinespring dilation [61] and by the vector obtained from the embedded columns of e , arranged as the matrix e . The resulting positive functional ϕ_x on $\mathfrak{C}$ has norm d . The group and matrix operators act on these embedded columns by right multiplication. The regular-normal central summand of the group algebra bidual, mapped normally into the crossed-product bidual, therefore supplies canonical right multipliers by matrices over the regular group von Neumann algebra. The vector functional is supported on the corresponding right- e corner. This support assertion concerns the embedded matrix frame; angular multiples generating the dilation need not themselves have regular group support.

Source translation acts on ϕ_x by an inner automorphism of this fixed corner. Indeed, in the vector formula for the translated map, e is replaced by

$$\pi_0(\gamma)^* e = e \pi_0(\gamma)^*.$$

The right side uses the same canonical right corner unitary for every label and every dilation. Equality on crossed-product polynomials extends normally to the bidual. Extending that corner unitary by the identity on its complementary projection, if desired, gives a unitary of the bidual. In particular every central summand is preserved by these inner automorphisms.

The positive subfunctionals of ϕ_x encode exactly the corresponding equivariant subordinate maps. To verify this, note that the matrix vector is cyclic: its group and matrix translates span the embedded input, and applying angular functions generates the minimal dilation. A dominated positive functional is represented by a positive contraction in this cyclic commutant. It commutes with the matrix index and group representation; compressing weighted angular multiplication gives a subordinate equivariant map on the original input. Conversely, a subordinate map gives the dominated form in the same dilation. This is the covariant form of the CP domination correspondence [2, Theorem 1.4.2]. Its coefficients between group and matrix translates of the cyclic vector give uniqueness. This correspondence also intertwines source conjugation by the corner unitaries just described.

Fix the central support z of one functional representation from the second family. The summand $z\mathfrak{C}^{**}$ has separable predual, because that cyclic representation acts on a separable Hilbert space. The components ϕ_x^z form a measurable norm-separable family. Here is the measurable-selection detail. In the Polish norm topology of the positive normal ball of $z\mathfrak{C}^{**}$ of radius d , the condition $0 \leq \theta \leq \phi_x$ is Borel: test it on a countable norm-dense family of positive elements of $\mathfrak{C}$. Put $n(x) = \sup\{\|\theta\| : 0 \leq \theta \leq \phi_x, \theta \text{ is } z\text{-normal}\}$. This function is upper semianalytic. For each rational q , the Jankov–von Neumann selection theorem gives, on $\{n > q\}$, a universally measurable dominated normal functional $\theta_q(x)$ with norm greater than q [42]. Choose measurable rational $q_m(x) < n(x)$ with $n(x) - q_m(x) < 1/m$, treating $n = 0$ separately. The maximal normal component ϕ_x^z dominates each such selection, so positivity gives

$$\|\phi_x^z - \theta_{q_m(x)}(x)\| = n(x) - \|\theta_{q_m(x)}(x)\| < 1/m.$$

Thus ϕ_x^z is norm measurable on the completed boundary probability space. No simultaneous measurable choice of the second family's central supports is required.

The distance between ϕ_x^z and ϕ_y^z is invariant under the diagonal source action, since both are conjugated by the same corner unitary. Double ergodicity makes this distance essentially constant. A measurable map into a separable metric space cannot have a strictly positive constant distance at almost every pair: cover its range by countably many balls of diameter smaller than that constant and take one with positive preimage measure. Thus ϕ_x^z is constant almost everywhere. If it were nonzero, the encoding would give a nonzero source-invariant, target-equivariant positive map into the uninduced finite complementary commutant. Its

invariant finite trace would produce an invariant flag probability, which is impossible by target contraction. Hence $\phi_x^z = 0$ almost everywhere.

For each fixed label of the second family, the two functionals are therefore disjoint at almost every first label. Orthogonality is the Borel condition that the distance equals the sum of the norms; the norm is tested on a countable dense set in the separable algebra. Fubini gives the first assertion for almost every independent pair.

Now reinduce and consider the two seminorms in (5.6). If a nonzero scalar functional were bounded by both, the parallel sum of the two squared forms would be nonzero. Construct it by taking the direct sum of the two form completions and quotienting by the closed span of pairs $(h, -h)$ coming from the common algebraic domain. This subspace is reducing for smooth angular multiplication, base multipliers, and the common target covariances, including their adjoints. Angular multiplication descends, and compression at constant input vectors gives a decomposable equivariant CP map dominated by both original maps. It is nonzero: if every constant input vanished, its angular multiples would vanish, although those vectors generate the quotient. Disintegration gives forbidden subordinate maps on a positive-measure set of base points. This contradiction proves the form assertion. Finally, source covariance transports the generic conclusion over the single transverse orbit. $\square$

Remark 5.6 (A consequence for sharp measurements). If the two measurements in Lemma 5.5 are sharp, the ranges $P_x(U)P_y(V)\mathcal{H}$, for disjoint angular patches U, V , have dense sum. A vector orthogonal to all these ranges would, by regular approximation of the spectral measures, have the same spectral vector for the two PVMs. Pairing with that vector gives a common functional bounded by the two forms, contrary to the lemma.

Proposition 5.7 (Sharpness at isolated types). *At an isolated source vertex, every candidate is sharp, and all candidate choices agree.*

Proof. Take fast and slow paths towards the same source vertex x . Suppose that two angular effects on disjoint matched target patches have a nonzero ordered product. Choose smooth buffered angular cutoffs with bounded complementary thickness. Write A for the fast cutoff and L for its one-ended limiting effect. Let B_1 be the slow cutoff, used on slow-link columns, with its start patch disjoint from that of A . The finite corner-frame identity and exhaustion of fixed windows then give bounded columns f, f', b' for which the following conjugate-linear functional is nonzero. For $b(v) = \sum_i h_i(v)b_i$, with smooth h_i and bounded columns b_i , it is the slow limit of

$$(5.7) \quad \sum_i \left\langle D_{h_i(q^+)B_1} [b'(ay)^* X_a(y) f'(y)], b_i(ay)^* X_a(y) (Lf)(y) \right\rangle.$$

Here the inner product is linear in its second argument and q^+ is the image of the ordinary index q , that is, the reverse angular variable in the matched component. Smooth boundary tests on the large-gap windows are extended through the simple top singular line in the normalized proximal matrix chart.

We show that this functional is bounded by the fast uncut form at x . At fixed slow time, replace Lf by its defining fast weak limit along t . Trace cyclicity pairs $X_t X_a^* b_i(ay)$ with the transverse product of the cut fast column $D_A X_t f$ and the adjoint of the first column of (5.7). Its indices are p, q^{-1} . Transfer h_i to the pairing column by evaluating it at the image of the inverse product qp^{-1} , which approaches q^+ .

The transfer error has a uniform quantitative form. Normalized product formulas are smooth with bounded derivatives on the buffered patches, by Proposition 5.1. Replace the normalized matrix of q by its rank-one top truncation. The two angular tests agree exactly on that truncation. Smooth division, or the first-order Taylor formula in a matrix chart, writes their difference as a finite sum of entries of the rank-one defect times symbols with uniformly bounded seminorms, with the original cutoffs retained. The slow cut column multiplied by this defect tends to zero in Hilbert norm, since the slow path has a unique vertex type and

the complementary thickness is fixed. The small-factor assertion of Proposition 5.1 makes the product error tend to zero.

After transfer, the remaining cut product has a uniform Hilbert norm. The squared norm of the summed pairing column has fast limit equal to the fast uncut angular form on $\sum_i h_i U_a^* b_i$. The split element a fixes x , so parabolic invariance identifies this with the form on b at x . Extensions of the transferred symbols away from the tested windows can be chosen to have exactly the prescribed boundary values on every rank-one limit. Cauchy–Schwarz therefore gives the desired form bound.

Independent contracted invariance at the slow reverse end permits simultaneously replacing b' and every b_i by $U_n b'$ and $U_n b_i$, for a reverse-radical element n . The functional itself is unchanged. The bound just obtained is then the fast form on $\sum_i h_i U_n b_i$, hence the form on b at $n^{-1}x$; the slow translation may be removed because it fixes x . These opposite radical moves cover an open Bruhat cell. All constants may depend on the fixed move, which is harmless. The form bounds extend to arbitrary Hilbert coefficient columns by continuity. If the slow path is nonmixed, Lf is merely Hilbert, but every end column used before the link limit is operator bounded; the one-bounded-input extension in Lemma 4.4 is exactly what is used.

Choose two generic moved labels in that cell. The same nonzero functional is bounded by both of their forms, contradicting Lemma 5.5. Thus candidate products at the same source vertex are supported on equality of target values. For two unital POVMs E, F with this property, regular approximation gives $E(S)F(T) = 0$ for disjoint Borel sets. Unitality then gives $E(S) = E(S)F(S) = F(S)$ and $E(S)^2 = E(S)$. Hence the candidates are equal PVMs, as asserted. $\square$

5.4. Incidence range containments. Let F be a source face, let $z \sim F$ mean that z is incident to F , and fix a positive stratum of a mixed source face path towards F , or of a pure vertex path F . Denote its normalized image-face marginal by Ψ_F . The scalar mass of the unnormalized stratum is denoted $c_F > 0$. For an open patch O of target type $p(i)$, let

$$\mathcal{I}_i(O) = \{x : \text{the target face } x \text{ admits an incident type-}p(i) \text{ vertex in } O\}.$$

For an archimedean highest-line module V , put

$$R_x = \text{the orthogonal projector onto } \text{span}\{n(v) : v \sim x, \text{type}(v) = p(i)\},$$

and define

$$(5.8) \quad L_z = \overline{\text{ran}} \int n(v)n(v)^* \otimes \Phi_z(dv).$$

Proposition 5.8 (Incidence containment). *The following inclusions hold:*

- (i) *The support space of $\Psi_F(\mathcal{I}_i(O))$ is contained in $\overline{\sum_{z \sim F} \text{ran } \Phi_z(O)}$.*
- (ii) *If the path is mixed and $p(i)$ is archimedean, in every even highest power the support of $\int R_x \otimes \Psi_F(dx)$ is contained in $\overline{\sum_{z \sim F} L_z}$.*

In the proof of (i), a stronger individual-pair statement holds. On a compact full-data window that sends a buffered transverse end patch V_1 into O , if $f \in \ker \Phi_z(O)$ and $b \in \overline{\text{ran } \Phi_z(V_1)}$, then $E(\text{window})J(\bar{b}, f) = 0$. On a mixed path f may be any Hilbert column; on a nonmixed path f is required to be relatively bounded in the sense of Lemma 4.4. In both cases b may be any vector in the stated closed end range.

Proof. We prove (i) by annihilating the common kernel

$$\mathcal{C} = \bigcap_{z \sim F} \ker \Phi_z(O).$$

First choose the domain on which this calculation will be made. If the path is mixed, take any $f \in \mathcal{C}$ and operator-bounded columns $f_l \rightarrow f$ in Hilbert norm. If it is a nonmixed vertex path supported in G_0 , put $N_0 = \mathcal{N}^{G_0}$. When i lies in another component, every type- i

label is incident to F , and ordinary fullness gives $\mathcal{C} = 0$, proving the inclusion. Otherwise Lemma 4.4(ii) makes $\mathcal{C}$ invariant under N_0 . Take f in its dense relatively bounded part and, by part (iii) of that lemma, choose operator-bounded $f_l \rightarrow f$ satisfying

$$E_{N_0}(f_l f_l^*) \leq E_{N_0}(f f^*) \in N_0.$$

The approximants need not belong to $\mathcal{C}$. In either case, for each $z \sim F$, $\langle f_l, \Phi_z(O) f_l \rangle \rightarrow 0$.

Fix a compact full-data window that sends a buffered transverse end patch V_1 into O . Such windows cover the image event in (i). Fix also $z \sim F$ and an operator-bounded column b_0 . A vertex path s^{-1} toward z , with a smooth angular cut in V_1 and fixed complementary thickness, realizes its cut end effect by an auxiliary weak limit. Exhausting these cuts and varying b_0 gives a total set in $\overline{\text{ran}} \Phi_z(V_1)$.

Keep the window, angular cuts, thickness, and bounded testing columns b'', f'' fixed. For each fixed main time a and approximation index l , insert the auxiliary s -limit. Trace cyclicity pairs

$$(5.9) \quad X_s(ay) X_a(y) f_l(y)$$

with the product of cut versions of

$$X_s(ay) b_0(ay), \quad b''(ay)^* X_a(y) f''(y).$$

If the two product indices are q, h , their observed inverse product is $h^{-1} q^{-1}$. Transversality and the buffer put its pertinent image in O once the auxiliary gap is sufficiently large. The effective norm of the pairing column in this inner limit is therefore at most

$$\langle f_l, \Phi_z(O) f_l \rangle^{1/2}.$$

The main split translation fixes z , because it acts trivially on the Levi chamber refinements incident to F . The bound is thus independent of the main time.

Proposition 5.1 gives a uniform Hilbert bound for the other, cut product. Its constant depends on the fixed window, thickness, buffered patches, and the bounded columns b_0, b'', f'' . It does not depend on $\|f_l\|_\infty$: f_l appears only in the pairing column. For each fixed auxiliary cut and b_0 , its limiting end vector is fixed. Take the required main-link limits with l fixed, then let $l \rightarrow \infty$ against that vector. On a mixed path the tensor isometry justifies this passage. On a nonmixed path, the common conditional-expectation bound on f_l justifies it by Lemma 4.4. The displayed pairing bound tends to zero.

Now keep f fixed, exhaust the auxiliary thickness and angular cuts, and take the closed span of the resulting end ranges. The same lemma gives continuity in the entire Hilbert end variable on a nonmixed path; the mixed tensor isometry gives it on a mixed path. Consequently

$$E(\text{window}) J(\bar{b}, f) = 0 \quad (b \in \overline{\text{ran}} \Phi_z(V_1)).$$

At the fixed-time auxiliary insertion the approximating input f_l was operator bounded. The subsequent Hilbert-input passages are extensions of the limiting link; no finite-time product of two arbitrary Hilbert columns is used.

For the individual-pair assertion, fix a single z and repeat the calculation with $f \in \ker \Phi_z(O)$. In the nonmixed case take f relatively bounded. Its bounded truncations still have the same expectation bound and need not remain in the kernel; thus this argument applies even when i is outside the active component. Kernel invariance was needed only for density in the range-inclusion argument. This proves the individual-pair assertion for every Hilbert f on a mixed path, and every relatively bounded f on a nonmixed path, with b any vector in the stated closed end range.

Move the end occurrence also by the independent reverse radical. As $z \sim F$ and these moves vary, the resulting labels cover a transverse open cell, which is conull. Fullness and strong continuity on fixed input vectors make these end ranges total. The link is therefore annihilated on the window for every end vector.

We make the final frame step explicit, because the nonmixed case cannot use tensor independence. First exhaust the compact full-data windows to the event A consisting only of the chosen gap stratum and the image condition $\mathcal{I}_i(O)$. Monotone spectral continuity gives $E(A)J(\bar{b}, f) = 0$ for every end vector b . This exhausted event is invariant under the ignored fixed left Fourier shifts; the residual and reverse-window restrictions have been removed. For the finite constant column frame b_r of e , the frame identity of Proposition 4.3 is

$$c_F \langle f, \Psi_F(\mathcal{I}_i(O))f \rangle = \frac{1}{d} \sum_r \|E(A)J(\bar{b}_r, f)\|^2 = 0.$$

Divide by $c_F > 0$, use positivity, and extend from the relatively bounded common-kernel part by density. Orthogonal complements prove (i). An arbitrary Borel output subprojection may be imposed after the buffered ambient tests; no uniform estimate for a discontinuous unexhausted window has been asserted.

For (ii), write $S(h)$ for the Hilbert-normalized real module matrix at h^{-1} . If $(f_i) \perp L_z$ and (b_j) belongs to the amplified line range on a patch transverse to the output reverse window, the same computation gives

$$(5.10) \quad E(\text{window}) \sum_{ij} E(S_{ij}(h)) J(\bar{b}_j, f_i) = 0.$$

Indeed, the auxiliary fast symbol is $S(q)S(q)^*$ with its cutoffs, and

$$S(h)S(q)S(q)^* = \rho(h, q)S(qh)S(q)^*,$$

where the normalization ratio ρ and its needed derivatives are bounded on the transverse supports. Move $S(qh)^*$ to the full pairing column. Its squared symbol tends in the auxiliary limit to the line-projector effect defining L_z , with the same exact stabilizer translation as above. It therefore vanishes on the approximating kernel input in the limit.

The reverse radical may again move the end occurrence. Amplified fullness now makes the resulting end columns span the whole amplified end space. On a compact stratum window the surviving top block of $S(h)$ has range containing every incident highest line, by Lemma 4.1. Bounded local right inverses on its constant-rank image, or finite local spanning columns followed by output matrix multiplication, transfer the annihilation in (5.10) to the projector R_x . Compressing and using the mixed input independence from Proposition 4.3 proves (ii). This amplified argument is used only for real Hilbert modules; its finite-place replacement is given below. $\square$

5.5. Sharp flags and Coxeter compatibility. We next explain why the range containments force the measurements to be classical at each individual source flag. We do not yet claim commutation at arbitrary different source labels.

Lemma 5.9 (Propagation of incidence words). *Consider a finite word of operations, each replacing either one of a pair of vertices by an incident vertex of a prescribed type, and ending with dual types. Call the word successful if some sequence of choices ends at opposite vertices. Success is increasing in the initial Bruhat position. If a word fails at a source pair, the ordered candidate product at that pair is supported on target pairs for which the permuted word fails. The sets of words successful at the least position are unchanged by the type permutation p .*

Proof. The base failure rule is nonopposition of dual types, supplied by Proposition 5.4; at equal self-opposite vertices it follows by its continuity argument. Suppose all incident replacements of one or both source vertices satisfy a previously proved failure rule. If an incident pair at the target violates that rule, choose small neighborhoods of the violating vertices. Proposition 5.8 puts the corresponding output ranges inside sums of the incident source-label ranges. The prior failure rule makes those product ranges orthogonal. Local homogeneous sections ensure that nearby base faces still admit incident values in the chosen neighborhoods. Thus the original ordered product is supported where every such incident target pair obeys the rule. Induction on the word proves failure propagation. This argument

uses ordered products and range sums, not a simultaneous Borel calculus for noncommuting measurements.

For the combinatorics, extend a vertex to a chamber in an apartment. Changing that chamber without changing the vertex permits its maximal parabolic Weyl subgroup. An incidence move permits the corresponding successive maximal-parabolic blocks. The largest attainable Bruhat positions are obtained by successive ascent to block maxima, that is, by Demazure multiplication. They are increasing in the initial position by the Coxeter subword rule, and the apartment realizes the maxima. Ending opposite at dual vertex types is equivalent to reaching the longest chamber position after the endpoint extensions. These are the apartment and Bruhat descriptions of parabolic incidence [7, 64], and prove monotonicity of success.

If a word is successful at the least target position, it is successful at every target position. Were its inverse-permuted word to fail at a least source pair, failure propagation would force the unital ordered source product to have empty support. Its total product is the identity, so that is impossible. This gives inclusion of the least-position success languages in one direction. Iterating p , which has finite order on the type list, gives the cyclic chain of inclusions back to the original language. They are therefore all equalities. $\square$

Lemma 5.10 (An inequality-detecting word). *Every higher-rank component has a vertex type j and an incidence word that fails exactly when two initial type- j vertices are equal.*

Proof. For type A , take an end node. Move the second vertex to an incident vertex of the opposite end type. The word succeeds precisely when a hyperplane containing the second point can avoid the first point, which is equivalent to inequality; the apartment point–hyperplane test gives the same assertion in the building language.

In every other crystallographic diagram choose a self-opposite leaf j . Let k be its neighbor, write $W_{\hat{j}}$ for the subgroup omitting s_j , and consider round trips from type j to type k and back. Omitting k isolates j , so each round trip permits exactly the Weyl block

$$W_{\hat{j}}\{1, s_j\}W_{\hat{j}}.$$

Let r be the minimum number of occurrences of s_j in a reduced word for the longest element of this component. Starting at equality, $r - 1$ round trips cannot end opposite: an expression so obtained has at most $r - 1$ occurrences of s_j , and a reduced subword reaching the longest element would contradict the definition of r .

Starting instead at the least nontrivial double coset $W_{\hat{j}}s_jW_{\hat{j}}$, the first such block is already available. Choose a reduced longest word with exactly r occurrences of s_j and separate it into the intervening $W_{\hat{j}}$ blocks. The remaining $r - 1$ round trips realize it. Every unequal position is above that least nontrivial double coset, so success at all unequal pairs follows from Lemma 5.9. The self-opposite leaf exists in the non- A crystallographic list, including the nonreduced cases through their underlying Weyl diagram. $\square$

Proposition 5.11 (Sharp flag system). *All candidates are sharp and independent of every candidate choice. For each source vertex z , face F , and chamber C , write the resulting PVMs as P_z, P_F, P_C . Incident source vertex measurements commute and are supported on incidence of their target values. Every lexicographic face path, and every mixed face path, towards F observes exactly the vertices of $p(F)$ with these measurements. Incident source labels also commute across the two normal measured copies.*

Proof. Apply Lemma 5.10 in a target higher-rank component, and pull the least-position language back by Lemma 5.9. The inverse-permuted word fails at equality of source type $p^{-1}(j)$. Failure propagation therefore puts every ordered product of two candidates at that same source label on the target diagonal. The unital-POVM argument at the end of Proposition 5.7 makes them equal projections. Together with that proposition, this supplies at least one sharp type in every target component.

Extend any vertex path lexicographically to a full chamber path. Its original marginal is retained. If an already sharp vertex label is incident to that chamber, Proposition 5.8 puts the corresponding full-path marginal ranges inside that sharp PVM's ranges. Unitality and the same assertion for complements make the marginal equal to the sharp PVM. A sharp marginal intertwines the minimal dilation: equality in the CP Schwarz inequality on its projections gives the intertwining directly. Thus the tested candidate commutes with the sharp incident measurement, and their ordered product is supported on target incidence.

Let a remaining target type lie in the component of a sharp type. For two candidates at the same source label, suppose their target values u, v have different incidence families of the sharp type. Choose a sharp-type vertex w incident to u but not to v , with small buffered neighborhoods. The first range containment puts the ranges near u in the sum of the incident source sharp-label ranges near w . Every one of those ranges is orthogonal to the ranges near v , by the full-path incidence rule just proved. This works with two independent full-path choices for the tested candidate. Hence its self-product, and products between choices, are supported where the two sharp-type incidence families agree.

Those families determine the vertex. Their stabilizer contains its maximal parabolic and is proper, because the connected apartment contains nonincident pairs of the relevant types. Bruhat maximality of the maximal parabolic makes that stabilizer precisely the original vertex parabolic. Thus equal families imply $u = v$. Diagonal support again proves sharpness and uniqueness. Repeating in each component makes all candidates sharp.

We now treat an arbitrary mixed face path of source type I . Fix a positive target stratum with normalized image measurement Ψ_F . For $i \in I$, the type- i source vertex z_i incident to F is unique. Proposition 5.8(i) puts the support of $\Psi_F(\mathcal{I}_i(O))$ inside $P_{z_i}(O)$. A target face omitting $p(i)$ admits two distinct incident vertices of that type. Choose disjoint basic open patches O, O' around them. The effect $\Psi_F(\mathcal{I}_i(O) \cap \mathcal{I}_i(O'))$ is supported in both orthogonal projections $P_{z_i}(O), P_{z_i}(O')$, and hence is zero. Countably many such pairs exhaust the omitted-type event. Thus every $p(i)$, $i \in I$, occurs. The general upper bound of Proposition 4.10 forces the stratum type to equal $p(I)$. On this stratum $\mathcal{I}_i(O)$ is precisely the event that its type- $p(i)$ vertex lies in O . The same support containment and unitality identify its marginal with P_{z_i} . These sharp marginals put the joint face measurement in their common multiplicative domain, giving P_F . This argument uses mixing, but no order among the diverging coordinates.

For lexicographic paths, the full-path marginals are now uniquely identified and commute by the sharp-marginal argument. Their joint laws give P_F, P_C and the stated incidence support. We also verify that a lexicographic path towards F has no additional observed vertex type. Extend it to a full path and use persistence and the gap count. An additional sharp marginal would be invariant under the source face parabolic of F as well as under its own maximal parabolic. These generate the entire relevant simple factor, because the extra source type is not retained by F . For a positive subpiece, its scalar mass times the sharp marginal is its entire angular effect: a POVM dominated by a PVM and having scalar mass cI equals c times that PVM, by restriction to a projection and its complement. Normalizing the positive subpiece would therefore produce a source-invariant boundary measurement. Its finite invariant trace is forbidden by flag contraction. Thus precisely $p(F)$ is observed. In the three-slot system, the two normal copies are marginals of one common full-path dilation. Their sharp incident marginals therefore commute as well. $\square$

Proposition 5.12 (Coxeter compatibility and ordered galleries). *The permutation p preserves the Coxeter matrix. Along a source chamber gallery, the ordered product of the chamber PVMs is supported on target galleries of the permuted type, allowing stutters. More precisely, smooth functions vanishing on that target gallery manifold annihilate the ordered product, with arbitrary smooth multipliers. At finite places the corresponding assertion holds for locally constant tests by compact clopen partitions.*

Proof. Adjacent source chambers share their sharp common panel. Insert its projection between the corresponding consecutive spectral slots. Differences of the two common-panel coordinates therefore annihilate the ordered product, including after multiplication by arbitrary smooth functions expressed by summably separated expansions. The common-panel equations define the gallery space regularly: it is a successive fibre product of the submersions from chambers to panels. They consequently generate its local smooth vanishing ideal, including at stutters. Smooth partitions of unity prove the asserted annihilation. In totally disconnected coordinates, finite clopen partitions insert the common panel projection exactly and give the same conclusion for locally constant tests.

Fullness gives a nonzero opposite-target restriction of the product at some generic, hence opposite, source chamber pair. Source covariance transports this assertion over the opposite orbit. Insert all intermediate PVM identities along any reduced longest source gallery, and use compact partitions of the intermediate target chamber spaces. Nonzero opposite endpoint support requires a gallery of the permuted type connecting opposite target chambers. Both ends use the full spherical Weyl system of the fixed ambient group G . Their longest elements therefore have the same Coxeter length. The permuted word must therefore itself be a reduced longest word; no stutter or nonminimal step is possible.

An alternating rank-two word of Coxeter length m_{ij} is reduced and extends to a reduced longest word. Its permuted segment is also reduced, so $m_{ij} \leq m_{p(i)p(j)}$. Cycling the finite permutation p gives equality. This proves Coxeter compatibility. $\square$

5.6. Exact residue links. Take a mixed lex face path missing one type in a higher-rank component, or one type in each of several higher-rank components. Fix one such component. Its source rank-one Levi boundary parametrizes matched opposite-panel completions at the two source ends. Denote this common Levi flag by z . Refining each of the two opposite source faces by that same flag of their common Levi gives its start and ordinary-end missing vertices. This is the transverse parabolic identification of the residues, with the ordinary-end types relabeled by opposition. Write P_z and Q_z for the sharp missing-vertex PVMs at these start and ordinary-end vertices, respectively. Their common source parameter is transported by $z \mapsto sz$ under the simultaneous Levi action. Frame the two sharp endpoint faces separately by compact transports, and identify the reference opposite panels by transverse projection. The residual output coordinate g is then a rank-one target projectivity from the end pencil to the start pencil.

Proposition 5.13 (Residue graph equation). *On compact output windows the mixed link satisfies*

$$(5.11) \quad E(dg)J(\overline{Q_z}(dq) \otimes P_z(du)) \quad \text{is supported on} \quad u = gq.$$

The equation means annihilation by smooth joint tests vanishing on the graph, with locally constant tests in finite-place angular variables; conjugate spectral measures conjugate their projections. It holds with bounded Borel output parameters after the angular identities have been established. The joint law is invariant under simultaneous source Levi transport, and exact input translations may be made before any further limit.

Proof. We first work with an archimedean target component. There are two steps, both using ordered spectral slots.

Projection onto the reverse pencil. On the ordinary-end Hilbert space, use the slots in the order: reverse panel, end missing vertex at z , and the observation at the start label, possibly moved by the reverse radical. Denote the value in the last slot by v . A source gallery shortened from the longest by the missing reflection connects the matched chambers. Its target gallery has the same shortened type by Proposition 5.12. The shortened Weyl position is $s_i w_0 = w_0 s_i^*$; it has all the other respective left and right descents, so saturation by the parabolics omitting those missing reflections cannot reach w_0 . Thus the missing vertices are nonopposite.

A vertex transverse to a panel of the dual missing type has exactly one incident completion of that panel whose missing vertex is nonopposite to it. This is the ordinary rank-one parabolic projection. The ordered gallery equation therefore says that the end missing vertex is exactly $\text{pr}(v)$ on a transverse window. Integrating the intermediate gallery slots by their identity projections preserves the equation. Other components can be completed arbitrarily.

The line equation. Use (5.10) at the single start vertex z and move only its end occurrence by the independent reverse radical. Sharpness permits taking the start column to be the projector $1 - n(P_z)n(P_z)^*$ applied to an arbitrary column, and the end column to be the moved line projector on a transverse patch. Thus, after bounded output localization, the link is annihilated by

$$(1 - n(u)n(u)^*)S(h)n(v).$$

Its two angular inputs commute because they act on the two separate input tensor slots. The identity consequently holds with arbitrary smooth joint angular multipliers, by summable separation. On the stratum, the matrix action is the residual projectivity after transverse projection, so the equations are precisely the line graph $u = g \text{pr}(v)$. They generate its smooth vanishing ideal on compact transverse windows: the highest-line embedding is immersive, so the derivative in the image-vertex argument has full injective rank. Smooth division is uniform on such windows. Output coordinates may be Borel parameters with locally uniform angular derivative bounds, since separation is performed only in the angular variables.

We now combine the two steps in their necessary order. Preapply at the end a reverse-panel cut followed by a moved-vertex cut, on a buffered transverse rectangle, and leave the start vector arbitrary. Insert the Q_z spectral slot next to the reverse-panel slot; they commute. The shortened-gallery equation replaces its value by $\text{pr}(v)$, including after smooth multiplication in the independent start and output variables. Integrate out that slot. Pull the sharp reverse-panel projection through J , using Proposition 4.3. The line equation then supplies $u = g \text{pr}(v) = gq$. This proves (5.11) on those ordered ranges. In particular, we have not applied a discontinuous joint symbol to an arbitrary pair of noncommuting end measurements.

These ordered ranges are total. Their closed join is target invariant. It is fixed by a split source subgroup escaping in every ambient component, fixing the two panels and their respective completions, and normalizing the reverse-radical moves. Its projection is therefore scalar by the invariant-mass and mixing rule of Proposition 4.3. Some such ranges are nonzero when also varying z : moved start labels then cover generic vertices, and patch fullness applies to a nonzero reverse-panel projection on a uniform transverse target patch. Source Levi transitivity on the completion parameter transports nonzeroness to every z . Hence the join is the whole end space, and the equation holds everywhere.

Finite target places. At a finite target place, replace the Hilbert line argument by the individual-pair assertion of Proposition 5.8. Choose scalar clopen start and end patches. On a buffered output window carrying the end patch strictly inside the start image patch, a start vector in the kernel of the corresponding P_z projection and an end vector in the cut range of the same start-label measurement give zero. Since P_z is sharp, finite clopen partitions imply the exact transverse image substitution. Move the end occurrence by the reverse radical as before. The action on transverse values is the residual projectivity after projection to the reverse pencil.

The shortened-gallery substitution holds in this component by compact clopen partitions and insertion of the common sharp panel projection. Thus the reverse-panel cut followed by the moved-vertex cut again imposes $u = gq$, including arbitrary locally constant angular multipliers. The same total-range argument applies. This proves the finite-place equation without tensoring the complex Hilbert input with a representation over a nonarchimedean field.

For mixed target coordinates, use finite partitions in the clopen slots and summable smooth separated expansions in the real slots. Individual stratum restrictions are made by Borel

output projections after buffered tests in the ambient compactification; no uniform transverse estimate for an arbitrary discontinuous mask in an unbounded-overlap problem is required.

Finally, a source Levi element centralizes the chosen face path. Exact cocycle composition, made before limiting, shows that its simultaneous action at the two inputs changes the array only by base reindexing and a common scalar phase. The tested forms are therefore invariant. If a genuine action is needed, use the conjugate-multiplier regular amplification from Section 4. It tensors the forms by identity and cancels the phase defect at each input, without altering any finite-trace mixing assertion. This proves the stated transport properties. $\square$

Corollary 5.14 (Fibrewise links and polynomial fullness). *Disintegrate over the two sharp endpoint faces and use genuine source amplifications if necessary. For product-almost every pair of endpoint fibres, the mixed isometry (4.1), the graph equation (5.11), simultaneous source Levi invariance, and exact input translations hold. On almost every individual fibre, for every even highest power in an archimedean target module, the line ranges over z span at least the Hilbert fibre tensored with the span of all ordinary incident highest lines.*

Consequently, a homogeneous polynomial relation with linear Hilbert-vector coefficients that holds spectrally at every residue label holds at every ordinary residue value. The assertion includes algebraic subspace membership expressed by Pluecker equations and chartwise relations after the denominator procedure in the proof.

Proof. The sharp face marginals intertwine the link dilation, so its spectral span disintegrates over the product of the two endpoint spectral measures. The strongly continuous source Levi representations commute with those face PVMs and therefore disintegrate as well. One may obtain continuous fibre representations by disintegrating a countable dense convolution algebra of the group and then using its nondegenerate integrated representations, as in the standard direct-integral construction [4, Theorem 1.G.6].

Before removing a null set, choose countably many dense residue labels, compact windows, local graph generators, angular Fourier tests in real charts, fixed-level clopen tests in finite charts, and total input vectors. All the corresponding identities then hold on one conull set. Strong source transport and the uniform angular bounds extend them to every residue label and smooth or locally constant angular test there. The stabilizer invariance at a standard label also disintegrates, first on a countable dense subgroup and then by strong continuity, so the transported family is well defined on the residue quotient. Once a vector identity holds on an individual fibre, any bounded Borel output multiplier is applied directly through that fibre's spectral calculus. Uniformly summable angular expansions then give tests with Borel output parameters. No countable norm-dense class of all bounded Borel functions is used.

Disintegrate the amplified incidence containment of Proposition 5.8 on the same determining family. On almost every fibre its closed span contains the Hilbert fibre tensored with the span of all ordinary incident highest lines, and this holds separately for every even highest power. A polynomial with linear Hilbert-vector coefficients that vanishes on all spectral line ranges is orthogonal to their span. It is therefore orthogonal to each ordinary incident highest line, proving polynomial fullness. Norm powers homogenize different even degrees.

For a relation known only on a chart $\{\Delta \neq 0\}$, first clear a common homogeneous denominator and call the resulting numerator F_{num} . Apply global polynomial fullness to ΔF_{num} : it vanishes spectrally on the chart by the relation, and on $\{\Delta = 0\}$ by the extra factor. Divide by Δ on the original chart to recover the relation there. Pluecker equations and common local denominators put the stated algebraic membership tests in this form. $\square$

6. RECONSTRUCTION AT FINITE-PLACE PANELS

The exact residue link relates the two endpoint observations at each source label. When the target panel is nonarchimedean, we show that these relations force each endpoint family to be a commuting law of homogeneous analytic boundary maps. The same maps then describe translated links and give the upper-speed estimate. The proof uses compact-open

fixed algebras and Lie algebras over the prime local field; it will also show that the source panel has the same residue characteristic as the target.

6.1. The exact link and the frame algebra. Fix a mixed lexicographic face path missing one type in a non-isolated component. The same construction will subsequently be used with one missing type in each of several components. Write B_S, B_T for the source and target panel boundaries. Let S be the effective root-generated source panel group, and let T be the effective target adjoint point group, including the finite-index overgroup needed for changes of frame in Convention 3.6. In this section T is a group over a finite extension of a prime local field. Its prime-field Lie algebra is simple. The group S is simple after its ineffective center has been removed, and T embeds faithfully in the automorphism group of its prime-field Lie algebra. In particular, the centralizer in T of an open subgroup of its root-generated group is trivial.

We first retain the whole ordinary endpoint Hilbert spaces, denoted by H' at the start and I at the end. They include the genuine source amplification of Lemma 4.6, when that amplification is needed. The sharp endpoint-face observations and their decompositions are retained. Separate compact frames, chosen measurably from these observations, identify the two target residues with B_T . For $z \in B_S$, the missing-vertex observations in these frames are PVMs P_z on H' and Q_z on I . We have an isometry

$$(6.1) \quad J : \bar{I} \otimes H' \longrightarrow \mathcal{K}$$

and an output projectivity PVM E on T . Equation (5.11) says precisely that

$$E(dg)J(\overline{Q_z}(dq) \otimes P_z(du)) \quad \text{is supported on } u = gq.$$

We use here its exact ordered-test meaning established in Section 5: clopen graph substitutions are valid on compact projectivity windows, with bounded output cuts retained. This is stronger than a relation between the supports of two marginals.

There is an actual source Levi cover $\tilde{S} \rightarrow S$ acting strongly continuously on the ordinary endpoint spaces. It commutes with the sharp endpoint-face observations, fixes their frame calculi, and moves the residue labels by $z \mapsto sz$. Its ineffective finite center therefore acts trivially on the algebras generated by the residue observations, even if it acts nontrivially on the Hilbert spaces. Only this algebra action is asserted to descend to S . All these properties, including the exact input reindexing before a further limit, are properties of the link already proved in Sections 4 and 5.

Our goal is to recover a fixed homogeneous analytic homeomorphism $\tau : B_S \rightarrow B_T$ such that, on their respective spectral spaces, the endpoint families are evaluations of maps

$$z \mapsto a\tau(z), \quad z \mapsto b\tau(z),$$

with measurable target projectivity values a, b . The graph equation will then identify the output projectivity as ab^{-1} . Both commutation of the endpoint families and this homogeneous form are conclusions to be proved.

We put all external target frames into one algebra. For a locally constant scalar function f on the compact space B_T , define the paired decomposable operator

$$(6.2) \quad F_{z,f}(t) = (f(tP_z), f(tQ_z)), \quad t \in T,$$

on $L^2(T; H') \oplus L^2(T; I)$. The notation means spectral evaluation: $f(tP_z) = \int f(tu)P_z(du)$, and likewise at the end. Let D_* be the von Neumann algebra generated by these operators. A countable dense set of source labels and countable determining clopen tests suffice, since the original observations are continuous in the label on each fixed test. Thus D_* has separable predual.

In the intended ordinary-map model, the start test in Equation (6.2) would be $f(ta\tau(z))$; its frame and projectivity variables occur only through $x = ta$. We will prove that the actual paired algebra has precisely one such intrinsic T -coordinate: D_* is $L^\infty(T)$ with the frame

action given by left translation. This will establish commutation and reduce the homogeneous form of the endpoint maps to source equivariance.

Left frame translation defines a continuous action α of T :

$$(\alpha_a d)(t) = d(a^{-1}t), \quad \alpha_a(F_{z,f}) = F_{z, f \circ a^{-1}}.$$

The actual source transport defines a continuous action β of S on D_* with

$$\beta_s(F_{z,f}) = F_{sz,f}.$$

These actions commute. Continuity is in the point-ultraweak topology; the spatial implementations and boundedness of the generators also give the corresponding strong continuity on fixed vectors.

The first step is to understand the fixed algebra of a compact open subgroup of T . The graph equation produces Hilbert–Schmidt intertwiners between its frame evaluations. The nonzero eigenspaces of their compact positive products supply enough finite-dimensional representations to give the following atomic decomposition.

Proposition 6.1 (Compact-open fixed algebras). *For every compact open subgroup $U < T$, the algebra D_*^U is a countable product of finite matrix algebras. Moreover,*

$$(6.3) \quad D_*^T = \mathbb{C}1,$$

and the source action β is nontrivial.

Proof. A decomposable operator fixed by left translation by U is essentially constant on each left coset. Thus there are normal evaluations

$$\pi_t^+(d) = d_t^+ \in B(H'), \quad \pi_r^-(d) = d_r^- \in B(I), \quad Ut, Ur \in U \backslash T, \quad d \in D_*^U.$$

They are restrictions of the normal coordinate evaluations on

$$\ell^\infty(U \backslash T; B(H') \oplus B(I)).$$

They are jointly faithful, and $U \backslash T$ is countable.

Identify $\bar{I} \otimes H'$ with $\text{HS}(I, H')$ by $\bar{\xi} \otimes \eta \mapsto |\eta\rangle\langle\xi|$. For fixed Ut, Ur , put

$$\Omega_{t,r} = \{g \in T : Utg = Ur\}.$$

For every $\zeta \in \mathcal{K}$, the Hilbert–Schmidt operator $K = J^*E(\Omega_{t,r})\zeta$ satisfies

$$(6.4) \quad d_t^+ K = K d_r^-, \quad d \in D_*^U.$$

Here is the word-transfer justification of Equation (6.4). First take a word in generators with real-valued clopen tests and average that word over U . At frame u_0t , the adjoint of the exact graph identity transfers each start test to the corresponding end test at frame u_0tg . On a compact g -window, take a finite partition into relatively clopen pieces such that each end function $q \mapsto f(u_0tgq)$ in this finite list is independent of g within a piece, uniformly for $u_0 \in U$. Indeed the action and the tests are continuous on the compact set of parameters, while a clopen test has finite image. Within a piece every factor therefore intertwines simultaneously. Transferring a word successively gives the reversed order on the conjugate Hilbert space, which under the Hilbert–Schmidt identification is right multiplication by the end word in its ordinary order. All the output cuts remain in place.

On $\Omega_{t,r}$, write $tg = u_g r$ with $u_g \in U$. Haar averaging in u_0 absorbs u_g , so the averaged end word has value d_r^- and the averaged start word has value d_t^+ . This proves Equation (6.4) for averaged words on compact windows. Linear combinations and adjoints treat complex tests. Kaplansky density approximates a bounded element of D_* by bounded generator polynomials, with arbitrarily small norm slack. Normal Haar averaging approximates every element of D_*^U ultraweakly. The evaluations on this fixed algebra are normal, and multiplication by a fixed Hilbert–Schmidt operator is weakly continuous on bounded sets. Thus the identity extends to all $d \in D_*^U$. Exhausting T by compact windows removes the temporary localization. The

endpoint frames cause no extra evaluation step: their sharp decompositions are retained throughout this calculation.

For fixed t , the sets $\Omega_{t,r}$ partition the output variable as Ur varies. Since J is an isometry, J^* maps onto $\text{HS}(I, H')$. The linear span of all the resulting K is therefore dense in Hilbert–Schmidt norm. The same statement holds with r fixed and t varying, because right multiplication by a fixed g permutes the left cosets. Equation (6.4) and its adjoint give

$$KK^* \in \pi_t^+(D_*^U)', \quad K^*K \in \pi_r^-(D_*^U)'.$$

The nonzero eigenspaces of these compact positive operators are finite-dimensional reducing spaces for the indicated representations. Together they span the ordinary evaluation spaces. For example, a vector orthogonal to all the ranges of the K would be orthogonal to the ranges of a Hilbert–Schmidt dense family, and hence would vanish.

Restriction to these reducing spaces supplies a jointly faithful family of normal finite-dimensional representations of D_*^U . The central support of each such representation is a finite sum of central atoms, and its algebra on those atoms is a finite sum of finite matrix algebras. Joint faithfulness makes the join of these supports equal to one. Separability of the predual makes the resulting product countable. This proves the first assertion without any restriction on the endpoint Hilbert-space multiplicities.

If $d \in D_*^T$, its two evaluations are independent of t, r . Equation (6.4) then holds across every Hilbert–Schmidt operator. If $AK = KB$ for every rank-one map $K : I \rightarrow H'$ on nonzero Hilbert spaces, testing $K = |\eta\rangle\langle\xi|$ gives $A = \lambda 1$ and $B = \lambda 1$ with the same scalar. This proves Equation (6.3).

Finally suppose that S acted trivially on D_* . Source covariance and transitivity on B_S would make $F_{z,f}$ independent of z . For countably many labels and tests, evaluate these explicit decomposable identities at one common generic external frame; the transformed tests still generate $C(B_T)$. Continuity in the source label then makes the original framed P_z independent of z on the whole start space. The actual source Levi cover commutes with the sharp base observation and its frame calculus, so unframing gives the same statement for the original missing vertex.

Choose a chamber completing the source face, fixing completions in the other components. Its full observation is the commuting product of its canonical vertex observations. The retained vertices are fixed by the actual source Levi factor, and the missing vertex has just been shown fixed. Hence the full chamber observation is fixed by this Levi factor as well as by the minimal source parabolic. These groups generate the larger parabolic allowing the omitted simple root. The resulting unital target-equivariant full-boundary observation contradicts Lemma 4.8. The assertion concerns the original observations and so descends through the identity amplification. In a many-panel link, adjoin the other fixed chamber completions before applying the same argument. Thus the source action is nontrivial in every case used here. $\square$

Corollary 6.2 (Separation of archimedean types). *A connected noncompact simple real source cannot be matched to a finite-place target panel. Consequently the type permutation preserves the archimedean and nonarchimedean subsets of the type list.*

Proof. A continuous action of a connected group fixes every central atom of an atomic von Neumann algebra. To see the relevant discreteness, a normal state supported on one atom takes values zero and one on its distinct translates, so that atom has an open stabilizer. On each finite matrix block the action is a continuous homomorphism into a projective unitary group. A noncompact simple real Lie algebra has no nonzero homomorphism into a compact Lie algebra: simplicity would make it injective, and a positive definite invariant inner product would then make the source Lie algebra compact. A homomorphism with zero differential from a connected Lie group is trivial. Thus the source would fix every D_*^U pointwise. As compact open U decreases to the identity, the normal averages $\int_U \alpha_u(d) du$ converge ultraweakly to d . Their ranges therefore generate D_* . This contradicts Proposition 6.1. The archimedean subset

of the finite type list is mapped into itself; finiteness and bijectivity of the type permutation give the assertion for both subsets. $\square$

6.2. Faithful corners and finite block actions. Both panel groups are now nonarchimedean. The preceding fixed algebras may have infinitely many matrix blocks. Passing to a minimal corner will instead give a sequence of finite-dimensional fixed algebras. We will show that their block stabilizers shrink uniformly; this will identify the corner with the translation algebra of a compact open subgroup of T . Recovering the full frame algebra will then require identifying its center and removing its remaining matrix fibres.

Denote the residue characteristics temporarily by p_S, p_T . Choose small torsion-free logarithmic compact open subgroups $V < S$ and $U < T$, respectively pro- p_S and pro- p_T , in faithful adjoint matrix coordinates. Choose a minimal projection $e \in D_*^U$. We may shrink V to fix e . In fact an atomic central block has an open stabilizer, by the normal-state test in the preceding proof. The image of that compact profinite stabilizer in the automorphism group of its finite matrix block is both profinite and a closed Lie subgroup, and hence is finite. Its open kernel fixes the entire block. Choose a still smaller logarithmic subgroup inside that kernel. Put

$$(6.5) \quad D_0 = eD_*e, \quad D_0^U = \mathbb{C}e.$$

Lemma 6.3 (Automorphisms fixing a covering family of corners). *Let A be a von Neumann algebra and let (r_j) be projections with join one. The group of automorphisms fixing every r_jAr_j pointwise is abelian.*

Proof. An automorphism in question fixes r_i, r_j and their central supports. Work on the common central support of these two projections. The rectangular space $X = r_iAr_j$ is the full selfdual equivalence bimodule between its endpoint corners: the ultraweak spans of XX^* and X^*X are those corners on this support. The restriction of the automorphism to X is right linear, preserves the right inner product x^*y , and has its inverse as adjoint. It is also left linear.

Adjointable right-module operators on this equivalence bimodule are left multipliers by the left corner. One can verify this directly: finite sums of the rank-one maps $x \mapsto yz^*x$ approximate its identity strongly; applying an adjointable operator to these sums produces left multipliers, and their bounded strong limit remains in the left corner. Left linearity forces the resulting multiplier to be central. Preservation of the inner product makes it unitary. Thus the automorphisms act on every r_iAr_j by commuting central unitary multipliers. These multipliers are themselves in corners fixed pointwise by every automorphism in the group, so composition of the actions on each rectangular space is commutative.

It follows that the commutator θ of any two such automorphisms fixes every r_iAr_j . For $a \in A$, $r_i(\theta(a) - a)r_j = 0$ for all i, j . Since the ranges of the r_j span the representation space, this gives $\theta(a) = a$. $\square$

Lemma 6.4 (Kernel on the corner). *The kernel of the V -action on D_0 has zero Lie algebra.*

Proof. This kernel is a closed normal subgroup of V , so its Lie algebra is an ideal in the simple prime-field algebra $\mathfrak{s}$. If it were nonzero, it would be all of $\mathfrak{s}$, and an open subgroup $V_0 < V$ would fix D_0 pointwise. The source and target actions commute, so V_0 would also fix $\alpha_t(e)D_*\alpha_t(e)$ pointwise for every $t \in T$. The join of the projections $\alpha_t(e)$ is nonzero and T -fixed; it is one by Proposition 6.1. A countable subfamily has the same join. Lemma 6.3 now shows that every commutator from V_0 acts trivially on D_* . There is a nonidentity such commutator because $\text{Lie}(V_0) = \mathfrak{s}$ is nonabelian. The kernel of the S -action is normal in the effective simple group S . It would therefore be all of S , contrary to Proposition 6.1. $\square$

Choose a decreasing sequence of compact open normal subgroups $U_n \triangleleft U$ with intersection $\{1\}$, and set

$$D_n = D_0^{U_n}.$$

The union of these increasing algebras generates D_0 ultraweakly, by normal averaging and continuity of the U -action. The finite group U/U_n acts ergodically on D_n . For a nonzero projection $a \in D_n$, its average is $\lambda_a e$ and

$$a \leq [U : U_n] \lambda_a e, \quad \lambda_a \geq [U : U_n]^{-1}.$$

An orthogonal family of nonzero projections consequently has at most $[U : U_n]$ members. A von Neumann algebra with such a bound is finite-dimensional: otherwise repeated splitting of nonminimal projections or its infinite family of atoms gives arbitrarily large orthogonal families. Thus every D_n is finite-dimensional.

Lemma 6.5 (Block stabilizers). *The simple blocks of D_n form a transitive U -set. The stabilizer $L_n < U$ of a chosen block acts on it by an irreducible projective unitary representation. If $G_n < V \times U$ is the stabilizer of the same block for the commuting source and target actions, then G_n is closed, its projection onto V is surjective, and*

$$(6.6) \quad [K_n, K_n] \text{ acts trivially on } D_n, \quad K_n = \{v \in V : (v, 1) \in G_n\}.$$

Proof. A union of block orbits would give an invariant central projection, so ergodicity gives transitivity. An element of the chosen block fixed by L_n extends, by U -transport, to an invariant element of D_n . Therefore the fixed algebra of L_n on that block is scalar. In a matrix algebra its automorphism action is implemented projectively, and this scalar-commutant condition is precisely irreducibility. The action on the finite set of blocks is continuous, so G_n is closed. For each $v \in V$, transitivity supplies $u \in U$ carrying the block moved by v back to the chosen one. Hence $G_n \rightarrow V$ is onto.

An element of K_n fixes every block label: its permutation commutes with the transitive U -action and fixes one label. On the chosen block its projective implementer centralizes the projective representation of L_n . Such a projective centralizer is abelian. Indeed, for an implementer A and projective representation implementers W_l , write

$$AW_l A^* = \chi_A(l) W_l.$$

The representation multipliers cancel, making χ_A a scalar character. The map $[A] \mapsto \chi_A$ is a homomorphism into the abelian character group. Its kernel is trivial by Schur's lemma. Thus commutators of K_n act trivially on this block. Commutation with U and transitivity give the assertion on all blocks. $\square$

Corollary 6.6 (Separation of residue characteristics). *The source and target residue characteristics are equal. The type permutation preserves each residue-characteristic subset of the type list.*

Proof. If $p_S \neq p_T$, a closed subgroup of the product of a pro- p_S group and a pro- p_T group is the product of its projections. Pass to finite quotients: a subgroup of a finite p_S -group times a finite p_T -group has projected groups with no common nontrivial quotient, so is their product. The assertion for closed subgroups follows by inverse limits. Consequently $V \times 1 \subset G_n$ for every n . Lemma 6.5 makes $[V, V]$ act trivially on all D_n and hence on D_0 . The closure of this commutator subgroup has full Lie algebra $[\mathfrak{s}, \mathfrak{s}] = \mathfrak{s}$; for example, take the Lie algebra of the compact abelian quotient by its closure. This contradicts Lemma 6.4. Applying the assertion at every type proves the last statement. $\square$

6.3. A uniform logarithm lemma. From now on both prime-field Lie algebras $\mathfrak{s}, \mathfrak{t}$ are over the same $\mathbb{Q}_p$. The finite-level block stabilizers vary with n , so a subgroup-dependent logarithm constant would not suffice. We prove the needed uniform statement.

Lemma 6.7 (Uniform saturation in a logarithm chart). *Let W be a sufficiently small logarithmic compact open subgroup of a linear p -adic Lie group. There is an integer $c \geq 0$, depending only on W , such that for every closed subgroup $H < W$,*

$$(6.7) \quad \exp(p^c M_H) \subset H, \quad M_H = \text{span}_{\mathbb{Z}_p} \{\log h : h \in H\}.$$

The assertion includes $p = 2$ and subgroups of smaller Lie dimension.

Proof. In dimension zero, a sufficiently small W is trivial and $c = 0$ suffices. Assume henceforth that the ambient Lie algebra has positive dimension. Choose a lattice Λ in that Lie algebra $\mathfrak{w}$ on which matrix exponential and logarithm are inverse, with $W = \exp \Lambda$. Shrink it so that

$$(\mathrm{Ad}(w) - 1)\Lambda \subset p\Lambda \quad (w \in W).$$

Write $N = \dim_{\mathbb{Q}_p} \mathfrak{w}$ and $M = M_H$. As a submodule of the finite free $\mathbb{Z}_p$ -module Λ , M is finitely generated, free, and closed in its rational span. Choose finitely many $h_1, \dots, h_a \in H$ whose logarithms generate M . Such a subfamily exists because $\mathbb{Z}_p$ is Noetherian.

For $h \in H$, conjugation by h and h^{-1} preserves M . Thus $A = \mathrm{Ad}(h) - 1$ is an endomorphism of M . Its characteristic polynomial on the fixed ambient space has the form

$$x^N + a_{N-1}x^{N-1} + \dots + a_0, \quad a_j \in p\mathbb{Z}_p.$$

Cayley–Hamilton applied on M , rather than just on the ambient lattice, gives

$$(6.8) \quad A^N M \subset pM, \quad A^k M \subset p^{\lfloor k/N \rfloor} M.$$

Set

$$b = \max \left(0, \sup_{k \geq 1} \{v_p(k) - \lfloor k/N \rfloor\} \right) < \infty.$$

The logarithm series and Equation (6.8) give

$$\log(1 + A)M \subset p^{-b}M.$$

In the fixed matrix logarithm chart, $\log \mathrm{Ad}(h) = \mathrm{ad}(\log h)$. Extending over the chosen generators of M therefore proves

$$[M, M] \subset p^{-b}M.$$

Choose $c \geq b + 2$ and put $L = p^c M$. Then $[L, L] \subset p^2 L$.

We check the BCH estimate at every prime. In Dynkin’s formula a homogeneous term of degree m has a denominator of the form $km \prod_i r_i! s_i!$, where the k blocks are nonempty and $\sum_i (r_i + s_i) = m$. Its p -valuation is at most $m - 1 + v_p(m)$: the factorial terms contribute at most $m - k$, and $v_p(k) \leq k - 1$. A bracket of length m in L lies in $p^{2(m-1)}L$. For $m \geq 3$ the resulting term therefore lies in $p^{m-1-v_p(m)}L \subset pL$, with its exponent tending to infinity. The degree-two term is $[X, Y]/2 \in pL$, including when $p = 2$. Thus BCH converges, preserves L , and satisfies

$$(6.9) \quad \mathrm{BCH}(X, Y) \equiv X + Y \pmod{pL}.$$

Every nonlinear term contains both variables. The same estimates therefore give, for $r, s \geq 0$,

$$\mathrm{BCH}(X, Y) - X - Y \in p^{r+s+1}L \quad (X \in p^r L, Y \in p^s L).$$

In particular $\exp L$ and $\exp(p^j L)$ are groups.

At level j , the elements $h_i^{p^{c+j}} \in H$ have logarithms $p^{c+j} \log h_i$ spanning $p^j L / p^{j+1} L$. Given $x \in L$, choose a product of these powers at level zero whose logarithm agrees with x modulo pL . Its inverse times $\exp x$ has logarithm in pL . Correct this error with a product at level one, and continue. Equation (6.9) and its refined estimate improve the error by one level at every step. The correcting factors tend to one, and their successive products in H converge to $\exp x$. Closedness of H proves Equation (6.7). Every constant was chosen before H , as required. $\square$

We use the usual analytic structures on closed subgroups and quotients of p -adic Lie groups, and the fact that continuous homomorphisms between them are analytic; see [31, Propositions 2.2–2.3]. In particular a surjective homomorphism between compact p -adic Lie groups has surjective differential. Indeed, the image of a sufficiently small exponential open subgroup is a compact finite-index subgroup of the target and is therefore open. Naturality of exponential puts its logarithm in the differential image, which consequently spans the target Lie algebra.

6.4. Hausdorff limits and equality of panel dimensions. We now use the uniform logarithm estimate to control the block stabilizers. The compact metrizable group $V \times U$ has a compact space of closed subgroups in the Hausdorff topology. Every limit G_∞ of the G_n still projects onto V : for fixed $v \in V$, choose $(v, u_n) \in G_n$ and pass to a convergent subsequence of (u_n) . The key point is that a nonzero source-only Lie kernel in such a limit would force one fixed open source subgroup into all sufficiently late stabilizers. Its commutators would then fix the whole corner, contrary to Lemma 6.4. The next argument applies to every choice of a block at every level.

Lemma 6.8 (The limiting source-only kernel). *For any sequence of levels tending to infinity, any choices of blocks at those levels, and any Hausdorff limit G_∞ of their stabilizers, the subgroup $G_\infty \cap (V \times 1)$ has zero Lie algebra.*

Proof. Its Lie algebra is invariant under $\text{Ad}(V)$, because $G_\infty \rightarrow V$ is onto. It is thus an ideal of the simple algebra $\mathfrak{s}$. If nonzero it equals $\mathfrak{s}$, and G_∞ contains $V_0 \times 1$ for an open subgroup $V_0 < V$. Set $d = \dim_{\mathbb{Q}_p} \mathfrak{t}$. Choose finitely many $v_1, \dots, v_a \in V_0$ such that

$$(6.10) \quad \sum_{i=1}^a \text{ran}((\text{Ad}(v_i) - 1)^d) = \mathfrak{s}.$$

For this choice, start with a noncentral sufficiently small element of a split torus inside V_0 . Its adjoint action is semisimple over a splitting field, so the nonzero image of $\text{Ad}(v) - 1$ is unchanged upon taking a positive power. The span of its conjugates by V_0 is invariant under this open subgroup, hence under $\mathfrak{s}$. It is a nonzero ideal and equals $\mathfrak{s}$. Finitely many conjugates suffice and give Equation (6.10).

Pass to the chosen convergent subsequence, denoting its levels again by n . Hausdorff convergence gives

$$(v_{i,n}, w_{i,n}) \in G_n, \quad (v_{i,n}, w_{i,n}) \longrightarrow (v_i, 1).$$

Let

$$M_n = \text{span}_{\mathbb{Z}_p} \log G_n, \quad M_V = \text{span}_{\mathbb{Z}_p} \log V.$$

Surjectivity of $G_n \rightarrow V$ gives the exact equality $\text{pr}_{\mathfrak{s}} M_n = M_V$. All these modules lie in fixed logarithm lattices. Put

$$\chi_{i,n}(X) = \det(X1 - (\text{Ad}(w_{i,n}) - 1)|_{\mathfrak{t}}), \quad B_{i,n} = \chi_{i,n}(\text{Ad}(v_{i,n}) - 1).$$

The polynomials $\chi_{i,n}$ have integral coefficients and converge coefficientwise to X^d . Conjugation by $(v_{i,n}, w_{i,n})$ preserves M_n . Therefore $\chi_{i,n}(\text{Ad}(v_{i,n}, w_{i,n}) - 1)$ preserves M_n , and Cayley–Hamilton annihilates its target component exactly. For every $x \in M_V$ choose a lift $(x, y) \in M_n$ and apply this polynomial. It follows that

$$(6.11) \quad \left(\sum_i B_{i,n} M_V \right) \times \{0\} \subset M_n, \quad B_{i,n} \longrightarrow (\text{Ad}(v_i) - 1)^d.$$

Choose a basis of the fixed lattice M_V and concatenate the matrices $B_{i,n}$. Equation (6.10) gives a nonzero full-rank square minor in the limiting concatenation. The corresponding integral determinants at all sufficiently late levels have one fixed finite valuation, say a . The adjugate identity then yields

$$p^a M_V \times \{0\} \subset \left(\sum_i B_{i,n} M_V \right) \times \{0\} \subset M_n$$

uniformly at those levels. Apply Lemma 6.7 in the fixed product logarithm group $V \times U$. With a constant c independent of n , it follows that

$$\exp(p^{c+a} M_V) \times 1 \subset G_n.$$

After increasing a if needed, the displayed source exponential is a fixed open subgroup $V_1 < V$. Equation (6.6) says that $[V_1, V_1]$ fixes D_n at every sufficiently late level of this subsequence.

These levels tend to infinity and their algebras exhaust D_0 . Thus $[V_1, V_1]$ fixes D_0 . Its closure has full Lie algebra $[\mathfrak{s}, \mathfrak{s}] = \mathfrak{s}$, contradicting Lemma 6.4. $\square$

Proposition 6.9 (Dimensions and shrinking stabilizers). *Matched finite-place panel groups have equal Lie dimension over their common prime field. The target stabilizers of the blocks of D_n shrink to the identity uniformly over all choices of blocks. The corner D_0 is abelian and, as a U -algebra, is $L^\infty(U)$ with translation and Haar probability.*

Proof. Let G_∞ be any limit. Its Lie projection onto $\mathfrak{s}$ is surjective. By Lemma 6.8, its Lie projection into $\mathfrak{t}$ is injective. Consequently

$$(6.12) \quad \dim_{\mathbb{Q}_p} \mathfrak{s} \leq \dim_{\mathbb{Q}_p} \mathfrak{t}.$$

Use the same missing-root construction at every type. Source and target panel dimensions come from the same finite list, and Corollaries 6.2 and 6.6 keep each type cycle over a single prime field. Iterating Equation (6.12) around that cycle forces equality at every step. Only effective prime-field Lie dimensions enter this argument; no ineffective compact factor or complex Hilbert-space dimension is counted.

Both projections of $\text{Lie}(G_\infty)$ are now isomorphisms. The target-only subgroup $G_\infty \cap (1 \times U)$ has dimension zero. A compact zero-dimensional analytic subgroup is finite, and U is torsion-free, so

$$(6.13) \quad G_\infty \cap (1 \times U) = \{1\}.$$

Suppose a neighborhood of one failed to contain every sufficiently late target block stabilizer. Choose a smaller clopen neighborhood, levels tending to infinity, offending blocks, and stabilizer elements u_n outside it. Along a subsequence their G_n converge to G_∞ and $u_n \rightarrow u \neq 1$. Then $(1, u) \in G_\infty$, contradicting Equation (6.13). This proves uniform shrinking for arbitrary block choices.

Fix m . For all sufficiently large n , the stabilizer of every block of D_n lies in U_m . It therefore fixes the image of D_m in that block pointwise. Its projective representation is irreducible by Lemma 6.5, so that image is scalar. The block representations are jointly faithful, and therefore D_m is abelian. Exhaustion makes D_0 abelian.

Write the spectrum of D_n as the finite transitive U -space X_n . The inclusions give surjective equivariant maps $X_{n+1} \rightarrow X_n$. Averaging a faithful normal state of D_0 over U gives a faithful invariant state whose restriction to each X_n is the uniform probability. Thus

$$D_0 = L^\infty(X, \nu), \quad X = \varprojlim X_n,$$

with these uniform marginals. The compact group U acts transitively on X : the sets of elements carrying the first n coordinates of one point to those of another are nonempty nested closed subsets of U . Their intersection is nonempty. The stabilizer of a point belongs to all its finite-level stabilizers, so uniform shrinking makes it trivial. Hence X is the translation U -torsor, and ν is Haar probability. $\square$

6.5. The homogeneous center and elimination of matrix fibres.

Proposition 6.10 (The full frame algebra). *There is a T -equivariant normal isomorphism*

$$D_* \cong L^\infty(T)$$

with the left translation action. In these coordinates the commuting source action has the form

$$(6.14) \quad (\beta_s f)(x) = f(x\eta(s)), \quad \eta : S \longrightarrow T,$$

where η is a continuous homomorphism with isomorphic $\mathbb{Q}_p$ -differential.

Proof. By Proposition 6.9, e is an abelian projection. On its central support z_e the algebra is type I, and the map

$$(6.15) \quad Z(D_*)z_e \longrightarrow eD_*e, \quad a \longmapsto ae$$

is a normal isomorphism. In the type I decomposition of this central summand, e has rank one in every factor, and full central support makes the scalar field faithful, which proves the assertion. The map is U -equivariant because U fixes e . Countably many T -translates of e have join one, so their central supports cover one and D_* is type I. Its center has a positive-measure U -invariant chart isomorphic to the translation space U . Its T -action is ergodic by Equation (6.3).

Use the standard Borel point realization for a continuous action of a locally compact second countable group on a separable measure algebra [46]. Let X realize the center, and let $\varphi : U \rightarrow X$ be the measurable isomorphism onto the chart. Equivariance holds almost everywhere for each group element. Fubini in the two Haar variables gives a point $x_0 \in X$ with

$$\varphi(u) = ux_0 \quad \text{for Haar-almost every } u \in U.$$

Explicitly, choose a generic u_0 such that $\varphi(vu_0) = v\varphi(u_0)$ for almost every v , and put $x_0 = u_0^{-1}\varphi(u_0)$. Let $L = \text{Stab}_T(x_0)$. If $h \in L \cap U$, compare u and uh in a common conull injectivity set of φ ; their images agree, so $h = 1$. A subgroup meeting a neighborhood of one only at one is discrete and closed. The orbit map $T/L \rightarrow X$ is a Borel injection and has a Borel inverse on its image. Countably many translates of the chart cover the center modulo null sets, so this orbit is conull. The measure class is quotient Haar class, as follows on the chart and its translates. We have obtained

$$Z(D_*) \cong L^\infty(T/L).$$

The measurable automorphisms of T/L commuting with left translation are exactly right translations from $N_T(L)/L$. For such an automorphism F , the function $g \mapsto g^{-1}F(gL)$ is essentially left invariant and is therefore essentially constant, say nL . Thus $F(gL) = gnL$ almost everywhere. Right- L consistency gives $n^{-1}Ln \subset L$, and the same argument applied to F^{-1} gives equality. Conversely a normalizer coset defines such an automorphism. The injection from $N_T(L)/L$ into the measure-class automorphism group is Borel, with Borel inverse onto its image. Indeed its action is measurable on a countable determining collection of sets, and the Borel inverse follows from injectivity between standard Borel spaces. The commuting source action consequently gives a measurable homomorphism

$$\eta : S \longrightarrow N_T(L)/L,$$

where the point action uses inverse right translation so that the algebra convention is Equation (6.14). The normalizer is closed. Measurable homomorphisms between locally compact second countable groups are continuous, so η is continuous and analytic over $\mathbb{Q}_p$.

Its differential is injective. Otherwise its kernel has positive Lie dimension; a corresponding small subgroup of V acts trivially on the center. Since V fixes e , Equation (6.15) would make that subgroup act trivially on eD_*e , contradicting Lemma 6.4. Because L is discrete and $\dim \mathfrak{s} = \dim \mathfrak{t}$, the closed Lie subgroup $N_T(L)$ has the full dimension of T and is open. For each $l \in L$, conjugation from $N_T(L)$ into the discrete group L is continuous. The element l therefore centralizes an open subgroup. The effective open-centralizer assertion from Section 3 forces $l = 1$. Thus $L = \{1\}$, the center is the T -torsor, and $d\eta$ is an isomorphism.

It remains to remove matrix fibres. The type I decomposition and ergodicity of the center give

$$D_* \cong L^\infty(T; B(\mathcal{V}))$$

with one separable Hilbert space $\mathcal{V}$ of constant dimension. In a measurable trivialization the action is

$$(\alpha_a d)(x) = c(a, x)(d(a^{-1}x)), \quad c(ab, x) = c(a, x)c(b, a^{-1}x),$$

where c takes values in $\text{Aut}(B(\mathcal{V}))$. No unitary lift of this automorphism cocycle is needed. Changing variables in its cocycle identity and applying Fubini allows one to freeze a generic starting point x_0 and define

$$b(x) = c(xx_0^{-1}, x).$$

The identity becomes

$$c(a, x) = b(x)b(a^{-1}x)^{-1} \quad \text{for almost every } (a, x).$$

Changing the fibre trivialization by b removes the cocycle. Continuity of the action extends the equality from almost every a to every a . Pure translation has fixed algebra the constant fields $B(\mathcal{V})$. Equation (6.3) therefore forces $\dim \mathcal{V} = 1$. $\square$

6.6. Ordinary homogeneous panel maps.

Proposition 6.11 (Recovery of the original panel observations). *The family $(P_z)_{z \in B_S}$ commutes on the original start space, and the family $(Q_z)_{z \in B_S}$ commutes on the original end space. There is a homogeneous analytic homeomorphism $\tau : B_S \rightarrow B_T$ and measurable target projectivity observations a, b on their respective spectra such that these families are evaluations of*

$$(6.16) \quad z \longmapsto a\tau(z), \quad z \longmapsto b\tau(z).$$

In the independent input tensor product, the output projectivity is sharp and equals ab^{-1} :

$$(6.17) \quad E(W)J = J1_W(ab^{-1}) \quad \text{for every Borel } W \subset T.$$

The assertions hold coordinatewise on many-panel links.

Proof. The commutativity of D_* makes the commutators of its paired generators vanish at almost every external frame. Use a common conull frame set for countably many labels and clopen tests. At one such frame the transformed tests still generate $C(B_T)$, so the original P_z commute, as do the Q_z , for the determining labels. Continuity in the source label extends commutation to every fixed pair of labels.

In the intrinsic coordinates $D_* = L^\infty(T)$ of Proposition 6.10, the spectral values of the generators give a boundary-valued measurable function $F(x, z)$. Countable determining observations and continuity in measure give a jointly measurable version. More explicitly, use a faithful probability in the Haar class and approximate the compact label space by finite simple label choices whose errors in measure are summable. The almost-everywhere limit of the resulting jointly measurable approximations represents each fixed label; the countable boundary tests determine the boundary value. Frame covariance yields

$$F(ax, z) = aF(x, z)$$

almost everywhere, and Fubini gives $F(x, z) = xF_0(z)$. The commuting source action then gives

$$(6.18) \quad F_0(sz) = \eta(s)F_0(z)$$

in the ordinary measure class on the source boundary.

We explain why $d\eta$ identifies the original boundaries. Regard each effective adjoint group over its coefficient field as its restriction of scalars to $\mathbb{Q}_p$. The resulting Lie algebra is the prime-field Lie algebra used above. In characteristic zero, all derivations of a semisimple Lie algebra are inner, and the identity component of its algebraic automorphism group is the corresponding adjoint algebraic group. Conjugation by $A = d\eta$ therefore gives an isomorphism of these adjoint algebraic groups over $\mathbb{Q}_p$. Differentiating the homomorphism identity gives

$$(6.19) \quad A \operatorname{Ad}(s) = \operatorname{Ad}(\eta(s))A, \quad s \in S.$$

In faithful adjoint coordinates the algebraic isomorphism on the root-generated point groups is consequently exactly η .

An algebraic group isomorphism carries rational parabolics and their unipotent radicals to the corresponding groups. The rational parabolics of a restriction of scalars are precisely the restrictions of the original rational parabolics. For completeness, after a finite separable splitting extension the restriction of scalars is a product indexed by the embeddings of the coefficient field. A parabolic of a product is a product of parabolics; descent permutes the conjugate factors. Choosing one factor identifies the descent condition with a parabolic

over the original field. The same argument applies if the effective almost-simple group itself has several absolutely simple factors in one Galois orbit. Minimal rational parabolics thus recover the original rank-one boundary, with no splitness assumption and no prior equality of coefficient fields. This produces a homogeneous analytic homeomorphism $\tau : B_S \rightarrow B_T$ intertwining S with $\eta(S)$.

Write $B_S = S/P$. By Equation (6.18), the measurable lift

$$s \mapsto \eta(s)^{-1} F_0(sP)$$

is essentially left invariant, hence essentially constant. Right- P consistency makes this constant fixed by $\eta(P)$. A minimal parabolic fixes exactly its own point on a rank-one boundary: its unipotent radical fixes that point and acts transitively on the complementary open Bruhat cell, which has more than one point. Therefore this constant is the parabolic point defining τ , and $F_0 = \tau$ almost everywhere. Both the original generators and the functions $x \mapsto f(x\tau(z))$ are continuous in measure in z . The generator identity consequently holds in D_* for every fixed label, not only for almost every label.

We now remove the external frame without using point evaluation on arbitrary elements of a diffuse L^∞ algebra. Let Ω^+ be a standard spectrum of the commutative algebra of the original start observations, with a faithful normal probability class. The normal representation of $D_* = L^\infty(T)$ by its start decomposable fields takes values in $L^\infty(T_{\text{ext}} \times \Omega^+)$. It is pullback by a measurable map $h(t, \omega)$ into the intrinsic T variable. The explicit generator identities read

$$t p_z(\omega) = h(t, \omega) \tau(z)$$

for almost every (t, ω) , simultaneously for countably many labels and tests. Choose one generic external frame t_0 and put $a(\omega) = t_0^{-1} h(t_0, \omega)$. Then $p_z(\omega) = a(\omega) \tau(z)$ for the determining labels. Each map $a(\omega) \tau$ is continuous on the compact source boundary. Bounded convergence of its spectral evaluations and the original label continuity extend the equality to every fixed label. The end representation gives b in the same way. Only these countably many measurable generator identities were evaluated at t_0 ; no normal evaluation on all of $L^\infty(T)$ is asserted.

On the independent input tensor product, a and b commute. Substitution of Equation (6.16) in the exact graph link gives

$$a\tau(z) = g b\tau(z)$$

for countably densely many labels simultaneously. Products of the input spectral restrictions for these labels are now legitimate, since all these input observations commute. The map τ is onto and the target action is faithful, so the common zero set of these equations is exactly $g = ab^{-1}$. On compact sets in T^3 , finite clopen partitions and successively more determining labels give the corresponding graph substitution. Exhausting the compact sets and then using normal spectral calculus gives Equation (6.17). Equivalently, for a clopen output cut its range annihilates the complementary input graph cut and conversely; the two identities give the displayed intertwining, and monotone approximation extends it to Borel cuts.

The map from target projectivities to continuous homogeneous boundary maps is continuous and injective, and its inverse onto its image is Borel. Thus the recovered homeomorphism observations and their inverses are measurable in the usual map spaces. The argument uses just the selected coordinate of the link. Other measured coordinates remain in the coefficient spaces, so the same conclusion holds separately on each coordinate of a many-panel link. Commutation between different coordinates or different measured copies will be established in Section 12; it has not been used here. $\square$

6.7. Exact translated laws and uniform upper speed. For each finite-place effective panel group R in the fixed type list, choose a norm on its prime-field adjoint Lie algebra, with $|p| = p^{-1}$ on $\mathbb{Q}_p$. Let ℓ_R be its nonnegative rank-one Cartan gap in the module-absolute-value convention of Section 3. There is a constant $d_R > 0$, depending only on this rooted panel type, such that

$$(6.20) \quad \log \|\text{Ad}(r)\| = d_R \ell_R(r) + O(1), \quad r \in R,$$

with uniform error. On a dominant split translation the largest adjoint eigenvalue has logarithm a fixed positive multiple of the single simple-root coordinate. The conversion from the module absolute value over the coefficient field to the prime-field norm changes this fixed multiple. More explicitly, if k is the coefficient field and $r_{\max} \in \{1, 2\}$ is the largest relative root multiple in this rank-one group, then

$$d_R = \frac{r_{\max}}{[k : \mathbb{Q}_p]}.$$

Indeed $|a|_k = |a|_p^{[k:\mathbb{Q}_p]}$. This prime-field conversion is distinct from the k -root multiplicities in the Haar character 2ρ . Bounded centralizer parts and the compact Cartan factors change operator norms by bounded factors, proving Equation (6.20) for all representatives. The symmetry of the adjoint roots makes d_R unchanged by inversion or opposition. These constants belong to one list on source and target types.

Proposition 6.12 (Uniform finite-place upper speed). *After the inner limit of a translated panel link, let $t \geq 0$ be the total missing-root coordinate of the source translation, including all slower offsets in that coordinate. The translation is taken on the permissible cocharacter mesh of Section 3. If g is the target residual projectivity, then*

$$(6.21) \quad d_T \ell_T(g) \leq d_S t + O_{\text{tight}}(1).$$

Here tightness is on each fixed ordinary input vector, uniformly in the slower translations with this total coordinate as center. The assertion holds coordinatewise, and simultaneously as an upper-tail estimate, on many-panel links.

Proof. Use the convention in which the translated start observation is $P_z^{(t)} = P_{s_t z}$ on its original input space and the end observation is Q_z . Thus s_t is the effective start-label transport; it may be the inverse of the actual completing displacement, and $\ell_S(s_t) = t$ in either orientation. The permissible mesh ensures that s_t belongs to the root-generated group S , so $\eta(s_t)$ is defined. The exact translated-input identity of Proposition 5.13 and Proposition 6.11 now give

$$a\tau(s_t z) = g b\tau(z).$$

Since $\tau(s_t z) = \eta(s_t)\tau(z)$, this identifies

$$(6.22) \quad g = a\eta(s_t)b^{-1},$$

where s_t is deterministic with total missing gap t . The Hilbert-space transport uses the actual source element or its actual cover; only its effective label action occurs in Equation (6.22).

Crucially, a and b in this equation are the observations on the original independent input spaces. Source covariance moves the entire translation dependence into the middle factor. All slower translations with nonzero missing coordinate have already been combined in s_t ; replacing a completing time t_0 by $t_0 + u$ replaces this factor by $\eta(s_{t_0+u})$. Other split translations have zero coordinate in this residue and act trivially on its observations. Any remaining bounded effective actions range in a fixed compact set and can be included in the compact multipliers. The sharp endpoint bases and their compact frame functions retain their fixed input laws throughout these translations. These statements hold after the inner limit for each fixed choice of the slower parameters.

With $A = d\eta$, Equation (6.19) gives

$$\text{Ad}(\eta(s_t)) = A \text{Ad}(s_t) A^{-1}, \quad \log \|\text{Ad}(\eta(s_t))\| \leq d_S t + C.$$

The same estimate holds for s_t^{-1} . Fix an input vector $\xi \in \bar{I} \otimes H'$ and $\varepsilon > 0$. The commuting map observations a, b have a joint scalar spectral measure on this vector. Choose compact sets containing both observations except for squared input norm at most $\varepsilon \|\xi\|^2$. On that joint cut, $\|\text{Ad}(a)\|$ and $\|\text{Ad}(b^{-1})\|$ have fixed bounds independent of t and all slower parameters. Submultiplicativity, Equation (6.22), and Equation (6.20) yield

$$d_T \ell_T(g) \leq d_S t + C_{\xi, \varepsilon}$$

on this cut, with a constant independent of those parameters. The exact graph intertwining and the isometry of the link show that the excluded output squared norm is at most the excluded input squared norm. This proves Equation (6.21), including for input vectors that are not simple tensors.

For finitely many panel coordinates, apply the same argument to each coordinate's ordinary input laws and sum their exceptional output masses; no commutation of the different input map systems is needed for this upper-tail union bound. All constants are independent of the slower parameters. Consequently the estimate persists through further prescribed iterated limits with nonnegative total coordinate centers. $\square$

At a finite-place non-isolated type i , define the comparison constant $s_i = d_S/d_T$. Opposition leaves it unchanged, and its product around every type cycle is one, because the d_R are drawn from a single type list. Proposition 6.12 is the finite-place upper-speed input for the simultaneous height calculation. The isolated types left by Corollary 6.2 are archimedean and will be treated by the separate estimates below.

7. LENGTH MULTIPLIERS AND RAY LIFTS

This section supplies the length estimate needed for the isolated boundary factors. We construct ray lifts with uniform density and intertwining bounds (Proposition 7.7), then combine them with property (T) averaging to prove addition of rescaled spherical length laws (Proposition 7.13). Addition will let us compare repeated source segments in Section 8, where it rules out total superlinear growth and supplies the length window used to construct the isolated heights.

The isolated components requiring the present estimate are real Lie factors, as established in Sections 5 and 6. Their symmetric spaces are quaternionic hyperbolic spaces of dimension at least two and the Cayley hyperbolic plane. Fix one such space X . Normalize its metric so that $r(g) = d(o, go)$ is the logarithm of the dilation in the indivisible restricted root. The argument uses an entire ambient factor; a rank-one residue inside a larger factor does not supply this axis.

All other measured coordinates, including all finite places, remain in the coefficient algebra. Fourier indices remain distinct even when their projections to $\text{Isom}(X)$ coincide. The analytic estimates use only this real projection, whereas the property-(T) averaging below uses the whole ambient product or its lattice. No projected discreteness or one-factor counting assumption enters these estimates.

Constants may depend on X and the specified smooth functions, but not on the number of Fourier indices, scalar twists, or coefficient-algebra dimensions. The analytic estimates are stated homogeneously in the operator and Hilbert norms, so arbitrary finite coefficient traces are allowed there. When an assertion is stated uniformly along a sequence, every displayed Hilbert bound is part of its hypotheses. For a fixed finite algebra and fixed matrix size, an operator bound implies the corresponding Hilbert bound. All varying finite factors used for spherical laws below have normalized trace, and the quotient base has probability measure.

7.1. Coordinates and a uniform smooth calculus. Write the horospherical group as $N_b = V \times Z$, with multiplication

$$(v, z)(v', z') = (v + v', z + z' + \tfrac{1}{2}[v, v']), \quad \delta_R(v, z) = (Rv, R^2z).$$

The inner products can be chosen so that $\langle J_z v, w \rangle = \langle z, [v, w] \rangle$ and $J_z^2 = -|z|^2 \text{id}$. The horizontal layer is bracket-generating: if z is orthogonal to all brackets, then $J_z = 0$, and the displayed identity forces $z = 0$. Set $Q = \dim V + 2 \dim Z$. In half-space coordinates $(n, \lambda) \in N_b \times (0, \infty)$, the metric and volume, with a fixed normalization of Haar measure, are

$$(7.1) \quad ds_X^2 = \frac{d\lambda^2}{\lambda^2} + \frac{|dv|^2}{4\lambda^2} + \frac{|dz - \frac{1}{2}[v, dv]|^2}{4\lambda^4}, \quad dp = \frac{dn d\lambda}{\lambda^{Q+1}}.$$

These are the Iwasawa coordinates for the indicated symmetric spaces; their two root spaces are precisely V and Z . In the conventions of [23, Section 1, Proposition 1], set $a = \lambda^2$ and divide the metric tensor by four to obtain Equation (7.1). In particular a vertical dilation has length $|\log R|$. The H-type description and the boundary geometry are developed in [20, 21, 51].

The horizontal term in (7.1) determines the canonical horizontal conformal class on ∂X . The chart changes induced by isometries preserve this class [51, Lemma 9.6]; the metric comparison in Lemma 7.1 below verifies this directly in coordinates. Choose a smooth global representative and average its pullbacks over the maximal compact stabilizer of o . All the averaged metrics lie in the same conformal class, so their average is a metric in that class and is compact-invariant. We use this metric throughout and let $d_{\partial X}$ be its Carnot–Carathéodory distance. In a buffered horospherical chart this distance is comparable to $(|v|^4/16 + |z|^2)^{1/4}$, where $(v, z) = \log(n^{-1}n')$.

Here and below a scale box means a translate of the dilation by R of a fixed bounded box in exponential coordinates, with a fixed larger box available for smooth cutoffs. On the compact boundary we use finitely many such coordinate charts. Bounded overlap and bounded adjacency refer to constants for these fixed larger boxes. A normalized C^m bound is the ordinary C^m bound after translation and dilation to the fixed box. Auxiliary variables ranging in a fixed compact set may be appended.

Lemma 7.1 (Distance and changes of scale). *The following assertions hold with constants uniform on buffered compact coordinate charts and, for every integer $m \geq 0$, in normalized C^m norms.*

(i) *If $(v, z) = \log(n^{-1}n')$, then*

$$(7.2) \quad \cosh d((n, \lambda), (n', \lambda')) = \frac{\sqrt{(\lambda^2 + \lambda'^2 + |v|^2/4)^2 + |z|^2}}{2\lambda\lambda'}.$$

(ii) *Changes between the chosen compactly rotated horospherical boundary charts preserve normalized bounds on boxes of every scale $0 < R \leq 1$. The same assertion holds in the adjoined height variable $0 \leq \lambda \leq CR$. The conversion between the outward endpoint and e^{-r} , and finite half-space coordinates, has the form*

$$(7.3) \quad n = n_0(e^{-2r}U, e^{-4r}W), \quad \lambda = e^{-r}b,$$

where U, W, b are smooth in the endpoint and e^{-r} , and b is bounded above and below by positive constants.

(iii) *Let g have length L , with pole g^- and image endpoint g^+ . An input annulus of size $q = e^{-u}$, where size includes height, maps to an output annulus of size $q' = e^{-L+u}$, for $0 \leq u \leq L$. On these annuli its normalized derivatives are bounded. A smaller boundary box of radius $\delta \leq cq$, whose height is at most $c\delta$, has image radius at most $Ce^{2u-L}\delta$. After normalization at input scale δ and output scale $e^{2u-L}\delta$, all prescribed derivatives, including the height derivatives, are uniformly bounded. Exterior pieces $q \geq c$ have an output upper bound Ce^{-L} .*

Proof. For completeness, the distance formula can be checked directly from the metric. Translate and dilate the first point to $(0, 1)$, put $A = \lambda^2 + 1 + |v|^2/4$, $C = A^2 + |z|^2$, and $c = \sqrt{C}/(2\lambda)$. In the orthonormal frame of (7.1), the three components of the gradient of c are

$$\frac{\lambda A}{\sqrt{C}} - \frac{\sqrt{C}}{2\lambda}, \quad \frac{Av + J_z v}{2\sqrt{C}}, \quad \frac{\lambda z}{\sqrt{C}}.$$

Since $J_z v \perp v$ and $|J_z v|^2 = |z|^2|v|^2$, the sum of their squared norms is $c^2 - 1$. Thus $\operatorname{arcosh} c$ has gradient norm one away from its unique zero $(0, 1)$. It is proper. Its negative gradient curves reach that zero in time $\operatorname{arcosh} c$; conversely its gradient bound bounds its variation by the length of any curve. It is therefore the distance from $(0, 1)$, proving (i).

One convenient Weyl inversion, in these coordinates [56, Section 2.4], is

$$(7.4) \quad j(v, z, \lambda) = \left(\frac{(-A_0 + J_z)v}{B_0}, -\frac{z}{B_0}, \frac{\lambda}{\sqrt{B_0}} \right), \quad A_0 = \lambda^2 + |v|^2/4, \quad B_0 = A_0^2 + |z|^2.$$

The identity $j\delta_R = \delta_{R^{-1}}j$ follows by substitution. On a compact annulus defined by $(|v|^4/16 + |z|^2)^{1/4} + \lambda \asymp 1$, the denominator is bounded away from zero. All derivatives of the displayed formula are therefore bounded, including at height zero, and its height ratio is positive and bounded. These statements apply to the Iwasawa groups of the symmetric spaces under consideration; the inversion is their Weyl action. No assertion about arbitrary nonsymmetric two-step groups is needed.

We spell out the coordinate issue in (ii). In a finite Bruhat cell the Gauss factors of a compact rotation times a horospherical translation are smooth, with bounded derivatives on buffered compact sets. Write them as $n'ma_b\bar{n}$. Passing $\bar{n}$ across $a_{\log \lambda}$ replaces its negative-root coordinates by λY_{-1} and $\lambda^2 Y_{-2}$. At o the first of these tangent directions has no component in the central horospherical tangent space. Taylor's formula consequently gives horizontal coordinate $O(\lambda)$ and central coordinate $O(\lambda^2)$, with smooth quotients, before the outer dilation by $b\lambda$. After that dilation the two errors are $\lambda^2 U$ and $\lambda^4 W$. This proves (7.3), including bounded derivatives and a positive height ratio.

Boundary transitions are contact maps: a nonzero central image of a horizontal tangent would, by (7.1), turn growth of order λ^{-2} for the squared metric into growth of order λ^{-4} , contradicting invariance of the metric. After the contact conclusion, comparison of the leading λ^{-2} terms shows that the horizontal derivative is a positive dilation followed by an orthogonal map. Thus the boundary action is horizontal conformal. After recentring both charts, their central component therefore has no linear horizontal term. Substitute $(v, z, \lambda) = (RV, R^2 Z, Rh)$ in Taylor's formula. The horizontal component is divisible by R , the central component by R^2 , and the height by R , with bounded derivatives of every prescribed order. The same argument applied to the inverse gives (ii). The stronger errors in (7.3) also remain bounded after this substitution.

In compactly rotated pole and image charts, g is $\delta_{e^{-L}}j$, up to compact isotropy. Rescale the input annulus by q^{-1} and the output by $(q')^{-1}$. The resulting map is exactly the restriction of j to a fixed buffered annulus. The derivative and height-ratio conclusions follow from (7.4). Applying (ii) to a box of relative radius δ/q gives the last assertion of (iii). A finite cover away from the pole proves the exterior assertion. $\square$

The horizontal metric also fixes the exact volume convention used later. Let $\mathcal{V}$ be the horizontal bundle and give $T\partial X/\mathcal{V}$ the quotient metric induced by the surjective bracket map

$$\bigwedge^2 \mathcal{V} \longrightarrow T\partial X/\mathcal{V}.$$

The product of the horizontal and quotient volume densities is a smooth density on ∂X , independent of a splitting of its tangent bundle. An isometry of X acts on $\mathcal{V}_x$ by a dilation $d > 0$ followed by an isometry. Naturality of the bracket makes its induced map on the quotient a dilation d^2 followed by an isometry. Its Jacobian for this density is consequently

$$d^{\dim V} d^{2 \dim Z} = d^Q.$$

The density is compact-invariant. Since the maximal compact group is transitive on the boundary, its normalization to mass one is exactly the compact angular probability. Thus the horizontal dilation has angular Jacobian equal to its Q th power. Here $Q = \dim V + 2 \dim Z$ is the homogeneous dimension; the ordinary boundary dimension is $\dim V + \dim Z$.

Lemma 7.2 (Triangle defect). *There is a constant C such that, if $x, y \in X$ and $b = d(o, x) + d(o, y) - d(x, y) > 0$, their outward endpoints satisfy*

$$d_{\partial X}(x^+, y^+) \leq C e^{-b/2}.$$

Consequently, if $r(g) + r(h) - r(gh) \geq b > 0$, then $d_{\partial X}(g^-, h^+) \leq C e^{-b/2}$.

Proof. Both radii are at least $b/2$, by the triangle inequality. Choose a compactly rotated finite chart containing both endpoints in its buffered interior. Finitely many such choices cover the compact space of endpoint pairs, so their constants are uniform. Let $\delta = d_{\partial X}(x^+, y^+)$. The heights of x, y are comparable to their radial exponentials, and their boundary coordinate errors are at most $Ce^{-\min(d(o,x), d(o,y))}$, by (7.3). If δ is at most a fixed multiple of this quantity the assertion follows. Otherwise the homogeneous separation of their current-point horizontal coordinates is comparable to δ , and (7.2) gives

$$d(x, y) \geq d(o, x) + d(o, y) + 2 \log \delta - C.$$

Rearranging proves the assertion. For the last statement apply it to $x = g^{-1}o$, $y = ho$. $\square$

Lemma 7.3 (Smooth separation and cells). *Fix the dimensions of the row and column coordinate spaces and a bound on the number of auxiliary compact variables. For every prescribed summable power N , there are an integer m and a constant C with the following properties.*

- (i) *A smooth symbol on the product of two fixed buffered coordinate boxes, with C^m norm at most A , has a separated Fourier expansion with coefficients bounded by $CA(1 + |\nu| + |\nu'|)^{-N}$. Unary factors have absolute value at most one and are supported in the respective larger boxes. This also holds for unary symbols, with an arbitrary external parameter in their coefficients.*
- (ii) *If the pairs of boxes supporting a kernel form a graph of degree at most b , and the normalized extensions in (i) have bounds A , the kernel is a Schur multiplier of norm at most $C_b A$. Its matrix amplifications have the same bound. The number of boxes is irrelevant.*

Proof. Multiply an extension by fixed cutoffs and view it as a smooth function on a fixed torus. Integration by parts with $(1 - \Delta)^k$ gives (i) whenever $2k \geq N$; choose N larger than the dimension plus two. At a later use requiring derivatives of the unary factors, increase N by the required derivative order. Thus every assertion here uses a finite, specified number of derivatives, chosen before any scale tends to infinity.

For a fixed Fourier term and a matching of row and column boxes, the kernel is a sum of disjoint rectangular rank-one kernels and has Schur norm bounded by the coefficient bound times the overlap constant. A graph of degree b is the union of at most $2b - 1$ matchings: greedily color its edges, first for finite subgraphs and then pass to a subsequence of the finite colorings. Split boundedly overlapping supports by assigning each point to one of its boundedly many boxes; the assignment is a row or column multiplier of norm one. Summing the matchings and the absolutely convergent Fourier series proves (ii). Each rank-one estimate is a factorization estimate and is unchanged under matrix amplification. $\square$

7.2. Uniform radial multipliers. Completely bounded radial approximations on these rank-one groups are classical [22]. The fixed profiles and uniform coefficient-algebra estimates used below require the explicit smooth separation argument that follows. The search for a uniform fixed-profile radial Schur estimate was motivated in part by Ozawa's bounded-degree hyperbolic-graph estimates [50, Theorem 1]. Those graph-metric estimates are not applied here; the exact continuous kernel is treated by the Carnot-cell argument below.

Proposition 7.4 (Radial cutoffs). *Let $f \in C^\infty([0, \infty))$ be constant on a neighborhood of zero and on a neighborhood of infinity. There are $R_f, C_f > 0$ such that the kernel*

$$(x, x') \longmapsto f(d(x, x')/R)$$

has Schur multiplier norm at most C_f for every $R \geq R_f$. Consequently, if a countable group Γ has any homomorphism into $\text{Isom}(X)$, η is a normalized scalar cocycle, and B is a finite tracial von Neumann algebra, then

$$D_{f(r/R)} : B \bar{\otimes} L_\eta(\Gamma) \longrightarrow B \bar{\otimes} L_\eta(\Gamma), \quad \sum_h Y_h u_h \longmapsto \sum_h f(r(h)/R) Y_h u_h$$

has completely bounded norm at most C_f . The constants are independent of Γ, η, B and of the homomorphism.

Proof. Subtract the constant value of f near infinity. Choose $0 < a < b$ such that f is constant on $[0, 2a]$ and vanishes on $[b, \infty)$. Write $s = -\log \lambda$, $s' = -\log \lambda'$. Since $|s - s'| \leq d(x, x')$, only a fixed number of offset bands between height blocks of length R can contribute. At a fixed offset the block pairs are a matching. Simultaneous dilation permits us to work with $|s|, |s'| \leq C_b R$ in each pair.

Take smooth radial cutoffs $\chi_j(n^{-1}n')$ at radii $\epsilon_j = e^{-j}$, equal to one on a smaller ball, and choose $J = \lceil C'_b R \rceil$, with $C'_b > C_b + b/2 + 2$. Formula (7.2) then implies that $\chi_{-J} = 1$ throughout the support of $f(d/R)$. The annuli $\chi_j - \chi_{j+1}$, $-J \leq j < J$, and the innermost ball χ_J cover the support of the kernel. On each annulus set

$$H_j = \log \frac{\max(\epsilon_j, \lambda, \lambda')^2}{\lambda \lambda'}.$$

The following estimates are uniform in j, R : the kernel $f(d/R) - f(H_j/R)$, restricted to that annulus, has Schur norm $O(R^{-1})$; the same holds on the innermost ball; and the height kernel $f(H_j/R) - f(H_{j-1}/R)$ has Schur norm $O(R^{-1})$. We verify the details, since an uncontrolled height transition would invalidate the assertion.

Put $(v, z) = \delta_{\epsilon_j}(V, Z)$. On an annulus (V, Z) lies in a fixed compact set separated from zero. Each ratio involving only one height and the fixed scale ϵ_j is adjoined as an independent compact unary coordinate before Fourier separation; derivatives in $s/R, s'/R$ keep these auxiliary coordinates fixed. If $\lambda, \lambda' \leq \epsilon_j$, put $\alpha = \lambda/\epsilon_j$, $\beta = \lambda'/\epsilon_j$. Equation (7.2) gives

$$2 \cosh d = e^{H_j} \sqrt{P}, \quad P = (\alpha^2 + \beta^2 + |V|^2/4)^2 + |Z|^2.$$

Here P is bounded above and below by positive constants. If $\lambda \geq \epsilon_j \geq \lambda'$, instead put $\alpha = \epsilon_j/\lambda$, $\beta = \lambda'/\epsilon_j$; the same formula holds with

$$H_j = \log(\lambda/\lambda'), \quad P = (1 + (\alpha\beta)^2 + \alpha^2|V|^2/4)^2 + \alpha^4|Z|^2.$$

Again P and all its derivatives are bounded and $P \geq 1$. The reversed mixed case is identical. The four cuts made so far are rectangular height cuts; their Schur norms are one.

If both heights exceed ϵ_j , distance differs by a bounded amount from $|s - s'|$. Where $|s - s'| \leq aR$, both arguments of f are in its constant interval for sufficiently large R . Outside this interval use smooth sign cutoffs in $(s - s')/R$. On the positive piece, for example, take the larger height as the normalizing scale. The formula for P is the preceding one with the ratio of the smaller height to the larger height in place of $\alpha\beta$. In the bounded variables $s/R, s'/R$ that ratio is $e^{-R|s/R - s'/R|}$, separated from its nonsmooth transition at zero. Every fixed derivative is uniformly bounded, because $R^m e^{-aR}$ is bounded. The sign cutoffs themselves have uniformly bounded derivatives and hence bounded Schur norm by smooth separation.

In each of these cases, when $H_j \geq aR$, write explicitly

$$d - H_j = \frac{1}{2} \log P + \log \left(\frac{1 + \sqrt{1 - 4e^{-2H_j}/P}}{2} \right).$$

The last function is evaluated in a compact interval strictly inside its domain of smoothness. It has bounded derivatives in the compact ratio variables just described and in $s/R, s'/R$. Where H_j is smaller the multiplier difference is zero, after a smooth transition still inside the constant interval of f . Thus

$$f(d/R) - f(H_j/R) = \frac{d - H_j}{R} \int_0^1 f'(H_j/R + t(d - H_j)/R) dt$$

has normalized C^m norm at most C_m/R . Use adjacent ϵ_j -cells for n, n' . Their relative coordinates (V, Z) have uniform bounds. Lemma 7.3 gives the required Schur estimate. The innermost ball requires only the large-height cases: J was chosen larger than all the height exponents. Past the exterior cutoff the distance is at least bR .

For the height increments use

$$H_j = s + s' - 2 \min(j, s, s').$$

Split each of s, s' into $(-\infty, j-1]$, $[j-1, j]$, and $[j, \infty)$. If either is in the first interval, the increment vanishes. If both are in the last interval, H_j and H_{j-1} differ by exactly two; their scaled f -difference has C^m norm $O(R^{-1})$ in $s/R, s'/R$. If exactly one variable lies in the middle interval, use its unit-scale coordinate and the other variable divided by R , with the same conclusion. If both lie in the middle interval, both arguments of f are in its constant interval and the difference is zero. These are rectangles, so no triangular projection on height indices occurs.

Finally write $\phi_j = f(H_j/R)$. The exact telescoping identity is

$$\sum_{j=-J}^{J-1} \phi_j (\chi_j - \chi_{j+1}) + \phi_J \chi_J = \phi_{-J} \chi_{-J} + \sum_{j=-J+1}^J (\phi_j - \phi_{j-1}) \chi_j.$$

Our choice of C'_b also gives $H_{-J} > bR$, so $\phi_{-J} = 0$. Each ball kernel has uniformly bounded Schur norm: its two horizontal entries belong to adjacent scale cells and its normalized symbol is smooth. There are $O(R)$ increments and $O(R)$ annular errors, each costing $O(R^{-1})$. Summing the fixed number of height-block bands proves the Schur bound.

Restrict the kernel to the orbit points $h^{-1}o$. The regular matrix of the Fourier multiplier is its entrywise multiplication kernel, up to the scalar phases already present in the twisted regular representation. The factorization proof of the Schur bound is unchanged by these phases or by tensoring with B . Coincident orbit points cause no difficulty. This proves the last assertion, including complete boundedness. $\square$

7.3. One-sided localization. Let $D = B \bar{\otimes} L_\eta(\Gamma)$, with its product trace τ . Write $Y_h \in L^2(B)$ for the Fourier coefficients of Y . For a bounded scalar function φ on Γ , D_φ denotes coefficient multiplication on $L^2(D)$; it need not be bounded on D . Phases in the coefficient formula for YY^* have absolute value one. For finite Fourier sums that formula is

$$(YY^*)_g = \sum_h \eta(gh, h^{-1}) \overline{\eta(h, h^{-1})} Y_{gh} Y_h^*.$$

All coefficient identities below are first understood for such sums.

Lemma 7.5 (Unary Plancherel estimate). *Suppose $\varphi_{i,\nu} : \Gamma \rightarrow \mathbb{C}$, with i in a countable set, satisfy*

$$\sup_h \sum_i |\varphi_{i,\nu}(h)|^2 \leq b$$

for every ν . Let T_g be an arbitrary family of indices, and associate $i(g, t)$ to each $t \in T_g$, with at most b occurrences of any given i for a fixed g . Suppose

$$z_{g,t} = \sum_\nu c_{g,t,\nu} [Y(D_{\varphi_{i(g,t),\nu}} Y)^*]_g, \quad |c_{g,t,\nu}| \leq A a_\nu, \quad \sum_\nu a_\nu < \infty.$$

Then

$$(7.5) \quad \left(\sum_g \sum_{t \in T_g} \|z_{g,t}\|_{2,B}^2 \right)^{1/2} \leq bA \left(\sum_\nu a_\nu \right) \|Y\|_\infty \|Y\|_2.$$

There is an identical estimate when the localized factor is the first factor of YY^ . In particular coefficients, admissibility of a cell, and the choice of finitely many companion cells may depend on g without affecting the estimate.*

Proof. For one ν , discard the admissibility restriction after using the multiplicity bound. Plancherel and the operator norm bound for the uncut factor give

$$\begin{aligned} \sum_{g,t} \| [Y(D_{\varphi_{i(g,t),\nu}} Y)^*]_g \|_2^2 &\leq b \sum_i \| Y(D_{\varphi_{i,\nu}} Y)^* \|_2^2 \\ &\leq b \| Y \|_\infty^2 \sum_i \| D_{\varphi_{i,\nu}} Y \|_2^2 \leq b^2 \| Y \|_\infty^2 \| Y \|_2^2. \end{aligned}$$

Minkowski's inequality sums the absolutely summable frequencies. Taking adjoints proves the other orientation. Notice that no operator norm for a localized column was used. $\square$

Lemma 7.6 (Angular column estimate). *There are constants $c, C > 0$ and an integer m , depending only on X , with the following property. Put $\delta = e^{-s}$, $s \geq 1$. Let $(f_C)_C$ be smooth masks supported on buffered boundary boxes of radius δ , with fixed overlap and normalized C^m bounds. If $Y \in D$ has Fourier support in $\{r \geq l\}$, where $l \geq Cs + C$, then, with $A_C = D_{f_C(h^+)} Y$,*

$$(7.6) \quad \sum_{C,D'} \| A_C A_{D'}^* \|_2^2 = \left\| \sum_C A_C^* A_C \right\|_2^2 \leq C \| Y \|_\infty^2 \| Y \|_2^2.$$

The constants can depend on the stated mask bounds, but not on their number. The assertion holds with arbitrary finite tracial coefficient algebras and with any scalar twist.

Proof. We estimate the coefficients of $A_C A_{D'}^*$ at a fixed shift g . Put $L = r(g)$. Their weights in the coefficient formula for $Y Y^*$ are

$$(7.7) \quad f_C((gh)^+) \overline{f_{D'}(h^+)}.$$

Only pairs with both h and gh in the long support matter. Their polar heights are at most e^{-l} . We describe the smooth extensions of these weights; all coordinate boxes below come from fixed covers independent of g .

First suppose that the input box is separated from g^- by at least a sufficiently large multiple of δ , and its pole distance is $q = e^{-u}$. If $2u \leq L + C_1$, Lemma 7.1(iii) says that the map from this box to the output boundary, including the extra coordinate $e^{-r(h)}/\delta$, has normalized derivatives bounded by a constant. Its image meets only boundedly many output boxes. Extend on a fixed larger box, multiply by its buffered cutoff, and Fourier-expand there. The weight in (7.7) is now a unary expansion in h , with coefficients decaying faster than any prescribed fixed power. The basis functions depend on D' , the fixed coordinate chart, and the frequency, and not on g or C . If $2u > L + C_1$, inversion gives output pole distance e^{-L+u} ; unless the output box is itself a pole box, the inverse map has the same bounded-magnification property. Use the reversed unary estimate in that case. A box intersecting the transition $2u = L + O(1)$ admits either construction, with a larger fixed buffer. Starting from an output box gives the same alternatives. Bounded L is included by increasing the constants.

Thus every pair except pairs of boxes lying within $C_2\delta$ of the two respective poles is handled by Lemma 7.5. To justify the multiplicity used there, a map with bounded magnification sends one buffered δ -box into a ball of radius $C\delta$, which meets a bounded number of reference boxes. The height variable changes this radius by at most Ce^{-l} . This proves the required multiplicity for each g and initial box, without a restriction on the number of group elements in it.

There remain only boundedly many exceptional pairs for each g . If such a pair is compatible with the long supports, the two current-point pole sizes have product comparable to e^{-L} , by (7.4), whereas each is at most $C\delta$. Consequently

$$(7.8) \quad L \geq 2s - C_3.$$

Expand each mask as its value at its pole plus its difference from that value. The constant term costs at most $C \| Y Y^* \|_2$. Terms with just one nonconstant mask are covered by Lemma 7.5; any additional constant term created by this expansion is treated with the first term.

For the product of the two differences, partition the input current-point pole coordinate, including height, into smooth unit-width annuli with size e^{-u} , $0 \leq u \leq L/2 + C_4$. Use the analogous output annuli on the remaining part. A smooth transition at the scale $e^{-L/2}$ has bounded normalized derivatives on both descriptions, by (7.4). Exterior annuli mean a fixed finite cover away from the pole.

On an input annulus, the two differences, after substitution, have product seminorm bounded by

$$(7.9) \quad C_m \min(1, e^{s-u}) \min(1, e^{s-L+u}) \leq C'_m e^{-|s-u|}.$$

Here is the derivative justification for both factors. On scales smaller than a mask box, its difference from its pole value is bounded by the relative scale, by Taylor's formula; the same bound holds for every normalized derivative on the smaller box. On scales larger than the mask box by a fixed factor, the mask is zero on the long support, because its current-point height is much smaller than δ . It can therefore be replaced by zero on that part before subtracting its pole value, giving a bounded constant symbol. At comparable scales its given seminorm bound suffices. The output factor has exactly the same three descriptions after the normalized inversion, with scale e^{-L+u} . The collar substitutions have bounded derivatives including the normalized height by Lemma 7.1(ii). These constructions give smooth extensions on buffered boxes; equality is required only on the two long supports. Finally, if $u \leq s$, use (7.8) in the second minimum; if $u \geq s$, use the first minimum. This proves the last inequality in (7.9).

At a fixed integer depth u , use the reference boundary boxes of radius e^{-u} , and append the height divided by e^{-u} as an auxiliary unary variable. Only boundedly many such boxes meet the annulus about the given pole. Their basis functions are chosen before g ; the coordinates of the pole merely change the expansion coefficients. The number of original exceptional pairs is also bounded. Lemmas 7.3 and 7.5 therefore give a bound $Ce^{-|s-u|}\|Y\|_\infty\|Y\|_2$ in the square sum of all coefficients using this depth. Sum over u in norm; the geometric series has a uniform sum. The reversed annuli give the same bound. Combining the nonexceptional and exceptional estimates proves the inequality in (7.6). Its equality is the trace identity $\tau(A_C A_{D'}^* A_{D'} A_C^*) = \tau(A_C^* A_C A_{D'}^* A_{D'})$, summed over the masks.

For general Y , choose bounded Fourier-polynomial approximations in the strong-star topology and apply a smooth radial cutoff equal to one on $\{r \geq l\}$ and zero on $\{r \leq l/2\}$. Proposition 7.4 bounds these approximants uniformly. Increase the constant in the hypothesis on l to allow this slack. Each masked column converges in L^2 , so each product of two masked columns converges in L^1 , by tracial Cauchy-Schwarz. For a fixed finite family of pairs, the preceding estimate bounds their products in the Hilbert direct sum of $L^2(D)$. Weak compactness gives a weak L^2 limit, and the L^1 convergence identifies it with the desired products. Lower semicontinuity proves the bound for this finite family; exhaustion proves it for all pairs. In particular the diagonal terms put every masked column in L^4 , so the trace identity in (7.6) remains valid. This also explains why neither infinite Fourier support nor a nondiscrete projected group changes the proof. $\square$

7.4. The ray lift and its densities. Let $\mathcal{K}_X = L^2(X, dp)$, and let T_g be translation on $\mathcal{K}_X$. Define

$$\Delta : D \longrightarrow B(\mathcal{K}_X) \bar{\otimes} D, \quad \Delta(bu_h) = T_h \otimes bu_h.$$

This is a normal unital homomorphism. Indeed the unitary on $\mathcal{K}_X \otimes \ell^2 \Gamma$ that sends $\zeta \otimes \delta_h$ to $T_h^{-1} \zeta \otimes \delta_h$ conjugates $T_g \otimes u_g$ to $1 \otimes u_g$, with the same scalar twist. Tensoring with the coefficient representation proves normality in the displayed algebra. The right action of D on $\mathcal{K}_X \otimes L^2(D)$ acts on its second factor. Choose real smooth functions ψ and F , with $\text{supp } \psi \Subset (1, 2)$, F compactly supported on N_b and invariant under the compact isotropy of a ray, such that $\int \psi^2 = \int F^2 = 1$. For $d_1 \geq 1$, a basepoint x , and an ideal endpoint v , use a

chart sending v to infinity and put

$$(7.10) \quad \theta_{x,d_1}(p, v) = d_1^{-1/2} \psi \left(\frac{\log(\lambda_p/\lambda_x)}{d_1} \right) F(\delta_{\lambda_p^{-1}}(n_x^{-1}n_p)).$$

Changing that chart by a horospherical translation, dilation, or compact ray isotropy leaves the formula unchanged. It is therefore equivariant in x, p, v . The substitution $n_p = n_x \delta_{\lambda_p} q$, $t = \log(\lambda_p/\lambda_x)/d_1$ shows that $\int \theta_{x,d_1}(p, v)^2 dp = 1$. Its support stays a bounded distance from the ray from x to v , at depths between $d_1 - O(1)$ and $2d_1 + O(1)$. The coefficient isometry is

$$\mathbb{V}_{d_1} Y = \sum_h \theta_{o,d_1}(\cdot, h^+) \otimes Y_h u_h.$$

At a zero-displacement index choose any fixed direction; none of the long-support assertions below depends on that choice.

For a column $\xi \in \mathcal{K}_X \otimes L^2(D)$, its left and right densities are the positive $L^1(D)$ elements representing the vector functionals for the actions $\Delta(D)$ and D^{op} , respectively. Thus

$$\langle \xi, \Delta(x)\xi \rangle = \tau(\rho_\xi^L x), \quad \langle \xi, \xi x \rangle = \tau(\rho_\xi^R x), \quad \rho_\xi^R = \xi^* \xi.$$

The density on the whole amplified left algebra is $\xi \xi^*$, with its product semifinite trace. The following proposition puts all three densities in trace L^2 .

Proposition 7.7 (Ray lift). *Consider any sequence of finite tracial coefficient algebras, twists, and countable group homomorphisms into $\text{Isom}(X)$. Let $d_1 \rightarrow \infty$, $l/d_1 \rightarrow \infty$, and let $Y \in D$ have Fourier support in $\{r \geq l\}$. Then the following statements hold. For (ii), assume that $\|x\|_\infty, \|x\|_2, \|Y\|_\infty, \|Y\|_2$ remain bounded; its convergence is uniform under these bounds. For (iii), assume only the three norm bounds stated there.*

(i) *For all sufficiently large indices, the vector $\mathbb{V}_{d_1} Y$ has left density on D , for the action Δ , and right density on D^{op} , both in L^2 . Their norms are at most $C\|Y\|_\infty\|Y\|_2$. Also $\|\mathbb{V}_{d_1} Y (\mathbb{V}_{d_1} Y)^*\|_{2, \text{Tr} \otimes \tau}$ has this bound.*

(ii) *If $\text{supp } \hat{x} \subset \{r \leq a\}$, $a/d_1 \rightarrow 0$, then*

$$\|\mathbb{V}_{d_1}(xY) - \Delta(x)\mathbb{V}_{d_1} Y\|_2 \rightarrow 0, \quad \|\mathbb{V}_{d_1}(Yx) - (\mathbb{V}_{d_1} Y)x\|_2 \rightarrow 0.$$

The lift in these formulas uses the same d_1 ; the products have support in $\{r \geq l - a\}$.

(iii) *Suppose $Z \in L^2(D)$ is another column with Fourier support in $\{r \geq l\}$, and $x \in D$ is bounded. Suppose that for every contributing triple $h \in \text{supp } \hat{Y}$, $hk \in \text{supp } \hat{Z}$, $k \in \text{supp } \hat{x}$, one has*

$$d_{\partial X}(h^+, (hk)^+) \leq e^{-c'}, \quad c > 0, \quad l'/d_1 \rightarrow \infty.$$

Then

$$(7.11) \quad \langle \mathbb{V}_{d_1} Z, (\mathbb{V}_{d_1} Y)x \rangle - \langle Z, Yx \rangle \rightarrow 0.$$

The assertion holds uniformly with $\|x\|_\infty, \|Y\|_2, \|Z\|_2$ bounded.

All assertions remain valid for finite matrix versions and for measurable fields, with the indicated operator bounds uniform and the indicated Hilbert bounds taken in the integrated tracial norms.

We first prove the right-density bound in (i), then establish the overlap estimate and the left-density bound, and finally prove (iii) followed by (ii).

Proof of the right-density assertion. Write $K_{d_1}(v, w) = \int \theta_{o,d_1}(p, v) \theta_{o,d_1}(p, w) dp$. Disintegrate this integral at radial depth $s = d(o, p)$. Its range is $[d_1 - C, 2d_1 + C]$. With measure ds/d_1 , the resulting kernels $k_{d_1,s}(v, w)$ are supported on adjacent boundary boxes of radius e^{-s} , with normalized C^m bounds independent of d_1, s , for every fixed m .

Here are the normalization details. The polar volume is $J(s) ds d\nu$, where $J(s) = c(\sinh s)^{\dim V} (\sinh 2s)^{\dim Z} \asymp e^{Qs}$ for $s \geq 1$. The shadow of one ray has angular volume $O(e^{-Qs})$. Formula (7.10) contributes d_1^{-1} , removed by the choice of ds/d_1 . Thus no volume

factor remains. To check derivatives, choose smooth compact frames near one endpoint, translate back along its ray by depth s , and rescale neighboring directions by e^{-s} . Their negative-root coordinates have sizes e^{-s} and e^{-2s} ; conjugation by the split translation makes both bounded. The positive-root coordinates contract and the remaining factors stay compact. Lemma 7.1(ii), followed by (7.10) on this fixed bounded set, therefore bounds every prescribed normalized derivative. The same argument applies to the polar Jacobian after its factor e^{Qs} is removed.

Separate $k_{d_1,s}$ on adjacent boxes. At each frequency the right density is a sum $\sum_{C,D'} t_{C,D'} A_C^* B_{D'}$, where $(t_{C,D'})$ is a scalar matrix of bounded degree and bounded operator norm, and both mask families satisfy the hypotheses of Lemma 7.6. The column form of tracial Hölder gives

$$\left\| \sum_{C,D'} t_{C,D'} A_C^* B_{D'} \right\|_2 \leq \|t\| \left\| \sum_C A_C^* A_C \right\|_2^{1/2} \left\| \sum_{D'} B_{D'}^* B_{D'} \right\|_2^{1/2}.$$

Equation (7.6) bounds this by $C\|Y\|_\infty\|Y\|_2$. Fourier coefficients are summable; choose their decay order larger than the polynomial derivative growth of the masks. Integration in ds/d_1 , over an interval of bounded total mass, proves the right-density bound. Finally

$$\|\mathbb{V}_{d_1} Y (\mathbb{V}_{d_1} Y)^*\|_2^2 = \|(\mathbb{V}_{d_1} Y)^* \mathbb{V}_{d_1} Y\|_2^2$$

by the semifinite trace, first for finite-rank columns and then by monotone approximation. This proves the additional assertion in (i). $\square$

Lemma 7.8 (Common-endpoint overlap). *For a displacement g of length L , put*

$$w_{d_1}(g, v) = \int_X \theta_{o,d_1}(p, v) \theta_{go,d_1}(p, v) dp.$$

There are smooth annular partitions about g^+ , at integer depths $0 \leq j \leq L$, with an exterior piece and an innermost ball, such that on each piece

$$(7.12) \quad \begin{aligned} \|w_{d_1}(g, \cdot) - a_{d_1}(j, L)\|_{C_{\text{normalized}}^m} &\leq C_m/d_1, \\ a_{d_1}(j, L) &= \frac{1}{d_1} \int_{s>j} \psi(s/d_1) \psi((s-2j+L)/d_1) ds. \end{aligned}$$

The constants are uniform in $L, d_1 \geq 1$. Further,

$$(7.13) \quad |\partial_j a_{d_1}(j, L)| \leq C/d_1, \quad a_{d_1}(0, L) = 1 + O(\min(1, L/d_1)).$$

Noninteger endpoints of the last annulus are absorbed in its fixed buffer. When L is bounded, a fixed finite cover suffices.

Proof. We first prove uniformity in all angular derivatives, including at the two ends of the annular decomposition. By a compact rotation put $go = (0, \epsilon)$, $\epsilon = e^{-L}$, and $g^+ = 0$. For a finite endpoint $v = (V, W)$, define

$$p(v) = |V|^2/4, \quad D_h(v) = (h^2 + p(v))^2 + |W|^2, \quad P_h(v) = \left(\frac{(h^2 + p(v) + J_W)V}{D_h(v)}, \frac{W}{D_h(v)} \right).$$

Formula (7.4) shows that $\delta_{\sqrt{D_1(v)}} L_{P_1(v)^{-1} j L_{v^{-1}}}$ fixes o and sends v to infinity. In this chart the displaced point is (n, e^β) , with

$$(7.14) \quad n = \delta_{\sqrt{D_1(v)}} (P_1(v)^{-1} P_\epsilon(v)), \quad e^\beta = \epsilon \sqrt{D_1(v)/D_\epsilon(v)}.$$

All products in this formula are products in the real two-step group N_b ; no nonassociative scalar multiplication is used.

Put $v = \delta_R u$, $R = e^{-j}$, $t = \epsilon/R$, on a fixed buffered annulus for $u = (V, W)$, and write

$$\Delta_R(u) = (1 + R^2 p(u))^2 + R^4 |W|^2,$$

$$Q_R(u) = \left(\frac{R^2(1 + R^2 p(u) + R^2 J_W)V}{\Delta_R(u)}, \frac{R^4 W}{\Delta_R(u)} \right).$$

Direct substitution in (7.14) gives

$$(7.15) \quad \beta = 2j - L + B(R, t, u), \quad B = \frac{1}{2} \log(\Delta_R/D_t), \quad \delta_R n = m(R, t, u) = \delta_{\sqrt{\Delta_R}}(Q_R^{-1} P_t).$$

For $R, t \in [0, 1]$, the denominator D_t is bounded away from zero on the buffered annulus, and $\Delta_R \geq 1$. Thus B, m have bounded derivatives of every fixed order. As $R, t \rightarrow 0$, $m \rightarrow P_0(u)$, uniformly with derivatives; this limit stays away from the identity. Consequently m is uniformly separated from the identity if both j and $L - j$ exceed a fixed constant. For the innermost ball take $j = L, t = 1$: $D_1 \geq 1$, so all upper bounds remain valid, including at $u = 0$. The exterior follows from finitely many compact charts away from the pole, with $\beta = -L + O_{C^m}(1)$, $n = O_{C^m}(1)$.

Let $H(m) = \int F(q) F(m^{-1}q) dq$. Then H is smooth and compactly supported, and $H(e) = 1$. Haar substitution in (7.10) gives the exact formula

$$(7.16) \quad w_{d_1}(g, v) = \int \psi(t) \psi(t - \beta/d_1) H(\delta_{e^{-td_1}} n) dt.$$

On a normalized annulus it becomes

$$\frac{1}{d_1} \int \psi(s/d_1) \psi((s - 2j + L - B(u))/d_1) H(\delta_{e^{j-s}} m(u)) ds.$$

Replace B by zero using the mean-value formula. This costs $O_m(d_1^{-1})$ after angular differentiation: the extra factor is d_1^{-1} , and all derivatives of the accompanying H factor are bounded on the relevant support. Indeed, in the interior annuli, compact support of H and separation of m from the identity force $s - j \geq -C$; near $j = 0$, the first ψ forces $s - j \geq d_1 - C$; near $j = L$, either of the shifted ψ 's in the mean-value formula forces the same inequality. The interval of integration has length $O(d_1)$.

It remains to replace $H(\delta_{e^{j-s}} m(u))$ by $\mathbf{1}_{s > j}$. On the interior annuli, with $r = s - j$, every angular derivative satisfies the integrable bound

$$|\partial_u^\gamma (H(\delta_{e^{-r}} m(u)) - \mathbf{1}_{r > 0})| \leq C_\gamma \left(\mathbf{1}_{[-C, 0]}(r) + e^{-r} \mathbf{1}_{[0, \infty)}(r) \right).$$

For negative r this follows from the support of H ; for positive r it follows from Taylor's formula at the identity, because horizontal and central coordinates and all their u -derivatives acquire factors e^{-r} and e^{-2r} . Integration against ds/d_1 gives $O_m(d_1^{-1})$. In the two end bands the ψ supports already imply $r \geq d_1 - C$, so the same conclusion follows without a lower bound on m . This proves (7.12) on all pieces.

Finally differentiation under the integral gives

$$\partial_j a_{d_1}(j, L) = -\frac{1}{d_1} \psi(j/d_1) \psi((L - j)/d_1) - \frac{2}{d_1^2} \int_j^\infty \psi(s/d_1) \psi'((s - 2j + L)/d_1) ds.$$

This is $O(d_1^{-1})$. At $j = 0$ the integral is $\int \psi(t) \psi(t + L/d_1) dt$; the fundamental theorem of calculus in L^2 proves the last assertion. $\square$

Proof of the left-density assertion in Proposition 7.7. At a shift g , the left density has the coefficients of YY^* weighted, term by term, by the overlap of the two rays based at o, go , towards $(gh)^+, gh^+$. There are no such coefficients unless $L = r(g) \leq 4d_1 + C$: both ray functions are supported within distance $2d_1 + C$ of their basepoints.

We may replace the second endpoint by $(gh)^+$, with an error whose coefficient L^2 norm is at most

$$(7.17) \quad C e^{C'd_1 - l} \|Y\|_\infty \|Y\|_2.$$

To verify this uniform bound, regard h in compact polar coordinates with the additional unary variable $e^{l-r(h)} \in [0, 1]$. Equation (7.3) gives smooth agreement of its current-point direction and ideal endpoint at height zero. Acting by g , whose length is at most $4d_1 + C$, preserves this vanishing and multiplies each fixed derivative bound by at most $e^{C_m d_1}$. One way to see the latter bound directly is to compose at most $4d_1 + C + 1$ isometries of length at most one and differentiate a fixed finite number of times; compact coordinate transitions have bounded derivatives, and the resulting bounds grow at most exponentially. The half-space charts needed at interior depth at most Cd_1 have denominators bounded below by $e^{-C_m d_1}$, as is also explicit in (7.4). Formula (7.10) and the volume on its support have the same fixed-order exponential bounds. The overlap difference therefore has a unary smooth extension with seminorm at most $C_m e^{C_m d_1 - l}$, on each of a fixed finite set of angular charts and the normalized height interval. Fourier separation and Lemma 7.5, with the other factor uncut, give (7.17). There is no summation over the number of shifts in $\{r \leq 4d_1 + C\}$.

For the resulting common-endpoint coefficients, apply Lemma 7.8. At each annular depth, the error is a unary symbol of seminorm $O(d_1^{-1})$ on a bounded number of reference boxes around the pole for that g . Thus its coefficient L^2 norm is at most $Cd_1^{-1} \|Y\|_\infty \|Y\|_2$, by Lemma 7.5. Telescope the constants $a_{d_1}(j, L)$ against single smooth balls. Their increments have the same $O(d_1^{-1})$ bound and are treated by the same unary estimate. The baseline simply multiplies the uncut coefficient by a bounded scalar. There are at most $C(1 + d_1)$ levels because $L \leq 4d_1 + C$. Summing these norms and (7.17) proves the asserted uniform bound for the left density.

We record the stronger short-coefficient conclusion needed below. If $a = o(d_1)$, then on the shifts $r(g) \leq a$ the baseline differs from one by $O(a/d_1)$, and only $C(1 + a)$ levels occur. The left-density coefficients restricted to these shifts therefore differ from those of YY^* by a quantity bounded in L^2 by

$$(7.18) \quad C \left(\frac{1+a}{d_1} + e^{C'd_1-l} \right) \|Y\|_\infty \|Y\|_2.$$

This tends to zero when both displayed norms remain bounded, in the scale regime of Proposition 7.7. For cross densities of two bounded long columns the corresponding statement follows by polarization; the sums and differences of the columns retain the same long support. In particular,

$$(7.19) \quad \langle \mathbb{V}_{d_1} Z, \Delta(x) \mathbb{V}_{d_1} Y \rangle - \langle Z, xY \rangle \longrightarrow 0$$

uniformly when the operator and Hilbert norms of the long columns Y, Z , and the Hilbert norm of x , are bounded, with $\text{supp } \hat{x} \subset \{r \leq a\}$. Explicitly, polarization and Cauchy–Schwarz bound the absolute value of the difference by

$$C \left(\frac{1+a}{d_1} + e^{C'd_1-l} \right) \|x\|_2 (\|Y\|_\infty + \|Z\|_\infty) (\|Y\|_2 + \|Z\|_2).$$

The Cauchy–Schwarz step pairs the restricted density coefficients with the Fourier coefficients of x . In particular it does not replace $\|x\|_2$ by an operator bound on an algebra whose trace mass could vary.

For infinite Fourier support, the bounded approximants with slacked radial cutoffs used after (7.6) apply here as well. The density bounds yield weakly convergent L^2 subsequences, whose trace pairings identify their limits with the vector densities. This proves (i) in full generality. $\square$

Proof of the pairing and intertwining assertions in Proposition 7.7. First prove (iii). The kernel $K_{d_1}(v, w)$ satisfies $K_{d_1}(v, v) = 1$. For each fixed derivative order m , its derivatives on fixed angular charts are bounded by $C_m e^{C_m d_1}$: use smooth compact frames, the depth restriction $d(o, p) \leq 2d_1 + C$, and the same fixed-order coordinate and volume bounds as in (7.17). On a pair of adjacent boxes of radius e^{-d_2} , Taylor’s formula on the diagonal and the

weighted coordinate scaling imply

$$(7.20) \quad \|K_{d_1} - 1\|_{C_{\text{normalized}}^m} \leq C_m e^{C_m d_1 - d_2}.$$

For derivatives of positive order, at least one normalized angular derivative supplies a factor e^{-d_2} ; the zero-order estimate uses the equality on the diagonal. Central derivatives supply the stronger factor e^{-2d_2} .

Choose d_2 with $d_1 \ll d_2 \ll l'$. The assumed support condition makes every contributing pair adjacent at scale e^{-d_2} . Insert a smooth partition at that scale and separate $K_{d_1} - 1$ on each adjacent pair. At one frequency, Cauchy–Schwarz for the columns of cut vectors gives

$$\left| \sum_{C,D'} t_{C,D'} \langle Z_C, Y_{D'} x \rangle \right| \leq \|t\| \|x\|_\infty \left(\sum_C \|Z_C\|_2^2 \right)^{1/2} \left(\sum_{D'} \|Y_{D'}\|_2^2 \right)^{1/2}.$$

The matrix norm is bounded by its degree times its largest coefficient, which tends to zero by (7.20). Overlap bounds control the two sums by $C\|Z\|_2^2, C\|Y\|_2^2$. Choose the finite derivative order in advance sufficiently large for a summable Fourier expansion. Since $d_2/d_1 \rightarrow \infty$, the sum of the frequency bounds still tends to zero. This proves (7.11). The proof only needs $Y, Z \in L^2$; Fourier truncation of those two vectors and continuity extend it with x remaining bounded.

For right multiplication by a short element we verify the support condition. If $r(h), r(hk) \geq l - a$ and $r(k) \leq a$, then $d(ho, hko) \leq a$. In a compact finite boundary chart the heights of these two points are comparable to their radial exponentials. Their horizontal separation, measured by a homogeneous gauge, either is of that small height order or, by (7.2), satisfies

$$d(ho, hko) \geq r(h) + r(hk) + 2 \log d_{\partial X}(h^+, (hk)^+) - C.$$

The collar errors are absorbed in the first alternative. A finite chart cover, or a compact rotation placing the pair in one such chart, gives $d_{\partial X}(h^+, (hk)^+) \leq C e^{-c(l-2a)}$ with fixed $c > 0$. Since $a/d_1 \rightarrow 0$ and $l/d_1 \rightarrow \infty$, assertion (iii) applies. It also applies with xx^* , supported in $\{r \leq 2a\}$, and with the long column Yx . Expanding the square of $\mathbb{V}_{d_1}(Yx) - (\mathbb{V}_{d_1}Y)x$, its three terms tend respectively to $\|Yx\|_2^2, \|Yx\|_2^2, -2\|Yx\|_2^2$. This proves the right assertion in (ii).

For the left assertion use (7.19) first with x^*x , then with the cross pair $Z = xY$ and x . The support of x^*x lies in $\{r \leq 2a\}$, and xY has support in $\{r \geq l - a\}$. The same expansion of the squared norm now proves $\|\mathbb{V}_{d_1}(xY) - \Delta(x)\mathbb{V}_{d_1}Y\|_2 \rightarrow 0$.

All proofs used trace identities, Plancherel, and constants uniform in the coefficient algebra. They therefore apply to matrix algebras and, after integration, to measurable fields. To make any support-preserving approximation needed here, first use bounded Fourier-polynomial approximation and then smooth radial cutoffs with fixed slack. Proposition 7.4 gives uniform operator bounds; doubling a short support and halving a long support preserves all scale separations in the statement. $\square$

Corollary 7.9 (Stability under Hilbert-small changes). *If Y is a long column as above with $\|Y\|_\infty$ and $\|Y\|_2$ uniformly bounded and $x - x' \rightarrow 0$ in $L^2(D)$, with x, x' uniformly bounded, then*

$$\|\Delta(x - x')\mathbb{V}_{d_1}Y\|_2 \rightarrow 0, \quad \|(\mathbb{V}_{d_1}Y)(x - x')\|_2 \rightarrow 0.$$

The analogous statement holds for bounded elements in the amplified left algebra when their difference tends to zero in its semifinite L^2 norm.

Proof. Pair the appropriate density from Proposition 7.7 with $(x - x')^*(x - x')$, or with the reversed product on the right. Tracial Cauchy–Schwarz and

$$\|(x - x')^*(x - x')\|_2 \leq \|x - x'\|_\infty \|x - x'\|_2$$

give, with $z = x - x'$, the homogeneous bound

$$(7.21) \quad \max\{\|\Delta(z)\mathbb{V}_{d_1}Y\|_2^2, \|(\mathbb{V}_{d_1}Y)z\|_2^2\} \leq C\|Y\|_\infty\|Y\|_2\|z\|_\infty\|z\|_2.$$

In the amplified case use the density $\mathbb{V}_{d_1} Y (\mathbb{V}_{d_1} Y)^*$ and the same inequality with the semifinite Hilbert norm of z . For measurable fields, integrate this inequality. Uniform operator bounds and Cauchy–Schwarz on the base give the product of the integrated Hilbert norms of Y and z . Thus integrated Hilbert-small changes suffice; no pointwise error bound is required. $\square$

7.5. Admissible representations and spherical laws. Fix an isolated measured factor T of G , its symmetric space X_T , and a base point o_T . Write

$$r_T(h) = d(o_T, h_T o_T), \quad h \in \Gamma,$$

with distance normalized by the indivisible simple root. The projection $h \mapsto h_T$ need not have discrete image. In this Section D_f denotes the Fourier multiplier with symbol f . All finite factor traces are normalized.

Definition 7.10. An admissible representation for T consists of a finite factor

$$D = B \bar{\otimes} L_\eta(\Gamma)$$

and a normalized projective representation $\Theta : \Gamma \rightarrow \mathcal{U}(D)$ such that

$$D \cap \Theta(\Gamma)' = \mathbb{C}1$$

and the representation

$$(7.22) \quad \gamma \mapsto \text{Ad } \Delta_T(\Theta(\gamma)) \quad \text{on} \quad L^2(B(L^2(X_T)) \bar{\otimes} D, \text{Tr} \otimes \tau)$$

has no invariant vector. Here $\Delta_T(u_h) = \lambda_{X_T}(h_T) \otimes u_h$, with the identity action on B . The scalar multiplier of Θ is not required to equal η .

Keep the section $d_0 : Y = G/\Gamma \rightarrow G$ and cocycle c from Section 3. For $a \in G$, put

$$X_a^\Theta(y) = \Theta(c(a, y)).$$

If S is an isolated source factor and $a_t \in S$ is a positive split translation of length t , the spherical law at t , on scale $R > 0$, is the probability measure

$$(7.23) \quad P_{\Theta, R}(t)(E) = \int_Y \|D_{1_{\{r_T/R \in E\}}} X_{a_t}^\Theta(y)\|_2^2 dy \quad (E \subset [0, \infty] \text{ Borel}).$$

The value $+\infty$ is used only in weak limits. Addition on $[0, \infty]$ is the usual extended addition.

Lemma 7.11 (Uniform short approximation). *Let $R_n \rightarrow \infty$, and let Θ_n be projective representations in finite tensors as above. Suppose*

$$(7.24) \quad \|D_{1_{\{r_T > \varepsilon R_n\}}} \Theta_n(\gamma)\|_2 \rightarrow 0 \quad (\gamma \in \Gamma, \varepsilon > 0).$$

There are $a_n = o(R_n)$ and uniformly operator-bounded fields $Z_{n,s}$, with Fourier support in $\{r_T \leq a_n\}$, such that

$$(7.25) \quad \sup_{s \in C} \|X_s^{\Theta_n} - Z_{n,s}\|_{L^2(Y; L^2(D_n))} \rightarrow 0$$

for every compact $C \subset G$. Consequently a compact family of source translations before or after a segment does not change any limiting rescaled length law of that segment.

Proof. Choose a compact $K \subset G$ for which $|\{y : d_0(y) \notin K\}| < \varepsilon$. Uniformly for $s \in C$, outside a set of measure at most 2ε , both $d_0(y)$ and $d_0(sy)$ belong to K . There

$$c(s, y) \in \Gamma \cap K^{-1}CK,$$

a finite set. Apply (7.24) to this finite set. Diagonalization over compact exhaustions of G , followed by Proposition 7.4, gives (7.25). More explicitly, use a fixed smooth function equal to one on $[0, 1]$ and zero on $[2, \infty)$, at a cutoff $b_n = o(R_n)$ tending sufficiently slowly to infinity relative to the required finite lists, and put $a_n = 2b_n$. The completely bounded multiplier norm gives the uniform operator bound. The normalized factor traces also give $\|X_s^{\Theta_n}(y)\|_2 = 1$; Plancherel bounds the integrated Hilbert norms of the short approximants by

the supremum norm of the fixed cutoff. Thus both kinds of bounds required in Proposition 7.7 are uniform in these applications.

Here is the spectral assertion needed in the last sentence. Multiplication by an element supported in $\{r_T \leq a_n\}$ sends the Fourier support of a radial interval into its a_n -neighborhood, on either side, because

$$(7.26) \quad |r_T(gh) - r_T(h)| \leq r_T(g), \quad |r_T(hg) - r_T(h)| \leq r_T(g).$$

Take two closed rescaled intervals at positive distance. A short multiplier cannot connect their radial cuts for large n . For smooth radial cuts of a uniformly bounded column the errors from replacing an actual transporter by its short approximation are Hilbert-small: one factor is Hilbert-small and the other is uniformly operator-bounded. The radial multiplier bound supplies the latter bound for every cut used here. Finite spectral partitions therefore show that multiplication by the actual unitary transporter intertwines the limiting rescaled length measurement. Approximation of continuous functions on $[0, \infty]$ removes the partitions. Reindexing Y preserves its probability measure, and the scalar phases in a product of transporters cancel from all these Gram forms. $\square$

Lemma 7.12 (Averaging with a uniform gap). *There are a compactly supported, absolutely continuous, symmetric probability m_0 on G and $0 < c_0 < 1$ such that its convolution operator is positive in every unitary representation and has norm at most c_0 on the orthogonal complement of the invariant vectors.*

Proof. Let (C, ε) be a compact Kazhdan pair for the whole locally compact product G ; property (T) provides such a pair [5]. Choose a relatively compact neighborhood V of the identity with positive Haar measure, and a symmetric compactly supported probability density ν bounded below on $V \cup CV$ and their inverses. For a unit vector ξ , writing $g = (gv)v^{-1}$ and averaging over $v \in V$ bounds

$$\sup_{g \in C} \|\pi(g)\xi - \xi\|^2 \leq C_1 \int_G \|\pi(h)\xi - \xi\|^2 d\nu(h).$$

Indeed the two terms in the triangle inequality are integrated over CV and V^{-1} , where the density has a uniform positive lower bound. The Kazhdan inequality thus gives a uniform upper bound $1 - \delta$ for the self-adjoint averaging operator $\pi(\nu)$ on the complement of the invariant vectors. The lazy average $(1 + \pi(\nu))/2$ is positive and has norm at most $1 - \delta/2$ there. Convolve this probability on both sides with a symmetric compactly supported probability density χ . Its operator is

$$\pi(\chi) \frac{1 + \pi(\nu)}{2} \pi(\chi),$$

which is positive and has the same uniform upper bound. The resulting probability m_0 is absolutely continuous. $\square$

Proposition 7.13 (Addition in distribution). *Let $R_n \rightarrow \infty$, let Θ_n be admissible for T , and suppose (7.24) holds. Fix an isolated source factor and a positive split axis in it. For any $t_n, u_n \geq 0$, if the individual spherical laws converge weakly to P_t, P_u , then*

$$(7.27) \quad P_{\Theta_n, R_n}(t_n + u_n) \longrightarrow P_t * P_u.$$

The same statement permits pre- and post-translations in fixed compact subsets of G .

Proof. *The inserted path and the order of limits.* First pass to a subsequence on which the sum laws converge. Insert a fixed translation $s \in G$ between the two positive axial segments. Away from the Bruhat wall corresponding to backtracking, Cartan multiplication expresses $a_{t_n} s a_{u_n}$ as $a_{t_n + u_n}$ with pre- and post-translations in a compact set depending on s . This includes a segment whose length stays bounded. To verify the assertion when both lengths diverge, write s in the open opposite-radical-Levi-radical cell for the axis: the two radical factors contract at the respective ends, and the Levi factor has a fixed split component. The omitted wall is a proper lower-dimensional Bruhat cell and has Haar measure zero. The

components of s outside the chosen source factor are fixed compact displacements. The exceptional wall in that factor has Haar measure zero, also after taking its product with the complementary factors. Lemma 7.11 therefore identifies the uncut law of the inserted product with the limiting sum law, for almost every s .

The Gram computation. Write $A_n(y) = X_{at_n}^{\Theta_n}(y)$ and $B_n(y) = X_{au_n}^{\Theta_n}(a_{u_n}^{-1}y)$. For smooth radial functions f, g , constant near the two ends of the compactified interval, put

$$(7.28) \quad A_{n,f} = D_{f(r_T/R_n)} A_n, \quad B_{n,g} = D_{g(r_T/R_n)} B_n.$$

Consider the columns

$$(7.29) \quad A_{n,f}(sy) X_s^{\Theta_n}(y) B_{n,g}(y).$$

They are uniformly operator-bounded. Their mixed Gram forms are pairings between $A_{n,f}^* A_{n,f}$ and the translate of $B_{n,g} B_{n,g}^*$ under the induced trace action

$$F(y) \longmapsto \text{Ad } \Theta_n(c(s, s^{-1}y)) F(s^{-1}y).$$

Its invariant space consists of constant scalar fields: an invariant equivariant field lifts to a constant field on G , whose value commutes with $\Theta_n(\Gamma)$. More explicitly, the lifted field is left invariant, hence essentially constant by Haar invariance; its residual right Γ -equivariance imposes precisely that commutation relation. Conversely a scalar constant gives an invariant field. The matrix products appearing in the Gram forms have uniformly bounded integrated Hilbert norms, since each radial piece has both a uniform operator bound and a uniform Hilbert bound. Average s with m_0^{*k} . Lemma 7.12 shows that the difference from the product of the scalar Gram forms is $O(c_0^k)$, uniformly in n .

Take first a Hilbert limit in n , for each fixed k , and then a Hilbert limit in k . Equivalently, form the two successive Hilbert ultraproducts of the spaces with measures $dy dm_0^{*k}(s)$. Polarization of the preceding calculation gives an isometry

$$(7.30) \quad J : L^2(P_t) \otimes L^2(P_u) \longrightarrow \mathcal{K}, \quad f \otimes g \longmapsto (A_{n,f}(sy) X_s^{\Theta_n}(y) B_{n,g}(y))^{\lim}.$$

The Fourier length in the output has its own spectral measurement E on $\mathcal{K}$. For each fixed k , the measure m_0^{*k} has compact support. All short-approximation errors may therefore be sent to zero in n before increasing k . Dominated convergence identifies the uncut output law for that fixed average with the subsequential sum law. No uniform approximation over the growing supports of all convolution powers is asserted or needed.

Positive finite input windows. We prove that its restriction to the range of J is addition. An output interval strictly above the sum of two input intervals is excluded by (7.26) and short approximation of $X_s^{\Theta_n}$. It remains to exclude a strict deficit. Localize the two input lengths in smooth windows bounded below by $\varepsilon' R_n$, and localize the output in a window shorter than their sum by εR_n , with $\varepsilon, \varepsilon' > 0$. All window boundaries may be enlarged slightly. Write $A = A_{n,f}$, $B = B_{n,g}$. In testing the output cut, trace cyclicity gives pairings of $A(sy)^*$ against

$$X_s^{\Theta_n}(y) B(y) C(y, s),$$

where C is uniformly operator-bounded and has the Fourier support of the inverse shortened-output window. In particular one may take C to be the adjoint of the smooth squared output cut of the product itself.

Choose, by Lemma 7.11, scales

$$a_n \ll d_n \ll R_n.$$

On each compact set of s 's replace $X_s^{\Theta_n}$ by its short approximation. Lemma 7.2 implies that every contributing pair of long indices has directions at distance at most $Ce^{-c\varepsilon R_n}$. Proposition 7.7 therefore permits insertion of the two ray lifts and then intertwines the short left multiplication through the lift. The result, with an error tending to zero, is

$$(7.31) \quad \tau(\zeta_n(sy)^* \Delta_T(X_s^{\Theta_n}(y)) \xi_n(y) C(y, s)), \quad \xi_n = \mathbb{V}_{d_n} B, \quad \zeta_n = \mathbb{V}_{d_n} (A^*).$$

The errors are uniform for s in a fixed compact set. Replacing the short approximation by the actual transporter on the lifted vector is legitimate because the left density has uniformly bounded tracial L^2 -norm: for a bounded Hilbert-small change z ,

$$(7.32) \quad \|\Delta_T(z)\xi_n\|^2 \leq \|z\|_\infty \|z\|_2 \|\rho_{\xi_n}^L\|_2.$$

The integrated version follows from Cauchy–Schwarz on Y .

The exact trace inequality for (7.31) is

$$(7.33) \quad \left| \tau(\zeta_n(sy)^* \Delta_T(X_s^{\Theta_n}(y)) \xi_n(y) C(y, s)) \right|^2 \leq \|C(y, s)\|_2^2 (\text{Tr} \otimes \tau) \left[\zeta_n(sy) \zeta_n(sy)^* \text{Ad} \Delta_T(X_s^{\Theta_n}(y)) (\xi_n(y) \xi_n(y)^*) \right].$$

Both density fields on the right have uniformly bounded trace-Hilbert norm, by Proposition 7.7; they are independent of s apart from the displayed transport. The induced amplified action has no invariant vector, by admissibility and the same equivariant-field lifting argument. Its m_0^{*k} -average is therefore bounded by Cc_0^k . Integrate (7.33), first let n tend to infinity, and then let k tend to infinity. The shortened-output cut vanishes.

Zero and infinite input values. Suppose first that one input value is zero and the other is a finite value v . Restrict the first input to $[0, \delta]$ and the second to $[v - \delta, v + \delta] \cap [0, \infty)$, on the rescaled length axis. The short intermediate transporter changes length by $o(1)$ on this scale. The forward and reverse triangle inequalities therefore restrict the output to $[v - 2\delta - o(1), v + 2\delta + o(1)] \cap [0, \infty)$. Every closed output window separated from v is excluded by choosing δ sufficiently small. This includes $v = 0$, where the upper triangle bound suffices. The cut replacements have the same Hilbert-small errors as before.

For an input value $+\infty$, it suffices to exclude output in a fixed finite interval $[0, M]$. If the other input is finite, choose a bounded neighborhood $[0, B]$ containing it and restrict the infinite input to $[N, \infty]$, with $N > M + B + 1$. The reverse triangle inequality excludes the output window, after the short intermediate replacement. If both input values are infinite, restrict both to $[N, \infty]$, where $2N > M + 1$. The connecting output index then has a deficit at least $(2N - M - o(1))R_n$ from the sum of the two long input lengths. The lifted shortening argument just proved applies: it used a positive lower length bound and a uniform deficit, but no upper bound on either long input. Use smooth tail cutoffs constant near infinity, whose multiplier norms are bounded by Proposition 7.4. Hold N, M fixed through the limits in n and then k ; afterward exhaust the finite output intervals and increase the finite lower input cutoffs. This excludes every finite output value at an infinite sum.

Identification of the output measurement. We have proved that a product of supported smooth input functions has no output in a closed window disjoint from the sums of its input supports. Characteristic functions of interval partitions whose boundaries have zero input measure are obtained by Hilbert approximation with such smooth functions. The isometry J preserves this approximation.

Here is an exact spectral argument that does not assume in advance that E preserves the range of J . Write $a(s, t) = s + t$, a continuous map on the compact square, and let P_{in} be multiplication by Borel subsets of that square on $L^2(P_t) \otimes L^2(P_u)$. Fix a closed output set F . Product test functions supported on rectangles whose closures lie in the open set $a^{-1}(F)^c$ have dense span in $\text{ran } P_{\text{in}}(a^{-1}(F)^c)$, by regularity of the product measure and Hilbert approximation with smooth interval cutoffs. For each such rectangle the compact sum support misses F . A finite cover of F by the excluded output windows and the rectangle support conclusion therefore give

$$E(F)JP_{\text{in}}(a^{-1}(F)^c) = 0, \quad E(F)J = E(F)JP_{\text{in}}(a^{-1}(F)).$$

Choose closed sets $K_j \uparrow F^c$; for example use positive distance at least $1/j$ from F in a compatible compact metric. Apply the same conclusion to every K_j . Since $a^{-1}(K_j)$ is disjoint from $a^{-1}(F)$, it gives

$$E(K_j)JP_{\text{in}}(a^{-1}(F)) = 0.$$

Strong convergence of these spectral projections yields $E(F^c)JP_{\text{in}}(a^{-1}(F)) = 0$. Combining the two identities proves

$$E(F)J = JP_{\text{in}}(a^{-1}(F)).$$

The monotone-class theorem extends this equality to all Borel sets, and spectral integration then gives

$$E(\varphi)J = JM_{\varphi(s+t)} \quad (\varphi \in C([0, \infty])).$$

On $J(1 \otimes 1)$, the output law is the limiting sum law identified in the first paragraph. This proves (7.27). Compact pre- and post-translations are covered by Lemma 7.11. Every convergent subsequence of the sum laws has the same limit, so compactness of the space of probability measures on $[0, \infty]$ proves convergence of the original sequence. $\square$

8. LINEAR SUBSEQUENCES AND HEIGHTS FOR ISOLATED COMPONENTS

At an isolated type, Section 5 has already supplied sharp boundary measurements $x \mapsto P_x$. We now construct a height at each source label: a positive injective self-adjoint operator that commutes with P_x and transforms by the inverse boundary Jacobians. The construction will take an inverse power of the modulus of a horizontal derivative of $x \mapsto P_x$. Its main analytic requirement is a square-integrable derivative on a dense, invariant domain.

The first part of the section controls Fourier lengths. We prove that, along a sequence of long source intervals, every fixed input retains a positive amount of mass below a linear length bound. The addition theorem of Section 7 and property (T) are used to rule out total superlinear growth. This relatively weak upper-window conclusion is sufficient: lattice packing costs one factor of the length, which is canceled by averaging over a source stabilizer interval of comparable length. The resulting estimate for ordered products at opposite source labels controls small horizontal differences. We then construct the closed derivative modulus and prove that its kernel is zero.

The source ray belongs to an entire simple factor. The isolated factors are real and locally isomorphic to $\text{Sp}(n, 1)$, $n \geq 2$, or $F_{4(-20)}$. Complementary factors may be archimedean or nonarchimedean. We retain a compact window in those complementary coordinates in each lattice count. The horizontal metrics and angular probabilities are the compatible compact-invariant choices made in Section 7.

We use the admissibility and probability-law conventions of Definition 7.10 and Section 7. Thus all varying finite factor traces are normalized, the base $Y = G/\Gamma$ has probability measure, and extra finite matrices carry the unnormalized matrix trace with fixed matrix size. Every application of the ray estimates below has both the stated operator bounds and bounded integrated Hilbert norms. A lower effect bound will always concern a fixed Hilbert vector; it is not a bound uniform over a moving unit ball.

Here are the precise earlier geometric conclusions used. The complementary regular realizations and all-factor mixing are supplied by Lemma 3.4, Proposition 3.5, and Lemma 3.7. The gap count in Proposition 4.10 gives full-gap escape on complete vertex-origin lexicographic chains. The angular conclusions of Propositions 5.4, 5.7, and 5.11 give sharp, strongly continuous pure-vertex measurements, their source and measured covariance, nonconstancy, a permutation of isolated types, and tightness of every complementary length on a pure isolated ray. We also use scalarity of limiting rescaled one-column effects from Lemma 4.5: their measured invariance places them in $\mathcal{N}$, and their face-parabolic and complementary source invariances make them scalar by all-factor mixing. The compact-change argument for those effects uses bounded approximations of the fixed transporters and has arbitrary fixed slack in radial windows. These are conclusions of the preceding constructions, not additional assumptions on the lattice or its projected actions.

8.1. An admissible model for the original module.

Lemma 8.1. *For the two-slot model of Section 3 there is a finite tensor $D = B \bar{\otimes} L_\eta(\Gamma)$, with B diffuse, and an admissible representation $\delta : \Gamma \rightarrow \mathcal{U}(D)$ for each isolated measured factor,*

obtained from the source action by a fixed finite matrix corner equivalence. It has a sequence γ_n whose Fourier laws escape in every measured simple factor. Rescaled continuous radial laws, scalar effects, and complementary tightness are preserved by this corner equivalence.

Proof. The complementary Λ -factor in the two-slot regular model is diffuse. In a sufficiently large finite matrix amplification choose a projection in that coefficient factor with the same trace as the regular-model projection e . A partial isometry identifying the projections gives the required unital tensor-coordinate representation. Scalar relative commutant follows from Proposition 3.5.

Choose a full lexicographic chain starting at a vertex and adding one new simple gap at each stage. By Proposition 4.10, all its target gaps diverge. Apply this to the finite matrix of unit columns, and choose successively finite path times at which the integrated mass in successively larger Cartan windows is small. At each selected time, first make the sum of the errors for the finite matrix and the finitely many prescribed windows smaller than a number tending to zero. Choose a base point where that nonnegative sum is no larger than its integrated bound. Each individual error is then small at the same base point. The resulting lattice cocycle values give γ_n .

A fixed matrix corner equivalence preserves this escape. Approximate its entries and their adjoints in trace Hilbert norm by uniformly bounded finite Fourier matrices. Errors are uniformly small on the uncut unitary columns. Each fixed Fourier shift changes every Cartan coordinate by a bounded amount. It therefore preserves escape, and it intertwines rescaled continuous radial tests in the limit. After moving the test through the approximating matrix, its uncut Gram factor converges to the Gram of the partial isometry. The same argument, with slack in an unscaled window, preserves complementary tightness.

For amplified admissibility, finite-rank compressions of $\lambda_{X_T}(h_T)$ tend to zero when $r_T(h) \rightarrow \infty$. This is also seen by embedding $L^2(X_T)$ into a multiple of $L^2(T)$, since the maximal compact stabilizer is compact, and testing compactly supported vectors. Compress $\Delta_T(\delta(\gamma_n))$ between finite-rank operators in the translation coordinate. Plancherel and the escaping Fourier law make its tracial L^2 -norm tend to zero. Pairings for conjugation between finite-rank, bounded trace-Hilbert tensors consequently tend to zero, by Cauchy–Schwarz; approximation extends this to all trace-Hilbert vectors. An invariant vector would have constant nonzero self-pairing along this sequence. $\square$

Lemma 8.2 (Entropy lower bound). *Suppose an isolated source type i , with factor S , is matched to the measured type $p(i)$, with factor T . Let Q_S and Q_T be the homogeneous dimensions of their boundaries. For the representation δ of Lemma 8.1, complementary lengths are uniformly tight on the pure source ray, and, for every $0 < c < Q_S/Q_T$,*

$$(8.1) \quad P_{\delta,t}(t)([0, c]) \longrightarrow 0.$$

Both assertions hold uniformly under source pre- and post-translations in compact sets.

Proof. Complementary tightness is the pure-ray conclusion of Section 5, transported through the fixed corner equivalence. We prove the lower bound before that equivalence. Complementary regularity supplies a nonzero constant frame column f for $\pi_0(\Gamma)$ together with W_Λ . If e_j are the finitely many identity columns in the measured regular model, the frame inequality, applied to $W_h e_j$, gives

$$(8.2) \quad \sum_{\gamma \in \Gamma} \|D_{\{h\}} \pi_0(\gamma) f\|_2^2 \leq C_f \quad (h \in \Gamma).$$

Indeed the left side is the sum, over γ , λ , and the finite column index j , of the squared coefficients against $W_h W_\lambda e_j$. Summing first over (γ, λ) is exactly the complementary frame bound.

Let Ω_t be a source Cartan shell of fixed positive thickness about length t in S , with all compact rotations and with complementary coordinates in a fixed compact neighborhood.

Cartan integration gives $|\Omega_t| \asymp e^{Q_S t}$. Restrict both section values to a compact set K . For fixed y and γ , the times with transporter γ lie in

$$K\gamma d_0(y)^{-1},$$

so their Haar volume is at most $|K|$. Integrating (8.2) therefore bounds the averaged mass at each measured index by a constant independent of that index and of t .

For a fixed complementary compact window, the number of measured lattice indices with $r_T(h) \leq c't$ is at most $Ce^{Q_T c't}$. To see this without assuming a discrete projection to T , thicken the indices on the right by a small lattice-disjoint neighborhood in the full ambient group. Cartan length changes by a bounded amount and the complementary window remains compact; Haar integration then gives the stated bound. It follows that the restricted low-length mass averaged over Ω_t , divided by $|\Omega_t|$, is at most

$$(8.3) \quad Ce^{(Q_T c' - Q_S)t}.$$

Choose $c < c' < Q_S/Q_T$. The lost section sets have arbitrarily small measure, uniformly in the translating time, and the norm of the uncut translated constant column is unchanged. Complementary tightness similarly removes the restriction to the fixed measured compact window. Finally Lemma 4.5 makes every limiting low-length effect scalar, and the compact-change argument identifies its value uniformly throughout the thick shell, with slack from c to c' . A subsequence violating (8.1) would thus have positive averaged mass in (8.3), a contradiction. The corner-change argument of Lemma 8.1 finishes the proof. $\square$

8.2. Tightening a Kazhdan window. For a sequence of cutoffs $L_n \geq 1$, a bounded sequence $x_n \in D_n$ is called tight at scale L_n if

$$(8.4) \quad \lim_{C \rightarrow \infty} \lim_{n \rightarrow \mathcal{U}} \|D_{1_{\{r_T > CL_n\}}} x_n\|_2 = 0,$$

where $\mathcal{U}$ is a free ultrafilter. Ultraproducts in this Subsection are tracial ultraproducts. Passage to an ordinary subsequence will be made after all countably many requirements have been fixed.

Lemma 8.3. *The tight elements form a von Neumann subalgebra Q_0 of $\prod_{\mathcal{U}} D_n$, and Q_0 is a factor. If each Θ_n is relatively irreducible, the ultraproduct representation Θ also has scalar relative commutant.*

Proof. Smooth radial truncations have a uniform operator bound by Proposition 7.4. Thus tight elements admit Hilbert approximation by bounded sequences supported in $r_T \leq CL_n$. The triangle inequalities (7.26) prove closure under multiplication; inversion preserves length and gives closure under adjoint. The tightness condition is closed in Hilbert norm on every bounded operator ball. These facts prove von Neumann closure.

All coefficient-unitary sequences belong to Q_0 . If z is central in Q_0 , its representing sequence commutes asymptotically, uniformly, with those unitaries: otherwise choose a violating unitary at each index. The Hilbert closed convex hull of the coefficient conjugation orbit gives conditional expectation onto its commutant. Consequently z is represented, up to a Hilbert-null sequence, on the group factor alone.

All fixed group labels also belong to Q_0 , since $L_n \geq 1$. Conjugation by those labels acts on the orthogonal complement of the identity Fourier coefficient without invariant vectors. In fact the squared moduli of an invariant vector's coefficients would be constant on conjugacy classes; ICC forces all nonidentity coefficients to vanish. Scalar cocycle phases do not change this argument. A fixed Kazhdan pair for Γ gives a uniform spectral gap for all these conjugation representations, even as the cocycles vary. Approximate commutation therefore forces z to be scalar.

Apply the same Kazhdan inequality to $\text{Ad } \Theta_n$ on $L^2(D_n) \ominus \mathbb{C}1$. Relative irreducibility says exactly that this representation has no invariant vector. An ultraproduct element commuting with every $\Theta(\gamma)$ must accordingly be scalar. $\square$

Lemma 8.4 (Kazhdan tightening). *Fix a finite Kazhdan set $\mathcal{F} \subset \Gamma$. For every $\varepsilon > 0$ there is $\varepsilon_K > 0$ with the following property. Suppose Θ_n are admissible and*

$$(8.5) \quad \|D_{1_{\{r_T > L_n\}}} \Theta_n(\gamma)\|_2 \leq \varepsilon_K \quad (\gamma \in \mathcal{F}).$$

After passage to a subsequence there are unitaries u_n with $\limsup_n \|u_n - 1\|_2 < \varepsilon$ such that every fixed element of $u_n^ \Theta_n(\Gamma) u_n$ is tight at scale L_n . Admissibility is preserved.*

Proof. The Kazhdan gap first supplies an invariant finite right module close to the original copy of the tight algebra. Its dimension is close to one, but we need it to be exactly one in order to obtain a conjugating unitary. We will exclude any excess dimension by applying the ray lift to a column orthogonal to the tight algebra; the resulting almost invariant amplified state contradicts admissibility. We then align the unitary with the identity.

A nearby invariant module. Put $D_{\mathcal{U}} = \prod_{\mathcal{U}} D_n$ and use Lemma 8.3. In the Jones basic construction $\langle D_{\mathcal{U}}, e_{Q_0} \rangle$, with canonical semifinite trace [40], one has

$$(8.6) \quad \|\Theta(\gamma) e_{Q_0} \Theta(\gamma)^* - e_{Q_0}\|_2^2 = 2\|\Theta(\gamma) - E_{Q_0} \Theta(\gamma)\|_2^2.$$

The right side is $O(\varepsilon_K^2)$: a smooth cutoff equal to one on $[0, L_n]$ produces a tight approximation. Its outer radius may be a fixed larger multiple of L_n , chosen so that the multiplier scale always exceeds the threshold in Proposition 7.4. This does not increase the error beyond the tail in (8.5). The Hilbert-space Kazhdan inequality [5] makes the projection of e_{Q_0} to the invariant Hilbert subspace $O(\varepsilon_K)$ -close to it. This projection is the Hilbert limit of convex combinations of its conjugates, and is therefore a positive contraction z . Put $p_0 = 1_{[1/2, 1]}(z)$. Spectral rounding gives

$$(8.7) \quad \|p_0 - e_{Q_0}\|_2 \leq 2\|z - e_{Q_0}\|_2 = O(\varepsilon_K).$$

For completeness, split the left squared norm into $\text{Tr}(e_{Q_0}(1 - p_0)) + \text{Tr}((1 - e_{Q_0})p_0)$; the two terms are bounded by four times the corresponding terms with $(1 - z)^2$ and z^2 . This proves the inequality without a commutation assumption. The projection p_0 commutes with $\Theta(\Gamma)$, and its trace d' is close to one.

Take ε_K small enough that $d' < 2$. A right Q_0 -module isometry onto $p_0 L^2(D_{\mathcal{U}})$ has initial projection $q \in M_2(Q_0)$, of trace d' . Let its two basis entries be the row V . Initially these entries lie in $L^2(D_{\mathcal{U}})$, and

$$(8.8) \quad Vq = V, \quad E_{M_2(Q_0)}(V^*V) = q, \quad \Theta(\gamma)V = Vb_\gamma,$$

where b_γ is a projective unitary representation in $qM_2(Q_0)q$. The positive L^1 -element $VV^* = \sum_i V_i V_i^*$ is invariant under Θ . Its spectral projections belong to the scalar relative commutant; hence

$$(8.9) \quad VV^* = d'1.$$

This also proves that every V_i is bounded. Thus

$$(8.10) \quad e_1 = (d')^{-1} V^* V \leq q$$

is a projection of trace one and commutes with all b_γ .

Excluding excess module dimension. The projection q has trace d' , whereas e_1 has trace one. Thus $e_1 = q$ will force $d' = 1$ and turn the invariant module into one unitary copy of Q_0 . Suppose instead that $e_1 \neq q$, and put

$$H_1 = e_1 - q/d'.$$

Then $H_1 \neq 0$, $E_{M_2(Q_0)} H_1 = 0$, and H_1 commutes with b_γ . This is the nonzero column whose long Fourier mass will contradict the amplified gap. Orthogonality to Q_0 implies that representing columns of H_1 have vanishing Fourier mass below every fixed multiple of L_n . Indeed test orthogonality against the smooth positive radial cutoff of the same bounded column; this cutoff belongs to Q_0 and is equal to one on the desired lower window.

Here and below the finite matrix trace only changes constants. Choose bounded representatives G_n of H_1 with exact support $r_T \geq l_n$, and short approximations to q, b_γ with support $r_T \leq a_n$, so that

$$(8.11) \quad L_n \ll a_n \ll d_n \ll l_n, \quad \|G_n - H_{1,n}\|_2 \rightarrow 0.$$

All approximations are uniformly operator-bounded. Since these are elements of fixed 2×2 matrix algebras over normalized finite factors, they also satisfy

$$\|x_n\|_2 \leq \sqrt{2} \|x_n\|_\infty.$$

Thus both norm bounds in Proposition 7.7 and Corollary 7.9 apply to every short or long column used here. One explicit diagonal choice is as follows. For the first j group labels choose a fixed multiple $M_j L_n$ that approximates their tight columns within $1/j$. Require the long column to have mass less than $1/j$ below $j^2 M_j L_n$, and pass to indices satisfying this and the finitely many algebraic approximations. Then use $a_n = M_j L_n$, $d_n = j M_j L_n$, $l_n = j^2 M_j L_n$, after harmless fixed slack in the cutoffs. This construction produces an ordinary subsequence.

Lift the coisometric row $V/\sqrt{d'}$ to exact coisometric rows W_n . This is obtained by polar decomposition, cutting a small spectral neighborhood of zero, and filling the vanishing trace defects in the finite factors. The corrections tend to zero in Hilbert norm: away from zero polar correction is uniformly continuous, and the discarded support has trace tending to zero. Put $e_{1,n} = W_n^* W_n$ and

$$\Theta'_n(\gamma) = W_n^* \Theta_n(\gamma) W_n.$$

These are exact projective representations in the $e_{1,n}$ -corners. Choose projection lifts q_n and corner-unitary lifts $b_{\gamma,n}$. The identities in the ultraproduct give

$$(8.12) \quad \|\Theta'_n(\gamma) - e_{1,n} b_{\gamma,n}\|_2 \rightarrow 0.$$

Neither the representatives $b_{\gamma,n}$ nor their short cuts need satisfy an exact representation law.

Let $\xi_n = \mathbb{V}_{d_n} G_n$. Proposition 7.7, first for short approximations and then using the density error (7.32), gives

$$(8.13) \quad \Delta_T(b_{\gamma,n})\xi_n - \xi_n b_{\gamma,n} \rightarrow 0, \quad \Delta_T(q_n)\xi_n - \xi_n \rightarrow 0, \quad \xi_n q_n - \xi_n \rightarrow 0$$

in Hilbert norm. These statements use only the ordinary identities $b_\gamma H_1 = H_1 b_\gamma$ and $q H_1 = H_1 q = H_1$. Put $\xi'_n = \Delta_T(e_{1,n})\xi_n$. The left density of ξ_n has Fourier support in $\{r_T \leq 4d_n + C\}$, since the two ray tubes can overlap only when their base points are at that distance. Consequently

$$\langle \Delta_T(G_n)\xi_n, \xi_n \rangle = 0$$

for large n . The identity $e_{1,n} = G_n + q_n/d' + o_{L^2}(1)$ and the uniform density bound now yield

$$(8.14) \quad \|\xi'_n\|^2 = (d')^{-1} \|\xi_n\|^2 + o(1) \rightarrow (d')^{-1} \|H_1\|_2^2 > 0.$$

Using (8.12) and (8.13), we also obtain

$$(8.15) \quad \Delta_T(\Theta'_n(\gamma))\xi'_n - \xi'_n b_{\gamma,n} \rightarrow 0.$$

Extend $b_{\gamma,n}$ by the identity off q_n . This does not change the last assertion, since $\xi'_n q_n - \xi'_n \rightarrow 0$.

The normalized left vector states defined by ξ'_n on the semifinite corner with unit $\Delta_T(e_{1,n})$ are now almost invariant in predual norm. Right multiplication by the extended $b_{\gamma,n}$ does not change such a state; the error in (8.15) changes it by at most twice the vector error times the vector norm. Their positive density square roots are almost invariant Hilbert unit vectors by the Powers–Størmer inequality [58], whose short trace proof we include:

$$(8.16) \quad \|A^{1/2} - B^{1/2}\|_2^2 \leq \|A - B\|_1 \quad (A, B \geq 0).$$

To verify this inequality, put $S = A^{1/2}$, $T = B^{1/2}$, and $D = S - T$. The trace duality inequality and cyclicity give

$$\|A - B\|_1 \geq \text{Tr}(\text{sgn}(D)(S^2 - T^2)) = \text{Tr}(|D|(S + T)).$$

Subtracting $\text{Tr}(D^2)$ from the last expression leaves $2 \text{Tr}(D_+T) + 2 \text{Tr}(D_-S) \geq 0$. All products are trace-class by Hilbert–Schmidt Cauchy–Schwarz; finite spectral truncation gives the same calculation in the semifinite setting. Conjugation on this corner is unitarily equivalent, via $\Delta_T(W_n)$, to the original amplified conjugation. Admissibility and the fixed Kazhdan gap contradict these almost invariant unit vectors. This proves $e_1 = q$.

Recovering a near-identity unitary. Taking traces gives $d' = 1$. A matrix equivalence over the factor Q_0 reduces q to a single unit, and the row V becomes an actual unitary u with $u^*\Theta(\Gamma)u \subset Q_0$ and $p_0 = ue_{Q_0}u^*$. Moreover

$$(8.17) \quad \|E_{Q_0}u\|_2^2 = \text{Tr}(e_{Q_0}p_0).$$

Align the polar part of $E_{Q_0}u$ by a right unitary from Q_0 ; partial polar isometries extend to unitaries in a finite factor. Since $E_{Q_0}u$ is a contraction, the aligned unitary satisfies

$$(8.18) \quad \|u - 1\|_2^2 \leq 2 - 2\|E_{Q_0}u\|_2^2 = \|p_0 - e_{Q_0}\|_2^2.$$

Lift u by unitaries and diagonalize the countably many tightness conditions. Choosing ε_K sufficiently small gives the requested bound. Inner conjugacy preserves relative irreducibility; applying Δ_T to the conjugating unitary proves preservation of amplified admissibility. $\square$

8.3. Composition of admissible kernels.

Lemma 8.5 (Exact insertion). *Let $\delta : L_\alpha(\Gamma) \rightarrow B_0 \bar{\otimes} L_\beta(\Gamma)$ be a trace-preserving embedding whose group representation is relatively irreducible and has a sequence escaping in Fourier probability in every ambient simple factor. Let Θ be admissible for an isolated measured factor T . There is an admissible projective representation $\Theta \otimes \delta$, with source multiplier α , whose spherical Fourier laws are the exact compositions of the spherical Fourier kernels of δ and Θ . The final measured cocycle is the measured cocycle of Θ .*

Proof. Write ξ for the source multiplier of Θ , and put

$$T_h(y) = \Theta(c(h, h^{-1}y)).$$

The genuine section-cocycle identity gives

$$\begin{aligned} T_g(y)T_h(g^{-1}y) &= \nu(g, h)(y)T_{gh}(y), \\ \nu(g, h)(y) &= \xi(c(g, g^{-1}y), c(h, h^{-1}g^{-1}y)). \end{aligned}$$

Thus ν is a normalized scalar-function-valued cocycle for the action on Y . In the finite twisted crossed product for this action use shift unitaries b_h with multiplier β/ν . Tensor this crossed product with the final finite algebra D . Then

$$(8.19) \quad \Psi(w_h) = T_h b_h, \quad \Psi(w_g)\Psi(w_h) = \beta(g, h)\Psi(w_{gh}).$$

Orthogonality of the shifts makes Ψ trace-preserving on the twisted group algebra. It extends normally: trace preservation gives equality of every even moment of the absolute value of a group polynomial, hence equality of its operator norm; the resulting trace-preserving isometric map extends by bounded strong approximation. Apply $\text{id}_{B_0} \otimes \Psi$ to δ .

Here is the kernel formula. If $\delta(w_\gamma) = \sum_h a_{\gamma, h} \otimes w_h$, then, for a Borel set E of final Fourier lengths, its mass after insertion is

$$(8.20) \quad \sum_{h \in \Gamma} \|a_{\gamma, h}\|_2^2 \int_Y \|D_{1_{\{r_T \in E\}}} \Theta(c(h, h^{-1}y))\|_2^2 dy.$$

Distinct h 's are orthogonal in the middle shift coordinate. Integration over a source spherical transporter gives the same formula for spherical laws. The change $y \mapsto h^{-1}y$ identifies the inner integral with the spherical kernel at the translation $h \in G$. In particular all inserted kernels are probability kernels.

It remains to prove admissibility and factoriality of the coefficient tensor, rather than assume them. Use $\Psi(w_h)$ as a right Fourier basis for the middle crossed product. Its coefficient Hilbert space is

$$\mathcal{E} = L^2(Y; L^2(D)).$$

Left multiplication by $\Psi(w_g)$ acts on this coefficient space by

$$(8.21) \quad (\mathcal{U}_g k)(y) = \text{Ad } T_g(y) k(g^{-1}y),$$

and shifts the basis from h to gh , with multiplier $\beta(g, h)$. Right multiplication only shifts the basis from h to hg , with multiplier $\beta(h, g)$. The coefficient action is the restriction of the G -induced trace action of $\text{Ad } \Theta$. Its invariant space is $\mathbb{C}1$. On $\mathcal{E} \ominus \mathbb{C}1$, coefficients tend to zero along sequences escaping in every simple factor, by Howe–Moore and the product invariant-space decomposition of Section 3.

To apply this decay to Fourier-distributed unitaries, take bounded finite basis tests $b k \Psi(w_h)$ and $b' k' \Psi(w_{h'})$, and write $A = \sum_i a_i w_i$ for one of the escaping unitaries of δ . The conjugation pairing is bounded, up to the fixed norms of b, b' , by

$$(8.22) \quad \sum_{i,j: ih=h'j} \|a_i\|_2 \|a_j\|_2 |\langle k', \mathcal{U}_i k \rangle|.$$

The relation determines j uniquely from i . Its sum without the last factor is at most $\sum_i \|a_i\|_2^2 = 1$, by Cauchy–Schwarz. Outside fixed large compact windows in every factor, the last coefficient is uniformly small. The remaining terms tend to zero by Fourier escape and Cauchy–Schwarz; fixed shifts in the relation do not change escape. Thus the conjugation pairing tends to zero on the nonscalar coefficient summand. On the scalar coefficient summand it is precisely the conjugation representation of δ , whose invariant space is scalar. Density proves scalar relative commutant for the inserted representation. The center of its ambient finite tensor lies in that commutant, so the ambient tensor is a factor.

For the amplified assertion replace $\mathcal{E}$ by

$$\mathcal{E}_{\text{amp}} = L^2(Y; L^2(B(L^2(X_T)) \bar{\otimes} D, \text{Tr} \otimes \tau))$$

and replace T_g by $\Delta_T(T_g)$. This induced coefficient representation has no invariant vector, by admissibility of Θ . Finite-rank trace-Hilbert coefficient tests and the calculation (8.22) now apply on the entire amplified space. They prove absence of invariant vectors there. The coefficient algebra of the resulting tensor consists of B_0 , the middle finite crossed product, and the coefficient factor of D ; its final group factor is still the original measured factor of Θ . This proves all assertions. $\square$

8.4. A linear subsequence.

Lemma 8.6. *Let Θ_n satisfy the assumptions of Proposition 7.13. Suppose $t_n \rightarrow \infty$ and for some $a, \alpha > 0$,*

$$P_{\Theta_n, R_n}(t_n)([a, \infty]) \geq \alpha.$$

If $u_n/t_n \rightarrow \infty$, then

$$(8.23) \quad P_{\Theta_n, R_n}(u_n) \longrightarrow \delta_\infty.$$

This assertion is uniform under compact pre- and post-translations.

Proof. Fix an integer m . For large n , $u_n = mt_n + (u_n - mt_n)$ with nonnegative remainder. Pass to a subsequence on which the laws of these segments converge. Repeated use of Proposition 7.13 makes the limit at u_n equal to $P^{*m} * Q$, where $P([a, \infty]) \geq \alpha$ and Q is supported in $[0, \infty]$. For a finite M ,

$$(8.24) \quad (P^{*m} * Q)([0, M]) \leq \mathbb{P}\{\text{Bin}(m, \alpha) \leq M/a\}.$$

The right side tends to zero as $m \rightarrow \infty$. Diagonal extraction over the countably many m 's justifies taking them after the subsequential limit. This proves (8.23). Compact changes

are covered by Lemma 7.11; a failure of uniformity would give a violating deterministic sequence. $\square$

Proposition 8.7 (Positive mass at linear speed). *For each isolated source type i , with matched measured type $p(i)$, there exist $C < \infty$, $\alpha > 0$, and $T_n \rightarrow \infty$ such that, in the original regular module model, every fixed $\xi \in \mathcal{H}$ satisfies*

$$(8.25) \quad \liminf_{n \rightarrow \infty} \inf_{T_n/2 \leq t \leq T_n} \|D_{1_{\{r_{p(i)} \leq CT_n\}}} U_{a_t} \xi\|^2 \geq \alpha \|\xi\|^2.$$

The assertion also holds in normal Hilbert amplifications and after compact source pre- and post-translations. It makes no assertion uniform over a moving Hilbert unit ball.

Proof. Use the representation δ of Lemma 8.1. Suppose first that, on a pure isolated source type i , its matched length is totally superlinear:

$$(8.26) \quad P_{\delta,t}(t)([0, C]) \rightarrow 0 \quad (C < \infty).$$

The contradiction will use the entire permutation cycle through i . Composing the original representation around

$$i \mapsto p(i) \mapsto \cdots \mapsto i$$

returns to the same source and measured type. Under the assumption (8.26), its first stage increases length faster than linearly; the entropy bound prevents the later stages from losing that increase. We will insert this cycle before an admissible representation that reaches a large Fourier threshold in nearly the least possible source time. The composition would reach the same relative threshold in a strictly shorter time.

Choosing a nearly earliest witness. Fix numbers $D'_n \rightarrow \infty$. For each n , minimize over all admissible representations with both source and measured type equal to i , all cutoffs $L \geq 1$, and times $t \geq 1$, subject to

$$(8.27) \quad \|D_{1_{\{r_i > L\}}} \Theta(\gamma)\|_2 \leq \varepsilon_K \quad (\gamma \in \mathcal{F}),$$

$$(8.28) \quad P_{\Theta, D'_n L}(t)([1, \infty]) \geq \frac{1}{2}.$$

Here L controls the Fourier tails of the fixed Kazhdan set, and $D'_n L$ is the output threshold to be reached. Choose a witness t_n at most twice the infimum, and write its cutoff as L_n and its output scale as $R_n = D'_n L_n$.

The feasible set is nonempty. Use the identity group embedding into $L_\eta(\Gamma)$, with any fixed cocycle, and take L containing the finite Kazhdan set in the measured length. Relative irreducibility is ICC; amplified admissibility follows from all-factor escape of lattice elements and translation coefficient decay. For this embedding the transporter length on a pure axis differs from the source length by a bounded amount outside arbitrarily small section sets. Thus sufficiently large t meets (8.28).

Tightening the witness and repeating increments. Apply Lemma 8.4, choosing ε_K so small that the resulting conjugacies preserve at least 1/4 mass in (8.28). This preservation is uniform on moving transporters: for every unitary x , $\|u_n^* x u_n - x\|_2 \leq 2\|u_n - 1\|_2$, and the same bound holds after integration over the section cocycle. Rename the tightened representations Θ_n . Each fixed element is tight at scale L_n , hence short on scale R_n .

After extraction $t_n \rightarrow \infty$. Otherwise Lemma 7.11, with its proof applied to tightness on scale L_n , would make all bounded-time transporters short on scale $R_n = D'_n L_n$, contradicting the retained 1/4 mass. Lemma 8.6 therefore gives

$$(8.29) \quad u_n/t_n \rightarrow \infty \implies P_{\Theta_n, R_n}(u_n) \rightarrow \delta_\infty.$$

Shortening the witness by a cycle of insertions. Now insert copies of δ before Θ_n , once for each step around the finite permutation cycle through i , using Lemma 8.5. The first insertion, viewed from the source, is the totally superlinear one in (8.26). Choose $m_n \rightarrow \infty$ sufficiently slowly that

$$(8.30) \quad t'_n = t_n/m_n \rightarrow \infty, \quad P_{\delta, t_n}(t'_n)([m_n, \infty]) \rightarrow 1.$$

Such a choice follows directly from total superlinearity: for each integer m , choose a threshold after which the ratio of measured length to source length exceeds m^2 with probability at least $1 - 1/m$, and require t_n/m to exceed that threshold.

Every subsequent insertion retains length much larger than t_n , by Lemma 8.2. Its complementary coordinates remain tight. To justify this assertion for the random middle indices in (8.20), first restrict the complementary coordinates to a fixed compact window with arbitrarily large probability. Cartan decomposition then writes those indices as pure translations with compact pre- and post-factors. The entropy estimate is uniform on these windows. After the finitely many insertions, the terminal kernel Θ_n is tested on its source type i at lengths divided by t_n tending to infinity in probability. Equation (8.29) and its compact uniformity show that the final measured length divided by R_n tends to infinity in probability.

For a fixed source group element the preceding finite chain of kernels instead gives a fixed probability measure on the countable set of terminal input labels. Its mass can be restricted to a finite set with arbitrarily small loss. Tightness of the terminal fixed-label laws at scale L_n , and the uniform section approximation for each such label, prove tightness at scale L_n for every fixed group element of the composite representation. Hence, for a fixed sufficiently large C_1 , and after extraction, the composite satisfies (8.27) with cutoff $C_1 L_n$. Its final escape also gives (8.28) with threshold $C_1 R_n$, at time t'_n . The ratio of these new scales is still D'_n . Admissibility is preserved by every insertion. Thus this is a feasible witness in the very same minimization, and

$$t_n \leq 2 \inf t \leq 2t'_n,$$

contrary to $t'_n/t_n \rightarrow 0$.

From a scalar subsequence to the fixed-vector conclusion. Total superlinearity is therefore impossible. There are $C_0, \alpha_0 > 0$ and $T_n \rightarrow \infty$ for which the scalar law has mass at least α_0 below $C_0 T_n$. This is uniform for $t \in [T_n/2, T_n]$, after enlarging C_0 slightly. Indeed, a violating sequence t_n would, by Proposition 7.13 applied to the fixed representation δ on scale T_n , give a convolution limit for $T_n = t_n + (T_n - t_n)$. The mass of a nonnegative sum below C_0 cannot exceed the mass of either summand below C_0 . Weak-limit boundary issues are removed by the stated slack in the cutoff.

Lemma 4.5 identifies all these limiting effects as scalar. The corner transfer in Lemma 8.1 identifies their scalar values in the original regular model. It follows that the same lower bound holds on every fixed bounded column, uniformly over the indicated times. Effect boundedness and Hilbert density extend it to every fixed ξ , proving (8.25). Tensor density gives normal Hilbert amplifications. The compact perturbation statement follows from the same short approximation and rescaled-effect argument. $\square$

8.5. Complementary localization and the opposite-pair estimate. Fix an isolated source factor S and its matched target factor T . The two factors occur in copies of the same ambient group G ; the arbitrary right group Λ supplies no geometric factor. Write $\hat{G}_S$ and $\hat{G}_T$ for the respective complementary products. We work first with one measured copy. Its pure-vertex family is denoted by $x \mapsto P_x$, for $x \in B_S = \partial S$, with values in the projection-valued measures on $B_T = \partial T$. The sharpness and covariance from Section 5 give

$$\sigma_s P_x = P_{sx}, \quad \beta_h P_x(f) = P_x(f \circ h^{-1}),$$

and the complementary source factors fix this family. Let d_S, d_T be the compact Carnot distances, let dx be compact angular probability on B_S , and let Q_S, Q_T be the homogeneous dimensions.

Our next goal is to control how the measurements change when their source labels approach one another. We will integrate the squared norms of ordered products $P_y(g)P_x(f)$, where f, g are smooth target functions with separated supports. Lattice packing gives a bound proportional to $1 + L$ when measured Fourier length is at most L . The stabilizer of an opposite source pair has a split parameter. For suitable $T_n \rightarrow \infty$ and $A_0 < \infty$, its positive segment of length comparable to T_n retains positive Fourier mass below $L = A_0 T_n$ for each fixed input

vector, by Proposition 8.7. Averaging over that segment and dividing by T_n removes the length factor from the packing bound. Unfolding then gives the integrated opposite-pair estimate used for horizontal differences in the next subsection.

Every lattice count in this argument keeps a compact window in $\widehat{G}_T$. The projection of Γ to T need not be discrete. We first construct an isometry that permits this complementary window while preserving the source action and the angular measurements.

Lemma 8.8 (Complementary localization). *There is a base-decomposable isometry*

$$V : \mathcal{H} \longrightarrow \ell^2 \otimes L^2(\widehat{G}_T) \otimes \mathcal{H}$$

such that, for $s \in S$ and measured labels (h, λ) ,

$$\begin{aligned} VU_s &= (1 \otimes 1 \otimes U_s)V, \\ VP_x(f) &= (1 \otimes 1 \otimes P_x(f))V, \\ VW_{(h,\lambda)} &= (1 \otimes L_{\hat{h}} \otimes W_{(h,\lambda)})V. \end{aligned}$$

Here $L_q\zeta(p) = \zeta(q^{-1}p)$, and $\hat{h}$ is the complementary projection of h . Consequently, for every compact $C \subset \widehat{G}_T$, the positive contraction

$$W_C = V^*(1 \otimes 1_C \otimes 1)V$$

commutes with the base multipliers, U_s , W_Λ , and all the $P_x(f)$. For a compact exhaustion, $W_C \rightarrow 1$ strongly.

Proof. Use the regular corner from the complementary regular realization in Section 3 and amplify its left Fourier action by $\widehat{\Delta}(w_{(h,\lambda)}) = L_{\hat{h}} \otimes w_{(h,\lambda)}$. In the rectangular trace Hilbert space with corners $\widehat{\Delta}(e), e$, source transport acts by

$$Z(y) \longmapsto \widehat{\Delta}(X_s(s^{-1}y))Z(s^{-1}y)X_s(s^{-1}y)^*.$$

The scalar projective phases cancel. We show that every nonzero projection $d_2 \in \mathcal{N}^S$ contains the right support of a nonzero invariant column.

Choose a unit vector $\psi \in L^2(\widehat{G}_T)$ with strictly positive translation coefficient $q \mapsto \langle L_q\psi, \psi \rangle$, bounded below on each compact set. For example, choose an everywhere positive square-integrable function; its coefficient is positive and continuous. Test the preceding action on $\widehat{\Delta}(d_2)(\psi \otimes d_2)$. Reindexing the base and using $X_s(y)d_2(y)X_s(y)^* = d_2(sy)$ expresses its overlap with its s -translate as the sum of squared Fourier coefficients of $X_s(y)d_2(y)$ weighted by $\langle L_{\hat{h}}\psi, \psi \rangle$. The total unweighted mass is $\|d_2\|_2^2$. Complementary tightness for a pure isolated type makes these overlaps bounded below by a positive constant, uniformly in $s \in S$. The uniformity follows first on a Cartan ray and then on all of S by Cartan decomposition and the compact-transport approximation used above; bounded Cartan parameters form a strongly continuous compact family.

The element of least norm in the closed convex hull of this orbit is therefore nonzero and invariant. Its right support is contained in d_2 . Polar decomposition in the rectangular affiliated-operator space gives an invariant partial-isometry column; its initial projection belongs to $\mathcal{N}^S$. Repeat in the orthogonal complement of the initial projection. Separability gives a countable exhaustion. Place the resulting columns in separate multiplicity coordinates. Their direct sum V_Δ satisfies $V_\Delta^*V_\Delta = e$ and the source intertwining relation. In particular it is bounded, and its integrated rectangular trace Hilbert norm squared is $(\text{Tr} \otimes \tau)(e) = d$.

We now remove the left amplification. For a Fourier output index k and parameter $P \in \widehat{G}_T$, use the unitary change of variables

$$F_k(P) \longmapsto \widetilde{F}_k(p) = F_k(\hat{k}p).$$

Left multiplication by $\widehat{\Delta}(w_h)$ becomes ordinary left multiplication, while right multiplication by w_h^* acquires $L_{\hat{h}}$ in the new variable. This proves the source and right action relations for the resulting isometry V .

The same change of variables leaves diagonal angular Fourier tests unchanged. To pass such a test through V_Δ on a defining pure path, first trim the extra Hilbert coordinate to finite rank and approximate the remaining column by bounded finite Fourier columns in integrated trace Hilbert norm. A fixed left Fourier shift does not change the limiting image direction. Moving each such shift through the angular test leaves its uncut Gram factor. The approximation error tends to zero on bounded input columns because $V_\Delta^* V_\Delta = e$; Hilbert density then handles every input. Thus compression of the constant-amplified limiting angular measure by V is P_x . Both measures are sharp. For a spectral projection E with $V^* E V = P$, the equality $\|(EV - VP)\xi\|^2 = 0$ proves exact intertwining. This proves the angular relation. The assertions about W_C now follow directly from the displayed intertwining and monotone exhaustion of $L^2(\widehat{G}_T)$. $\square$

Definition 8.9 (Controlled angular masks). Fix an integer $m \geq 1$ and positive constants specifying support diameter, separation, and overlap. A controlled family at scale $0 < \delta \leq \delta_0$ consists of smooth functions $\{f_A\}_A, \{g_D\}_D$ on B_T and a set $\mathcal{R}$ of ordered pairs (A, D) . Each support lies in a ball of radius $a\delta$; each family of centers has boundedly many points in every ball of radius $b\delta$, for each fixed b ; and

$$\text{dist}_T(\text{supp } f_A, \text{supp } g_D) \geq c\delta \quad ((A, D) \in \mathcal{R}).$$

After translation and graded dilation to a unit chart, derivatives through order m are bounded by one. Changing this last bound only changes the constants by scalar rescaling. We permit any fixed enlargement of the diameter, overlap, or separation parameters. Set

$$p_\delta(x, y) = \sum_{(A, D) \in \mathcal{R}} P_x(f_A)^* P_y(|g_D|^2) P_x(f_A).$$

This is a positive operator. Bounded overlap and spectral calculus give $0 \leq p_\delta(x, y) \leq C_0 1$, with C_0 depending only on the controls, without any commutation assumption between P_x and P_y .

Lemma 8.10 (Packing with a complementary window). *There is an integer m_0 , depending only on the fixed boundary charts and the geometric support, separation, and overlap parameters, such that for each compact $C \subset \widehat{G}_T$, controlled masks of order $m \geq m_0$ satisfy*

$$(8.31) \quad \sum_{h \in \Gamma: r_T(h) \leq L, \hat{h} \in PC^{-1}} W_h p_\delta(x, y) W_h^* \leq C_C (1 + L) \delta^{-Q_T} 1$$

for every $P \in \widehat{G}_T$, $L \geq 1$, and source opposite pair (x, y) . The constant is independent of P, L, δ, x, y and the number of masks.

Proof. We prove a bound $C_C \delta^{-Q_T}$ in each unit length shell. Choose a sufficiently small relatively compact identity neighborhood U such that hU , $h \in \Gamma$, are disjoint. Right thickening changes $r_T(h)$ by a bounded amount and changes the forward direction h^+ by $O(e^{-r})$ in a shell $r \leq r_T(h) < r + 1$. It enlarges PC^{-1} to a translate of a fixed compact set. Cartan integration therefore gives, for a ball of radius $R \geq e^{-r}$,

$$(8.32) \quad \#\{h : r \leq r_T(h) < r + 1, \hat{h} \in PC^{-1}, h^+ \in B(z, R)\} \leq C_C (1 + R e^r)^{Q_T}.$$

Indeed the angular ball has measure at most $C(R + e^{-r})^{Q_T}$, the radial shell has Jacobian at most $C e^{Q_T r}$, and the complementary factor has bounded Haar volume independent of its left translate P . This argument counts full lattice points, not points in a possibly nondiscrete projection.

We record the two boundary estimates used below. Put $R_0 = C' e^{-r} / \delta$. A set at distance at least $c\delta/2$ from h^- has image under h contained in $B(h^+, R_0)$. Moreover, if $\text{supp } f_A$ is within $c\delta/2$ of h^- , then on an annulus at distance $\rho \geq R_0$ from h^+ ,

$$(8.33) \quad \|f_A \circ h^{-1} - f_A(h^-)\|_{C^m(\rho\text{-normalized charts})} \leq C_m \frac{R_0}{\rho}.$$

Here one may enlarge C' by a fixed factor. Use the normalized Carnot charts of Lemma 7.1. To see these estimates, put the two polar endpoints at zero and infinity. The Cartan element is a graded dilation of ratio e^{-r} in the chart at the attracting endpoint. Chart inversion sends an annulus of radius ρ to an annulus of radius comparable to ρ^{-1} , with uniformly bounded derivatives after graded normalization. Thus the inverse image of the annulus is within $O(e^{-r}/\rho)$ of the repelling endpoint. Taylor's formula for a mask whose derivatives are normalized at scale δ proves (8.33). Every nonconstant normalized derivative has the same bound: its chain-rule factors contain at least one positive-degree dilation factor. Compact polar rotations preserve the chosen horizontal metric, so all constants are uniform in h .

If R_0 is bounded below by a fixed positive number, total shell counting already gives $C_C \delta^{-Q_T}$. Assume otherwise. From (8.32), for either one of the individual PVMs P_x, P_y , we obtain

$$(8.34) \quad \sum_h P_z(1_{B(h^+, bR_0)}) \leq C_{C,b} \delta^{-Q_T} 1 \quad (z = x \text{ or } y),$$

where the sum is over the shell and complementary window. This is just the pointwise counting inequality integrated against a single PVM.

Conjugation by W_h composes the masks with h^{-1} . For those A whose supports are at distance at least $c\delta/2$ from h^- , the rightmost masks are supported in $B(h^+, R_0)$. Bounded overlap gives

$$\begin{aligned} \sum_{A,D} \|P_y(g_D \circ h^{-1}) P_x(f_A \circ h^{-1}) \xi\|^2 \\ \leq C \sum_A \|P_x(f_A \circ h^{-1}) \xi\|^2 \leq C \|P_x(1_{B(h^+, R_0)}) \xi\|^2. \end{aligned}$$

Summing in h uses (8.34) for P_x .

Only boundedly many remaining A occur for each h . Separation implies that all their paired g_D are far from h^- . Summing the g_D squares reduces their contribution to

$$C \sum_{h,A \text{ remaining}} \|P_y(1_{B(h^+, R_0)}) P_x(f_A \circ h^{-1}) \xi\|^2.$$

Subtract the scalar $f_A(h^-)$. The scalar term is controlled by (8.34) for P_y . The remainder in an enlarged pole ball is controlled by that equation for P_x , using its uniformly bounded amplitude. Decompose the rest into annuli of radii $\rho_j = 2^j R_0$. On each annulus, (8.33) gives normalized size $O(2^{-j})$.

Here is the summation detail that preserves the order of the two PVMs. Fix a bounded-overlap global cover by cells of scale ρ_j . Only boundedly many cells meet any one of these annuli. Smooth expansion on each normalized cell gives

$$F_{h,A,j} = \sum_{E,\nu} c_{h,A,E,\nu} \varphi_{E,\nu}, \quad |c_{h,A,E,\nu}| \leq C_m 2^{-j} (1 + |\nu|)^{-m},$$

where $F_{h,A,j}$ is the annular remainder, $\varphi_{E,\nu}$ is supported in an enlargement of E , and the functions $\varphi_{E,\nu}$ do not depend on h or A . For fixed ν they have a uniformly bounded square sum over E . The expansion follows by a smooth cutoff to a unit cube and its Fourier series; taking m above the chart dimension makes the weights summable. Cauchy-Schwarz over the bounded number of cells for a fixed h, A and then (8.34) for P_y give

$$\begin{aligned} \sum_{h,A,E} |c_{h,A,E,\nu}|^2 \|P_y(1_{B(h^+, R_0)}) P_x(\varphi_{E,\nu}) \xi\|^2 \\ \leq C_m 2^{-2j} (1 + |\nu|)^{-2m} \delta^{-Q_T} \sum_E \|P_x(\varphi_{E,\nu}) \xi\|^2 \\ \leq C_m 2^{-2j} (1 + |\nu|)^{-2m} \delta^{-Q_T} \|\xi\|^2. \end{aligned}$$

In the first inequality the vector $P_x(\varphi_{E,\nu}) \xi$ is independent of h . Sum first over frequencies and then over annuli in the Hilbert norm of the (h, A) -indexed family. The sums $\sum_\nu (1 + |\nu|)^{-m}$

and $\sum_j 2^{-j}$ converge. This proves the required shell bound. Summing the $O(1+L)$ shells proves (8.31). $\square$

Let $B_S^{(2)}$ be the space of ordered opposite pairs. In rank one this is $B_S \times B_S$ with the diagonal removed. Let dm be its S -invariant measure. In compact charts,

$$(8.35) \quad dm(x, y) \asymp d_S(x, y)^{-2Q_S} dx dy.$$

For completeness, first use horospherical Haar measure dn in a source boundary chart. Write the smooth positive pair density there as $F(n, n') dn dn'$. Simultaneous horospherical translation gives $F(n, n') = F(e, n^{-1}n')$. A split dilation scales each Haar variable by t^{Q_S} and preserves the pair measure, so

$$F(e, \delta_t w) = t^{-2Q_S} F(e, w).$$

The density is smooth and strictly positive on a compact homogeneous unit annulus. It is therefore comparable to the homogeneous gauge to the power $-2Q_S$. On buffered compact charts, compact angular probability has a smooth density bounded above and below relative to dn , and the Carnot distance is comparable to that gauge. Finitely many such charts give (8.35).

Proposition 8.11 (Integrated opposite-pair estimate). *There is a dense linear subspace $\mathcal{E} \subset \mathcal{H}$ such that, for each fixed set of mask controls and each $\xi \in \mathcal{E}$, there is a finite constant C_ξ with*

$$(8.36) \quad \int_{B_S^{(2)}} \langle \xi, p_\delta(x, y) \xi \rangle dm(x, y) \leq C_\xi \delta^{-Q_T}$$

for all controlled families and $0 < \delta \leq \delta_0$. The subspace can be chosen invariant under the source and target actions and under every bounded operator commuting with all $P_x(f)$. Constants may change with the vector and the fixed controls, but not with δ or the chosen masks.

Proof. Fix an opposite pair (x_0, y_0) and let $J_0 = MA < S$ be its compact-by-split stabilizer. Temporarily write $p_\delta = p_\delta(x_0, y_0)$. We first place a Parseval frame below this positive effect, then apply Fourier packing to its complementary localization. The stabilizer average will cancel the length factor, after which we can unfold the estimate over all opposite pairs.

Choose complementary Parseval frame columns $b_j \in H$ for the joint action π_0, W_Λ ; thus

$$\sum_{\gamma, \lambda, j} |W_\lambda \pi_0(\gamma) b_j\rangle \langle W_\lambda \pi_0(\gamma) b_j| = 1$$

in the strong sense. Such a finite frame of orbit generators is part of the complementary regular realization. Let $f \geq 0$ be bounded and compactly supported on G , with

$$(8.37) \quad \int_{J_0} f(ag) da \leq 1 \quad (g \in G).$$

Using the induction section d_0 , define the fixed Hilbert vectors

$$v_{\gamma j}(o') = f(d_0(o')\gamma)^{1/2} p_\delta(o')^{1/2} \pi_0(\gamma) b_j.$$

The vectors depend on the masks and on f , which remain fixed throughout the later limit in n . Since J_0 fixes the reference pair, U_a commutes with p_δ . At each base point, reindexing γ by the section cocycle and using (8.37) gives the following inequality of positive operators on the fibre H :

$$\int_{J_0} \sum_{\lambda, \gamma, j} |W_\lambda(U_a v_{\gamma j})(o')\rangle \langle W_\lambda(U_a v_{\gamma j})(o')| da \leq p_\delta(o') \quad \text{for almost every } o'.$$

Its global rank-one consequence on $\mathcal{H} = L^2(Y; H)$ is

$$(8.38) \quad \int_{J_0} \sum_{\lambda, \gamma, j} |W_\lambda U_a v_{\gamma j}\rangle \langle W_\lambda U_a v_{\gamma j}| da \leq p_\delta.$$

Indeed the reindexed scalar is $f(a^{-1}d_0(o')\gamma')$, where $\gamma' = c(a, a^{-1}o')\gamma$; its integral is at most one. The remaining sum is the Parseval identity compressed by $p_\delta^{1/2}$. We retain this pointwise positive inequality for the Fourier-row tests below. It also gives the displayed inequality on the global Hilbert space by first applying

$$\left| \int_Y \langle \xi(y), v(y) \rangle dy \right|^2 \leq \int_Y |\langle \xi(y), v(y) \rangle|^2 dy$$

and then summing and integrating. The projective phases disappear in these pointwise squared coefficients.

We next identify the Fourier norm to which packing applies. Let V_Δ be the column in the proof of Lemma 8.8, before the change of variables that removes its left amplification. At complement parameter P and base point o' , an adjoint identity row is $V_\Delta(P, o')^* \mathbf{1}_{\text{row}}$. For a fibre input $z \in H$, its pairing with $W_h W_\lambda z$ reads the corresponding output row of the (h, λ) Fourier coefficient of $V_\Delta z$:

$$\langle V_\Delta(P, o')^* \mathbf{1}_{\text{row}}, W_h W_\lambda z \rangle.$$

This uses the right-adjoint convention for W . Summing the squared coefficients over rows and λ is therefore the Fourier Hilbert norm in the measured h coordinate. After the change of variables in the localization lemma, $P = \hat{h}p$. Thus

$$p \in C \iff \hat{h} \in PC^{-1}.$$

This is precisely the complementary window in the packing inequality.

Test (8.31) on these adjoint rows, insert the pointwise frame inequality underlying (8.38), and integrate in P, o' . The integrated sum of their squared trace norms is $(\text{Tr} \otimes \tau)(e) = d$. The preceding coefficient identity and Parseval therefore give, with $D_{\leq L}$ the measured Fourier projection onto $r_T(h) \leq L$,

$$(8.39) \quad C_C(1+L)\delta^{-Q_T} \geq \int_{J_0} \sum_{\gamma, j} \|D_{\leq L} 1_C V U_a v_{\gamma j}\|^2 da.$$

The coefficient calculation is first made for bounded columns of finite Fourier support. Approximation and the finite integrated squared norm of the rows extend it to the fixed Hilbert vectors above. Positivity permits all sums and integrals by Tonelli's theorem.

We now choose the stabilizer interval so that its length cancels the factor $1+L$. Proposition 8.7 provides $T_n \rightarrow \infty$, $A_0 < \infty$, and $\eta_0 > 0$ such that, with fixed slack in A_0 , the cutoff $r_T \leq A_0 T_n$ has lower limiting mass at least η_0 times the squared norm of each fixed input, uniformly for stabilizer lengths in $[T_n/2, T_n]$. The assertion passes to the constant amplification by tensor density. Since 1_C commutes with the constant-amplified U_a , the inputs here are the fixed vectors $1_C V v_{\gamma j}$. Restrict (8.39) to that stabilizer segment, set $L = A_0 T_n$, and divide by T_n . Its Haar measure is comparable to T_n , with the M factor compact. Fatou's lemma, first for finite sums of (γ, j) and then by monotone exhaustion, yields

$$(8.40) \quad \sum_{\gamma, j} \langle v_{\gamma j}, W_C v_{\gamma j} \rangle \leq C'_C \delta^{-Q_T}.$$

The constant is independent of the density f satisfying (8.37).

Undo induction by lifting at $g = d_0(o')\gamma$ with operator conjugation by $\pi_0(\gamma)^*$. Since W_C commutes with U_S , its uninduced field W_C^0 depends only on the complementary coordinate $g' \in \hat{G}_S$. Since the pure-vertex family is fixed by complementary source factors, its uninduced effect depends only on the inverse-translated pair. These two operators commute. Unfolding the sum over γ in (8.40) therefore integrates

$$\sum_j \langle b_j, W_C^0(g') p_\delta^0(x, y) b_j \rangle$$

against the density induced by f on $J_0 \backslash G \cong B_S^{(2)} \times \widehat{G}_S$. Every compactly supported quotient density between zero and one can be obtained by choosing, on finitely many local sections, a nonnegative compactly supported fibre density of integral one and using a partition of unity. Exhausting the quotient and applying Tonelli's theorem gives

$$(8.41) \quad \int_{\widehat{G}_S} \int_{B_S^{(2)}} \sum_j \langle b_j, W_C^0(g') p_\delta^0(x, y) b_j \rangle dm(x, y) dg' \leq C'_C \delta^{-Q_T}.$$

The same argument applies to any fixed frame translate $\pi_0(\gamma_0)W_{\lambda_0}b_j$: its full orbit is again a Parseval family, up to irrelevant scalar phases.

We spell out why this gives a dense supply of constant inputs for (8.36). Restrict o' to a measurable set on which $d_0(o')$ lies in a compact subset K of G . Haar measure on the section image is restricted Haar, up to the fixed covolume normalization. Projection of its restriction to K onto $\widehat{G}_S$ is dominated by a finite constant times Haar measure: the fibre in S lies in a fixed compact set of finite Haar volume. For a bounded scalar function a supported on this set, put

$$\xi(o') = a(o')W_C(o')^{1/2}\pi_0(\gamma_0)W_{\lambda_0}b_j.$$

Commutation with the angular effects moves $W_C^{1/2}$ into the positive pairing. Changing the pair variable by the S coordinate of $d_0(o')$ preserves dm . The preceding Haar domination and (8.41) now prove (8.36) for this fixed ξ . Finite sums are handled by Cauchy–Schwarz. Such vectors span a dense space: compact section restrictions exhaust Y , the frame translates span H , bounded simple functions are dense in the base variable, and $W_C^{1/2} \rightarrow 1$ strongly along a compact exhaustion.

Finally, let $\mathcal{E}$ be the maximal linear domain of vectors for which (8.36) holds for every fixed finite set of mask controls of sufficiently high order. The dense supply just obtained is contained in this domain. A bounded operator commuting with all $P_x(f)$ commutes with p_δ and preserves the bound, with constant multiplied by its squared norm. A fixed source transformation preserves dm and reindexes the pair. A fixed target transformation changes scale, support, separation, and normalized derivatives by fixed constants because it is a smooth conformal map of the compact horizontal boundary. Thus it maps a controlled family into another permitted controlled family. These observations prove all the asserted domain invariances. $\square$

8.6. Horizontal differentiation and an injective height. We retain the matched isolated factors S, T and their boundaries B_S, B_T . Their homogeneous dimensions are denoted by Q_S, Q_T . The Carnot distances are d_S, d_T , and dx is compact-rotation probability on B_S . For $R = S, T$, let $\mathcal{V}_R \subset TB_R$ be the horizontal bundle, with its compact horizontal metric. Its iterated brackets span TB_R . The action is horizontally conformal; write $\text{dil}(a, x) > 0$ for its horizontal dilation. Thus

$$(8.42) \quad \text{dil}(ab, x) = \text{dil}(a, bx) \text{dil}(b, x).$$

All the isolated factors in the present setting have $Q_T > 2$.

Let $\mathcal{E} \subset \mathcal{H}$ be the maximal invariant linear domain defined by the estimates in Proposition 8.11; that Proposition proves its density. In particular, (8.36) holds for every $\xi \in \mathcal{E}$, uniformly over each fixed collection of normalized mask controls. Constants below may depend on ξ and these controls. The domain is invariant under the source and measured symmetries and under every bounded operator commuting with all the P_x . We use the strong continuity, sharpness, covariance, and nonconstancy of the pure-vertex family $x \mapsto P_x$ established previously.

Definition 8.12. For $F \in C^\infty(B_T \times B_T)$, its ordered evaluation is

$$\mathcal{K}_{x,y}(F) = \iint F(w, v) P_y(dw) P_x(dv).$$

More explicitly, on finitely many coordinate rectangles expand F as an absolutely summable series of products of smooth functions; replace $g(w)f(v)$ by $P_y(g)P_x(f)$ and sum in operator norm. Smooth Fourier expansion with sufficiently many derivatives gives absolute summability, with the sum bounded by a fixed sufficiently high smooth seminorm of F . Approximation in that seminorm by finite separated smooth sums proves independence of the coordinate expansion.

A smooth kernel has horizontal order at least one if it vanishes on the diagonal. It has horizontal order at least two if, in addition, its derivative in the second variable vanishes there on $\mathcal{V}_T$. These conditions are invariant under the coordinate changes used below.

The next estimate uses $\mathcal{K}_{x,y}$ in its displayed order. No commutation between P_x and P_y is needed.

Lemma 8.13. *Fix a compactly contained source horospherical chart, a bounded nonnegative source cutoff χ , and a smooth nonnegative function ϑ of compact support on its nilpotent coordinate group. Set $y = x\delta_\epsilon w$ in that chart, where δ_ϵ is Carnot dilation, and use the measure $d\nu_\epsilon = \chi(x)\vartheta(w) dx dw$. For ϵ sufficiently small, the following conclusions hold.*

(i) *If F has horizontal order at least one, then, for each $\xi \in \mathcal{E}$,*

$$\|\mathcal{K}_{x,y}(F)\xi\|_{L^2(d\nu_\epsilon; \mathcal{H})} \leq C_{F,\xi} \epsilon^{Q_S/Q_T}.$$

(ii) *If $Q_S = Q_T = Q > 2$ and F has horizontal order at least two, then the same norm is $o(\epsilon)$. More precisely it is bounded by a constant times*

$$\begin{cases} \epsilon^{Q/2}, & 2 < Q < 4, \\ \epsilon^2(1 + |\log \epsilon|), & Q = 4, \\ \epsilon^2, & Q > 4. \end{cases}$$

The constants are uniform when a sufficiently large fixed finite collection of the indicated smooth seminorms of F is bounded.

Proof. Under the change from w to y , the density of ν_ϵ with respect to $dx dy$ is at most $C\epsilon^{-Q_S}$ and is supported where $d_S(x, y) \leq C\epsilon$. Since the invariant pair measure has density comparable to $d_S(x, y)^{-2Q_S}$, the pullback of any separated target mask effect satisfies

$$(8.43) \quad \int \langle \xi, p_\delta(x, y)\xi \rangle d\nu_\epsilon \leq C_\xi \epsilon^{Q_S} \delta^{-Q_T}.$$

Its bound by a constant times $\|\xi\|^2$ also follows from the uniform operator bound for p_δ in Definition 8.9.

Here are the details needed to apply this estimate to a kernel. Choose a smooth dyadic decomposition of the complement of the target diagonal. The piece at scale δ is supported where $c\delta \leq d_T(w, v) \leq C\delta$. Cover each boundary by cells of radius $c'\delta$ with c' small relative to c . The pairs of cells meeting this support have bounded degree: each cell is paired with at most a fixed number of other cells. Their enlarged supports are separated, and their centers have uniformly bounded density at scale δ .

Suppose F has horizontal order at least m , where $m = 1$ or 2 . On each such product of cells, every fixed finite collection of normalized derivatives of the kernel piece is bounded by $C\delta^m$. For $m = 1$, this is Taylor expansion at the diagonal. For $m = 2$, the constant and horizontal linear terms vanish; the remaining horizontal quadratic terms and the nonhorizontal linear terms both have Carnot degree at least two. Taylor expansion in the rescaled coordinates also gives the same estimate after each of the specified normalized derivatives. Compactness makes the bounds uniform in the cell centers.

Expand the rescaled kernel in a Fourier series on slightly larger coordinate boxes. Integration by parts in both variables, to an order larger than the box dimensions and the required mask orders, gives a summable family of separated terms

$$\delta^m c_{A,D;k,l} g_{D,k}(w) f_{A,l}(v).$$

The coefficients decrease faster than the polynomial growth of the normalized mask seminorms in k, l . For a fixed pair of frequencies, the leftmost supports have bounded overlap. The spectral calculus for P_y , followed by Cauchy–Schwarz for the bounded number of terms at a cell, therefore gives

$$\left\| \sum_{A,D} c_{A,D;k,l} P_y(g_{D,k}) P_x(f_{A,l}) \xi \right\|^2 \leq C_{k,l} \sum_{A,D} \|P_y(g_{D,k}) P_x(f_{A,l}) \xi\|^2.$$

Coefficients can be absorbed into the masks; duplicating an index when necessary changes density and overlap by only the bounded degree. Thus (8.43), followed by the summation over frequencies, bounds this annular piece in root mean square by

$$(8.44) \quad C_{F,\xi} \delta^m \min\{1, \epsilon^{Q_S/2} \delta^{-Q_T/2}\}.$$

This computation uses only one PVM for the leftmost overlap and the ordered positive effect for the remaining factors.

There is also an estimate $C_F \delta_0^m \|\xi\|$ for the entire kernel piece supported where $d_T(w, v) \leq C\delta_0$. Indeed repeat the cell expansion at scale δ_0 ; only boundedly many cells meet a given cell, and the constant mask bound applies without separation.

Put $a = Q_S/Q_T$ and choose a dyadic δ_0 comparable to ϵ^a . For $m = 1$, the triangle inequality and (8.44) give

$$C_{F,\xi} \left(\delta_0 + \epsilon^{Q_S/2} \sum_{\delta \geq \delta_0} \delta^{1-Q_T/2} \right) \leq C'_{F,\xi} \epsilon^a,$$

because $Q_T > 2$. When $Q_S = Q_T = Q$ and $m = 2$, take $\delta_0 \asymp \epsilon$ instead. Summing $\epsilon^{Q/2} \delta^{2-Q/2}$ gives exactly the three rates in the statement, including the $C\epsilon^2$ diagonal remainder. Each rate is $o(\epsilon)$. $\square$

Lemma 8.14. *For every matched isolated pair, $Q_S = Q_T$. Write their common value as Q . If X is a smooth horizontal source field, $f \in C^\infty(B_T)$, and $\xi \in \mathcal{E}$, then the distributional derivative*

$$D_X(f)\xi = X(x \mapsto P_x(f)\xi)$$

belongs to $L^2(B_S, dx; \mathcal{H})$. Differentiation is linear in f, ξ , and is linear over smooth source functions in X .

Proof. For $F(w, v) = f(w) - f(v)$, ordered evaluation is $P_y(f) - P_x(f)$. To extract a horizontal derivative, use a smooth compactly supported weight $\psi_i(w)$ whose horizontal first moments are the i th coordinate vector. Its absolute value is dominated by a fixed smooth nonnegative weight of the kind used in Lemma 8.13. The expressions

$$\frac{1}{\epsilon} \int \psi_i(w) (P_{x\delta_\epsilon w}(f) - P_x(f)) \xi \, dw$$

are consequently bounded in local L^2 by $C\epsilon^{Q_S/Q_T-1}$. For a smooth compactly supported test field, change variables from x to $x\delta_\epsilon w$ in its adjoint pairing. First-order Taylor expansion of the test field and the smooth measure density shows that the limit is the distributional horizontal derivative: the adjoint test is $-X\varphi - (\operatorname{div}_{dx} X)\varphi$.

If $Q_S/Q_T > 1$, all these derivatives vanish. Brackets of horizontal fields then annihilate the same distribution, because $[X, Y]u = X(Yu) - Y(Xu)$. Bracket generation implies that every coordinate derivative vanishes, so $P_x(f)\xi$ is constant on the connected boundary. Density of $\mathcal{E}$ and strong continuity make P_x constant, a contradiction. Hence $Q_S \leq Q_T$. The matching is a permutation of the finite collection of isolated types; going around each cycle forces equality throughout that cycle.

With equality, the displayed difference quotients are bounded in L^2 . Weak compactness and their identified distributional limit give $D_X(f)\xi \in L^2$ in each chart. A finite cover and a

smooth partition of unity give the global statement. The linearity assertions follow directly from distributional differentiation. $\square$

We next specify the operator domain. This is necessary because an individual vector $P_x(g)\xi$ need not belong to the constant-vector estimate domain $\mathcal{E}$.

Lemma 8.15. *Set $\mathcal{K} = L^2(B_S, dx; \mathcal{H})$ and*

$$\mathcal{C} = \text{span}\{x \mapsto a(x)P_x(g)\xi : a \in L^\infty(B_S), g \in C^\infty(B_T), \xi \in \mathcal{E}\}.$$

For real f and real horizontal X , the rule

$$(8.45) \quad T_X(f) \left(\sum_j a_j P_\bullet(g_j) \xi_j \right) = \sum_j a_j P_\bullet(g_j) D_X(f) \xi_j$$

defines a densely defined symmetric operator on $\mathcal{C}$. Furthermore, if real smooth g_j, f_j satisfy $\sum_j g_j d_{\text{hor}} f_j = 0$, then

$$(8.46) \quad \sum_j P_\bullet(g_j) T_X(f_j) = 0 \quad \text{on } \mathcal{C}.$$

All the operators in (8.45) commute on $\mathcal{C}$ with scalar parameter multipliers and smooth angular multipliers. Every real smooth angularly weighted component $P_\bullet(g)T_X(f)$ is symmetric on $\mathcal{C}$.

Proof. Write $\Delta_{x,y}f = P_y(f) - P_x(f)$. Sharpness gives the exact identity

$$(\Delta_{x,y}f)P_x(g) - P_y(g)\Delta_{x,y}f = \mathcal{K}_{x,y}((f(w) - f(v))(g(v) - g(w))).$$

The right side has horizontal order two. Its norm on any $\xi \in \mathcal{E}$, divided by ϵ , tends to zero in the averages used to recover a horizontal derivative.

Fix $\xi, \eta \in \mathcal{E}$ and real g . In the pairing of the left term with η , move $\Delta_{x,y}f$ to η by adjunction. For the other term, keep the bounded quotient applied to ξ and move $P_y(g)$ to the test vector. Strong continuity permits replacing $P_y(g)$ there by $P_x(g)$: its difference on a fixed test vector tends to zero in L^2 by dominated convergence, whereas the averaged quotients stay bounded in L^2 . The weak derivative limit therefore gives

$$(8.47) \quad \langle P_x(g)D_X(f)(x)\xi, \eta \rangle = \langle \xi, P_x(g)D_X(f)(x)\eta \rangle \quad \text{for almost every } x.$$

One obtains the pointwise assertion by first inserting arbitrary bounded scalar parameter tests. Complex weights follow by linearity and adjunction.

Apply (8.47) with products of angular tests. If $u = \sum_j a_j P_\bullet(g_j)\xi_j$ and $v = \sum_k b_k P_\bullet(h_k)\eta_k$, the formal right side of (8.45) satisfies

$$\langle T_X(f)u, v \rangle_{\mathcal{K}} = \langle u, T_X(f)v \rangle_{\mathcal{K}}.$$

In particular, if an expression for u represents zero, its formal image is orthogonal to every $v \in \mathcal{C}$. This space is dense, since it includes bounded simple parameter fields with values in $\mathcal{E}$. The image is therefore zero. This proves both well-definedness and symmetry, and hence closability. The identical calculation with one more real angular weight proves the final symmetry assertion.

For cancellation, consider

$$F(w, v) = \sum_j g_j(w)(f_j(w) - f_j(v)).$$

It vanishes on the diagonal, and its second-variable horizontal derivative there is $-\sum_j g_j d_{\text{hor}} f_j = 0$. Lemma 8.13 gives an $o(\epsilon)$ bound. Taking the same weak derivative limit and replacing left multipliers at y by those at x gives $\sum_j P_x(g_j)D_X(f_j)(x)\xi = 0$ for every $\xi \in \mathcal{E}$. Commuting the additional multipliers in (8.45) proves (8.46) on $\mathcal{C}$. The asserted commutation on this domain also follows from that rule. $\square$

For a real smooth horizontal target covector field $\theta = \sum_j g_j d_{\text{hor}} f_j$, define

$$T_X(\theta) = \sum_j P_\bullet(g_j) T_X(f_j) \quad \text{on } \mathcal{C}.$$

Equation (8.46) makes this independent of the chosen expression. Such expressions exist globally: finitely many smooth coordinate functions have horizontal differentials spanning the bundle, and their Gram operator has a smooth inverse on that bundle.

Lemma 8.16. *There is a positive self-adjoint measurable field l_x on $\mathcal{H}$, affiliated to $\mathcal{A}_1$, commuting strongly with P_x , and invariant under the complementary source factors, such that, for every fixed source element $a \in S$ and measured lattice element h ,*

$$(8.48) \quad \sigma_a(l_x) = \text{dil}(a, x) l_{ax}, \quad \beta_h(l_x) = \text{dil}(h^{-1}, P_x) l_x$$

for almost every x . The field is not identically zero. Here the target dilation is evaluated by the spectral calculus of P_x , using the T -coordinate of h .

Proof. Choose finite smooth Parseval horizontal source fields $(X_\alpha)_{\alpha=1}^{n_S}$ and finite smooth Parseval horizontal target covectors $(\theta_r)_{r=1}^{n_T}$. For completeness, start with finitely many smooth spanning sections. Their positive Gram operator is invertible on the horizontal bundle; multiplication by its inverse square root gives the required Parseval system. Smooth coordinate differentials provide the target spanning sections, so each θ_r has an expression of the form used above. Define the finite column

$$D : \mathcal{C} \longrightarrow \mathcal{K}^{n_S n_T}, \quad Du = (T_{X_\alpha}(\theta_r)u)_{\alpha, r}.$$

Every component is symmetric on $\mathcal{C}$ by Lemma 8.15. If $u_n \rightarrow 0$ and $Du_n \rightarrow v$, testing each component against $\mathcal{C}$ gives $v = 0$. Thus D is closable. Let $\overline{D}$ be its closure and put $l = |\overline{D}| = (\overline{D}^* D)^{1/2}$ on $\mathcal{K}$.

Multiplication by a scalar parameter unitary preserves $\mathcal{C}$ and commutes with D , acting diagonally on the output. The same is true for $P_\bullet(e^{itg})$, with g real smooth. Their inverses have the same property. These identities extend to the closed graph, and consequently to the closed quadratic form $u \mapsto \|\overline{D}u\|^2$. Therefore l strongly commutes with the parameter multiplication algebra and the angular spectral algebra. In particular $l = \int^\oplus l_x dx$ is decomposable and l_x strongly commutes with P_x almost everywhere.

If V is a unitary commuting with every P_x , invariance of $\mathcal{E}$ and distributional differentiation give $D_X(f)V\xi = VD_X(f)\xi$. This identity extends to $\mathcal{C}$ and then to the closed graph. Applying it to the base unitaries and the unmeasured group actions proves affiliation to $\mathcal{A}_1$; applying it to the complementary source factors proves their invariance. Separability permits a common determining set for these almost-everywhere commutation assertions.

We give the norm argument for covariance to keep track of closures. The Parseval systems realize isometric embeddings of the horizontal bundles into trivial finite-dimensional bundles. Their Gram matrices are the corresponding orthogonal projections. Linearity in X and (8.46) say exactly that Du belongs to the range of these source and spectral target projections. Thus an isometry between horizontal bundles preserves the sum of squares of the column entries, even if the Parseval systems are redundant. In particular, any two finite Parseval systems give the same squared form on the same intrinsic module core. Their closed squared forms, and hence their operators l , agree. The maximal estimate domain $\mathcal{E}$ is also unchanged by replacing the controlled Carnot atlas: a finite refinement and the smooth changes of rescaled coordinates transfer every mask control with a fixed change of its constants, in both directions. Thus the construction depends only on the PVM family and the chosen compact horizontal metrics. The cell expansions used to prove the estimates, and the localizing isometry used to prove density of $\mathcal{E}$, do not enter its definition.

Differentiate the identity $\sigma_a P_x = P_{ax}$ on constant-domain vectors. The chain rule sends a source vector X at x to $da_x X$ at ax . This map is the product of $\text{dil}(a, x)$ and a horizontal isometry. Similarly, differentiating $\beta_h P_x(f) = P_x(f \circ h^{-1})$ replaces a target covector by its

pullback under h^{-1} . Horizontal cancellation justifies writing this pullback in the chosen target coframe. Its norm factor, evaluated spectrally, is $\text{dil}(h^{-1}, P_x)$; its other factor is a horizontal isometry. The preceding projection identities justify dropping those isometries when taking the sum of squared column norms.

These identities hold on the full core $\mathcal{C}$. Indeed source transport reindexes its scalar coefficients, transforms $\mathcal{E}$ into itself, and transports $P_x(g)$ by covariance; measured transport replaces g by $g \circ h^{-1}$ and also preserves $\mathcal{E}$. On $\mathcal{K}$, source change of variables includes the square root of the smooth Radon–Nikodym derivative of dx . This is an allowed scalar multiplier and commutes with D , so it contributes no additional derivative. For each fixed group element, all dilation factors and inverses are bounded on the compact boundaries. Source and target transforms therefore give invertible maps of the graph norms. The identities extend to the closed quadratic forms. The uniqueness of the positive operator associated to such a form proves (8.48); spectral target dilations commute strongly with l , as already proved.

Finally, if $l = 0$, every column component vanishes on $\mathcal{C}$. Spanning by the Parseval fields and covectors, followed by horizontal cancellation, gives $D_X(f)\xi = 0$ for every smooth f , horizontal X , and $\xi \in \mathcal{E}$. The bracket argument in Lemma 8.14 would make P_x constant. Hence $l \neq 0$. $\square$

Proposition 8.17 (Height at an isolated type). *For a matched isolated pair the homogeneous dimensions agree, $Q_S = Q_T = Q$. There is a family of positive injective self-adjoint operators b_x , affiliated to $\mathcal{A}_1$, for every $x \in B_S$, which strongly commute with P_x and are invariant under the complementary source factors. They satisfy the exact identities*

$$(8.49) \quad \sigma_a(b_x) = \text{dil}(a, x)^{-Q} b_{ax}, \quad \beta_h(b_x) = \text{dil}(h^{-1}, P_x)^{-Q} b_x.$$

These are the inverse angular-Jacobian laws. The same construction passes to the normal copies used in the amplified models.

Proof. We first replace the field in Lemma 8.16 by an everywhere covariant version. Fix $x_0 \in B_S$ and let J be its parabolic stabilizer. Pull the field back along $g \mapsto gx_0$ and set

$$F(g) = \text{dil}(g, x_0) \sigma_g^{-1} (l_{gx_0}).$$

This is a measurable field of positive self-adjoint operators. One can test measurability through their bounded resolvents, or their closed graph projections. Haar disintegration over S/J and (8.48) give $F(ag) = F(g)$ for almost every g , for each fixed $a \in S$. Fubini, followed by the nonsingular change of variables $(a, g) \mapsto (ag, g)$, shows that $F(g_1) = F(g_2)$ for almost every pair. Thus F has a single essential value L_0 . A countable determining family of bounded matrix coefficients of graph projections suffices for this argument.

For each $k \in J$ the definition and (8.42) give

$$F(gk) = \text{dil}(k, x_0) \sigma_k^{-1} F(g).$$

Consequently $\sigma_k L_0 = \text{dil}(k, x_0) L_0$. We may therefore define, independently of the representative g ,

$$l_{gx_0} = \text{dil}(g, x_0)^{-1} \sigma_g L_0.$$

This agrees with the old field almost everywhere and satisfies its source covariance for every g, x . Its other properties hold as well. To see the target law explicitly, commute β_h past σ_g in the pullback. Since $\sigma_g P_{x_0} = P_{gx_0}$, its spectral dilation becomes the fixed operator $\text{dil}(h^{-1}, P_{x_0})$ on $F(g)$. The essential value L_0 therefore satisfies the target law at x_0 , and transport proves it everywhere. Angular commutation, affiliation, and complementary source invariance pass to L_0 and transport in the same way.

Let $e_0 = \text{supp } L_0$. The target dilation is bounded, positive, invertible, and commutes with L_0 . Taking supports in its covariance law therefore gives $\beta_h(e_0) = e_0$ for all h , so $e_0 \in \mathcal{N}$. Source stabilizer covariance gives $\sigma_k(e_0) = e_0$ for $k \in J$. In addition, e_0 is fixed by every complementary source factor. Choose a split sequence in J escaping in S , and multiply it by sequences escaping in each complementary factor. Every resulting element fixes e_0 , while

the all-factor mixing established in Section 3 makes its translates converge weakly to $\tau(e_0)1$. Hence $e_0 = \tau(e_0)1$. It is nonzero because the derivative column was nonzero; as a projection it must be 1. This use of complementary source invariance is essential to the application of mixing.

All transported l_x thus have zero kernel. Define $b_x = l_x^{-Q}$ by the spectral calculus. It is densely defined, positive, self-adjoint, injective, and affiliated to $\mathcal{A}_1$. Taking inverse powers in the two covariance laws, using strong commutation with the spectral dilation, proves (8.49). The compact horizontal Jacobian is the Q th power of the dilation, so these are precisely the asserted inverse-Jacobian laws.

We verify the normal-copy assertion at the level of domains. For an ampliation $\mathcal{H} \otimes \mathcal{L}$, fix an orthonormal basis (e_n) of the separable multiplicity space. If $\Xi = \sum_n \xi_n \otimes e_n$ belongs to its maximal estimate domain, positivity of every pair effect implies $\xi_n \in \mathcal{E}$ for each n : each coordinate energy is bounded by the total energy. Conversely, every finite sum with $\xi_n \in \mathcal{E}$ belongs to the amplified estimate domain. The finite-coordinate projections commute with the PVMs, preserve that domain, and commute with its weak derivatives. For a generator $aP_\bullet(g)\Xi$ of the amplified module core, those projections therefore converge in both the input norm and the derivative-column norm. Its projected vectors lie in $\mathcal{C} \odot \text{span}\{e_n\}$, and the same holds for finite sums of generators. Thus this algebraic tensor core is graph-dense in the amplified core. On it the derivative column is exactly $D \otimes 1$. Their closures agree, and the spectral calculus gives $l^{\text{amp}} = l \otimes 1$ and $b^{\text{amp}} = b \otimes 1$. Unitary identification with a normal copy preserves these identities. In particular no mixing assertion on an enlarged commutant is required. $\square$

The isolated branch now supplies its own-label height calculus: b_x is positive and injective, commutes strongly with P_x , and obeys (8.49) in every required normal copy. Section 10 uses these identities to transport imaginary powers through the Fourier links and establish the centered length laws. The closed derivative column above is the construction of this height; the common graph domain used to compare heights at different labels is constructed later from the Haar-height isometry in Section 11.

9. UPPER SPEEDS AT REAL PANELS

This section treats a non-isolated source position and its matched target position when both places are archimedean. The finite-place argument is in Section 6. The differentiation argument of Section 8 concerns isolated axes and is not used to supply the estimate at a real panel.

9.1. The estimate and the order of limits. Fix one measured coordinate of the exact graph relation (5.11); this coordinate may also be one of the two measured coordinates of a two-leg link. Begin with a mixed many-panel lexicographic face path that selects every isolated gap and omits one gap in each higher-rank component. Let S and T be the noncompact simple rank-one Levi groups of an omitted source position and its matched target position. Compact and ineffective factors are discarded. Source actions below may be lifted to the simply connected covering group. Write e_i for the coordinate vector dual to the omitted simple root. Thus translation by te_i has simple gap t , rather than root-coroot parameter t .

Proposition 9.1 (The real-panel upper estimate). *There are positive constants d_R , determined by each real rank-one panel type R and unchanged at the opposite type, with the following property. Complete the mixed path by translations in the same split chamber along its missing coordinates, taking their times to infinity successively and more slowly than the original path. Additional slower translations along already selected isolated coordinates are allowed. If t is the completing simple gap at the selected panel, and $\ell(g)$ is the residual target Cartan gap, then*

$$(9.1) \quad d_T \ell(g) \leq d_S t + O_{\text{tight}}(1).$$

Here a tight upper error means that the event on which $d_T \ell(g) - d_{St}$ tends to positive infinity has zero mass in the iterated limit, on every fixed input. One may add offsets fixed during all preceding path limits and then take further limits in those offsets; t in (9.1) is the resulting shifted gap. Gap signs are the eventual orientations on the chosen filters.

We record precisely the angular information available during this completion. At the first, mixed limit the residual missing gap is finite, and its Cartan gap differs by a bounded quantity from the corresponding inverse Fourier-index gap. A completing translation fixed at this first limit is an exact input precomposition of the link. It therefore retains all the previously sharp angular marginals. At each subsequent limit the growth and upper vertex-count argument of Section 5 applies to the enlarged path. If a previously selected gap is repeated, maximal vertex count on that face has already been attained, so the repetition introduces no additional gap. In the stated order of limits, the total split parameters still contract the relevant face radicals. Thus every resulting link is mixed and isometric, has its full sharp marginals, and satisfies (5.11) with the indicated input reindexing. The same reasoning applies to outer translations in the common split torus after a full path, including the offsets in Proposition 9.1. These assertions use the exact precomposition and angular statements of Sections 4 and 5; they require neither the height calibration nor the classicality conclusion proved later.

The structural classification in Section 3 leaves the orthogonal, unitary, and symplectic groups in the following table as the real rank-one panel groups. The exceptional rank-one group $F_{4(-20)}$ cannot occur as the noncompact simple factor of a panel Levi in a higher-rank component. Its complexification would be a component of a proper absolute Dynkin subdiagram, whereas no proper subdiagram of an irreducible Dynkin diagram has a component of type F_4 . Equivalently, in the real restricted-root list a multipliable simple root arising in such a Levi has double-root multiplicity 1 or 3, and not 7. The reduced low-rank isomorphisms are accounted for in the table's orthogonal line.

9.2. Optimal modules for a real rank-one panel. Let R be a noncompact rank-one panel factor, and let H_R be its split generator, normalized so that the simple restricted gap is one. For a finite-dimensional complex representation V of its real Lie algebra, extended to the complexification, write

$$V_t = \ker(d\pi(H_R) - t), \quad h(V) = \sum_{t>0} \dim V_t, \quad z(V) = \dim V_0.$$

For a nontrivial irreducible V , the ratio $\dim V/h(V)$ is the inverse proportion of positive-weight vectors. We will extract finite-dimensional source representations from the measured graph relation and compare their stage dimensions with those of a target module. A sharp ratio and a uniform gap away from its maximizers will bound the dimension lost when only maximizing source summands are retained. Their largest split weights then bound source growth. These are the different roles of the constants c_R and d_R below.

We allow representations of the simply connected covering group. Thus the weight lattices below are the full weight lattices, and spin representations are included. Isomorphic rank-one groups are named only once: $SU(1,1)$ uses the $SO(2,1)$ line and $Sp(1,1)$ uses the $SO(4,1)$ line, with the reduced restricted root in both cases.

Lemma 9.2 (Optimal panel modules). *Put $k = n + 1$, and let E denote the standard complex module for the group in the relevant line. Among the nontrivial irreducible complex Lie-algebra modules, the maximum*

$$c_R = \max_V \frac{\dim V}{h(V)}$$

and all modules attaining it are as follows:

R	c_R	maximizing modules
$SO(n, 1), n \geq 2$	k	E
$SU(n, 1), n > 2$	k	E, E^*
$SU(2, 1)$	3	$E, E^*, \text{Sym}^2 E, \text{Sym}^2 E^*$
$Sp(n, 1), n > 2$	k	E
$Sp(2, 1)$	$7/2$	$\bigwedge_0^3 E$.

In the $Sp(2, 1)$ row, $\bigwedge_0^3 E$ is the primitive exterior cube, the kernel of symplectic contraction $\bigwedge^3 E \rightarrow E$. There is a number $\varepsilon_R > 0$ such that every nontrivial irreducible outside the displayed list satisfies

$$\frac{\dim V}{h(V)} \leq c_R - \varepsilon_R.$$

Every displayed module has integral split weights and a nonzero split-weight-zero space. It has a preserved nondegenerate Hermitian form q , positive definite on V_0 , whose matrix in a compact-form Hilbert norm satisfies

$$Q = Q^* = Q^{-1}, \quad d\pi(H_R)^* = d\pi(H_R).$$

The form pairs V_t perfectly with V_{-t} and pairs it trivially with all other weight spaces. In particular, for $V_{>0} = \bigoplus_{t>0} V_t$ and $V_{\geq 0} = \bigoplus_{t \geq 0} V_t$,

$$V_{>0}^{\perp q} = V_{\geq 0}, \quad V_{\geq 0}^{\perp q} = V_{>0},$$

and the induced form on $V_{\geq 0}/V_{>0}$ is positive definite. These two spaces are algebraic parabolic stages. The largest split weight occurring among the maximizing modules is

$$d_R = \begin{cases} 2, & R = SU(2, 1) \text{ or } Sp(2, 1), \\ 1, & \text{otherwise.} \end{cases}$$

The same constants apply to the opposite orientation of the panel.

We use this complete representation-theoretic package first; its proof is given in Subsection 9.7. The next step converts the residue graph relation into compact operators preserving the two parabolic stages. Their finite-dimensional spectral ranges will make the ratio c_R available for a trace comparison.

9.3. Algebraic stages and compact graph operators. Choose a target maximizing module V whose largest split weight is d_T , and put $n_0 = \dim V$. If a compact diagram symmetry interchanges inequivalent modules, replace V by the direct sum of its diagram conjugates. Every summand has the same optimal ratio, so still

$$(9.2) \quad h(V) = n_0/c_T.$$

Choose the compact-form Hilbert norms and preserved Hermitian forms provided by Lemma 9.2, and denote their common form symmetry by Q , with $Q = Q^* = Q^{-1}$. In compact projectivity frames there are local, or Borel, lifts $\pi_T(g)$ of target projectivities such that

$$(9.3) \quad \pi_T(g)^* Q \pi_T(g) = Q, \quad \|\pi_T(g)\| = \exp(d_T \ell(g)).$$

Indeed the connected action lifts to the simply connected module. Compact Cartan factors act unitarily, and compact diagram factors permute the chosen summands. The forms can be chosen consistently under these permutations: on each irreducible their invariant forms are unique up to a real scalar, and positivity on the nonempty zero-weight space fixes its sign. For the coefficient calculations below, fix Borel lifts $\pi_T(g)$. Changing a covering lift acts by a scalar of modulus one on each irreducible summand, preserving the weight planes, the form and norm identities above, and the limiting top split-weight projector. The calculations use these fixed lifts and do not require a multiplication law for them.

For a Hilbert space H_0 , consider matrices in $\text{End}(V) \otimes H_0$. We use the total and column Hermitian forms, whose symmetries relative to the Hilbert norm are, respectively,

$$X \mapsto QXQ, \quad X \mapsto QX.$$

Their adjoints are denoted by $\sharp$ and $\sharp_{\text{col}}$. On the start or end of the link, let $\mathcal{L}(z) \subset \mathcal{D}(z)$ be the closed matrix subspaces whose columns are spectrally in the strictly positive and nonnegative split-weight planes of the corresponding P_z or Q_z in (5.11). The label z specifies the source panel endpoint. These are split-weight stages, not stages defined by positive eigenvalues of the preserved form. Parabolic invariance makes the stages depend only on the directed endpoint, independently of the opposite endpoint used to choose a split axis.

Lemma 9.3 (Stage geometry and polynomial tests). *The stages are mutual annihilators for either matrix form. Their column middle quotient $\mathcal{D}(z)/\mathcal{L}(z)$ is a positive Hilbert space, with its quotient topology in compact frames. On almost every sharp endpoint fibre, the closed span of $\mathcal{L}(z)$ over all labels is the whole matrix space. All the stage and projector relations used below are legitimate consequences of the polynomial fullness statement in Section 5.*

Proof. Opposite nonzero weights pair perfectly for the preserved form, and its zero-weight space is positive. Consequently the positive and nonnegative weight spaces are mutual annihilators and their middle quotient is positive. The assertion passes to spectral stages and Hilbert multiplicities. Right multiplication by Q preserves a column stage, so inserting that additional symmetry gives the same annihilator relation for the total form.

Here is the algebraic content of the required polynomial tests. In an individual compact face frame, restrict the surviving highest-line embedding of the ambient vertex to the rank-one Levi orbit. The complex stabilizer of this line is the parabolic consisting of the split centralizer and positive root groups. It embeds the corresponding complex parabolic variety projectively. The positive and nonnegative planes of an algebraic covering module are stabilized by that parabolic. So is its top split-weight plane: the split centralizer preserves it, and positive-root operators cannot raise the split weight beyond its maximum. The Plücker coordinates of all these planes are therefore rational, homogeneous coordinates on patches of this embedding. An orthogonal projector is rational in a spanning matrix and its conjugate, by Gram inversion. Thus each projector relation is a rational expression in the real coordinates of the highest line. Clear its denominators on a patch and multiply once more by a denominator that vanishes off that patch. The resulting polynomial vanishes there as well, so it is a global polynomial relation. Multiplication by norm powers makes it homogeneous of an even degree when necessary. This additional clearing factor is essential: no assertion about a rational expression on its zero-denominator set is being made.

Apply the fibrewise line-span statement of Section 5 to these homogeneous polynomial relations with linear Hilbert-vector coefficients. In particular, a vector orthogonal to every measured positive stage would be orthogonal to the positive plane at every ordinary flag. The latter planes span V : their span is a nonzero invariant subspace on each chosen irreducible summand. Tensoring with the coefficient Hilbert space and allowing every matrix column proves dense spanning. Countably many determining labels and patch equations suffice first; strong continuity of the label action then gives every label. $\square$

Disintegrate over the two sharp endpoint faces. The exact graph relation, its determining equations, and the strongly continuous source Levi action hold on almost every pair of fibres, as proved in Section 5. Use the genuine source amplification of Section 4 if a projective composition defect is present. We temporarily work on such a fibre pair, writing its end and start Hilbert spaces as I and H' , with isometry

$$J : \bar{I} \otimes H' \longrightarrow \mathcal{K}.$$

We identify $\bar{I} \otimes H'$ with $\text{HS}(I, H')$.

Lemma 9.4 (A compact graph operator). *For every $\varepsilon > 0$ there is a Hilbert–Schmidt operator*

$$D : \text{End}(V) \otimes I \longrightarrow \text{End}(V) \otimes H'$$

that carries both end stages to the corresponding start stages for every label, whose total adjoint carries the start stages back, and that reduces the scalar/traceless matrix splitting. If $A : I \rightarrow H'$ is its scalar block, then

$$(9.4) \quad a = \|A\|_2^2 > 0, \quad \|DD^\sharp - 1 \otimes AA^*\|_1 \leq \varepsilon a.$$

Proof. Set $C(g) = \text{Ad } \pi_T(g)$. On a relatively compact projectivity window U , take $v \in E(U)\mathcal{K}$ and define, with a, b indexing matrix entries rather than rows of V ,

$$D_{ab} = J^*E(C_{ab})v \in \text{HS}(I, H'), \quad A = J^*v.$$

The entry convention is explicitly

$$\langle \xi', D_{ab}\eta \rangle = \langle \bar{\eta} \otimes \xi', J^*E(C_{ab})v \rangle.$$

For end and start spectral stage projectors $F(u_-)$ and $L(u_+)$, pairing a coefficient of $(1-L)DF$ tests the link after multiplication by conjugated entries of

$$(1 - L(u_+))C(g)F(u_-).$$

These entries vanish by (5.11). Determine the equations first on countable labels and compact windows and then apply bounded output calculus. This proves the two stage inclusions; total adjunction and mutual annihilation prove the reverse inclusions. In particular the argument uses complex matrix entries and does not require a real structure on V .

Conjugation fixes scalar matrices and preserves traceless matrices. These two spaces are also orthogonal for the total form, since $\text{Tr}(QXQ) = \text{Tr } X$. Thus the splitting reduces D and $D^\sharp$, and its scalar block is A .

To obtain the approximation, first fix one compact window of nonzero compression. On it the norms of $C(g)$ and $C(g)^{-1}$ have a common finite bound. Subdivide this fixed window into finitely many Borel windows of arbitrarily small coefficient oscillation. At least one subwindow U still has nonzero compression. Choose a value C_0 on it; it is a total-form isometry with the already fixed norm bound. Put $b_U = \|J^*E(U)\| > 0$ and choose a unit $v \in E(U)\mathcal{K}$ nearly attaining this norm, with $\sqrt{a} = \|J^*v\| \geq b_U/2$. For a coefficient error f supported on U ,

$$\|J^*E(f)v\|_2 \leq b_U\|f\|_\infty \leq 2\sqrt{a}\|f\|_\infty.$$

There are a fixed finite number of matrix coefficients. Therefore, if the window oscillation is δ , $D = C_0 \otimes A + F$ with $\|F\|_2 \leq C\delta\sqrt{a}$. The form symmetries have norm one, and $C_0C_0^\sharp = 1$, so the ideal inequality $\|XY\|_1 \leq \|X\|_2\|Y\|_2$ bounds the difference in (9.4) by $C'(\delta + \delta^2)a$, where C' uses only the original compact window and the matrix size. Choose δ accordingly before the subdivision to obtain the asserted εa bound. $\square$

9.4. Finite-dimensional ranges and optimal source copies. Let W be a finite-dimensional nonzero spectral range of $DD^\sharp$, and let r be the dimension of its intersection with a positive endpoint stage. Our goal is to prove

$$\dim W \leq c_S r$$

and to control the dimension outside the optimal source-module copies by the defect $c_S r - \dim W$. We cannot apply the source representation count directly to W : it need not carry source transports preserving the column form. The first Riesz cut gives a range nondegenerate for the total form. A second cut preserves its entire positive middle quotient and makes the column form nondegenerate. This is the range on which we construct the finite-dimensional source representation.

Lemma 9.5 (The two Riesz cuts). *Let W be a nonzero-eigenvalue Riesz range of $DD^\sharp$, grouping conjugate eigenvalues together, and let F_0 be its total-selfadjoint Riesz projection. The integer $r = \dim(W \cap \mathcal{L}(z))$ is independent of z . There is a column-selfadjoint finite-rank projection e_1 with range $W_1 \subset W$ such that, for $V_z = e_1\mathcal{L}(z)$,*

$$(9.5) \quad \dim W - \dim W_1 = 2(r - \dim V_z).$$

The V_z span W_1 , are isotropic, and have positive perpendicular quotient. A finite-dimensional representation of the simply connected source group on W_1 preserves the column form and carries V_z onto V_{sz} .

Proof. Every resolvent away from the nonzero spectrum of a compact operator preserving a closed stage also preserves that stage. In fact the restriction of $\lambda - K$ to the stage is a scalar multiple of identity minus a compact operator, has Fredholm index zero, and is injective when the ambient operator is invertible. It is therefore onto the stage. Contour integration shows that F_0 preserves both stages. Because $DD^\sharp$ is total-selfadjoint, grouping conjugate eigenvalues makes F_0 total-selfadjoint. Its range is consequently nondegenerate, with the induced stages still mutual annihilators.

Let P_z^{stage} denote the Hilbert orthogonal projector onto the matrix stage $\mathcal{L}(z)$. The operator $F_0 P_z^{\text{stage}}$ is an idempotent with image $W \cap \mathcal{L}(z)$. It is norm-continuous in z : P_z^{stage} and its adjoint are strongly continuous, and F_0 has finite rank. Hence its rank is locally constant and therefore constant on the connected rank-one boundary. The same reasoning will apply to the second cut below.

Annihilation implies that $F_0^{\sharp\text{col}}$ preserves both stages as well. On the positive Hilbert quotient $\mathcal{D}(z)/\mathcal{L}(z)$, let f be the induced idempotent of F_0 . Its image has dimension

$$\dim(W \cap \mathcal{D}(z)) - \dim(W \cap \mathcal{L}(z)) = \dim W - 2r.$$

Column adjunction descends to Hilbert adjunction on this quotient, so $F_0 F_0^{\sharp\text{col}}$ induces ff^* . Choose e_1 to be the Riesz projection of $F_0 F_0^{\sharp\text{col}}$ onto all of its strictly positive real eigenvalues. Only finite-rank spectra away from zero occur. The projection is column-selfadjoint, preserves the stages, and has range in W . Passage to the quotient retains exactly the positive spectral subspace of ff^* , whose dimension is $\text{rank } f$. On W_1 the corresponding dimension is $\dim W_1 - 2\dim(e_1\mathcal{L}(z))$. Thus the second cut retains the whole original middle quotient: the nonzero spectral range of ff^* is $\text{ran } f$. The dimensions lost in W are twice those lost from its isotropic stage, proving (9.5).

Since e_1 is column-selfadjoint, W_1 is column-nondegenerate; $V_z = W_1 \cap \mathcal{L}(z)$ and its perpendicular in W_1 is $W_1 \cap \mathcal{D}(z)$. Positivity of the middle quotient is inherited from the ambient one. Applying e_1 to the dense stage span proves that the V_z span W_1 .

It remains to produce actual finite-dimensional transports. Denote the genuine, strongly continuous ambient source transport by U_s ; it is a column-form isometry. For s near the identity,

$$A_s = e_1 U_s|_{W_1}$$

is invertible, by finite rank and strong continuity. It maps V_z into V_{sz} for every z , and equal constant dimensions make this equality for every z . Its column adjoint is $e_1 U_{s^{-1}}|_{W_1}$ and carries V_{sz} back onto V_z . Consequently $B_s = A_s^{\sharp\text{col}} A_s$ preserves every V_z and is near identity. Its analytic inverse square root is column-selfadjoint and also preserves every V_z . Thus

$$A_s B_s^{-1/2}$$

is an exact column-form isometry transporting all these planes simultaneously. A multiplication rule for the compressions is not needed.

The pairs (s, A) such that A is a column-form isometry of W_1 and $AV_z = V_{sz}$ for all z form a closed subgroup of $S \times U(q|_{W_1})$. Closedness follows from continuity of the plane family. The preceding construction supplies local lifts; hence this Lie group projects onto connected S and its Lie algebra projects onto $\mathfrak{s}$. Its solvable radical maps to zero, since $\mathfrak{s}$ is simple.

A Levi subalgebra maps onto $\mathfrak{s}$, and a complementary simple ideal in that Levi subalgebra maps isomorphically onto $\mathfrak{s}$; these are the usual Levi decomposition and complete reducibility facts for finite-dimensional Lie algebras [44, Appendix B, Lemma B.1 and Theorem B.2]. Integrating this Lie-algebra section on the simply connected source group yields the asserted representation. $\square$

Lemma 9.6 (The defect and the good subspace). *For a range W of Lemma 9.5, put*

$$\delta_0 = c_S r - \dim W.$$

Then $\delta_0 \geq 0$. There is a sum $G_W \subset W_1$ of optimal source-module copies whose codimension in W is at most $C\delta_0$. Each copy takes its positive and nonnegative source weight planes into the measured positive and nonnegative matrix stages at every label. The constant depends only on the finite list of panel types.

Proof. Use the finite-dimensional source representation from Lemma 9.5. Fix a split endpoint z_0 , write h for the positive split-weight dimension of W_1 , and let r_0 be the negative index of the form on its zero-weight space. Nonzero opposite weight spaces pair perfectly. Since the isotropic V_{z_0} has positive perpendicular quotient, it exhausts the negative index, and

$$\dim V_{z_0} = h + r_0.$$

The source parabolic at z_0 preserves this plane, so in particular it has a split-weight decomposition. Its zero-weight part is isotropic in the zero-weight space and has dimension at most r_0 .

Let u be the dimension of the trivial summand of W_1 . Projection of V_z to this summand is independent of z , by transport, and is the whole summand because the planes span W_1 . At z_0 it is the image of the zero-weight part, so $u \leq r_0$. Complete reducibility and (9.5) give, with nontrivial irreducibles listed with repetitions,

$$(9.6) \quad \delta_0 = (c_S - 2)(r - \dim V_{z_0}) + (c_S r_0 - u) + \sum_{\sigma \neq 1} (c_S h(V_\sigma) - \dim V_\sigma).$$

Every term is nonnegative. Here $c_S > 2$, the first dimension difference is nonnegative, and $c_S r_0 - u \geq (c_S - 1)r_0$.

The uniform ratio gap in Lemma 9.2 now has a quantitative use. For a non-optimal σ , the ratio bound gives explicitly

$$c_S h(V_\sigma) - \dim V_\sigma \geq \frac{\varepsilon_S}{c_S - \varepsilon_S} \dim V_\sigma.$$

Thus its contribution to (9.6) controls its whole dimension. The dimension outside W_1 , the entire non-optimal and trivial dimension in W_1 , and r_0 are each at most $C\delta_0$.

Let O be the optimal isotypic sum and N its invariant complementary sum. This is an orthogonal decomposition for the preserved form. Indeed each optimal irreducible has a nondegenerate invariant Hermitian form, so its Hermitian contragredient is itself; invariant pairings with inequivalent non-optimal or trivial constituents vanish. Put $U = \text{pr}_O V_{z_0}$. It is parabolic-invariant and $\dim U_0 \leq r_0$.

We bound its negative-weight part explicitly. Choose a nonzero real simple restricted-root vector and its Cartan reflection. Their bracket lies in the split line: it lies in the centralizer and changes sign under the Cartan involution. After rescaling they form an $\mathfrak{sl}_2$ -triple with diagonal element $2H$, where H is the simple-gap-one generator. Every optimal weight for H is integral. In a finite-dimensional $\mathfrak{sl}_2$ -module, raising from weight $-j$ to zero is injective for each integer $j > 0$. Parabolic invariance therefore gives injections $U_{-j} \rightarrow U_0$. The finite optimal list has at most d_S distinct negative integral weights, whence

$$\dim U_{\leq 0} \leq (d_S + 1)r_0.$$

Since $\dim V_{z_0} \geq h(O)$ and $\dim N \leq C\delta_0$, the positive part of V_{z_0} has dimension at least $h(O) - \dim N - (d_S + 1)r_0$. Intersecting it with $O_{>0}$ loses at most another $\dim N$. Thus

$$\text{codim}_{O_{>0}}(O_{>0} \cap V_{z_0}) \leq 2 \dim N + (d_S + 1)r_0 \leq C\delta_0.$$

Moreover pairing $O_{\geq 0}$ against V_{z_0} only tests $U_{\leq 0}$, so

$$\text{codim}_{O_{\geq 0}}(O_{\geq 0} \cap V_{z_0}^\perp) \leq C\delta_0.$$

Write $O = \bigoplus_\sigma V_\sigma \otimes M_\sigma$ over the finite optimal list. For each basis vector of $(V_\sigma)_{>0}$ impose on $m \in M_\sigma$ the condition $v \otimes m \in V_{z_0}$; for each basis vector of $(V_\sigma)_{\geq 0}$ impose $v \otimes m \in V_{z_0}^\perp$. Each condition has rank at most the corresponding codimension just bounded. There are a fixed finite number of basis vectors, and their module dimensions are fixed. The resulting multiplicity subspaces therefore remove at most $C\delta_0$ dimensions from O . Their full-module sum G_W has the claimed codimension in W . Transport gives both plane conditions at all labels. Finally, $V_z \subset \mathcal{L}(z)$, and its perpendicular in W_1 equals $W_1 \cap \mathcal{D}(z)$ because e_1 is column-selfadjoint. Thus the nonnegative condition also lifts to the ambient measured stage. $\square$

The defect estimate now connects the finite-dimensional source ranges to the target trace. The compact graph operator is nearly a scalar multiplicity operator. Its spectral multiplicities, weighted by eigenvalue absolute value, have total at least $n_0^2 a$ up to a small error, while the corresponding positive-stage total is at most $n_0^2 a/c_T$ up to that error. Comparing these two quantities first proves equality of the panel ratios. The remaining defect is then too small to exclude every scalar vector from the optimal source copies.

Lemma 9.7 (Ratio equality and a scalar witness). *Matched real panel types satisfy $c_S = c_T$. On almost every endpoint fibre there is a nonzero vector η such that $V \otimes \eta$ is contained in the algebraic span of ranges of maps*

$$B : V_\sigma \longrightarrow V \otimes H'$$

over optimal source irreducibles V_σ , where B carries both source weight stages into the measured column stages at every label. The analogous assertion holds at the end of the link.

Proof. Sum the Riesz ranges W of $DD^\sharp$ with weight w equal to the absolute value of the associated eigenvalue, grouping conjugates with their common absolute value. Lidskii's theorem [60] and (9.4) give

$$\sum_W w \dim W \geq (n_0^2 - \varepsilon)a.$$

For the restriction to a fixed invariant positive stage, the sum of absolute eigenvalues, with their Riesz multiplicities r , is bounded above by its trace norm. The compression of $1 \otimes AA^*$ to that stage is positive. If P_z^{stage} is the Hilbert stage projector, its partial matrix trace is

$$\text{Tr}_{\text{End}(V)} P_z^{\text{stage}} = n_0 h(V) 1_{H'}.$$

Indeed each spectral positive-plane projector has rank $h(V)$, and there are n_0 matrix columns. The compression therefore has trace $an_0 h(V) = an_0^2/c_T$. Compressing the error in (9.4) does not increase its trace norm. Consequently

$$(9.7) \quad \sum_W wr \leq (n_0^2/c_T + \varepsilon)a.$$

These restriction multiplicities are exactly $\dim(W \cap \mathcal{L}(z))$: the resolvent argument in Lemma 9.5 also identifies the restricted Riesz projections. The inequality $\dim W \leq c_S r$ now gives $c_T \leq c_S$ as $\varepsilon \downarrow 0$. The real non-isolated type set is invariant under the type permutation: Corollary 6.2 preserves the real types, and Proposition 5.12 preserves the Coxeter components. Applying this inequality round each of its finite cycles gives equality on every matched pair.

With equality established, (9.7) and the lower bound above imply

$$\sum_W w\delta_0 \leq (c_S + 1)\varepsilon a.$$

The scalar block of $DD^\sharp$ is exactly AA^* . Put

$$S_W = W \cap (\mathbb{C}1_V \otimes H').$$

The nonzero such subspaces are precisely the eigenspaces $1_V \otimes \ker(AA^* - \mu)$, $\mu > 0$, so $\sum_W w \dim S_W = \text{Tr}(AA^*) = a$. If $G_W \cap S_W = \{0\}$ for every W , the quotient map $S_W \rightarrow W/G_W$ would be injective. Thus $\dim S_W \leq \text{codim}_W G_W$, and Lemma 9.6 would give

$$a \leq C \sum_W w \delta_0 \leq C(c_S + 1)\varepsilon a,$$

which is impossible for sufficiently small ε . Hence one G_W contains a nonzero scalar matrix $1_V \otimes \eta$. Projecting its optimal module copies to each matrix column gives the maps B in the statement. The columns of this scalar matrix are $v_i \otimes \eta$, proving the assertion for every vector of V . Interchanging start and end proves the other assertion.

If a genuine source amplification was used, apply a coefficient functional in the added Hilbert factor to the maps B and to the scalar witness. At least one coefficient of a nonzero witness is nonzero, and all plane equations are unchanged because the added factor is a multiplicity for those equations. This yields a witness on an original fibre. $\square$

9.5. Global section domains and their positive matrix action. The scalar witness is so far only nonzero on almost every endpoint fibre. We now pass from those witnesses to a dense global domain, and transfer the source matrix algebras to it. The two stage conditions make all cross pairings of sections source intertwiners. Positivity on source zero-weight spaces will then turn the transferred form symmetry into a strictly positive action. This action will be used on fixed vectors of its algebraic domain in the speed estimate.

Call a map B in Lemma 9.7 a *section*. In a global endpoint space H_0 , use the same compact frames at the sharp endpoint, denote the section space for V_σ by $\mathcal{S}_\sigma$, and put

$$\mathcal{E} = \sum_\sigma \sum_{B \in \mathcal{S}_\sigma} B(V_\sigma), \quad \mathcal{D} = \{\eta \in H_0 : V \otimes \eta \subset \mathcal{E}\}.$$

All sums defining $\mathcal{E}$ are algebraic. In particular no operator norm on a completed space is being asserted for the action constructed below.

Lemma 9.8 (Density of the scalar domain). *The scalar domain $\mathcal{D}$ is dense in each global endpoint space. Its defining section equations are decomposable over the sharp endpoint face, and the projection onto its closure disintegrates to the fibrewise closure projections. It is preserved by the applicable base and complementary right symmetries, by target translations after the compact-frame correction, and by source panel transports. It is also preserved by the split translations that fix all labels of the panel and escape in all factors.*

Proof. Fix countably many dense labels and bases of the finitely many optimal modules. Each section condition is a bounded linear equation on $\text{Hom}(V_\sigma, V \otimes H_0)$: multiply the image of a source-plane projector by the complement of the measured-plane projector. Thus $\mathcal{S}_\sigma$ is a closed common kernel. One can realize all conditions as the kernel of a single bounded map to a countable Hilbert sum by inserting summable positive weights. Strong continuity recovers the conditions at every label.

Choose bases $u_{\sigma,a}$ of V_σ and v_i of V . Algebraic membership $v_i \otimes \eta \in \mathcal{E}$ can be expressed using only one unknown section $B_{i,\sigma,a}$ per source basis element:

$$v_i \otimes \eta = \sum_{\sigma,a} B_{i,\sigma,a} u_{\sigma,a}.$$

Indeed finite sums of sections of a fixed type may be combined, including their scalar coefficients. The simultaneous equations for all i define another closed linear kernel in the product of H_0 and finitely many section spaces. The scalar domain is its coordinate image in H_0 . Kernel projections and range-support projections of bounded decomposable maps disintegrate, so the projection onto $\overline{\mathcal{D}}$ disintegrates to the fibrewise closure projections.

Lemma 9.7 makes this projection nonzero; no measurable choice of individual witnesses is necessary.

The defining equations commute with base multipliers and all complementary right symmetries. The closure projection consequently belongs to $\mathcal{A}_j$. An actual target translation changes compact angular frames by a projectivity depending on the sharp face, but not on the panel label. The corresponding matrix correction π_T is boundedly invertible for each fixed translation: its Levi projection is bounded in compact frames, and the same is true for its inverse. Correct each transported section by this matrix. The scalar columns remain in the span of the corrected sections by combining them with the entries of the inverse matrix. These entries are own-face spectral multipliers, which preserve the section equations. The inverse translation proves equality of the transported scalar domains. No correction is needed for base or complementary right symmetries. It follows that the closure projection is fixed by the measured right action and hence belongs to $\mathcal{N}$.

Source panel covariance follows directly as well: if B is a section, then $U_s B \sigma(s)^{-1}$ is a section with the labels reindexed. Since U_s acts only on the Hilbert coefficient, it preserves the scalar domain. Split translations that fix every panel label preserve its defining equations without reindexing. Choose such translations escaping in every factor, as in the original mixed face path. The closure projection $p \in \mathcal{N}$ is invariant under this sequence. Mixing in $\mathcal{N}$ therefore gives $\tau(p) = \lim \tau(p\sigma(a)p) = \tau(p)^2$. Since $p \neq 0$, it equals 1. This proves density, and also makes $\mathcal{E}$ dense in $V \otimes H_0$. $\square$

For the following algebraic argument, let H_{ax} range over the source conjugates of the normalized split generator. On an optimal source module write $\text{sgn}(d\sigma(H_{\text{ax}}))$ for the operator with values 1, 0, -1 on its positive, zero, and negative weight spaces. The generator H_{ax} specifies an ordered pair of opposite endpoints. Its positive and nonnegative stages belong to the attracting endpoint, and its negative and nonpositive stages belong to the repelling endpoint. Thus section equations at both endpoints can test all three sign spaces; no choice of split axis is inferred from a single endpoint.

Lemma 9.9 (The algebra generated by split signs). *On an optimal source irreducible, the conjugate split sign operators generate the full complex matrix algebra. An operator between two optimal source irreducibles that intertwines all these signs is either zero or an ordinary source-module isomorphism, unique up to scalar.*

Proof. Let $\mathcal{A}$ be the unital algebra generated by the signs on an irreducible module W . The connected source normalizes $\mathcal{A}$. Its Jacobson radical has a nonzero common kernel: a finite-dimensional nilpotent algebra annihilates a nonzero vector. That common kernel is source-invariant because the radical is characteristic. Irreducibility makes it all of W , so the radical is zero. The connected source cannot permute the finitely many central blocks of the semisimple algebra, and their images are invariant subspaces. There is consequently one block, and

$$W = E \otimes F, \quad \mathcal{A} = \text{End}(E) \otimes 1.$$

Derivations of a matrix algebra are inner. For each source Lie-algebra element choose the unique traceless implementer of its derivation on $\text{End}(E)$. These implementers form a Lie representation: the defect in a commutator is both scalar and traceless, hence zero. Subtracting this representation from the action on W leaves a representation in the commutant $1 \otimes \text{End}(F)$. This gives a tensor decomposition of the Lie representation.

Suppose first that the complexified source algebra is simple. Two nontrivial irreducibles of a simple complex Lie algebra do not have an irreducible tensor product. For completeness, if their highest weights are $\lambda, \mu \neq 0$, the tensor product contains the constituent of highest weight $\lambda + \mu$. In the Weyl dimension formula each positive coroot contributes

$$\frac{(a+r)(b+r)}{r(a+b+r)} \geq 1, \quad a = \langle \lambda, \alpha^\vee \rangle, \quad b = \langle \mu, \alpha^\vee \rangle, \quad r = \langle \rho, \alpha^\vee \rangle > 0.$$

There is a positive coroot pairing positively with both nonzero dominant weights, because the root system is irreducible. Its factor is strictly larger than one. Hence $\dim V_\lambda \dim V_\mu > \dim V_{\lambda+\mu}$, excluding irreducibility. In the displayed tensor decomposition, irreducibility of W therefore makes one factor trivial. The sign is nonscalar, so the E factor is nontrivial; a trivial F has dimension one by irreducibility. This proves $\mathcal{A} = \text{End}(W)$.

The remaining real-simple case is $\mathfrak{so}(3, 1)$, whose complexification has two simple summands. Its only optimal module is the Lorentz vector module. Its split weights are $1, 0, 0, -1$, so a split sign is the split generator itself. The linear span of its source conjugates is the image of a nonzero adjoint ideal in the real-simple algebra and is therefore the whole Lie-algebra image. The associative algebra it generates acts irreducibly and equals the full matrix algebra by the same radical and single-block argument, or by Burnside's theorem.

An intertwiner of all signs has invariant kernel and image for their full matrix algebras. A nonzero such map is therefore an isomorphism, and the space of such maps has dimension one. Conjugating by the source preserves this line, because it permutes the axes. Connected semisimplicity has no nontrivial one-dimensional characters, so the line is fixed pointwise. The map is an ordinary source intertwiner. $\square$

Proposition 9.10 (The positive algebraic action). *On $\mathcal{E}$ there is a well-defined action of the finite-dimensional algebra $\bigoplus_\sigma \text{End}(V_\sigma)$ given by*

$$R(X)(Bv) = BX_\sigma v.$$

It is adjoint for the target column form according to source-form adjunction. If $\mathbf{Q} = (Q_\sigma)_\sigma$ is the tuple of source form symmetries, then

$$(9.8) \quad q(x, R(\mathbf{Q})x) > 0 \quad (0 \neq x \in \mathcal{E}).$$

Proof. For two sections $B : V_\sigma \rightarrow V \otimes H_0$ and $C : V_\tau \rightarrow V \otimes H_0$, represent their cross pairing by $K : V_\tau \rightarrow V_\sigma$ using the nondegenerate source form:

$$q(Bv, Cw) = q_\sigma(v, Kw).$$

If w is positive at an endpoint and v is nonnegative, their images belong to mutually annihilating measured stages. Hence Kw is positive at that endpoint. Interchanging positive and nonnegative gives the corresponding nonnegative inclusion. Repeating at the opposite endpoint gives the negative and nonpositive inclusions, and hence exact preservation of the three split signs. Lemma 9.9 therefore makes K an ordinary source intertwiner. Schur's lemma shows that inequivalent types are orthogonal and, within a fixed type,

$$q(Bv, Cw) = q_\sigma(v, w) b_\sigma(B, C)$$

for a Hermitian multiplicity pairing b_σ .

This pairing is positive semidefinite. Choose a nonzero source zero-weight vector v ; its form is positive by Lemma 9.2. The vector Bv lies in the measured nonnegative stage, whose form is nonnegative. Thus $b_\sigma(B, B) \geq 0$, and the same argument applied to arbitrary linear combinations proves matrix positivity for every finite family of sections.

To justify descent rather than assume it, consider the algebraic evaluation map

$$T : \bigoplus_\sigma V_\sigma \otimes \mathcal{S}_\sigma \longrightarrow \mathcal{E}, \quad T(v \otimes B) = Bv.$$

Its pullback form is the direct sum of $q_\sigma \otimes b_\sigma$. For a finite collection of sections the radical is precisely the sum of V_σ tensored with the radicals of the positive semidefinite multiplicity forms. A radical vector remains radical after any further finite collection is added: Cauchy–Schwarz for a positive semidefinite form makes its pairing with every newly added vector zero. Its evaluation therefore pairs to zero with all of $\mathcal{E}$, and is zero by the density in Lemma 9.8 and ambient nondegeneracy. Conversely a vector evaluating to zero is in this radical, by pairing with the same finite module span. Thus the kernel of T is exactly the radical. It is invariant under every source matrix action, proving well-definedness and the adjoint assertion.

Finally $q_\sigma(v, Q_\sigma w)$ is the compact-form positive Hilbert pairing. The pullback of $q(x, R(\mathbf{Q})y)$ is consequently the tensor product of this positive pairing with b_σ , summed over types. Its only null vectors are the radicals just identified with $\ker T$. This proves the strict inequality (9.8) for every nonzero evaluated vector. $\square$

9.6. Section transport and the excess estimate. For $\eta \in \mathcal{D}$, let $Z_\eta(X)$ be the $n_0 \times n_0$ matrix with Hilbert-vector entries whose i th column is $R(X)(v_i \otimes \eta)$. In particular $Z_\eta(1) = 1_V \otimes \eta$. For fixed η , this is a linear map from a fixed finite-dimensional matrix algebra into a Hilbert space. Choosing finite section expressions for its columns gives a finite bound on each matrix unit; no bounded extension of R beyond its algebraic domain is implied.

Let J_t be a slower translated link as in Proposition 9.1. At the selected panel the exact graph relation is reindexed across its ends by the source split translation s_t , of simple gap t . Choose its orientation so that the formulas below use $\sigma(s_t)$; reversing the axis gives the opposite convention. Other completing shifts do not change this panel's indexing. All frames are taken at the previously sharp face slots, whose laws were retained by the exact precomposition.

Lemma 9.11 (Action exchange and its fixed-input bound). *Construct $D_t(v)$ from J_t on a compact projectivity window as in Lemma 9.4, without requiring its approximation property, and let $A_t(v) = J_t^* v$ be its scalar block. If η belongs to the end scalar domain, then $A_t(v)\eta$ belongs to the start scalar domain and*

$$(9.9) \quad D_t(v)Z_\eta(X) = Z_{A_t(v)\eta}((\text{Ad } \sigma(s_t)X_\sigma)_\sigma).$$

For fixed scalar-domain inputs at the two ends, weighting v by $e^{-2d_T \ell(g)}$ and cutting to $d_T \ell(g) > d_{st} + M$ bounds every paired matrix entry in this identity by $Ce^{-2M}\|v\|$. The constant is independent of t , M , and the location of the compact window.

Proof. Write D_t^{lm} for the block from input matrix column m to output matrix column l . The stage calculation in Lemma 9.4 shows that, for an end section B ,

$$D_t^{lm}(v)B\sigma(s_t)^{-1}$$

is a start section. This gives action exchange on each column block, with X_σ replaced by $\text{Ad } \sigma(s_t)X_\sigma$. Applying the statement with $X = 1$ to the scalar columns and using scalar/traceless reduction yields

$$D_t(v)(1_V \otimes \eta) = 1_V \otimes A_t(v)\eta.$$

Every output column is a sum of section evaluations, proving that $A_t(v)\eta$ is in the start scalar domain. The general column-block identity then gives (9.9) by summation.

To estimate a fixed matrix entry, pair its output against a fixed start scalar-domain test ξ . A Hilbert pairing with a column x can be written as a column-form pairing with Qx . Move the algebra action on the right side of (9.9) to this fixed test by Proposition 9.10. Expand that fixed test in finitely many section expressions and matrix units. On the finite optimal list,

$$\|\text{Ad } \sigma(s_t)\| \leq e^{2d_{st}} \quad (t \geq 0),$$

so the norm of the moved action on the fixed test is at most $C_\xi e^{2d_{st}}$. If

$$f_{t,M}(g) = e^{-2d_T \ell(g)} 1_{\{d_T \ell(g) > d_{st} + M\}},$$

then

$$\|A_t(E(f_{t,M})v)\|_2 \leq \|E(f_{t,M})v\| \leq e^{-2d_{st}-2M}\|v\|,$$

because J_t is an isometry. The remaining fixed input contributes only its Hilbert norm. This proves the bound. In particular the proof estimates the action on a fixed test vector and never the norm of an unbounded action on the moving vector $A_t(v)\eta$. $\square$

We spell out the conjugation convention needed to read this estimate as an output statement. Under $\bar{I} \otimes H' \simeq \text{HS}(I, H')$, pairing an entry

$$\sum_b (J_t^* E(C_{ab})v) Z_b$$

against ξ pairs v against

$$\sum_b E(\overline{C_{ab}}) J_t(\overline{Z_b} \otimes \xi).$$

Conjugating the vector test replaces J_t by its conjugate link $\bar{J}_t$. Thus Lemma 9.11, by duality and exhaustion of compact windows, says that the matrix with entries

$$(9.10) \quad [e^{-2d_T \ell(g)} \text{Ad } \pi_T(g)] \bar{J}_t(Z_\eta(X) \otimes \bar{\xi})$$

has norm at most Ce^{-2M} on the excess cutoff. Here the matrix coefficients of $\text{Ad } \pi_T(g)$ act by output spectral calculus. The normalized operator has norm at most one, since $\|\pi_T(g)^{-1}\| = \|\pi_T(g)\|$. Hence compact-window exhaustion is valid; the constant in the estimate has no window dependence.

Proof of Proposition 9.1. Take the iterated path limits in their stated order and then the limit $M \rightarrow \infty$. Equation (9.10) vanishes on the positive infinite-upper-excess summand. Put

$$b(g) = e^{-d_T \ell(g)} \pi_T(g).$$

Both b and b^* are contractions. Right multiplication of the vanishing matrix by Q uses

$$\pi_T(g)^{-1} Q = Q \pi_T(g)^*.$$

It changes normalized conjugation into left and right multiplication on ZQ by b, b^* . Further multiplication on the outside by b^*, b gives the vanishing matrix with both outside factors b^*b .

Compact Cartan decomposition identifies their limit. In the surviving Levi block, if $\pi_T(g) = k_1 \exp(\ell(g)H_T)k_2$ in compact frames, then

$$b^*b = k_2^* \exp(2\ell(g)(H_T - d_T))k_2$$

tends, on divergent residual gap, to the projector on the highest input split-weight plane. Denote this projector by Π ; as a function of the additional reverse angular vertex, it is the Q -image of the highest plane for the inverse projectivity. Compact factors commute with Q , so these descriptions agree.

All frame passages can be made on continuity sets of arbitrarily large fixed input mass. The sharp endpoint base laws are exactly the same throughout the slower translations, because those translations stabilize the bases. Approximate the Borel compact frames on such sets and then remove their complements. The additional reverse vertex is sharp by the total angular statements recalled at the beginning of the section. Thus Π is available as a sharp input spectral projector in the limit, without a continuity assumption on a moving Borel frame.

Suppose that the infinite-excess summand is nonzero. Fixed target shifts at either input alter inverse-index gaps only by bounded amounts, so they preserve this summand. The independent all-factor contracted symmetries preserve it as well, in the same order of limits as for the mixed link. The scalar-mass argument of Section 4 therefore makes its compression a positive scalar multiple of identity. Write its scalar mass as $m > 0$. The restricted link divided by $\sqrt{m}$ is an isometry and retains the sharp angular marginals. Indeed the angular subeffect at a sharp input projection P is supported on P , while its complementary cut is supported on $1 - P$; their sum is mI , so the first subeffect equals mP . This is the sharp-compression criterion of Proposition 4.3. We may therefore pull the additional reverse vertex, denoted Q_y , through the restricted conjugate link. Pulling the two outside copies of Π evaluates both at the same end slot, and removal of the scaled isometry and a fixed nonzero start test ξ gives

$$(9.11) \quad [\Pi(v)Z_\eta(X)Q\Pi(v)]_{v=Q_y} = 0.$$

The spectral multipliers in this formula act on its Hilbert-vector entries. In particular no section action has been moved through either projector.

To apply fibrewise polynomial fullness, we need this identity at every panel label with η and X fixed. Conjugate the completing axis by a compact source Levi rotation. This conjugates the entire path, moving fixed source transports into its two inputs, and preserves the same scalar excess mass. Its only effect on the section argument is a fixed conjugation of the source reindexing. The scalar domains are preserved by these transports by Lemma 9.8. To see explicitly that the data can be held fixed, the section transport is $B \mapsto U_k B \sigma(k)^{-1}$, whence $U_k R(X) U_k^{-1} = R(\text{Ad } \sigma(k) X)$. For prescribed η, X apply the unrotated argument to $\eta' = U_k^{-1} \eta$ and $X' = (\text{Ad } \sigma(k))^{-1} X$. Then

$$(1 \otimes U_k) Z_{\eta'}(X') = Z_{\eta}(X).$$

Thus Equation (9.11) holds at every rotated label for the same fixed η and X . Impose countably many dense labels first and extend by strong label continuity. Lemma 9.3, with its extra denominator clearing, now gives the vanishing at every ordinary residue value v for fixed η and X .

Fix such an ordinary value and take a constant nonzero $w \in \text{ran } \Pi(v)$ in the compact frame. Choose any nonzero $\eta \in \mathcal{D}$ at the end and set $X = \mathbf{Q}$. Pairing (9.11) with $w \otimes \eta$ gives

$$0 = \langle w \otimes \eta, R(\mathbf{Q})(Qw \otimes \eta) \rangle = q(Qw \otimes \eta, R(\mathbf{Q})(Qw \otimes \eta)).$$

The last quantity is strictly positive by (9.8), since $Qw \otimes \eta \neq 0$. This contradiction proves that the infinite-upper-excess mass is zero and establishes (9.1).

Every bound used fixed square-integrable scalar-domain data; these data are dense by Lemma 9.8, so the zero-mass conclusion extends to all fixed inputs. To check the offset assertion, absorb a displacement in this panel's split direction into the total parameter t before applying $\|\text{Ad } \sigma(s_t)\| \leq e^{2dst}$. For each fixed negative offset, that total parameter is eventually positive; offsets are exhausted only after the ordinary path limits. A split translation a in another completing direction fixes every label of this panel. It sends a section B to $U_a B$, so on $\mathcal{E}$ one has

$$R(X) U_a = U_a R(X).$$

Its unitary action therefore leaves unchanged the norms of the matrix actions moved to fixed tests in Lemma 9.11. This proves the stated uniformity for other completing translations and the subsequent offset limits. $\square$

The upper-speed proof is complete subject to the optimal-module package. We now verify that package without using any of the operator constructions above.

9.7. Proof of the optimal-module classification. We prove the weight bounds, the uniform separation, and the assertions about forms separately. The only nonsimple complexified Lie algebra that occurs is $\mathfrak{so}(3, 1)_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \oplus \mathfrak{sl}_2(\mathbb{C})$; it is included explicitly below.

Lemma 9.12 (Weight multiplicity comparisons). *Let V be a nontrivial irreducible module for a complex semisimple Lie algebra, let m_{μ} be its full-Cartan weight multiplicities, and let $\alpha_1, \dots, \alpha_r$ be absolute simple roots. Then*

$$m_0 \leq \sum_{j=1}^r m_{\alpha_j}.$$

For any root α and any weight μ satisfying $\mu(\alpha^{\vee}) > 0$,

$$m_{\mu} \leq m_{\mu-\alpha}.$$

The multiplicities are constant on absolute Weyl orbits.

Proof. The joint raising map

$$V[0] \longrightarrow \bigoplus_{j=1}^r V[\alpha_j], \quad v \longmapsto (E_{\alpha_j} v)_{j=1}^r$$

is injective. Indeed, a vector in its kernel is a highest-weight vector of weight zero: the simple raising operators generate the positive nilpotent algebra. The finite-dimensional module generated by such a vector is trivial. To see the last assertion directly, restrict to each simple-root $\mathfrak{sl}_2$: a highest vector of weight zero is also killed by its lowering operator. Thus all simple raising and lowering operators, and the Cartan algebra, kill the vector. A nonzero such vector would be an invariant vector in the nontrivial irreducible V , which is impossible.

For the second assertion, restrict V to the root $\mathfrak{sl}_2$ corresponding to α . In every irreducible $\mathfrak{sl}_2$ summand, lowering is injective on a positive $\alpha^\vee$ -eigenspace: only the lowest weight is killed, and that weight is nonpositive. The lowering map is therefore injective on $V[\mu]$ and takes it into $V[\mu - \alpha]$. Finally, Weyl-group representatives in the simply connected group give isomorphisms between the weight spaces in a Weyl orbit. $\square$

For a nonzero full-Cartan orbit $\mathcal{O}$, let $p(\mathcal{O})$ be the number of its weights that are strictly positive on H_R . For a proposed bound c , define its surplus by

$$s_c(\mathcal{O}) = cp(\mathcal{O}) - |\mathcal{O}|.$$

If $m_{\mathcal{O}}$ is its multiplicity, then the exact identity

$$(9.12) \quad ch(V) - \dim V = \sum_{\mathcal{O} \neq \{0\}} m_{\mathcal{O}} s_c(\mathcal{O}) - m_0$$

separates the full-Cartan zero weight from the other weights. A weight that is nonzero on the full Cartan but zero on H_R is already included in its nonzero orbit in this identity.

Lemma 9.13 (Orthogonal weight bound). *For every nontrivial irreducible complex module V of $SO(n, 1)$, with all covering-group modules allowed, $\dim V \leq (n + 1)h(V)$, with equality precisely for the standard module.*

Proof. First let $n + 1 = 2b + 1$ with $b \geq 2$. The absolute root system is B_b , in coordinates $e_1, \dots, e_b$, and the split functional extracts the first coordinate. The Weyl group acts by all signed permutations. If a nonzero weight has exactly j nonzero coordinates, symmetry gives

$$\frac{p(\mathcal{O})}{|\mathcal{O}|} = \frac{j}{2b}.$$

Consequently every nonzero orbit has positive surplus for $c = 2b + 1$. The short-root orbit has size $2b$, positive count one, and surplus one. The long-root orbit has size $2b(b - 1)$, positive count $2(b - 1)$, and surplus

$$(2b + 1)2(b - 1) - 2b(b - 1) = 2(b^2 - 1).$$

Write m_s and m_l for their multiplicities. There are $b - 1$ long absolute simple roots and one short one, so Lemma 9.12 gives

$$m_0 \leq (b - 1)m_l + m_s.$$

Substituting this bound in Equation (9.12) leaves the nonnegative expression

$$(2(b^2 - 1) - (b - 1))m_l + \sum_{\mathcal{O} \text{ neither root orbit}} m_{\mathcal{O}} s_{2b+1}(\mathcal{O}).$$

Every coefficient displayed here is strictly positive. Equality therefore permits only the short-root orbit and full-Cartan zero, with $m_0 = m_s$. The highest weight must be e_1 , which gives the standard module. Its weights are $0, \pm e_1, \dots, \pm e_b$, once each, so equality is attained. This reasoning also covers the half-integral weight lattice: those weights have no zero coordinates and hence strictly positive surplus.

Now let $n+1 = 2b$ with $b \geq 3$. The absolute root system is D_b . Its even signed permutations give the same fraction $j/(2b)$. If a zero coordinate is available, its sign can absorb the parity restriction. If every coordinate is nonzero, flipping the first and one other sign exchanges the two signs in the first coordinate, so that case also has positive fraction $1/2$. Thus

$$s_{2b}(\mathcal{O}) = (j-1)|\mathcal{O}| \geq 0.$$

The single root orbit has size $2b(b-1)$ and surplus $2b(b-1) > b$. Its multiplicity m_r satisfies $m_0 \leq bm_r$. Equation (9.12) therefore proves the bound. At equality the root orbit and full-Cartan zero are absent, and every remaining weight has only one nonzero coordinate. The highest weight is consequently ae_1 with an integer $a \geq 1$; half-integral weights cannot have this pattern. If $a > 1$, lowering by $e_1 - e_2$ produces the weight $(a-1)e_1 + e_2$ by Lemma 9.12. This has two nonzero coordinates and positive surplus, a contradiction. Hence $a = 1$, giving the standard module with weights $\pm e_i$.

For completeness, in D_2 the two root orbits $\{\pm(e_1 - e_2)\}$ and $\{\pm(e_1 + e_2)\}$ each have size two, positive count one, and surplus two for $c = 4$. Each contains one absolute simple root, so each surplus strictly exceeds its contribution to the zero-multiplicity bound. An explicit calculation also handles all modules for the nonsimple complexification. They are

$$\mathrm{Sym}^a \mathbb{C}^2 \otimes \mathrm{Sym}^b \mathbb{C}^2, \quad a, b \in \mathbb{Z}_{\geq 0}, \quad (a, b) \neq (0, 0).$$

Their split weights are $(a+b-2i-2j)/2$ for $0 \leq i \leq a$, $0 \leq j \leq b$. The dimension is $(a+1)(b+1)$, and the split-zero multiplicity is zero if a, b have different parity, and $\min(a, b)+1$ otherwise. In the latter case assume $a \leq b$. The zero proportion is $1/(b+1) \leq 1/2$, with equality possible only when $a = b = 1$: if $b = 0$ the representation is trivial, and if $b = 1$ equal parity forces $a = 1$. Since the positive and negative weight counts agree,

$$\frac{\dim V}{h(V)} = \frac{2}{1 - z(V)/\dim V} \leq 4,$$

with equality exactly for the four-dimensional standard module.

Finally, for B_1 , or $SO(2, 1)$, a covering-group irreducible has highest split weight $a \in \frac{1}{2}\mathbb{Z}_{>0}$ and the string $a, a-1, \dots, -a$. For integral a its ratio is $(2a+1)/a = 2 + 1/a \leq 3$, with equality only at $a = 1$. For strictly half-integral a the zero space is absent and the ratio is two. This proves the remaining case, including its spin modules. $\square$

Lemma 9.14 (Unitary weight bound). *For every nontrivial irreducible complex module V of $SU(n, 1)$, with $k = n+1 \geq 3$, one has $\dim V \leq kh(V)$. If $k > 3$, equality holds precisely for E, E^* ; if $k = 3$, it also holds for $\mathrm{Sym}^2 E$ and $\mathrm{Sym}^2 E^*$.*

Proof. Use the A_{k-1} coordinates with sum zero and split functional $\mu \mapsto \mu_1 - \mu_2$. Suppose that the distinct coordinate values of a nonzero weight have multiplicities $r_1, \dots, r_s$, where $s \geq 2$ and $\sum r_i = k$. Choosing the ordered first and second coordinates uniformly over a permutation orbit gives

$$\frac{p(\mathcal{O})}{|\mathcal{O}|} = \frac{k^2 - \sum_i r_i^2}{2k(k-1)} \geq \frac{1}{k}.$$

Indeed, merging parts increases their sum of squares, and among two positive parts $r, k-r$ that sum is largest at $r = 1$ or $k-1$. Thus $\sum r_i^2 \leq (k-1)^2 + 1$, with equality exactly for the partition $(k-1, 1)$. Equality in the orbit bound therefore means that exactly one coordinate is exceptional.

The root orbit consists of the $k(k-1)$ vectors $e_i - e_j$. There is one root of split weight two and $2(k-2)$ roots of split weight one, so its surplus for $c = k$ is

$$k(2k-3) - k(k-1) = k(k-2) > k-1.$$

If m_r is its multiplicity, then $m_0 \leq (k-1)m_r$. All other orbit surpluses are nonnegative. Equation (9.12) proves the bound and shows that equality forces $m_0 = m_r = 0$ and only one-exceptional-coordinate patterns.

A dominant highest weight with that pattern is $a\omega_1$ or $a\omega_{k-1}$, where a is a positive integer. The corresponding modules are $\text{Sym}^a E$ and its dual. Their weights are obtained from the monomial exponent tuples $(a_1, \dots, a_k)$, $a_i \geq 0$, $\sum a_i = a$, by subtracting a/k from each coordinate. If $a \geq 3$, the tuple $(a-1, 1, 0, \dots, 0)$ has three distinct coordinate values, and therefore positive surplus. If $a = 2$ and $k > 3$, the tuple $(1, 1, 0, \dots, 0)$ has two groups of sizes $2, k-2$, again giving positive surplus. These cases cannot attain equality. For $a = 1$, all weights have the required pattern and $\dim E = k$, $h(E) = 1$. For $a = 2, k = 3$, the only exponent patterns are permutations of $(2, 0, 0)$ and $(1, 1, 0)$; both have one exceptional coordinate and neither gives the full-Cartan zero weight. In that case $\dim \text{Sym}^2 E = 6$ and its split weights are $2, 1, 0, 0, -1, -2$, so $h = 2$. Dualizing reverses the split weights and preserves these conclusions. $\square$

Lemma 9.15 (Symplectic weight bound). *For every nontrivial irreducible complex module V of $Sp(n, 1)$, put $k = n + 1$. If $k \geq 4$, then $\dim V \leq kh(V)$, with equality exactly for E . If $k = 3$, then $\dim V \leq (7/2)h(V)$, with equality exactly for $\Lambda_0^3 E$.*

Proof. The absolute root system is C_k , its full weight lattice is $\mathbb{Z}^k$, and the split functional is $\mu \mapsto \mu_1 + \mu_2$. The Weyl group consists of all signed permutations. Suppose that a nonzero weight has j nonzero coordinates. The probability that just one of its first two coordinates is nonzero is $2j(k-j)/(k(k-1))$. In that event the split weight is positive with probability $1/2$. The probability that both are nonzero is $j(j-1)/(k(k-1))$. Conditional on their absolute values, at least one of the four sign choices has positive sum; exactly one does if and only if those absolute values agree. Consequently

$$(9.13) \quad \frac{p(\mathcal{O})}{|\mathcal{O}|} \geq \frac{j(k-j) + j(j-1)/4}{k(k-1)}.$$

For $k \geq 4$ the numerator is a strictly concave quadratic in j . At the endpoints $j = 1$ and $j = k$ its values are $k-1$ and $k(k-1)/4$, respectively. Thus the right side is at least $1/k$. Equality can occur only for $j = 1$, or for $k = j = 4$; in the latter case equality in the sign count further requires all absolute values to agree.

Let m_s and m_l denote the multiplicities of the short-root orbit $\{\pm e_i \pm e_j : i \neq j\}$ and the long-root orbit $\{\pm 2e_i\}$. Lowering $2e_1$ by $e_1 - e_2$ gives $e_1 + e_2$, so Lemma 9.12 gives $m_l \leq m_s$. Since there are $k-1$ short absolute simple roots and one long one,

$$m_0 \leq (k-1)m_s + m_l \leq km_s.$$

The short-root orbit has size $2k(k-1)$. Its positive roots for the split functional are $e_1 + e_2$ and the $4(k-2)$ roots having one index in $\{1, 2\}$ and the other outside that set. Thus its positive count is $4k-7$, and its surplus is

$$k(4k-7) - 2k(k-1) = k(2k-5) > k.$$

Together with Equation (9.13), this proves the bound and shows that equality requires $m_s = m_l = m_0 = 0$ and only zero-surplus nonzero orbits. A highest weight of the first possible kind is ae_1 . If $a > 1$, lowering by $e_1 - e_2$ produces $(a-1)e_1 + e_2$, which has strictly positive surplus. The other possible highest weight is (a, a, a, a) when $k = 4$. Lowering by $e_3 + e_4$ produces $(a, a, a-1, a-1)$. For $a = 1$ this has support two, and for $a > 1$ it has unequal nonzero absolute values; in either case its surplus is strictly positive. The only remaining highest weight is e_1 . The standard module has dimension $2k$ and two positive split weights, so it attains the bound.

For C_3 , use $c = 7/2$. The following table gives every possible nonzero orbit pattern, its size, its positive count, and its surplus. Letters denote positive integers, and distinct letters

denote distinct integers; permutations and all sign choices are included.

pattern	$ \mathcal{O} $	$p(\mathcal{O})$	$(7/2)p(\mathcal{O}) - \mathcal{O} $
$(a, 0, 0)$	6	2	1
$(a, a, 0)$	12	5	11/2
$(a, b, 0)$	24	12	18
(a, a, a)	8	2	-1
(a, a, b)	24	10	11
(a, b, c)	48	24	36

Here the first row has positive split weight exactly when its nonzero coordinate occupies one of the first two positions and has positive sign. In the second row, placing zero in the third position contributes one positive sign choice, while placing it in either of the first two positions contributes two, for a total of five. Distinct nonzero absolute values in the third and sixth rows prevent a zero sum of the first two coordinates, so exactly half the orbit is positive. In the fourth row the first two signs must both be positive, leaving two choices for the third sign. In the fifth row, the exceptional absolute value occupies the third position in eight orbit points, of which two are positive; in the other sixteen points the first two absolute values differ, so eight are positive. This gives all the entries without a generic-position assumption on the integers.

Write t_a for the multiplicity of the orbit (a, a, a) . Lowering this weight by $e_1 + e_2$ gives $(a - 1, a - 1, a)$, whose orbit multiplicity is at least t_a . For $a = 1$ this companion orbit is $(1, 0, 0)$, with surplus one; for $a > 1$ it has pattern $(a, a - 1, a - 1)$, with surplus eleven. The companion orbits are distinct as a varies. None is the short-root orbit $(1, 1, 0)$. Thus each negative contribution $-t_a$ in Equation (9.12) is paid for by a distinct companion orbit; the payment leaves at least $10t_a$ when $a > 1$.

Independently, the earlier lowering comparison of long and short roots gives $m_0 \leq 3m_s$. The short-root surplus is $11/2 > 3$, so its contribution pays for full-Cartan zero with strict excess unless $m_s = m_0 = 0$. Every other unassigned surplus is positive. The bound follows. At equality, all t_a with $a > 1$ vanish, and the only remaining orbits can be $(1, 1, 1)$ and $(1, 0, 0)$ with equal multiplicity. The module is nontrivial, so both occur. Since $(1, 1, 1) - (1, 0, 0) = e_2 + e_3$ is a positive root, its highest weight must be $(1, 1, 1)$. Its highest-weight multiplicity is one, so both orbit multiplicities are one.

This module is $\bigwedge_0^3 E$. Indeed, the standard weights are $\pm e_1, \pm e_2, \pm e_3$. In $\bigwedge^3 E$, choosing one signed weight of each index gives all eight $(\pm 1, \pm 1, \pm 1)$ weights once. Choosing an opposite pair and one other signed weight gives each $\pm e_i$ twice, because either of the other two indices can supply the opposite pair. Hence $\bigwedge^3 E$ has these eight weights and twelve sparse weight vectors. The invariant symplectic bivector gives an injective equivariant map $E \rightarrow \bigwedge^3 E$, $v \mapsto \Omega \wedge v$; on a symplectic basis vector its image is a sum of two distinct nonzero wedge-basis vectors. Removing this standard summand leaves the eight sign weights and the six sparse weights, each once. To verify irreducibility without assuming it, the possible dominant highest weights in this complement are $(1, 1, 1)$ and $(1, 0, 0)$. The former occurs once and, by lowering $e_1 + e_2$, its irreducible constituent already contains the sparse orbit. It therefore contains all fourteen available weight vectors. The complement is irreducible, has dimension fourteen and positive count four, as required. $\square$

The following argument uses convolution of orbital measures to force normalized-character decay, a method developed in [59, 32]. We give the form needed for the uniform gap in the present representation count.

Lemma 9.16 (Decay of the zero-weight proportion). *Let K_c be a compact connected semisimple Lie group, and let $A \in \mathfrak{k}_c$ have nonzero projection to every simple ideal. Suppose $\exp(TA) = 1$ for some $T > 0$. For its irreducible representations π , of dimension d_π , one has*

$$\frac{\dim \ker d\pi(A)}{d_\pi} \longrightarrow 0$$

as π leaves every finite subset of the irreducible dual.

Proof. First fix $x \in K_c$ such that $\text{Ad } x$ is not the identity on any simple ideal. Put

$$U = (1 - \text{Ad } x)\mathfrak{k}_c.$$

The span of $\text{Ad}(k)U$ over $k \in K_c$ is an $\text{Ad}(K_c)$ -invariant ideal. Its projection to every simple ideal is nonzero, and hence it is all of $\mathfrak{k}_c$. Choose $h_1, \dots, h_N \in K_c$ so that $\sum_j \text{Ad}(h_j)U = \mathfrak{k}_c$. Consider the analytic map

$$F : K_c^N \longrightarrow K_c, \quad F(k_1, \dots, k_N) = \prod_{j=1}^N k_j x k_j^{-1}.$$

Write $p_0 = 1$ and $p_j = \prod_{\ell \leq j} k_\ell x k_\ell^{-1}$. For variations $k_j(t) = k_j \exp(tX_j)$, right trivialization of the differential gives

$$dF(X_1, \dots, X_N)F^{-1} = \sum_{j=1}^N \text{Ad}(p_{j-1}k_j)(1 - \text{Ad } x)X_j.$$

Choose $k_j = p_{j-1}^{-1}h_j$ recursively. At that point the displayed differential is onto.

Equip the groups with invariant analytic metrics. In invariant frames the matrix B of dF is analytic, and $\det(BB^\top)$ is an analytic function on the connected manifold K_c^N , nonzero at the constructed point. Its zero set has Haar measure zero. On the remaining set, submersion coordinates make F a coordinate projection. A countable cover by such coordinates and Fubini's theorem show that the inverse image of any Haar-null subset of K_c is null there. The critical set is also null. Therefore the pushforward of product Haar measure under F is absolutely continuous. Equivalently, if ν_x is the conjugacy-orbit probability measure of x , then

$$\nu_x^{*N} = f_x dk, \quad f_x \in L^1(K_c).$$

For an irreducible unitary π , Schur's lemma gives

$$\int \pi(g) d\nu_x(g) = a_\pi I, \quad a_\pi = \frac{\chi_\pi(x)}{d_\pi}.$$

Taking normalized traces after N convolutions yields

$$a_\pi^N = \int_{K_c} f_x(g) \frac{\chi_\pi(g)}{d_\pi} dk.$$

Approximate f_x in L^1 by a function $f_{x,M} \in L^2$. Orthogonality of irreducible characters and Bessel's inequality give $\int f_{x,M} \chi_\pi dk \rightarrow 0$ as π leaves finite subsets; the conjugate characters form the same orthonormal family. Moreover,

$$\left| \int (f_x - f_{x,M}) \frac{\chi_\pi}{d_\pi} dk \right| \leq \|f_x - f_{x,M}\|_1,$$

because $|\chi_\pi| \leq d_\pi$. First letting π leave finite subsets and then letting M tend to infinity proves $a_\pi^N \rightarrow 0$, and therefore $a_\pi \rightarrow 0$.

For each simple ideal choose a nonzero eigenvalue $i\beta$ of $\text{ad } A$ on that ideal. Such an eigenvalue exists because the ideal has zero center and the projection of A is nonzero. If $\exp(tA)$ acts centrally on that ideal, then $t\beta \in 2\pi\mathbb{Z}$. The union of these exceptional parameter sets, for $0 \leq t \leq T$, has measure zero. For every other t , the preceding normalized-character decay applies. Averaging a representation over the periodic subgroup projects onto $\ker d\pi(A)$, so

$$\frac{\dim \ker d\pi(A)}{d_\pi} = \frac{1}{T} \int_0^T \frac{\chi_\pi(\exp(tA))}{d_\pi} dt.$$

The integrands are bounded in absolute value by one. Dominated convergence proves the assertion along every sequence leaving finite subsets, hence on the entire irreducible dual. The argument uses all simple ideals at once and therefore also applies when K_c has several simple factors. $\square$

Lemma 9.17 (Invariant forms on maximizing modules). *Every maximizing module in Lemmas 9.13–9.15 has the weight, form, and parabolic-stage properties in Lemma 9.2.*

Proof. We use a positive Hilbert norm invariant under a compact real form of the complexified group. In each standard realization, the split generator is Hermitian for this norm. Write an invariant Hermitian form as $q(v, w) = \langle Qv, w \rangle$, using consistent conventions for the two Hermitian products. Its invariance under $\exp(tH_R)$ implies

$$d\pi(H_R)^*Q + Qd\pi(H_R) = 0.$$

When the split generator is Hermitian, this says that Q takes V_t to V_{-t} . Nondegeneracy therefore pairs opposite weight spaces perfectly and all other pairs trivially. Each pair $V_t \oplus V_{-t}$ for $t > 0$ has signature $(\dim V_t, \dim V_t)$. It follows that if the negative index of q is $h(V)$, its restriction to V_0 is positive definite. We verify that index and the normalization $Q^2 = I$ in all the cases.

For the orthogonal standard module of dimension $N = n + 1$, take the Hermitian extension of the real quadratic form of signature $(N - 1, 1)$ and the Euclidean compact-form norm. Its symmetry is $Q = \text{diag}(1, \dots, 1, -1)$. A split generator interchanges one positive basis vector and the negative basis vector, with zero action on their orthogonal complement. Its eigenvalues are $1, -1$ and $N - 2$ zeros. Thus $h = 1$ is the negative index and the zero space is positive and nonzero. This includes the orthogonal low-rank names used above.

For the unitary standard module of dimension k , use the Hermitian form of signature $(k - 1, 1)$ and its usual positive compact-form norm. The same two-dimensional boost gives split weights $1, -1$, and zero with multiplicity $k - 2$. Again the negative index is $h = 1$. On the dual, use the inverse Hermitian form and the dual Hilbert norm; its matrix is the transpose inverse of Q , still a self-adjoint involution, with the same signature and opposite split weights.

For $SU(2, 1)$, the induced form on $\text{Sym}^2 E$ has positive index $\dim \text{Sym}^2 \mathbb{C}^2 + 1 = 4$ and negative index two: the negative vectors are the products of one vector from the positive two-plane with the negative basis vector. The induced symmetry is the restriction of $Q \otimes Q$ to the symmetric tensors, and hence is a self-adjoint involution for the induced compact-form norm. The induced split generator is Hermitian. Its split spectrum is $2, 1, 0, 0, -1, -2$, so $h = 2$ equals the negative index and the two-dimensional zero space is positive. The dual argument gives the identical conclusions for $\text{Sym}^2 E^*$.

For the standard $Sp(n, 1)$ module, of complex dimension $2k = 2n + 2$, group the orthonormal basis into symplectic pairs e_i, f_i . Give each of the first $k - 1$ pairs sign $+1$ and the last pair sign -1 . The preserved Hermitian form has signature $(2k - 2, 2)$, its symmetry is diagonal with these signs, and a split generator couples one positive pair to the negative pair. It has weights $1, -1$, each twice, and zero with multiplicity $2k - 4$. Thus its negative index is $h = 2$ and its zero space is positive. These are integral weights.

It remains to examine the primitive cube for $Sp(2, 1)$. Let $E = \mathbb{C}^6$ with compact-orthonormal basis e_i, f_i , $1 \leq i \leq 3$, and let

$$q(e_i, e_j) = q(f_i, f_j) = s_i \delta_{ij}, \quad (s_1, s_2, s_3) = (1, 1, -1), \quad \omega(e_i, f_j) = \delta_{ij}.$$

All other pairings between distinct displayed basis vectors vanish, with ω extended skew-symmetrically. Write $Ce_i = s_i e_i$ and $Cf_i = s_i f_i$. The Hermitian form q_3 on $\Lambda^3 E$ has signature

$$\left(\binom{4}{3} + 4 \binom{2}{2}, 2 \binom{4}{2} \right) = (8, 12).$$

The first number counts wedges with zero or two negative basis vectors, and the second counts wedges with one. Put $\Omega = \sum_i e_i \wedge f_i$ and $L(v) = \Omega \wedge v$. For each basis vector, $L(v)$ contains the two complete pairs of the other indices. Each term has signed squared norm equal to that of v , since the squared sign of a complete pair is one. The wedge basis supports

for distinct unpaired basis vectors are disjoint. Consequently

$$q_3(Lv, Lw) = 2q(v, w).$$

In particular L is injective and its image has signature $(4, 2)$.

Let $c : \bigwedge^3 E \rightarrow E$ be symplectic contraction, normalized as the adjoint of L for the positive compact-form norm. Explicitly it contracts each complete e_i, f_i pair with its wedge sign; equivalently,

$$c(u \wedge v \wedge w) = \omega(u, v)w - \omega(u, w)v + \omega(v, w)u.$$

This formula shows equivariance under the complex symplectic group, and a basis calculation gives $cL = 2I$. Since $(\bigwedge^2 C)\Omega = \Omega$, one has

$$\left(\bigwedge^3 C\right)L = LC.$$

It follows both that $L(E)$ reduces the self-adjoint involution $\bigwedge^3 C$ and that $q_3(Lv, w) = q(v, cw)$. Therefore

$$P = \bigwedge_0^3 E = \ker c = L(E)^{\perp_{q_3}}$$

is also the compact-form orthogonal complement of $L(E)$. Its signature for q_3 is $(4, 10)$. On P choose instead $q_P = -q_3|_P$, whose signature is $(10, 4)$ and whose symmetry is $-(\bigwedge^3 C)|_P$, a self-adjoint involution. The compact-form orthogonal decomposition is invariant under the compact real form, hence under its complexified Lie algebra; the split generator remains Hermitian on P .

The weight calculation in Lemma 9.15 gives the primitive split spectrum

t	2	1	0	-1	-2
$\dim P_t$	2	2	6	2	2.

Thus $h(P) = 4$ is exactly the negative index of q_P ; the general opposite-pair argument already proves positivity on the six-dimensional P_0 . One can also see its sign directly. Choose the split generator

$$H_R e_2 = e_3, \quad H_R e_3 = e_2, \quad H_R f_2 = -f_3, \quad H_R f_3 = -f_2, \quad H_R e_1 = H_R f_1 = 0.$$

It is Hermitian for the compact norm, preserves ω infinitesimally, and anticommutes with C . Define

$$a_1 = (e_2 + e_3)/\sqrt{2}, \quad a_2 = (f_2 - f_3)/\sqrt{2}, \quad b_1 = (e_2 - e_3)/\sqrt{2}, \quad b_2 = (f_2 + f_3)/\sqrt{2}.$$

The a_i have split weight one, the b_i have split weight minus one, and $q(a_i, b_j) = \delta_{ij}$ while each of their two spans is isotropic. For $z = e_1, f_1$ put $w_{ij}(z) = a_i \wedge z \wedge b_j$. These eight vectors span the zero space in the full cube and satisfy

$$q_3(w_{ij}(z), w_{kl}(z')) = -\delta_{il}\delta_{jk}\delta_{zz'}.$$

For each z , the positive direction is $w_{12} - w_{21}$; the three negative directions are w_{11}, w_{22} and $w_{12} + w_{21}$. Moreover,

$$\Omega = e_1 \wedge f_1 + a_1 \wedge b_2 - a_2 \wedge b_1, \quad L(z) = -(w_{12}(z) - w_{21}(z)).$$

Removing $L(E)_0$ therefore removes exactly the two positive directions, leaving six negative directions for q_3 , as asserted. Negating the form makes all of them positive.

In every case all listed weights are integral and the zero spaces just computed are nonempty. The statements about annihilators now follow from the perfect opposite-weight pairings and the positive zero space. To verify the parabolic assertion, use a positive multiple of H_R that is an integral cocharacter of the simply connected complexified group. A root space whose root is nonnegative on H_R maps a weight space V_t into $V_{t+\alpha(H_R)}$ and therefore preserves both $V_{>0}$ and $V_{\geq 0}$. The corresponding algebraic parabolic, generated by this Cartan and these root groups, preserves both spaces. This is the complex parabolic attached to the oriented rank-one boundary point. Its translates are algebraic subspaces in the appropriate Grassmannians,

hence give the asserted parabolic stages independently of the choice of opposite point. Their middle quotient identifies with V_0 and inherits its positive form. $\square$

Proof of Lemma 9.2. The bounds and their complete equality lists follow from Lemmas 9.13, 9.14, and 9.15; the form and integrality claims follow from Lemma 9.17. The displayed spectra give $d_R = 2$ in the two stated exceptional cases and $d_R = 1$ in the others.

It remains to obtain one uniform gap for all other irreducibles. Let K_c be the simply connected compact real form of the complexified Lie algebra and set $A = iH_R$ in that compact real form. In the coordinate realizations used above, A has a period after passing to the full covering-group weight lattice. Its projection to every simple ideal is nonzero. This is immediate in the simple cases; for $SO(3, 1)$ the two $\mathfrak{sl}_2$ components of the displayed split functional are both nonzero. Lemma 9.16 consequently gives $z(V)/\dim V \rightarrow 0$ as the highest weight leaves finite subsets. The split functional and its negative are Weyl conjugate in each of these realizations, so

$$h(V) = \frac{\dim V - z(V)}{2}, \quad \frac{\dim V}{h(V)} = \frac{2}{1 - z(V)/\dim V} \rightarrow 2.$$

Every tabulated c_R is greater than two. Choose τ with $2 < \tau < c_R$. Only finitely many irreducibles have ratio at least τ . After the maximizing modules are removed, the ratios of this finite set are all strictly smaller than c_R by the equality classification. The maximum of those remaining ratios and τ is therefore a number $b_R < c_R$. Taking $\varepsilon_R = c_R - b_R$ proves the uniform gap. Finally, reversing H_R interchanges positive and negative spaces, preserves all ratios, and preserves the largest absolute weight. Thus c_R and d_R are unchanged at the opposite orientation. $\square$

The representation-theoretic verification completes the real-panel upper estimate. Section 10 combines it with the other local branches and uses Haar volume to calibrate these one-sided bounds.

10. CENTERED HEIGHTS AND HAAR DENSITIES

The local arguments have supplied upper bounds for the missing panel gaps and positive heights at isolated types. We first use those isolated heights to prove tightness of their centered gaps. Together with the panel upper bounds, this permits a comparison of source Haar volume and target lattice packing on paths selecting one vertex in each component. The comparison fixes the panel speeds and makes all these tuple gaps tight. We then construct the remaining heights from their centered laws.

The resulting pair identities identify the full centered Cartan law. A second, angularly localized packing argument shows that inverse chamber heights turn the boundary measurements into ordinary chamber probability. The associated isometry and its exact height intertwining are the inputs to the common graph domains in Section 11 and to the real-pencil argument in Section 12.

10.1. Paths, line norms, and local-field conventions. For every non-isolated source position i , put

$$s_i = \frac{d_S}{d_T},$$

where S, T are its source and matched target panel groups and the constants are those of Sections 6 and 9. At an isolated position put $s_i = 1$. At non-isolated positions the constants satisfy the upper-speed estimates (6.21) and (9.1) at finite and archimedean places, respectively. All these constants have opposition symmetry $s_i = s_{i^*}$ and product one on every cycle of the type permutation p : the ratios of the d_R telescope, and the isolated constants are one.

We use mixed lexicographic paths with one missing high type per indicated component and all isolated types retained. A further completing translation is applied to the inputs

before the faster limit is taken. The sharp face marginals are unchanged by translations stabilizing those faces, although the residual projectivity may change. Adding a missing direction strictly refines the relevant inner stratum, by the gap-count argument of Section 4. Completing the chamber makes every target gap diverge. Repeating a selected direction by a slower translation preserves its image type and cannot add a type beyond the upper count.

Compact face frames may be chosen by Borel compact-transport sections. When a frame function is passed through a link or through a completing limit, first restrict to its compact continuity sets for the fixed input laws. Those laws are unchanged by the translations stabilizing the sharp faces. Their complements have arbitrarily small mass on each fixed testing vector. Countably many determining vectors and charts, followed by strong continuity, therefore justify the framed substitutions without assuming continuity of a global Borel section.

Offsets will always be held fixed during every ordinary path time and exhausted only afterwards. Thus even a negative offset in a selected positive direction does not undo that direction's divergence. The parameter t_i denotes its total value, including such offsets. At a finite place all split translations and offsets run on the fixed sufficiently divisible coweight mesh of Section 3.

Local type	Angular measurements and gap estimates	Covariance and domains
Finite-place panels Section 6	Each endpoint pencil is a commuting family of evaluations of homogeneous analytic homeomorphisms. The reduced output projectivity is ab^{-1} (Proposition 6.11). On the permissible cocharacter mesh, $d_T \ell_T(g) \leq d_{st} + O_{\text{tight}}(1)$, uniformly in slower translations with the same total gap t (Proposition 6.12).	The source acts through $\eta : S \rightarrow T$ with isomorphic differential over the common prime field $\mathbb{Q}_p$ (Proposition 6.10). The original endpoint observations and exact input translations are retained. This branch uses bounded spectral calculi. Its simultaneous upper-tail estimate uses a finite union bound over coordinates; it requires no commutation between different pencils.
Non-isolated archimedean panels Section 9	Sharp endpoint faces and missing-vertex measurements retain the ordered residue graph relation throughout the successive completing limits. The estimate is $d_T \ell(g) \leq d_{st} + O_{\text{tight}}(1)$ on every fixed input, including the stated subsequent offset limits (Proposition 9.1).	Source and measured-group covariance retain their compact-frame corrections. The scalar section domain $\mathcal{D}$ is dense and transport-invariant (Lemma 9.8). The finite-dimensional algebra acts on its section space with the positive form identity of Proposition 9.10; action exchange is estimated on fixed domain vectors (Lemma 9.11).
Isolated real factors Sections 7–8	The pure-vertex measurements are sharp, and complementary lengths are tight. A subsequence carries uniformly positive mass in a linear window on every fixed vector (Proposition 8.7). The integrated pair estimate gives $Q_S = Q_T = Q$ and square-integrable horizontal derivatives (Lemma 8.14).	The closed derivative construction gives positive injective heights b_x strongly commuting with their own P_x . They satisfy $\sigma_a(b_x) = \text{dil}(a, x)^{-Q} b_{ax}$ and $\beta_h(b_x) = \text{dil}(h^{-1}, P_x)^{-Q} b_x$. Closure of the module core and normal copying are proved before use (Lemma 8.15 and Proposition 8.17).

TABLE 1. The three local inputs to product calibration. Here P_z, Q_z are the endpoint projection-valued measures, t is the total completing source gap, ℓ is the residual target gap, d_R is the comparison constant for a non-isolated panel type, and Q is the homogeneous dimension of an isolated boundary. All tightness and linear-window assertions concern fixed Hilbert vectors.

At this stage, incident labels commute by Proposition 5.11; the stronger commutation of arbitrary labels is still to be proved. The isolated linear-window estimate becomes tightness of centered gaps in Lemma 10.4. The present section then supplies the Haar-height isometry and

exact height intertwining. The graph cores needed for unbounded locality, and commutation between the two normal copies at arbitrary isolated labels, are established in Section 11. Real-pencil commutation and simultaneous chamber commutation follow in Section 12.

An event of positive infinite excess or deficit is unchanged by fixed left or right target shifts: a fixed multiplication changes the Cartan vector by a bounded amount. The independent contracted invariances of the mixed link therefore make its compressed mass scalar. At a sharp angular slot its angular subeffect is that scalar times the sharp marginal: its compression is dominated by the relevant projection at every cut. This observation applies after retaining additional length or phase variables. It permits us to fix a purported positive mass in an iterated limit, work with bounded shift windows and continuous subtests with slack, and only then exhaust thresholds. No common good set of lattice indices for all rotations and offsets is being asserted.

Let λ_i be the weight of the nilradical determinant line at the standard vertex of type i . It is a positive multiple of the corresponding fundamental weight, and

$$\lambda_i(e_i) = 2\rho(e_i).$$

Indeed the positive roots omitted from that nilradical involve only the other simple roots and hence vanish on e_i . Superscripts S, T distinguish the source and target lists. Write $j(x, g)$ for the norm growth of the determinant line at x under g , and $k(y, x)$ for its normalized absolute pairing with the dual line at the opposite-type vertex y . The normalization is one at a standard compact-frame opposite pair. Consequently

$$k(gy, gx)j(y, g)j(x, g) = k(y, x).$$

The product of line cocycles with total weight 2ρ will be denoted by $j_{2\rho}$. It is the inverse Jacobian of the compact chamber probability.

Lemma 10.1 (Exact line normalization). *Give the determinant line and its dual homogeneous norms obtained from special-compact-invariant norms. At a finite place use the module absolute value, including its residue-degree normalization. For an actual Cartan centralizer representative z , the highest-line growth is exactly*

$$|\chi(z)| = \exp(\lambda_i(\mu(z))).$$

When the i -gap diverges, the normalized operator has only its highest-line block in the limit. If w is its limiting reverse dual vertex and v is transverse to w , then

$$(10.1) \quad j_T(v, h^{-1}) \exp(-\lambda_{p(i)}^T(\mu(h^{-1}))) \longrightarrow k_T(w, v).$$

The convergence is uniform with all needed derivatives on compact transverse real patches. In finite-place variables the normalized ratios are eventually locally constant on compact transverse boxes with a uniform level. Fixed multiplications on the two sides change the logarithmic top growth by the corresponding endpoint line cocycles.

Proof. Use the actual dominant translation vector of the centralizer, as fixed in Section 3; it records the exact module of each centralizer character. The remaining centralizer parts and their inverses are uniformly bounded on the weight spaces. Highest-weight strings separate every other block from the highest line by a positive multiple of the pertinent gap. After division by the exact highest-line growth those blocks therefore vanish. The compact factors are isometries for the chosen norms.

In compact frames the limit is the rank-one operator $\tilde{u} \otimes \tilde{w}$, with unit lifts of the extremal lines, up to a scalar of absolute value one. Evaluating it on the line at v gives its normalized pairing with w , proving (10.1). Multiplying on either side transforms the relevant extremal line; the norm needed to renormalize that line is exactly its j -factor. At a finite place one can alternatively evaluate on a compact transverse test line and divide by its reverse pairing. Once the gap is large, the ultrametric inequality makes the resulting norm identity exact. All denominators have a uniform lower bound on the buffered patches, so the locally constant

tests have uniform levels there. In particular there is no additional bounded but nonvanishing Cartan error in (10.1).

For the Jacobian assertion, reduce by compact frames to a minimal parabolic element. Its action on the opposite-root chart has determinant given by the product of positive-root modules, including double roots and their multiplicities. This is $\exp(2\rho)$ and gives the inverse compact-angular Jacobian. At an isolated real vertex the horizontal dilation gives the same Jacobian by its tangent grading. Thus its j is exactly the cocycle of the height already constructed in Section 8; equality of the homogeneous dimensions in Proposition 8.17 yields

$$\lambda_i^S(e_i) = \lambda_{p(i)}^T(e_{p(i)}) \quad \text{at isolated positions.}$$

□

At finite places a continuous real logarithmic character of a parabolic vanishes on its radical by contraction, on compact subgroups by compactness, and on root-generated semisimple directions. It therefore depends only on the split center, up to compact or finite-index changes. The selected coweight mesh is a full lattice in those split-center coordinates and determines such a character. This is the character convention used below; no real derivative on a building is needed.

10.2. Moving tests and ordered transport. We first record a uniform version of the smooth-test extension needed in the transfer argument. The coefficient traces and matrix sizes have the normalization fixed in Proposition 5.1. When writing a constant from the transverse-product estimate, we retain the homogeneous Hilbert bounds in (5.2), or its trace-mass factor after an operator-norm-only bound. In every application below the cut arrays are made from a fixed bounded-column family, so those factors are bounded.

Lemma 10.2 (Radial tests with moving centers). *On the buffered transverse supports of (5.1), one may add finitely many tests*

$$\varphi\left(\frac{\ell - b}{R}\right), \quad \varphi \in C_c^\infty(\mathbb{R}^q), \quad b \in \mathbb{R}^q, \quad R \geq 1,$$

where ℓ is a fixed finite list of logarithmic matrix norms, including inverse norms. At real places use Hilbert–Schmidt matrix norms, which are smooth away from zero; at finite places use the maximum norms fixed in Section 3. One may also add characters of these logarithmic norms with frequencies in a fixed bounded set. The transverse-product bounds are uniform in b, R and those frequencies, with constants depending on the fixed test functions and the original buffered supports. The same assertion holds in mixed real and finite-place coordinates, with compact clopen tests in the latter.

Proof. For a length-increasing substitution $p \mapsto up$, normalized matrix multiplication gives, componentwise,

$$\ell(up) = \ell(u) + \ell(p) + q(S_u, S_p).$$

Here S denotes projectively normalized matrix coordinates, and q has a smooth extension with uniform derivative bounds on the compact coordinate boxes. Its denominator is bounded below by transversality. For inverse norms use $(up)^{-1} = p^{-1}u^{-1}$, so the normalized product has the reversed order. At finite places q is bounded and locally constant with a uniform level on the same kind of buffered normalized boxes, by Lemma 10.1 and the local-field conventions.

Fourier inversion gives

$$\begin{aligned} \varphi\left(\frac{\ell(up) - b}{R}\right) &= \int_{\mathbb{R}^q} \widehat{\varphi}(\xi) \exp\left(\frac{i\xi \cdot (\ell(u) - b)}{R}\right) \exp\left(\frac{i\xi \cdot \ell(p)}{R}\right) \\ &\quad \cdot \exp\left(\frac{i\xi \cdot q(S_u, S_p)}{R}\right) d\xi, \end{aligned}$$

with the Fourier normalization absorbed in $\widehat{\varphi}$. The first factor is a scalar of modulus one; it contains the unrestricted center b . The second is a bounded unary multiplier on the

shorter argument. Its logarithmic argument is not restricted to a compact set and is not differentiated as a rescaled compact coordinate. Only the third factor is expanded in the compact normalized coordinates.

For each fixed derivative order r , that compact factor has seminorm at most $C_r(1 + |\xi|)^r$, uniformly in $R \geq 1$, the center, and the supported shifts. Smooth separated expansions on a fixed larger box therefore have a summable supremum bound of that polynomial size. In a finite-place slot one uses the fixed finite partition into clopen boxes instead. Apply the unary Gram–Plancherel estimate (5.2) at each frequency, retaining any designated shorter cutoff outside the expansion. Minkowski’s inequality and

$$\int_{\mathbb{R}^q} |\widehat{\varphi}(\xi)|(1 + |\xi|)^r d\xi < \infty$$

give the required uniform bound. The inequality holds because $\widehat{\varphi}$ is Schwartz. A bounded additional character frequency replaces ξ/R by $\theta + \xi/R$ in the compact factor, with the same polynomial estimate. Finitely many cutoffs use joint frequency variables and products of Schwartz transforms.

The compact expansion uses a common unary basis with coefficient bound $\sum_n \sup_u |c_n(u)| < \infty$, as in (5.4); a bound with the supremum and sum in the opposite order would not suffice. The frequency integral above retains exactly this stronger bound.

The argument uses only scalar unary masks and the coefficient-valued Gram estimate. It thus preserves the normalized-trace, fixed-matrix-size, and homogeneous Hilbert bounds of that estimate. In the bounded-overlap case, separate bounded radial and phase masks are already permitted directly. Finally, ordinary finite-support exhaustion is obtained by unshifted cutoffs in a proper finite list of logarithmic norms, equal to one on increasing Cartan balls. Their supports meet the discrete lattice in finitely many points. Their radii may increase as fast as a given test or approximation requires without changing the bound. $\square$

Choose a vertex tuple F selecting one type in each component, with type set I , and a compact-frame opposite tuple F^- . A tuple path has inverse split vector

$$L = \sum_{i \in I} t_i e_i.$$

For the height and radial-limit arguments below, take these parameters successively in any chosen order, starting at one selected vertex and adding the other selected coordinates singly. The mixed-face angular law itself also holds for simultaneous parameters, by Proposition 5.11. We also use full-chamber lexicographic paths through F , in any order, with opposite chamber through F^- ; then L includes all coordinates. The preceding angular statements and bounded complementary thickness hold for these paths, by sharpness, exact precomposition, and the total upper count. For an actual Fourier index h put

$$r_i(h) = \lambda_{p(i)}^T(\mu(h^{-1})) - \lambda_i^S(L).$$

The next lemma is first applied to the isolated heights of Proposition 8.17, yielding the isolated tightness in Lemma 10.4. Its later tuple applications use the heights of Proposition 10.7 and the auxiliary tight centered widths supplied by Proposition 10.6.

Lemma 10.3 (Ordered imaginary-power transfer). *Suppose that $b = b_{F,i}$ is a positive injective operator affiliated with $\mathcal{A}_j$, commuting with its own angular measurement, and satisfying*

$$\sigma_g b_{F,i} = j_S(x_i, g) b_{gF,i}, \quad \beta_h b_{F,i} = b_{F,i} j_T(P_{x_i}, h^{-1}).$$

In the isolated case take the height b_{x_i} of Proposition 8.17 and require only transversality of its end-input value v at the same positive source label with the reverse output dual value w . In the tuple case assume that b commutes with P_F , require the corresponding tuple transversality, and assume tight centered widths on an auxiliary mixed tuple path at F .

On buffered transverse patches, for a tuple or full main path and bounded frequencies $\theta_n \rightarrow \theta$ in the stated iterated parameter limit,

$$(10.2) \quad J(1 \otimes b^{i\theta}) = E(e^{i\theta_n r_i}) k_T(w, v)^{i\theta} J(\bar{b}^{i\theta} \otimes 1).$$

The bar conjugates the positive operator before its imaginary power is taken. A joint symbol in this formula means output multiplication followed by the adjacent end-input multiplication in the indicated order.

Proof. At fixed path parameters, move the start power of frequency θ_n through X_a . Since a^{-1} has split vector L , its source covariance produces $\exp(-i\theta_n \lambda_i^S(L))$. To read coefficient (h, r) , pair against $\zeta(ay)w_{(h,r)}$, where r comprises the unmeasured indices. Moving the end action through this right unitary changes ζ to

$$b^{-i\theta_n} j_T(P_{x_i}, h^{-1})^{-i\theta_n} \zeta.$$

Conjugation in the first input gives exactly the signs of (10.2). The remaining group, base, and coefficient multipliers commute with the power.

On this moved side take a bounded start column before any movement. The end column may be Hilbert: the finite-time bilinear estimate with one bounded input is uniform. Lemma 10.1 and uniform separated expansions replace the target factor by the pairing phase on transverse patches. Imaginary powers at θ_n converge in Hilbert norm on each fixed column to the power at θ .

We justify the original side without asserting a product estimate for two arbitrary Hilbert columns. Construct the end vector as a cut one-ended effect applied to a bounded column ζ_0 . Its defining auxiliary path is a pure vertex path at x_i in the isolated case, with smooth angular cut and compact complementary thickness. In the later tuple case it is a mixed path at F with bounded centered widths. Choose its image patch buffered transverse to the reverse main window. By sharpness the ranges of such vectors exhaust the corresponding angular patches.

First retain finitely many measured indices. Test the main link against bounded-column arrays, with smooth output tests at fixed thickness and sufficiently large prescribed gaps; allow powers of the retained phase as well. At a fixed main time the cut end vector can be replaced by its defining auxiliary weak limit. This remains valid when the other input is the Hilbert error from replacing a height-applied start vector by a bounded approximation: at finite measured support the remaining coefficients are bounded operators, and normality permits bounded tests in the full remaining tracial algebra.

More concretely, let f be bounded and let f_l be a bounded-column approximation to $b^{i\theta} f$. After trace rearrangement the start error is paired against a product of the form

$$(D_{\text{aux}} X_s(ay) \zeta_0(ay))(D_{\text{main}} T),$$

where T is the bounded-column testing array. Its intermediate directions are $q^{-1}o$ and ho . Transversality to the reverse tuple of a full chamber suffices for a tuple patch by opposite-face projection. In the tuple case the overlap widths are bounded; in the isolated case the auxiliary support is a single smooth vertex cone, with compact cuts in the other factors. Hence (5.1) and Lemma 10.2 bound this product uniformly. The error is at most that fixed bound times

$$\|b^{i\theta_n} f - f_l\|_2.$$

Common projective transport phases disappear from this tracial estimate. First choose l and then pass to the parameter limit; the error tends to zero as $l \rightarrow \infty$.

The finite-support tests can use the smooth radius cutoffs of Lemma 10.2. A phase may be computed from the logarithmic Hilbert norm of the weight matrix: at a divergent pertinent gap the difference from the logarithmic top norm tends to zero by rank one. At a finite place the top growth is computed exactly on a compact transverse input line for all sufficiently large gaps, after division by the bounded locally constant reverse pairing. Thus all phase/radius tests retain the uniform estimate before supports are removed.

At each main time the radii may be made sufficiently large for the moved-side expansions and the bounded start approximant. Their product error bound is independent of those radii. After the bounded start approximation, the end vector may remain Hilbert by the one-bounded-input estimate. Exhaust auxiliary thickness and transverse patches, then the main stratum using ambient continuous tests with slack and a final Borel output projection. Testing in the spectral span generated by these compact variables and the retained phase proves the vector equality. Additional phases not used in this equality need not be included in that span. A second measured leg can remain throughout in the coefficient algebra. This proves (10.2) with the stated order and domains. $\square$

Lemma 10.4 (Total transverse ranges and shrinking frequencies). *The closed sum of the ranges of transverse ordered angular rectangles for opposite source tuples is the whole endpoint space. The same holds for the isolated vertex-pair rectangles. In the isolated application of (10.2), the centered variable r_i is tight, including along the subsequent offset limits.*

Proof. The closed range sum is target invariant. It is also fixed by an all-factor escaping source subgroup fixing the relevant labels and normalizing the opposite radical moves. Its projection is therefore scalar by the link invariance argument. It is nonzero: simultaneous source transport to generic pairs, followed by patch fullness, supplies a nonzero ordered rectangle. This proves totality. The tuple assertion follows by the same argument on the open opposite-tuple orbit.

For the isolated height put $\theta_n \rightarrow 0$ in (10.2). Pull the sharp reverse angular variable through J . The pairing phase and both imaginary powers tend to one on each buffered rectangle. Hence the retained phase fixes J on its total range sum and therefore everywhere. If tightness failed, some fixed bounded-column input would retain a positive mass beyond thresholds tending sufficiently slowly to infinity. Averaging $|e^{itr} - 1|^2$ over a shrinking interval of t gives, by the elementary sinc formula, a uniformly positive value on that escaped mass. Choose a frequency in each such interval. These frequencies tend to zero and contradict the just-proved equality. Thresholds and testing vectors can be selected by countable exhaustion along the successive path filters. The same proof permits offsets held after all ordinary times, so the centered isolated widths are tight on those iterated paths too. $\square$

10.3. Two-volume calibration of tuple paths.

Lemma 10.5 (Coefficient integral bound). *Choose a finite Parseval frame (f_l) for the representation of π_0 and $W_{R'_j}$ on H : the full group translates of these constant Hilbert columns form a Parseval family. Let $b_0 \neq 0$ be a W_R -bounded end column. Up to the fixed Haar and module normalization, for a finite set $D \subset \Gamma_j$ the integral over $a \in G$ of the total measured coefficient probability at $h \in D$, summed over l , is at most*

$$|D| \|b_0\|_2^2.$$

The path probability comparisons can be integrated over bounded compact rotation and split-offset parameters by exact pre/post translation of these inputs, also through successive ultra-limits.

Proof. Unfold, for fixed y , the identity $ad_0(y) = d_0(y')\gamma$. At fixed h the integrated coefficient sum is, up to the fixed normalization, the sum of

$$|\langle W_{h,r}^* b_0(y'), \pi_0(\gamma) f_l \rangle|^2$$

over γ, r, l , at the independent base variables y, y' . Move W_r to the complementary frame index. Complementary Parseval and commutation up to scalar phases make the whole sum equal to one squared norm of $b_0(y')$. Integration proves the bound. In particular no extra factor comes from summing the unmeasured group indices.

A bounded set of compact rotations and permissible split offsets gives a compact family of translated inputs in Hilbert norm, preserving the operator bound of the bounded column. Approximate the Hilbert column first by bounded ones, and then both translated families by

finite compact nets. The one-bounded-input probability estimate controls the errors uniformly in path time. Exact translations therefore permit passage under these bounded parameter averages. Apply the argument successively for the different path times; no exchange with an arbitrary nonuniform ultralimit is needed. $\square$

Proposition 10.6 (Tuple calibration). *For every selected tuple position i ,*

$$(10.3) \quad 2\rho_T(s_i e_{p(i)}) = 2\rho_S(e_i), \quad \mu(h^{-1}) = \sum_{i \in I} s_i t_i e_{p(i)} + O_{\text{tight}}(1).$$

These assertions hold with the total parameters on the further offset limits specified above.

Proof. We first prove the high-coordinate upper bounds. Compare a tuple path a with a faster opposite many-panel lexicographic face path d . All the parameters of d are taken to their limits before those of a . The path d omits exactly the complementary selected high types i^* and retains full isolated directions. Thus da^{-1} has split vector $L + L_{\text{panel}}$, with the panel part faster. Cut the two columns to transverse start angular patches and compact complementary thickness. Also impose bounded centered isolated widths on the tuple column, using Lemma 10.4, and test a high excess

$$\alpha_{p(i)}(\mu(h^{-1})) > s_i t_i + M.$$

Trace cyclicity pairs the cut X_d column times the adjoint of the cut X_a column against $X_d X_a^*$, which, after base change and a harmless common phase, is the da^{-1} column. The product has a uniform Hilbert bound by the bounded-overlap case of (5.1). Transverse Cartan addition forces excess greater than $M - C$ in the pertinent residual quotient gap. Apply the local upper-speed estimate, (6.21) or (9.1), to the inverse path in the opposite orientation. Its missing-type bound is exactly the needed bound on the ordinary index product. Cauchy–Schwarz therefore makes this pairing tend to zero as $M \rightarrow \infty$.

Exhaust continuity cuts, complementary thickness, and angular restrictions. If positive infinite excess remained, its one-ended angular sublaw would be a positive scalar times P_F . Its ordered products with all transverse opposite many-panel patches could not vanish: simultaneous conjugation of the source tuple and opposite subfaces ranges over their open orbit, and fullness supplies nonzero products on conull such choices. The scalar mass is unchanged by these fixed conjugations, which only cost bounded shifts or tight perturbations. This contradiction proves all high upper bounds. The isolated ones already follow from centered isolated tightness. Both local upper-speed estimates use the total shifted parameters, so this reasoning also applies with the permitted further offsets.

Use the frame and bounded end column of Lemma 10.5. Around L integrate over a fixed thick shell, with independent compact rotations and offsets in a fixed box whose complementary coordinates are positive. At a real wall take an interior subbox; the Cartan density then gives the lower bound

$$c \exp(2\rho_S(L)).$$

At a finite place integrate over the disjoint selected Cartan double cosets in the permissible mesh. Each is bi-compact averaging times a constant comparable to $\exp(2\rho_S(L + \eta))$, by the root-coordinate index count. The same lower estimate follows. The mixed link isometricity, the just proved upper bounds, and complementary thickness retain a positive mass after sufficiently large fixed cuts. This can be done uniformly over the bounded translation parameters by the compact-net argument of the lemma.

Disjoint fixed neighborhood thickening of lattice points and Cartan integration bound the target count in the corresponding windows by

$$C \exp\left(\sum_{i \in I} t_i 2\rho_T(s_i e_{p(i)})\right).$$

At finite places include all Cartan entries in the window enlarged by a fixed constant; their multiplicity per unit mesh box is bounded. The coefficient integral bound gives the comparison

of these exponentials. Make i the fastest coordinate to obtain

$$2\rho_S(e_i) \leq 2\rho_T(s_i e_{p(i)}).$$

The lists are two copies of the same type list, and the speed ratios telescope around each permutation cycle. Multiplication around that cycle gives equality at every position.

Suppose next that a selected coordinate has positive mass of infinite deficit from its calibrated center. This sublaw again has scalar mass on the mixed link. Fix complementary and other upper widths large enough to retain a mass $\varepsilon > 0$, and then impose deficit greater than M . The same lower shell estimate and the coefficient integral bound give

$$\varepsilon c \exp(2\rho_S(L)) \leq C \exp(2\rho_S(L)) \exp(-M 2\rho_T(e_{p(i)})).$$

Here the cut widths are fixed before M is exhausted, so $c, C > 0$ do not depend on M . The final factor is the gain in the target packing bound. In simple coordinates every coefficient $2\rho_T(e_j)$ is positive, and integration up to each upper bound costs a constant times its endpoint exponential. There is consequently no polynomial length loss. The displayed inequality is impossible for large M , proving tightness of every deficit.

The same argument applies to any purported positive mass in a subsequent offset limit: offsets are fixed during all ordinary path times and are exhausted only afterwards. Finally, p preserves the Coxeter components by Section 5, and the tuple selects only one type in each component. Evaluating $\lambda_{p(i)}^T$ in (10.3), using $\lambda_{p(i)}^T(e_{p(i)}) = 2\rho_T(e_{p(i)})$, therefore proves tightness of each r_i . This evaluation requires no cross-weight Cartan identity. $\square$

10.4. Height solution lines and pair identities.

Proposition 10.7 (Tuple heights). *At each non-isolated tuple position there is a positive injective affiliated height $b_{F,i}$ commuting with P_F and satisfying*

$$(10.4) \quad \sigma_g b_{F,i} = j_S(x_i, g) b_{gF,i}, \quad \beta_h b_{F,i} = b_{F,i} j_T(P_{x_i}, h^{-1}).$$

At an isolated position the previous height b_{x_i} can be used and also commutes with P_F . On the tuple link,

$$(10.5) \quad J^* E(e^{i\theta r_i}) J = \chi(\theta) \bar{b}_{F^-,i}^{i\theta} \otimes b_{F,i}^{i\theta},$$

where χ is a scalar probability characteristic function. Heights may be chosen in the two-slot system and then copied normally to the system with two measured legs.

Proof. The compressed characteristic operators of the tight variable r_i commute with the sharp tuple coordinates. They lie in $\bar{\mathcal{A}}_j \otimes \mathcal{A}_j$, have independent contracted invariances, and transform under independent target shifts by the endpoint line phases of Lemma 10.1. At a given end and frequency t , a bounded solution B_t , commuting with the pertinent sharp vertex, has the tilted law

$$\beta_h B_t = B_t j_T(P_{x_i}, h^{-1})^{it}.$$

For two solutions of the same frequency, both orders of their products with adjoints have untilted target laws and the contracted invariances. They are in $\mathcal{N}$ and scalar. A nonzero solution is consequently a scalar multiple of a unitary, and the solution space has dimension at most one.

Normal slices of the characteristic operators give locally nonzero solutions near zero, strongly continuous together with their adjoints. Here tightness supplies strong convergence of the characteristic operators at zero. Normalize the nonzero slices to unitaries. Products of these local solutions give the corresponding solution line at every frequency, and the lines form a continuous projective one-parameter unitary group. They commute with P_F because the original compressed operators do.

The group consisting of those unitary solution lines has local continuous sections over $\mathbb{R}$ and circle fibers, hence is locally a two-dimensional Euclidean group. The locally Euclidean groups theorem makes it a Lie group [30, 47]. Its Lie-algebra extension over the one-dimensional abelian Lie algebra splits; integrating that section gives a continuous splitting over the simply

connected real line. Stone's theorem [62, Theorem B] now writes the resulting group as $b_{F,i}^{it}$ for the exponential of a selfadjoint affiliated operator. Thus $b_{F,i}$ is positive and injective. One-dimensionality of both end solution lines gives (10.5). After the genuine unitary groups are removed, its scalar coefficient is continuous and positive definite: in the operator-valued positive-definite matrix tests absorb the inverse unitary group factors into the testing vectors. Its value at zero is one.

We determine the source covariance rather than assuming its character. The source parabolic stabilizer of F normalizes these intrinsic solution laws and therefore acts on the height by a continuous positive character. Replace L by $L + mA_0$ along a selected split-center direction, with the integer m held after all ordinary path times. Before recentring, exact precomposition translates the characteristic law by $m \log(\text{char}(A_0))$ in the start slot. Recentring subtracts $m\lambda_i^S(A_0)$. Proposition 10.6 gives tightness even when these subsequent multiples are exhausted. A nonzero residual drift would contradict that tightness on each unit input. Hence the two characters agree on every selected split direction. Those directions span the split center of the tuple parabolic. A real logarithmic character vanishes on its radical, compact parts, and derived Levi factors, as explained above; the finite-place mesh determines it there as well. Compact transport now gives exactly (10.4) at all labels.

At an isolated position retain the previously constructed b_{x_i} . It solves the same one-dimensional laws, so its powers agree up to scalars with the tuple solution and commute with P_F . The reverse-end heights solve the same intrinsic laws at opposite positions. All constructions use countable determining labels and tests before taking Borel output cuts; strong continuity then supplies every fixed label. The normal-copy argument of Section 4, together with uniqueness of the solution lines, therefore permits these one-ended heights to be copied to the two-measured-leg system. Exact end-input identities are copied normally too. Later commutation comparisons use one common all-gap path in that system, rather than unrelated marginal limits. $\square$

Proposition 10.8 (Ordered pair law). *For opposite tuples Y, F , on transverse tuple patches,*

$$(10.6) \quad b_{Y,i^*}^{i\theta} P_Y(dw) b_{F,i}^{i\theta} P_F(dv) k_T(w, v)^{i\theta} = C(Y, F)^{i\theta} P_Y(dw) P_F(dv),$$

$$C(Y, F) = c k_S(y_{i^*}, x_i), \quad c > 0.$$

The statement also holds on the conjugate slot, with conjugation and the corresponding sign change. At an isolated position only the vertex-pair transversality is needed.

Proof. Apply (10.2) to a tuple path at constant frequency, and compress using (10.5). Cancel the independent start power and pull the sharp reverse tuple through J . At $Y = F^-$ this gives the left side of (10.6), multiplied by $\chi(\theta)$, equal to the unweighted ordered product.

This equality can be compared at both signs despite the ordering. Each height power commutes with its own spectral slot. Multiply outside the left and right slots by their inverse powers and insert the nonvanishing smooth pairing phase on the transverse patch. Applying the opposite-sign identity gives

$$\chi(\theta)\chi(-\theta) P_Y(dw) P_F(dv) = P_Y(dw) P_F(dv)$$

on such patches. Nonzero ordered rectangles exist by Lemma 10.4; hence $|\chi(\theta)| = 1$. Since χ is a probability characteristic function, it is the characteristic function of a point mass. Its scalar is a positive constant raised to the imaginary power. Simultaneous source transport and the pairing covariance give $C(Y, F) = c k_S(y_{i^*}, x_i)$.

For clarity about subsequent substitutions, write P, Q for the reverse and same-label tuple slots on a conjugate end input, and A_t, B_t for the corresponding conjugated height powers. The compressed equation reads

$$\chi(t) A_t P B_t Q k_T^{it} = P Q.$$

Thus the substitution used on a full path is $P B_t Q k_T^{it} = C^{it} A_t^{-1} P Q$, in exactly that order. The relevant total range sum may be multiplied on the left by the inverse reverse power and

remains total. Smooth multipliers and separated expansions on the buffered patches justify every localized identity. The isolated case follows from the first version of (10.2) in the same way. $\square$

10.5. The full centered law.

Proposition 10.9 (Sharp full-path heights). *On a full lexicographic path,*

$$(10.7) \quad E(e^{i\theta_n r_i})J = J\left(C(F^-, F)^{-i\theta} \bar{b}_{F^-, i^*}^{i\theta} \otimes b_{F, i}^{i\theta}\right) \quad (\theta_n \rightarrow \theta).$$

There is an invertible linear map A_ on the split-coordinate spaces satisfying*

$$(10.8) \quad \lambda_{p(i)}^T A_* = \lambda_i^S, \quad A_* e_i = s_i e_{p(i)}, \quad 2\rho_T A_* = 2\rho_S.$$

The centered Cartan vector $\Delta = \mu(h^{-1}) - A_ L$ has a tight sharp commuting law on input.*

Proof. Insert (10.6) into (10.2) on a full path. Pulling back the sharp reverse tuple first produces the ordered conjugate-end expression $\bar{P}_{F^-}(dw) \bar{b}_{F, i}^{i\theta} \bar{P}_F(dv)$. The ordered substitution in the proof of Proposition 10.8 gives (10.7) on transverse ordered end ranges acted on by the inverse reverse power. Those ranges are total. Letting frequencies tend to zero and applying the shrinking-frequency argument proves tightness on the full path too.

Apply this for every type using any tuples in the chamber. The determinant weights form bases, so the first equation in (10.8) determines an invertible A_* . To identify its column at e_i , choose a tuple prefix with i fastest, add its other selected types at successively slower scales, and then add the remaining chamber types one at a time. Dividing the centered law by that fastest time and using (10.3), with the subsequent translations handled by exact precomposition, gives $A_* e_i = s_i e_{p(i)}$. The last equality follows from the calibrated 2ρ values in (10.3). The output Cartan phases commute; (10.7) pulls their sharp law through the isometry, giving the asserted joint law of Δ . $\square$

The full-path law also controls the height groups before arbitrary-label commutation has been proved. Every height belonging to a chamber commutes with that chamber's angular measurements. This includes the angular measurements in either normal measured copy: they and the centered output phases belong to the same full-path spectral law, and (10.7) pulls them through the same isometry. More explicitly, write $\bar{U}_-(t) \otimes U_+(t)$ for the two endpoint height groups in that equation, with its scalar factor suppressed. If P is a chamber angular projection in either measured copy at the start input, then

$$[\bar{U}_-(t) \otimes U_+(t), 1 \otimes P] = \bar{U}_-(t) \otimes [U_+(t), P] = 0.$$

The reverse factor is unitary, so $[U_+(t), P] = 0$.

Two height groups belonging to that chamber may at this stage commute only up to a scalar. Indeed their tensor products in (10.7) commute, so tensor cancellation makes the component commutators reciprocal scalars. The one-parameter group laws make each scalar commutator a bicharacter, and strong continuity gives the form e^{icst} in the two real frequencies. The same argument applies across measured copies because both were retained in the common full-path law.

For the two copies of the same isolated height b_x , this scalar is one. On $H = K \otimes K \otimes K$ from Section 3, use the complex-linear unitary

$$\mathcal{F}(\xi_0 \otimes \xi_1 \otimes \xi_2) = \xi_0 \otimes \xi_2 \otimes \xi_1.$$

It fixes the trunk action and diagonal ρ , interchanges the two measured actions, and preserves the multiplier (3.5). It acts fibrewise on the induced space. Normal copying in Proposition 4.3 and the intrinsic closed derivative modulus in Proposition 8.17 therefore give $\mathcal{F}b_x^1 \mathcal{F}^* = b_x^2$ and the converse equality. If $U_j(t) = (b_x^j)^{it}$ and

$$U_1(s)U_2(t) = e^{icst}U_2(t)U_1(s),$$

conjugation by $\mathcal{F}$ and comparison with the relation with s, t interchanged give $e^{2icst} = 1$ for all $s, t \in \mathbb{R}$. Thus $c = 0$. The two copies of the complete same-label angle-height calculus consequently commute strongly. This interchange of measured slots is separate from the conjugate endpoint factor in the full-path law.

Write

$$2\rho_S = \sum_i a_i \lambda_i^S.$$

Equation (10.8) gives the same coefficients for the permuted target weights. At a chamber Z fix, by their type sets, tuple choices $F(i) \subset Z$ containing i . Define

$$D_Z^{it} = \left(\prod_i b_{F(i),i}^{ita_i/2} \right) \left(\prod_i^{\text{reverse}} b_{F(i),i}^{ita_i/2} \right).$$

The symmetric order cancels all scalar Weyl reordering phases, so this is a genuine strongly continuous unitary group. It defines a positive injective affiliated height D_Z . Use opposite tuple choices to define $D'_{Z'}$ at the other end. These heights commute with their chamber angles and have the covariance laws (10.4) with chamber cocycles. On independently compact-rotated full paths, spectral calculus in (10.7) gives

$$(10.9) \quad \exp(2\rho_T(\Delta)) = \text{const} \cdot \bar{D}'_{Z'} \otimes D_Z, \quad \text{const} > 0.$$

Compact rotations are exact input translations and preserve the line norms separately. A further fixed shift $L \mapsto L + \eta$ shifts the law relative to the old center by $A_*\eta$. These statements concern one common joint calculus, not separately chosen height marginals.

10.6. Weighted chamber probability.

Theorem 10.10 (Haar-height isometry). *Let B_S, B_T be the source and target chamber spaces with their compact probabilities. There is a constant $0 < c_0 < \infty$ and an isometry*

$$(10.10) \quad \begin{aligned} V : L^2(B_T; \mathcal{H}) &\longrightarrow L^2(B_S; \mathcal{H}), \\ (V(f \otimes \xi))(Z) &= c_0^{-1/2} D_Z^{-1/2} P_Z(f) \xi. \end{aligned}$$

The expression on the second line is initially interpreted through positive form integrals and then extended by the stated isometry.

Proof. The weighted packing estimate. Count inverse lattice indices in a fixed-width box around A_*L whose image chamber lies in a continuity patch $O \subset B_T$. Pack with fixed disjoint right neighborhoods of those inverse indices. In all-gap windows, right thickening perturbs the image chamber by $o(1)$ uniformly, by highest-line convergence, and changes the Cartan vector by a bounded amount. Cartan integration therefore gives

$$\limsup \frac{\#\{\text{such indices}\}}{\exp(2\rho_S(L))} \leq C|O|.$$

One may first enlarge angular patches by fixed slack and then shrink it by compact regular approximation. The finite-place count includes all double cosets in a fixed enlargement of the box, with bounded multiplicity per mesh box, as in Proposition 10.6.

Integrate both compact rotations and split shifts η in a bounded box, using the frame inputs $\bar{b}_0 \otimes f_l$ of Lemma 10.5. The coefficient bound and the Cartan density, normalized by $\exp(2\rho_S(L))$, show that a constant times $|O|$ bounds the weighted integral of limiting probabilities of

$$\Delta + A_*\eta \text{ in the fixed box,} \quad \text{image chamber in } O,$$

with weight $\exp(2\rho_S(\eta))$. Begin with smaller interiors and continuous tests. Exact input translations and the compact-net argument permit the bounded parameter integral before each limit. Equation (10.7) identifies these translated tests with the same sharp law, shifted by $A_*\eta$ about the old center; compact rotations introduce no extra line scalings.

Now exhaust the whole split space, using a sum on finite-place meshes and ordinary integration on real coordinates. Choose the fixed counting box sufficiently wide. For every Δ , the set of permissible offsets for which $\Delta + A_*\eta$ lies in a smaller box has uniformly positive sum-volume near $-A_*^{-1}\Delta$. Since A_* is invertible and $2\rho_TA_* = 2\rho_S$, this yields

$$\int_{\{\eta: \Delta + A_*\eta \text{ in the box}\}} e^{2\rho_S(\eta)} d\eta \geq ce^{-2\rho_T(\Delta)}.$$

Here and below this notation includes the indicated mesh sums. A full lattice has a bounded covering radius, which proves the assertion at finite places; the real subbox has a fixed positive volume. Only bounded offset windows are used before a path limit, so all chambers remain deep in the prescribed iterated order.

Positive integration thus bounds the expectation of $e^{-2\rho_T(\Delta)}$ with the angular cut. By (10.9), independence of the two compact rotations, and the product input $\bar{b}_0 \otimes f_l$, that positive form factors into an opposite factor and a forward factor, up to a fixed constant. No exponential moments have been presumed. Perform this step with positive spectral truncations first; any strictly positive bounded truncation of the averaged opposite factor is enough to divide. It follows that

$$(10.11) \quad \sum_l \int_{B_S} \langle f_l, D_Z^{-1} P_Z(O) f_l \rangle dZ \leq C'|O|.$$

Regular approximation extends the bound from continuity patches to all positive measurable tests.

The positive density map. For bounded $f \geq 0$ define the extended positive form

$$T(f) = \int_{B_S} D_Z^{-1} P_Z(f) dZ.$$

The height commutes with its angular slot, so this integral is positive. Its source covariance and the chamber Jacobian place it in the extended positive part of $(\mathcal{A}_j)^\sigma$. Indeed the inverse line factor is canceled by the change of angular probability under the source transport. After undoing induction a σ -fixed field is constant with value in the complementary commutant. Evaluation on the finite Parseval frame, summed over l , is precisely its faithful finite trace. Thus (10.11) makes T a bounded positive map

$$L^1(B_T) \longrightarrow L^1((\mathcal{A}_j)^\sigma).$$

It intertwines transport of chamber densities with the trace-preserving target action. This also justifies the claimed L^1 values of the unbounded positive form integral.

Why this map is scalar. Induce the finite algebra $(\mathcal{A}_j)^\sigma$ along the target lattice inclusion, with its induced finite trace. If $\mathcal{D}_g$ denotes transport of chamber densities, the induced equivariant map is represented by fields

$$g \longmapsto T(\mathcal{D}_{g^{-1}} f).$$

Their equivariance under the right lattice action follows from that of T . They are integrable over the quotient: $\mathcal{D}_{g^{-1}}$ is a positive L^1 isometry, and the norm of T gives a uniform bound on their L^1 norms. Let h_0 be the induced image of the constant density 1, and let α be the induced group action.

For every target group element s , averaging $\mathcal{D}_k \mathcal{D}_s 1$ over the special compact group gives 1, because that compact group is transitive on chambers and the density has total mass one. Equivariance therefore gives

$$h_0 = \int_K \alpha_k \alpha_s(h_0) dk.$$

Every integrand has the same spectral distribution as h_0 , since the induced action preserves the finite trace. For a threshold t which is not an atom of that distribution, the projection $p_t = 1_{(t, \infty)}(h_0)$ uniquely maximizes $\tau(p(h_0 - t))$ among projections. The maximum is the same for each integrand. Testing the displayed average against p_t gives equality in the average

of these upper bounds, so p_t is the maximizing spectral projection for almost every integrand. Take a countable dense set of atom-free thresholds; their spectral projections determine the whole positive L^1 element. Hence $\alpha_k \alpha_s(h_0) = h_0$ for almost every k . Strong L^1 continuity extends this to all k , and in particular $\alpha_s(h_0) = h_0$.

The span of the group transports of 1 is dense in $L^1(B_T)$. To see this directly, take a regular split contraction. It pushes chamber probability to a point mass away from its null repelling set. Convolution of these concentrating densities with continuous compact-rotation weights approximates continuous chamber densities uniformly, by lifting to the compact group modulo the chamber stabilizer. Continuous densities are dense in L^1 . The induced map therefore depends only on total integral. Take a common conull section value for a countable dense set of density tests. Because density transport is an onto L^1 isometry, this rank-one conclusion descends to the original map. Its value is fixed by both source and target actions, hence scalar. We obtain

$$T(f) = c_0 \int_{B_T} f(v) dv.$$

The trace bound makes c_0 finite, and positivity together with the injectivity of the heights makes it strictly positive.

Finally, for bounded angular tests, commutation of D_Z with P_Z gives the norm identity in (10.10) from this formula for T . Polarization handles finite sums of elementary tensors, and completion gives the claimed isometry. Positive truncation and separability give the measurable L^2 representative in that equation, without imposing an operator bound on $D_Z^{-1/2}$. $\square$

10.7. The height intertwiner.

Proposition 10.11 (Intertwining with ordinary chamber variables). *Fix F, i . For generic source chamber Z , let $Y(Z) \subset Z$ have the tuple type opposite to F . If v is an ordinary target chamber, with the relevant dual vertex used in the pairing, then*

$$(10.12) \quad V[k_T(v, P_{x_i})^{i\theta} b_{F,i}^{i\theta}] = [C(Y(Z), F)^{i\theta} b_{Y(Z),i*}^{-i\theta}] V.$$

Both bracketed operators are unitary. Moreover D_Z strongly commutes with each such own-coordinate height $b_{Y(Z),i}$.*

Proof. For any fixed spectral input value the nontransverse ordinary chamber set is null. Fubini therefore gives transversality on the full product of ordinary chamber probability and the input spectral measure. The powers in the two brackets have modulus one and commute with their own angular slots, proving unitarity.

Localize in the ordinary chamber variable and in the input P_F coordinates on buffered transverse rectangles; these give total domains. Before the factor $D_Z^{-1/2}$ in V , substitute at P_Z using (10.6). Chamber tests may be placed on the left because the $Y(Z)$ height commutes with them. Use separated smooth expansions, or finite clopen partitions at finite places, for the pairing tests, and truncate the D_Z height to bounded intervals.

The scalar Weyl relations obtained from (10.7) show that commuting $b_{Y(Z),i*}^{-i\theta}$ through $D_Z^{-1/2}$ costs at most a positive scalar, while its height truncations are only dilated. This scalar is constant on the generic chamber orbit: simultaneous source transport preserves the scalar commutator, and the tuple choices inside a chamber have fixed type sets. Remove the truncations using the isometry (10.10). We obtain (10.12) initially up to that positive scalar. Both bracketed operators are unitary and V is an isometry, so comparison of norms forces the scalar to equal one. For every real frequency this makes the Weyl coefficient vanish; hence the corresponding unitary groups, and therefore the positive affiliated heights, commute strongly. The same argument proves (10.12) exactly on the completed domains. $\square$

11. LOCALITY AND STRONG COMMUTATION AT ISOLATED TYPES

Fix an isolated source type i and its measured type $p(i)$. Let B_S and B_T denote their boundaries, equipped with compact-invariant probability measures. These are vertex boundaries; ordinary chamber variables in the other components will be integrated out without changing notation. Write $c_i > 0$ for the constant in Equation (10.6) at this type. The scale and height analysis of Sections 8 and 10 gives a common homogeneous dimension Q on the two sides. Set

$$\vartheta = \frac{2}{Q}, \quad K_S(x, z) = k_S(x, z)^\vartheta, \quad K_T(v, u) = k_T(v, u)^\vartheta.$$

In the groups admitted here, $Q > 8$, so $0 < \vartheta < 1/4$. For a source boundary point z , write

$$Q_z = (P_z, \log b_z), \quad B_z = b_z^\vartheta.$$

Here Q_z is the joint spectral measure of the indicated commuting pair at one label. Our first objective is strong commutation as z varies. We will then prove commutation between the two normal copies sharing the trunk slot.

The inputs from Section 10 are the exact ordered imaginary-power identity (10.6), the isometry (10.10), and the unitary-group intertwining (10.12), including the own-coordinate density commutation proved there. Each b_z is positive, has zero kernel, and has finite spectral values. Source transport gives strong continuity of bounded continuous own-coordinate spectral functions; when an unbounded operator varies with a label, we use its bounded resolvents.

The kernel construction below has four roles. Its bilinear formula makes the transported products $K_T(v, P_z)B_z$ depend linearly on a finite-dimensional source lift, and its inverse-square integrability supplies a dense common graph domain. On that domain, its fourth-order polynomial dependence turns a small off-diagonal estimate into exact locality. Finally, the embedded cone supplies enough commuting evaluation multipliers to eliminate every positive-order term in the resulting local finite-jet formulas. Graph cutoffs and individual cores then carry these exchange identities to strong commutation.

11.1. A finite-dimensional boundary kernel.

Lemma 11.1 (Polynomial boundary lifts). *For either boundary $B = B_S, B_T$ there are a finite-dimensional real vector space E , a nondegenerate symmetric bilinear form $[\cdot, \cdot]$ on E , and a smooth map $n : B \rightarrow E$ such that*

- (i) $n(B)$ spans E , and $(r, z) \mapsto rn(z)$ embeds $(0, \infty) \times B$ smoothly in E ;
- (ii) $[n(w), n(z)] = K(w, z)$, with $K(w, z) > 0$ for $w \neq z$;
- (iii) in a unipotent boundary chart, $n(z)$ is a polynomial of weighted degree at most four divided by a fixed smooth positive function;
- (iv) for a compact Carnot distance d ,

$$c d(w, z)^4 \leq K(w, z) \leq C d(w, z)^4;$$

- (v) there are positive constants a_1, a_2, a_{-2} such that, for every $u \in B$,

$$\int_B K(v, u) dv = a_1, \quad \int_B K(v, u)^2 dv = a_2, \quad \int_B K(v, u)^{-2} dv = a_{-2} < \infty.$$

Proof. We spell out the finite-dimensional representation used here. For $\mathrm{Sp}(n, 1)$, use the irreducible representation of absolute highest weight $e_1 + e_2$ in the root system C_{n+1} . Normalize the split generator so that its evaluations on the first two standard coordinates are both one. The highest restricted weight is then two, is attained on a single extremal weight, and has multiplicity one. The lowest restricted weight is minus two. For $F_{4(-20)}$, use the representation of highest short root. In the customary coordinates with short roots

$$\{\pm e_j\} \cup \left\{ \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4) \right\},$$

the split direction has $e_1(H) = 2$ and all other $e_j(H) = 0$. Again the extremal evaluations two and minus two are unique. The absolute vanishing root system is B_3 . These weight assertions also follow directly by maximizing the displayed linear functional on the Weyl orbit and its convex hull.

Here are also the determinant weights, which will identify the kernel exactly. In type C_{n+1} the roots of split value two are $2e_1, 2e_2, e_1 + e_2$, and those of split value one are $e_i \pm e_j$, $i = 1, 2, j = 3, \dots, n+1$. Thus

$$(m_\alpha, m_{2\alpha}, Q) = (4n - 4, 3, 4n + 2), \quad \sum_{\beta(H) > 0} \beta = (2n + 1)(e_1 + e_2).$$

For $F_{4(-20)}$ the roots of split value two are e_1 and $e_1 \pm e_j$, $j = 2, 3, 4$, and those of split value one are $\frac{1}{2}(e_1 \pm e_2 \pm e_3 \pm e_4)$. Hence

$$(m_\alpha, m_{2\alpha}, Q) = (8, 7, 22), \quad \sum_{\beta(H) > 0} \beta = 11e_1.$$

In either case the determinant of the positive nilradical has absolute highest weight $(Q/2)\lambda$, where λ is the highest weight chosen above. The quaternionic groups in the present class have $n \geq 2$; thus $Q \geq 10$ in that case and $Q = 22$ in the Cayley case.

We justify the required real representation. Let κ be split Chevalley conjugation on the complex Lie algebra and σ its given real-form conjugation. The types C_{n+1} and F_4 have no diagram automorphisms, so the complex automorphism $\sigma\kappa$ is inner, say $\text{Ad}(h)$. The highest-weight module has a Chevalley real form, with antilinear conjugation C intertwining κ . The map $J = \pi(h)C$ therefore commutes with the given real Lie algebra. Schur's lemma gives $J^2 = a \text{id}$. Since J preserves the unique complex top space for the real split generator, its square on that line is a positive real scalar. Thus $a > 0$, and rescaling J gives $J^2 = \text{id}$. Its fixed space is the required real module E . These highest weights lie in the root lattice, so the representations descend to the adjoint groups used here.

The longest Weyl element is -1 in both types. Self-duality therefore gives a nondegenerate invariant complex bilinear form. Its conjugate by J is a scalar multiple of itself; rescaling the form makes it J -real, so it restricts to a nondegenerate real pairing on E . Split-weight orthogonality says that the top line pairs only with the bottom line; nondegeneracy makes that pairing nonzero. The stabilizer of the top line contains the boundary parabolic and is proper. In rank one the Bruhat decomposition has only two cells, so this stabilizer is exactly that parabolic. Its projective orbit is the smoothly embedded boundary.

Choose a maximal-compact-invariant Euclidean norm on E . The boundary is a sphere, S^{4n-1} or S^{15} , and is simply connected. Its real line bundle consequently has a smooth global unit lift n . Connectedness of the group implies that all its action scalars on this lift are positive. Distinct ordered boundary pairs form one group orbit, so the pairing is nonzero with constant sign off the diagonal. Change its sign and scale to make it positive there and one at standard compact opposites. By irreducibility its transpose is a scalar multiple of itself, with scalar 1 or -1 ; positivity on both orders excludes -1 . The real pairing is therefore symmetric.

The determinant highest line computed above generates the Cartan component of $V_\lambda^{\otimes Q/2}$. Its lift is the tensor power $n(z)^{\otimes Q/2}$. Compact transitivity makes its unit normalization agree with that of the determinant line, and its invariant pairing is $[n(w), n(z)]^{Q/2}$. Normalizing at the same opposite pair removes the single constant. Consequently

$$k(w, z) = [n(w), n(z)]^{Q/2}, \quad K(w, z) = [n(w), n(z)].$$

This uses the entire highest-line representation, not only its split character.

In a chart based at the lowest line, the unnormalized lift is $\exp(X)n_-$, where X ranges over the graded positive nilradical. A term raising restricted weight from minus two to at most two has total graded degree at most four. Exponentiation is thus a polynomial of that weighted degree. Passing to a unit lift divides by its smooth positive norm. Write $q(X) = \exp(X)n_-$

for the unnormalized lift, and let δ_t multiply degree-one coordinates by t and degree-two coordinates by t^2 . Since the bottom weight is -2 ,

$$q(\delta_t X) = t^2 \pi(a_{\log t}) q(X).$$

Invariance of the pairing therefore makes its two-point kernel homogeneous of degree four. Its invariance under simultaneous left translation in the nilradical reduces it to the difference coordinates. It is positive away from zero. Compactness of the homogeneous unit sphere gives comparison with the fourth power of a homogeneous gauge; finitely many charts give the asserted compact distance comparison.

The highest-line orbit is an embedded projective orbit. The unit lift and its positive radial parameter consequently give the claimed cone embedding. Its span is a nonzero invariant subspace, hence all of E . Compact transitivity makes the three displayed integrals independent of u . The first two are finite and positive. For the last one, a ball of radius r has measure bounded by $C r^Q$ and the integrand is bounded by a constant times $d(v, u)^{-8}$. Dyadic annuli give

$$\int_{d(v, u) < 1} K(v, u)^{-2} dv \leq C \sum_{j \geq 0} 2^{8j} 2^{-Qj} < \infty.$$

Here $Q = 4n + 2 \geq 10$ for $\mathrm{Sp}(n, 1)$ and $Q = 22$ for $F_{4(-20)}$. $\square$

11.2. The common graph domain. On the target chamber-variable Hilbert space in Equation (10.10), define the positive decomposable operators

$$A_z(v) = K_T(v, P_z) B_z.$$

Their factors commute because they belong to the single Q_z calculus. All integrals in this Section use the fixed compact-invariant probabilities; no coefficient trace or matrix size is varied. Write I and I_S for the constant-function injections on the target and source chamber-variable spaces, respectively. Equation (10.12), with its frequency rescaled by ϑ , gives

$$(11.1) \quad V A_z^{\mathrm{it}} = C_z^{\mathrm{it}} V, \quad C_z(Z) = c_i^{\vartheta} K_S(y(Z), z) b_{y(Z)}^{-\vartheta}, \quad t \in \mathbb{R}.$$

The source point $y(Z)$ is the vertex of the ordinary source chamber Z at the opposite type. In particular, for fixed Z it is independent of z .

Lemma 11.2 (Same-variable commutation and norm identities). *For every $z, w \in B_S$ and $v \in B_T$, the operators $A_z(v)$ and $A_w(v)$ strongly commute. The domain*

$$\mathfrak{D} = \left\{ \xi \in \mathcal{H} : \int_{B_S} \|B_z \xi\|^2 dz < \infty \right\}, \quad \|\xi\|_{\mathfrak{D}}^2 = \|\xi\|^2 + \int_{B_S} \|B_z \xi\|^2 dz$$

is a Hilbert graph domain. Its vectors belong to every $\mathrm{dom} B_z$, and

$$(11.2) \quad \sup_z \|B_z \xi\|^2 \leq C \int \|B_x \xi\|^2 dx, \quad \|B_z \xi\|^2 + \|B_w \xi\|^2 \geq c_1 d(z, w)^8 \int \|B_x \xi\|^2 dx.$$

On $\mathfrak{D}$, the maps $z \mapsto A_z(v)$ and $z \mapsto B_z$ are linear in the coordinates of $n_S(z)$. The map $v \mapsto A_z(v)$ is linear in the coordinates of $n_T(v)$.

Proof. The range of V reduces each unitary group C_z^{it} : Equation (11.1) applies to both t and $-t$. The C_z strongly commute, since at each ordinary source chamber they are scalar multiples of the same operator $b_{y(Z)}^{-\vartheta}$. The spectral theorem transfers their resolvents to the target representation. Hence $A_z(v)$ and $A_w(v)$ strongly commute for almost every v . For fixed z , bounded resolvents of $A_z(v)$ depend strongly continuously on v , by bounded convergence in the Q_z calculus. Products of their bounded resolvents have the same continuity. Their commutation therefore holds for every v .

Positive spectral truncation in Equation (11.1) gives, including the value $+\infty$,

$$(11.3) \quad a_2^T \|B_z \xi\|^2 = c_i^{2\vartheta} \int_{B_S^{\mathrm{ch}}} K_S(y(Z), z)^2 \|b_{y(Z)}^{-\vartheta} (V I \xi)(Z)\|^2 dZ.$$

Here B_S^{ch} denotes the full chamber variable occurring in V ; its indicated vertex marginal is the ordinary probability on B_S . The left-hand side follows by integrating v and using Lemma 11.1. Integrating the right-hand side in z replaces the kernel square by the constant a_2^S . This identifies the graph norm with a closed weighted norm of $VI\xi$, and proves completeness. Boundedness of K_S gives the first estimate. For the second, the triangle inequality implies that at least one of $d(y, z), d(y, w)$ is at least $d(z, w)/2$. The sum of the two kernel squares therefore dominates $c d(z, w)^8$, uniformly in y , giving Equation (11.2).

If $\sum_j a_j n_S(z_j) = 0$, then $\sum_j a_j K_S(y, z_j) = 0$ for every y . All the source operators in this finite linear combination are defined on $VI\mathfrak{D}$ by Equation (11.3). Intertwining shows that $\sum_j a_j A_{z_j}(v)\xi = 0$ for almost every v . Continuity in v on this domain extends the equality to every v . Moreover,

$$B_z \xi = (a_1^T)^{-1} \int_{B_T} A_z(v) \xi \, dv, \quad \xi \in \mathfrak{D},$$

so the same relation holds for B_z . Linearity in $n_T(v)$ follows directly from the bilinear kernel representation. $\square$

Lemma 11.3 (Density, smooth stability, and cores). *The domain $\mathfrak{D}$ is dense in $\mathcal{H}$. For each compact subset $C \subset B_T \times \mathbb{R}$ there are $r \in \mathbb{N}$ and $C_C < \infty$ such that*

$$\|f(Q_z)\xi\|_{\mathfrak{D}} \leq C_C \|f\|_{C^r} \|\xi\|_{\mathfrak{D}}, \quad \text{supp } f \subset C.$$

There are $\chi_l \in C_c^\infty(B_T \times \mathbb{R})$ such that $\chi_l(Q_z)\xi \rightarrow \xi$ in $\mathfrak{D}$ for every $\xi \in \mathfrak{D}$. Finally, $\mathfrak{D}$ is a core for every B_z and $A_z(v)$.

Proof. Density. Let M_l be multiplication on the source chamber-variable space by $b_y^\vartheta 1_{\{b_y \leq l\}}$, and put

$$T_l = I^* V^* M_l I_S.$$

These are bounded operators. The constant-input formula in Equation (10.10) and the strong commutation of D_Z with its own-coordinate height give

$$\langle \eta, T_l \eta \rangle = c_0^{-1/2} \int \left\langle D_Z^{-1/2} \eta, b_y^\vartheta 1_{\{b_y \leq l\}} \eta \right\rangle dZ \geq 0.$$

All pairings are integrable by the isometry and boundedness of M_l . For nonzero η , they cannot vanish for every l : in the common D_Z, b_y calculus the union of the cuts $b_y \leq l$ is the identity, and both positive operators have zero kernel. Thus the positive operators T_l have zero common kernel and their ranges are total.

For $\zeta = V^* M_l I_S \eta$, the reducing-range intertwining gives

$$A_z \zeta = V^* C_z M_l I_S \eta = c_i^\vartheta V^* (K_S(y, z) 1_{\{b_y \leq l\}} I_S \eta).$$

Its norm is bounded uniformly in z . To estimate $B_z I^* \zeta$, test against $\eta' \in \text{dom } B_z$ and write

$$\langle I^* \zeta, B_z \eta' \rangle = \int_{B_T} \left\langle A_z(v) \zeta(v), K_T(v, P_z)^{-1} \eta' \right\rangle dv.$$

This identity is first made with joint spectral truncations of Q_z . The inverse-square bound in Lemma 11.1 permits their removal by Cauchy–Schwarz. It proves that $I^* \zeta \in \text{dom } B_z$ with a uniform bound. Integrating z proves $\text{ran } T_l \subset \mathfrak{D}$, and therefore density.

Smooth stability and graph cutoffs. The basic stability estimate concerns a bounded unary average

$$R_F = \int_{B_T} F(v, A_z(v)) \, dv.$$

For $\xi \in \mathfrak{D}$ and $\eta' \in \text{dom } B_w$, move $A_w(v)$ through $F(v, A_z(v))$, using Lemma 11.2, and pair against $K_T(v, P_w)^{-1} \eta'$. Cauchy–Schwarz gives

$$(11.4) \quad \|B_w R_F \xi\| \leq C \|F\|_\infty \|B_w \xi\|.$$

The adjoint domain criterion used above justifies this computation before the left-hand side is known to be finite. Integrating w gives a bound on $\mathfrak{D}$. Products of unary averages satisfy the iterated bound.

Choose $v_1, \dots, v_d$ such that the $n_T(v_j)$ form a basis of E_T . Allow each v_j to vary in a sufficiently small fixed neighborhood. The resulting inverse frame matrices are uniformly bounded. The evaluations

$$(K_T(v_j, u)e^{\vartheta s})_{j=1}^d$$

then recover $e^{\vartheta s}n_T(u)$ linearly. A compactly supported smooth function of (u, s) extends smoothly to a compact set in the ambient vector space away from zero. Compose such an extension with the linear recovery, and integrate the independent v_j over their neighborhoods. On fixed compact coordinate boxes, its Fourier expansion in the separate variables has absolutely summable coefficients after finitely many derivatives. Each separated term is a product of unary averages. Equation (11.4) therefore proves the stated smooth stability estimate.

For completeness, compact graph cutoffs can be constructed by this same procedure, without assuming a joint height law at different labels. Cut each of the basis evaluations smoothly from above at l , and subtract the piece on which all evaluations are smaller than $1/l$. Uniform frame inverses imply that the resulting function of $e^{\vartheta s}n_T(u)$ is supported in a bounded radial interval bounded away from zero. It is consequently a compact smooth function of (u, s) . Each upper unary cutoff tends strongly to the identity. Each simultaneous-small piece tends strongly to zero. The same convergences hold in the graph norm: in the proof of Equation (11.4), apply dominated convergence to the vector $A_w(v)\xi$, then integrate w . The only possible zero of an evaluation at a finite positive height is $P_z = v$; $P_z(\{v\}) = 0$ for almost every ordinary v , as follows by integrating against a faithful normal scalarization of P_z and using that the ordinary measure is nonatomic.

Individual cores. Let L be B_z or $A_z(v)$. Cut a vector in $\text{dom } L$ by compact smooth functions of its own Q_z variable that tend to one and have bounded height support. These cuts converge in the graph norm of L . For each fixed cutoff, approximate the uncut vector in Hilbert norm by vectors of the dense domain $\mathfrak{D}$, and apply that cutoff. Smooth stability keeps the resulting vectors in $\mathfrak{D}$, while boundedness of L on the cutoff gives convergence in its graph norm. A diagonal choice proves that $\mathfrak{D}$ is a core. $\square$

11.3. Exact locality and the exclusion of derivatives.

Lemma 11.4 (Real powers on transverse patches). *Fix $z \neq w$. On compact transverse angular patches, with the Q_z height restricted to a compact interval, the ordered replacement*

$$B_w \longmapsto c_i^\vartheta K_S(w, z) K_T(u', u)^{-1} B_z^{-1}$$

is valid, where the angular variables occur in the order $(u', u) = (P_w, P_z)$. The adjoint ordered replacement is valid as well. After division by $K_S(w, z)$ the norms of the replaced operators are bounded independently of w ; the fixed angular and height patches determine the bound.

Proof. To specify the ordered assertion, let $f \in C_c^\infty(B_T \times \mathbb{R})$, let ρ equal one on its angular support, and let χ' be supported in a patch transverse to $\text{supp } \rho$. For a smooth kernel on this transverse rectangle, write

$$\mathcal{M}_{w,z}(F) = \int F(u', u) P_w(du') P_z(du).$$

This denotes the operator-norm-convergent sum obtained from a summable separated smooth expansion of F ; it does not introduce joint spectral calculus for P_w, P_z . We will prove the bounded operator identity

$$(11.5) \quad B_w P_w(\chi') f(Q_z) = c_i^\vartheta K_S(w, z) \mathcal{M}_{w,z} \left(\frac{\chi'(u') \rho(u)}{K_T(u', u)} \right) B_z^{-1} f(Q_z).$$

In particular the entire range on the left belongs to the domain of B_w . Initially insert a fixed rightmost smooth angular-height test $f_0(Q_z)$ equal to one on $\text{supp } f$, with a slightly larger

transverse angular support and a fixed compact height interval. Equation (10.6) then gives, for a Schwartz function a and by Fourier inversion,

$$\begin{aligned} & a(\log b_w) P_w(\chi') f_0(Q_z) \\ &= \int \chi'(u') \rho_0(u) a\left(\log \frac{c_i k_S(w, z)}{k_T(u', u)} - s\right) P_w(du') Q_z(du, ds) f_0(Q_z), \end{aligned}$$

where ρ_0 is a fixed enlarged angular cutoff. In this formula the height on the right belongs to the single Q_z calculus, and the remaining ordered kernel is defined by the same separated expansions. Every fixed number of derivatives of $k_T(u', u)^{-it}$ on the transverse rectangle is bounded polynomially in $|t|$. The Schwartz Fourier transform therefore makes the integral and all required separated expansions absolutely summable.

The substituted real logarithm belongs to an interval $\log(c_i k_S(w, z)) + J$, where the compact interval J depends only on the fixed supports. Every smooth a supported outside this interval annihilates the left block. By spectral regularity this proves that $P_w(\chi') f_0(Q_z)$ has its $\log b_w$ spectrum in that interval. This is a bounded-operator support statement on the whole input range. It applies in particular after inserting any additional bounded oscillatory multiplier on that range, with a bound independent of its derivatives. We may now insert $a(s) = e^{\vartheta s}$ on the indicated interval using spectral truncations. This proves Equation (11.5). After division by $K_S(w, z)$, the remaining angular kernel is fixed and smooth and B_z^{-1} is bounded on the fixed input height interval. The operator norm bound is therefore uniform in $w \neq z$. Taking adjoints gives the corresponding ordered substitution with a fixed output test when the output is the transverse side. $\square$

Lemma 11.5 (Angle and height locality). *Let f, g be compact smooth functions of Q_z whose angular supports are disjoint. For every $w \in B_S$, $v \in B_T$, and $\xi, \eta \in \mathfrak{D}$,*

$$(11.6) \quad \langle g(Q_z) \eta, A_w(v) f(Q_z) \xi \rangle = 0.$$

For almost every v , the vector $A_w(v) f(Q_z) \xi$ is supported, in the full angle-height representation of Q_z , in $\text{supp } f$. The same support assertion holds for $B_w f(Q_z) \xi$.

Proof. Partition the middle angular variable P_w into finitely many smooth pieces so that each is transverse either to the angular support of f or to that of g . Such a partition exists because those two supports are disjoint and, in rank one, failure of transversality is equality. For $w \neq z$, apply Lemma 11.4 on the transverse side of each piece. After dividing the resulting scalar matrix element by $K_S(w, z)$, all remaining factors have fixed smooth separated expansions and bounded Q_z heights. As $w \rightarrow z$, strong continuity of the angular PVM makes that divided matrix element tend to zero: at $w = z$ the angular slots are in the same calculus, and the disjoint outer supports annihilate.

On the other hand, the original scalar matrix element is linear in $n_S(w)$ by Lemmas 11.2 and 11.3. Use a unipotent chart with z at its origin. After multiplication by a fixed smooth nonzero denominator, this is a polynomial of weighted degree at most four. Along the graded dilations of any fixed chart point, the preceding limit says that it is $o(t^4)$. Every homogeneous part of degrees zero through four must therefore vanish. The polynomial is zero throughout the chart, and continuity extends this to the whole boundary. This proves Equation (11.6), including $w = z$.

For fixed v , $A_w(v)$ strongly commutes with $A_z(v)$. The latter is the multiplier

$$m_v(u, s) = K_T(v, u) e^{\vartheta s}$$

in the Q_z representation. At every angle $u \neq v$, two different heights can be separated by disjoint intervals of this multiplier, after shrinking their angular neighborhoods. Combine this separation with Equation (11.6) and a finite smooth partition on compact sets. Strong commutation with the spectral cuts of m_v eliminates the blocks with separated heights, while Equation (11.6) eliminates the blocks with separated angles. The exceptional output projection $P_z(\{v\})$ vanishes for almost every v . Exhaustion of the complement of $\text{supp } f$

proves full angle-height locality. Finally integrate the operators $A_w(v)$ on $\mathfrak{D}$ and use $B_w = (a_1^T)^{-1} \int A_w(v) dv$. $\square$

The preceding locality statement still allows differential operators. The next two arguments eliminate them and also explain why no choice of a self-adjoint extension is hidden in the passage to strong commutation.

Lemma 11.6 (Finite jets from graph locality). *Fix $\xi \in \mathfrak{D}$ and a relatively compact coordinate chart in $B_T \times \mathbb{R}$. In the spectral representation of Q_z , write its variable as q . For $L = B_w$ and for a countable dense set of the operators $L = A_w(v)$ for which Lemma 11.5 holds, there is a common finite integer m such that, on a smaller chart,*

$$(11.7) \quad Lf(Q_z)\xi = \sum_{|\alpha| \leq m} \eta_{L,\alpha}(q) \partial^\alpha f(q), \quad f \in C_c^\infty,$$

where the coefficient vectors are measurable and locally square integrable in the spectral measure. The same assertion holds for any finite family of operators with these graph bounds and locality.

Proof. Let $\mathcal{Q} = B_T \times \mathbb{R}$, and realize the spectral representation as

$$\mathcal{H} = \int_{\mathcal{Q}}^{\oplus} \mathcal{H}_q d\nu(q), \quad (f(Q_z)\eta)(q) = f(q)\eta(q).$$

Here ν is a finite control measure obtained from a faithful normal scalarization; it is not assumed to be related to smooth volume on $\mathcal{Q}$. Fix a compact coordinate box and a smooth cutoff equal to one on a smaller box. Lemma 11.3, followed by the bound $\|L\eta\| \leq C_L \|\eta\|_{\mathfrak{D}}$, gives

$$\|Lf(Q_z)\xi\| \leq C_{L,\xi} \|f\|_{C^r}$$

for tests in the box. Take a Sobolev order $a > r + \dim(\mathcal{Q})/2$ with respect to ordinary coordinate volume. Multiplication by the fixed cutoff makes this a bounded map from H^a on a containing coordinate torus to $\mathcal{H}$. Increase to an integer b with $b - a > \dim(\mathcal{Q})/2$. The inclusion $H^b \hookrightarrow H^a$ is Hilbert–Schmidt: on the Fourier basis its squared singular values have convergent sum $\sum_{k \in \mathbb{Z}^{\dim \mathcal{Q}}} (1 + |k|^2)^{-(b-a)}$. Consequently the map $T_L : f \mapsto Lf(Q_z)\xi$, including the cutoff, is Hilbert–Schmidt from H^b to $\mathcal{H}$.

If (e_j) is an orthonormal basis of H^b , choose measurable versions $t_j(q) = (T_L e_j)(q)$. Tonelli gives

$$\int_{\mathcal{Q}} \sum_j \|t_j(q)\|^2 d\nu(q) = \sum_j \|T_L e_j\|^2 < \infty.$$

Outside one null set, $f \mapsto \sum_j \langle e_j, f \rangle_{H^b} t_j(q)$ is therefore a continuous $\mathcal{H}_q$ -valued distribution, and it represents $(T_L f)(q)$ for all smooth tests in a countable dense family. Continuity extends the equality to all tests. Choose that family to include dense tests in every box of a countable coordinate basis. Locality makes the distribution vanish on each such box not containing q , so its support is contained in $\{q\}$. A C^b seminorm bounds its H^b norm on the fixed box. Thus its distribution order is at most the same finite integer b , independently of q .

This is the distributional finite-jet mechanism underlying Peetre’s locality principle [52, 53]; compare the scalar point-support theorem [33, Theorems 2.3.3–2.3.4]. The preceding construction uses the actual control measure and does not require it to be absolutely continuous. For completeness, a point-supported vector distribution of order at most b annihilates any smooth function whose derivatives through order b vanish at the point. Multiply the Taylor remainder by a cutoff of radius ϵ about that point. All its derivatives through order b tend to zero uniformly as $\epsilon \downarrow 0$, whereas support at the point leaves its distribution value unchanged. This proves the finite-jet formula (11.7) with $m = b$. On a smaller box, cutoff monomials $x^\alpha/\alpha!$, $|\alpha| \leq b$, recover the coefficients by a triangular system with bounded polynomial entries in q . Each coefficient is therefore a finite combination of the vectors $Lf_\alpha(Q_z)\xi$ with

bounded measurable coefficients. In particular it is measurable and locally square integrable for ν .

A common finite order for the whole family is essential here. Choose finitely many source labels whose lifts span E_S , and choose a target lift basis from the common full-measure set on which locality holds for those source labels. All $A_w(v)$ on $\mathfrak{D}$ are linear combinations of this finite array, because their dependence on $n_S(w)$ and $n_T(v)$ is bilinear. All B_w are combinations of the same array by their ordinary- v averages. Apply the preceding construction to this finite array and take the largest Sobolev and jet order. Its formulas and locality then extend by finite linearity to every operator under consideration. This takes no supremum over an uncontrolled countable family of differential orders. $\square$

Lemma 11.7 (Vanishing of positive differential order). *Every expansion in Equation (11.7) has order zero. Consequently the Q_z and Q_w calculi strongly commute for all source labels z, w .*

Proof. Use a common upper order m and suppose $m > 0$. The highest homogeneous symbol is the vector polynomial

$$\sigma_L(q, \zeta) = \sum_{|\alpha|=m} \eta_{L,\alpha}(q) \zeta^\alpha.$$

It is obtained in Hilbert norm by applying L to $e^{it\psi}\chi$, removing the phase, and dividing by $(it)^m$, where ψ is real and smooth and χ is a compactly supported smooth cutoff. Terms with fewer derivatives vanish in this limit.

For $w \neq z$, take the angular support of χ in a small ball about u_0 . By Lemma 11.4, the P_w component outside a slightly larger ball has bounded norm after applying B_w to $e^{it\psi}\chi(Q_z)\xi$, independently of t . On the remaining component, $A_w(v) - K_T(v, u_0)B_w$ costs at most the variation of $K_T(v, \cdot)$ on that enlarged ball times the B_w output norm. After dividing by t^m and taking the limit, this gives

$$\|(\sigma_{A_w(v)}(q, d\psi) - K_T(v, u_0)\sigma_{B_w}(q, d\psi))\chi\|_2 \leq \text{var}(K_T)\|\sigma_{B_w}(q, d\psi)\chi\|_2.$$

Use arbitrary inner cutoffs and rational linear phases in the chart. Regularity of the spectral control measure turns the norm inequality for every such cutoff into the corresponding almost-everywhere pointwise inequality on the ball. Take a countable family of shrinking balls; their variations tend to zero and their centers tend to the tested angle. This gives the following polynomial identity without requiring a Lebesgue differentiation theorem for the spectral measure:

$$(11.8) \quad \sigma_{A_w(v)}(q, \zeta) = K_T(v, u(q))\sigma_{B_w}(q, \zeta)$$

almost everywhere, as a polynomial identity. For $w = z$ the operators already are multiplication operators, so their positive-order symbols are zero.

Strong commutation of $A_w(v)$ with $A_z(v)$ is now used a second time. On the support of a compact test, the smooth multiplier $m_v(q) = K_T(v, u(q))e^{\vartheta s(q)}$ is bounded. Locality places the output on this same compact support. Thus both sides of the commutation identity are defined, and substitution of Equation (11.7) into that identity is legitimate. The coefficients of derivatives of order $m - 1$ in the commutator give

$$D_{dm_v(q)}\sigma_{A_w(v)}(q, \zeta) = 0,$$

where $D_{dm_v(q)}$ denotes differentiation in the covariable in the direction of the covector $dm_v(q)$. All lower-order terms cancel or have smaller derivative order. For $v \neq u(q)$, the factor $K_T(v, u(q))$ in Equation (11.8) is positive. Hence

$$D_{dm_v(q)}\sigma_{B_w}(q, \zeta) = 0.$$

The functions $m_v(q)$ are the pairings of the embedded cone point $e^{\vartheta s}n_T(u)$ with the spanning vectors $n_T(v)$. Their differentials, with $v \neq u$, span the cotangent space of the angle-height manifold. Therefore σ_{B_w} is independent of the covariable. Homogeneity of positive degree

makes it zero; Equation (11.8) does the same for the other symbols. Decrease m and repeat until only order zero remains.

The coefficient formula concerns the map $f \mapsto Lf(Q_z)\xi$ for a fixed $\xi \in \mathfrak{D}$. Once its order is zero, it gives

$$L(F\chi)(Q_z)\xi = F(Q_z)L\chi(Q_z)\xi$$

for compact smooth tests supported in one coordinate chart. A finite partition of unity gives the same equality for arbitrary compact smooth F, χ . Take χ from the graph cutoffs of Lemma 11.3. Then $\chi(Q_z)\xi \rightarrow \xi$ in the graph norm of L , so closedness gives

$$LF(Q_z)\xi = F(Q_z)L\xi, \quad \xi \in \mathfrak{D}.$$

Since $\mathfrak{D}$ is a core for the self-adjoint operator L , another graph approximation extends this equality to every $\xi \in \text{dom } L$. Applying it to $(L - i)^{-1}\eta$ shows that $F(Q_z)$ commutes with the resolvent. Spectral regularity and bounded strong approximation by compact smooth functions now give commutation with every projection of Q_z . This proves strong commutation for $L = B_w, A_w(v)$. A finite basis of the evaluations $A_w(v)$ recovers the cone vector $B_w n_T(P_w)$, hence both P_w and $\log b_w$. The complete Q_w calculus therefore commutes with Q_z . $\square$

11.4. The resulting cone-map law.

Lemma 11.8 (Smooth angular bijections). *The common spectral law of the Q_z is a law of evaluations*

$$z \mapsto (\phi(z), \log h(z)),$$

where h is positive and smooth, $\phi : B_S \rightarrow B_T$ is a smooth diffeomorphism, and

$$(11.9) \quad Un_S(z) = c_i^{-\vartheta/2} h(z)^\vartheta n_T(\phi(z))$$

for a linear pairing isometry $U : E_S \rightarrow E_T$ carrying the positive cone onto the positive cone. The space $\mathcal{X}$ of these linear isometries is a smooth Lie coset, and the spectral law may be regarded as a PVM on $\mathcal{X}$.

Proof. Strong spectral continuity in z , supplied by compact source transport, gives versions measurable jointly in the spectral variable and the ordinary source label. For example, use a faithful normal scalarization of the common abelian algebra, approximate in measure on a countable dense label set, and take a fast almost-everywhere convergent subsequence of the resulting simple fields. For generic distinct z, w , their images cannot agree on a nonzero spectral projection e . Otherwise a nonzero vector $\eta \in e\mathcal{H}$ satisfies $P_z(f)\eta = P_w(f)\eta$ for every continuous f . On finite smooth sums $b(u) = \sum_j f_j(u)\xi_j$, the functional

$$b \mapsto \left\langle \eta, \sum_j P_z(f_j)\xi_j \right\rangle = \left\langle \eta, \sum_j P_w(f_j)\xi_j \right\rangle$$

is nonzero, by taking b constant equal to η , and is bounded by both sharp form norms in Equation (5.6). This contradicts the disjointness proved in Section 5. Exhaust transverse angular rectangles in Equation (10.6) and take a common conull set for rational frequencies. The resulting equality of scalar unitary characters gives, almost surely and for almost every ordinary pair,

$$(11.10) \quad h(z)^\vartheta h(w)^\vartheta K_T(\phi(z), \phi(w)) = c_i^\vartheta K_S(z, w).$$

Fubini permits two fixed anchors z_0, z_1 with distinct images almost surely, and a fixed generic countable dense set of source labels on which all pair identities and all identities with the anchors hold almost surely. Such a dense set can be selected by choosing one generic point in each member of a countable boundary basis. Fix a realization in this common conull set and put $r(z) = h(z)^\vartheta$. Adding the two anchor identities gives

$$r(z)(r(z_0)K_T(\phi(z), \phi(z_0)) + r(z_1)K_T(\phi(z), \phi(z_1))) = c_i^\vartheta (K_S(z, z_0) + K_S(z, z_1)).$$

The sum of the target kernels against their distinct images has a positive minimum on the compact target boundary. The two identities and boundedness of the source kernels give an upper bound for $h(z)^\vartheta$. Conversely, the sum of the source kernels against the two distinct anchors has a positive minimum, and boundedness of the target kernels gives a positive lower bound for $h(z)^\vartheta$.

On the chosen countable dense set, Lemma 11.1 and the two bounds on h give

$$c d_S(z, w) \leq d_T(\phi(z), \phi(w)) \leq C d_S(z, w),$$

where the constants may depend on this realization. Thus ϕ and its inverse on its image are uniformly continuous. Metric completion gives a continuous injective map on the whole boundary. The displayed sum of anchor pairings has a strictly positive denominator everywhere and recovers a continuous positive extension of the weight. Pair identities with almost every further source point show that these extensions give versions of the original fields.

A continuous injection between spheres implies that the source dimension is at most the target dimension. This conclusion holds at every isolated type. The type correspondence is a permutation of a finite list, so the inequalities around each permutation cycle are equalities. Brouwer's invariance of domain (see [63, Theorem 2 and Corollary 3] for the openness and dimension statements) now makes $\phi(B_S)$ open in B_T ; it is also compact and nonempty, hence all of B_T .

Define U on the spanning set $n_S(B_S)$ by Equation (11.9). It is well-defined linearly. Indeed a linear relation among the source lifts pairs to zero against every source lift. Equation (11.10) makes its proposed image pair to zero against every target lift, because ϕ is onto. Nondegeneracy and spanning force that image relation to be zero. The same equation proves that U preserves the pairing, and its image spans E_T , so it is an isomorphism. It carries the positive radial cone onto the positive radial cone. The smooth embedded cone descriptions now imply that ϕ , h , and the angular inverse are smooth.

Choose one such U_0 . Then $\mathcal{X}$ is the coset of the subgroup of the target orthogonal group preserving its positive cone, multiplied by U_0 . This subgroup is closed: preservation of the cone with its vertex follows by limits, and the same condition on inverse matrices gives equality rather than mere inclusion. The Closed Subgroup Theorem makes it a Lie group and $\mathcal{X}$ a smooth embedded coset. Choose a lift basis from the fixed generic dense source set. Its finitely many evaluation vectors determine U by a fixed linear inversion. Thus the passage from the original measurable map version to U is Borel, with no measurable choice of a continuously varying boundary map required. Conversely evaluation and angular inversion on this cone are continuous. The law on $\mathcal{X}$ recovers every fixed label by continuity in measure. $\square$

11.5. Two normal copies sharing the trunk. Use superscripts 1, 2 for the two measured copies. The same-label statement following Equation (10.7) in Section 10 gives strong commutation of Q_z^1 with Q_z^2 . It includes the mixed angle–height relations and the height groups; for the latter, interchange of the two normal-copy slots makes the possible scalar bicharacter vanish. Each copy separately has the $\mathcal{X}$ representation of Lemma 11.8.

To compare these two representations, we need a domain on which compact smooth functions of the first-copy matrix preserve the second-copy heights. We obtain it by averaging the available same-label commutation. The inverse-height estimate below plays the role of the inverse-kernel estimate used to construct the first graph domain. Its integrability again comes from fourth-order boundary geometry and $Q > 8$.

In the second calculus put

$$d_z = (b_z^2)^\vartheta, \quad H_2 = \left(\int_{B_S} d_z^2 dz \right)^{1/2}, \quad \mathfrak{B} = \text{dom } H_2.$$

The superscript on b_z^2 denotes the copy, not a square.

Lemma 11.9 (An inverse-height estimate and first-copy graph cuts). *In the second classical calculus,*

$$(11.11) \quad d_z \leq CH_2, \quad d_z + d_w \geq c_2 d(z, w)^4 H_2, \quad \int_{B_S} (H_2/d_z)^2 dz \leq C.$$

The domain $\mathfrak{B}$ is dense and is a core for all d_z and $A_z^2(v)$. Compact smooth functions of the first-copy $\mathcal{X}$ preserve $\mathfrak{B}$, with a finite differentiability bound on each compact set. There are compact smooth first-copy cutoffs converging to the identity in the graph norm of $\mathfrak{B}$.

Proof. The inverse-height estimate and the second-copy cores. The first two inequalities follow from Equation (11.2), using commutation in the second copy and continuity in the source label. In a realization of this calculus write $r_z = d_z/H_2$. If $r_z < \epsilon$ and $r_w < \epsilon$, the second inequality gives $d(z, w) \leq C\epsilon^{1/4}$. The sublevel set thus has ordinary measure at most $C\epsilon^{Q/4}$. Dyadic decomposition gives

$$\int r_z^{-2} dz \leq 1 + 4 \sum_{j \geq 0} 2^{2j} |\{z : r_z \leq 2^{-j}\}| \leq 1 + C \sum_{j \geq 0} 2^{(2-Q/4)j} < \infty.$$

The constants are uniform in the realization, proving the third estimate. Each second-copy cone map has bounded positive weights on the compact source boundary. Thus H_2 is finite and positive in every realization, and its bounded spectral cuts prove density of $\mathfrak{B}$. These cuts commute with each d_z , $A_z^2(v)$ and converge in their individual graph norms. Consequently $\mathfrak{B}$ is a core for each of them.

Unary averages and smooth matrix functions. For a bounded measurable unary field F , put

$$S_F = \int_{B_S} F(z, Q_z^1) dz.$$

For $\xi \in \mathfrak{B}$ and initially $\eta \in \text{dom } H_2$, same-label commutation gives

$$\langle S_F \xi, H_2 \eta \rangle = \int_{B_S} \langle F(z, Q_z^1) d_z \xi, (H_2/d_z) \eta \rangle dz.$$

Spectral truncation justifies this equality. The inverse-height bound and Cauchy–Schwarz bound its right-hand side by

$$C \left(\int_{B_S} \|F(z, Q_z^1) d_z \xi\|^2 dz \right)^{1/2} \|\eta\|.$$

The adjoint-domain criterion therefore gives $S_F \xi \in \mathfrak{B}$ and

$$(11.12) \quad \|H_2 S_F \xi\| \leq C \left(\int_{B_S} \|F(z, Q_z^1) d_z \xi\|^2 dz \right)^{1/2} \leq C \|F\|_\infty \|H_2 \xi\|.$$

Only the same-label commutation has been used.

Choose source-label neighborhoods $O_1, \dots, O_d$ such that $n_S(z_1), \dots, n_S(z_d)$ form a basis whenever $z_j \in O_j$, with uniformly bounded inverse basis matrices. Here $d = \dim E_S$. Fix smooth probability densities ρ_j supported in these neighborhoods. In fixed bases of E_S, E_T , the evaluation vectors recover the matrix by

$$U = [Un_S(z_1), \dots, Un_S(z_d)][n_S(z_1), \dots, n_S(z_d)]^{-1}.$$

Extend a compact smooth function on $\mathcal{X}$ to the ambient matrix space and compose it with this formula. The result is smooth in the separate pairs $(z_j, Un_S(z_j))$ on fixed buffered boxes. It has uniform compact support in the evaluation variables because the basis matrices and their inverses are uniformly bounded. A summable separated expansion, followed by independent integration against the ρ_j , expresses its first-copy calculus as a sum of products of the unary averages above. Equation (11.12) proves the required finite smooth-seminorm bound on $\mathfrak{B}$.

Compact cutoffs with a uniform graph bound. Choose $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near $[0, 1]$. For $l \geq 1$ define first-copy multipliers

$$t_{j,l}(U) = \int_{B_S} \rho_j(z) \chi(\|Un_S(z)\|^2/l^2) dz, \quad \chi_l(U) = \prod_{j=1}^d t_{j,l}(U).$$

Let $T_{j,l}$ be the first-copy spectral multiplier with scalar function $t_{j,l}$. Each integrand is a bounded function of Q_z^1 , since $Un_S(z) = c_i^{-\vartheta/2} (b_z^1)^\vartheta n_T(P_z^1)$. The graph bounds for $T_{j,l}$ depend on the fixed ρ_j and χ , and are independent of l . Applying the first inequality in Equation (11.12) to the field

$$\rho_j(z) \left[\chi(\|Un_S(z)\|^2/l^2) - 1 \right]$$

shows, by dominated convergence on the vectors $d_z \xi$, that $T_{j,l} \xi \rightarrow \xi$ in the graph norm of H_2 . The finite product has the same convergence, by telescoping and the uniform graph bounds.

Finally, χ_l is a compact smooth function on $\mathcal{X}$. Indeed, if its product integral is nonzero, some tuple of labels in the fixed neighborhoods has every evaluation norm bounded by a constant times l . The recovery formula then bounds $\|U\|$ by a constant times l . Bounded subsets of the pairing-isometry space have bounded inverses, since the inverse is obtained from the transpose using the two fixed nondegenerate pairing matrices. The cone-preserving condition is closed, so these closed bounded matrix windows are compact in $\mathcal{X}$. This proves the asserted compact graph cutoffs. $\square$

Lemma 11.10 (Cross-copy strong commutation). *The two normal $\mathcal{X}$ measurements strongly commute.*

Proof. We first establish locality in the first-copy $\mathcal{X}$ representation for $L = d_z, A_z^2(v)$ on $\mathfrak{B}$. If two matrices U, U' are distinct, some source evaluation separates them. After shrinking matrix neighborhoods and a neighborhood of that label, their entire evaluation images are disjoint for every label in this open set. Between smooth tests supported in these two matrix neighborhoods, the blocks of d_z and $A_z^2(v)$ vanish for those labels by same-label commutation. The scalar blocks are linear in $n_S(z)$ on $\mathfrak{B}$. A linear functional vanishing on the lifts of a nonempty open boundary patch vanishes on their whole span: in a unipotent chart it is a polynomial divided by a nonzero denominator, and polynomial uniqueness applies. Hence these blocks vanish for every z . Covering any pair of disjoint compact matrix supports by finitely many such separating neighborhoods proves locality.

Apply the proof of Lemma 11.6 on a compact chart of $\mathcal{X}$, now using the graph estimates of Lemma 11.9. Finite-dimensional linearity again gives a common finite upper order for a spanning family and thus for a countable dense list of labels and evaluations. We fix these countable lists, their rational coordinate phases, and a countable chart exhaustion at the outset; all pointwise polynomial assertions can then be made on one conull set. Suppose the current common highest order is positive. Same-label commutation says that the principal symbol of d_z is invariant under translations of the covariable by every covector in the span of the differentials of the evaluation $U \mapsto Un_S(z)$.

For $z \neq w$, both the z and the w evaluation spans contain the covector

$$(11.13) \quad \ell_{z,w}(\dot{U}) = [\dot{U}n_S(z), Un_S(w)] = -[Un_S(z), \dot{U}n_S(w)].$$

The equality is the derivative of the pairing-isometry identity. Thus the symbols of both d_z and d_w are invariant along this covariable line. Equation (11.11) also gives, on $\mathfrak{B}$,

$$\|L\psi\| \leq C_{z,w,L} (\|d_z\psi\| + \|d_w\psi\|)$$

for every operator L in the spanning family. Apply this estimate to localized oscillatory tests and take the leading-order limit. Arbitrary compact inner cutoffs and a countable determining set of linear phases imply the corresponding pointwise symbol estimate. In detail, squaring the norm bound gives an integral bound by twice the sum of the two squared symbol norms; arbitrary cutoffs give this bound pointwise for rational covariables, and polynomial continuity

extends it to every covariable. It bounds the polynomial $\sigma_L(q, \zeta + t\ell_{z,w})$ for all $t \in \mathbb{R}$ by a constant independent of t . A vector-valued polynomial bounded on the real line is constant on that line. Therefore every principal symbol is invariant along every $\ell_{z,w}$.

These covectors span the full cotangent space of $\mathcal{X}$. Indeed, if a tangent vector $\dot{U}$ is annihilated by all of them, then $[\dot{U}n_S(z), Un_S(w)] = 0$ for every distinct pair; by continuity the same holds for every pair. Nondegeneracy and the spanning of the two lift sets give $\dot{U} = 0$. Hence every principal symbol is invariant under all covariable translations. Positive homogeneity makes it zero. Descend through the finite orders to obtain order zero.

The compact graph cutoffs and the cores in Lemma 11.9 now give strong commutation with every first-copy spectral projection, exactly as in the last paragraph of Lemma 11.7. Second-copy basis evaluations recover its complete matrix coordinate, so this proves commutation of the two entire $\mathcal{X}$ PVMs. $\square$

Proposition 11.11 (Isolated-type commutation). *For each isolated type, all angle-height measurements Q_z strongly commute as z ranges over its source boundary. Their common law is the evaluation law of the smooth cone isometries in Lemma 11.8; in particular its angular maps are nonsingular smooth bijections. The two normal copies sharing the trunk slot commute with each other at every pair of labels. All these measurements retain the source and measured-group covariances, base decomposability, and complementary-group commutation of the original angle and height measurements.*

Proof. Lemmas 11.7, 11.8, and 11.10 give the first three assertions. Smooth diffeomorphisms of the compact boundaries preserve the ordinary measure class in both directions. Every constructed coordinate is obtained by the unique joint spectral calculus of the original measurements; its finite basis evaluation recovers that calculus. Thus their covariance and commutation identities extend to the joint law, first on a countable determining algebra and then by normality. No change of representation or enlargement of the original commutant is made in this passage. $\square$

12. HIGHER-COMPONENT PENCILS AND GLUING

We retain the source and target chamber spaces B_S, B_T , the type permutation p , and the sharp angular measurements constructed in Section 5. For a panel link, P_z and Q_z denote the start and end missing-vertex measurements in their separate compact frames, and g is the target projectivity from the end pencil to the start pencil. The two inputs of J in Equation (5.11) remain independent tensor factors. A cut on a vertex coordinate of a chamber measurement means its pullback under the corresponding flag projection.

The isolated positions, including commutation between their two normal copies, were treated in Section 11. At a finite place, Section 6 identifies each one-copy panel law with evaluations of homogeneous analytic homeomorphisms. We first establish the corresponding conclusion for the non-isolated archimedean positions. The Taylor argument below is used only at these positions. Afterwards the arguments joining the two copies and joining different panels apply at all places.

12.1. Real Taylor density for the angular pairing.

Lemma 12.1 (Density of pairing powers). *Let $\mathcal{V}$ be an archimedean target vertex variety and $\mathcal{V}^*$ its opposite-type dual variety. Use the nilradical determinant-line representation and normalized absolute pairing k_T of Section 10. Fix $v_0 \in \mathcal{V}^*$ and a compact set K_* of vertices transverse to v_0 . If O is any sufficiently small open neighborhood of v_0 transverse to K_* , there is a countable set $\mathcal{E} \subset \mathbb{R}$ such that*

$$\overline{\text{span}}^{\|\cdot\|_\infty} \{u \mapsto k_T(v, u)^{i\theta} : v \in O\} = C(K_*) \quad (\theta \notin \mathcal{E}).$$

The span is complex linear.

Proof. Use real coordinates x on the opposite unipotent cell determined by v_0 , and coordinates y on a dual unipotent cell with center $y = 0$ at v_0 . Normalize the algebraic highest-line lifts against the two fixed opposite centers. The pairing of these lifts is a real polynomial $F(y, x)$ with $F(0, x) = 1$. The homogeneous norms used to define k_T therefore give

$$(12.1) \quad k_T(v(y), u(x)) = a(y)b(x)|F(y, x)|, \quad a(y) > 0, \quad b(x) > 0.$$

After shrinking O , the polynomial F is positive on a neighborhood of $\{0\} \times K_*$. Thus its complex powers there use the branch defined by the real logarithm.

A central parabolic cocharacter gives positive integral weights to the nilpotent coordinates. Multiplying the cocharacter if necessary makes all weights integral. Write δ_t for the corresponding graded dilation. The cocharacter acts oppositely on the two cells, so

$$F(\delta_t y, \delta_{t^{-1}} x) = F(y, x).$$

For a multi-index α put

$$c_\alpha(s, x) = \frac{1}{\alpha!} \partial_y^\alpha F(y, x)^s \Big|_{y=0}, \quad s \in \mathbb{C}.$$

If $|\alpha|_{\text{gr}} = d$, then $c_\alpha(s, x)$ is a weighted homogeneous polynomial of degree d in x . Its coefficients are polynomial in s : in the finite Taylor jet at $y = 0$, the expansion of $(1 + (F - 1))^s$ uses only finitely many falling factorials in s . In particular, for each d only finitely many such coefficients occur.

We prove that these coefficients span all polynomials of degree d for a generic s . Complexify the real determinant-line cyclic representation. The wedge of the complexified parabolic nilradical is a highest vector, of weight the sum of its positive roots. Complete reducibility makes the module generated by that vector one irreducible highest-weight module; it is this cyclic module that we complexify. These are the usual highest-weight and parabolic constructions [37, 7]. As in Section 4, its highest-line orbit is the complex parabolic variety $X = G_{\mathbb{C}}/P_{\mathbb{C}}$, and the highest weight is strictly parabolic: its zero simple-coroot pairings are exactly those indexed by the Levi of $P_{\mathbb{C}}$. This assertion also covers a complex group regarded as real; its complexification and parabolic can have several factors.

Let h be the highest coordinate paired with the dual center. The principal open subset $\{h \neq 0\}$ of X is precisely the opposite unipotent cell. Indeed, on an opposite Bruhat cell indexed by a Weyl element w , the highest coordinate can be nonzero only when w fixes the highest weight. Strict parabolicity makes this equivalent to $w \in W_{P_{\mathbb{C}}}$, which is the open-cell index. Conversely the normalized unipotent lift has $h = 1$ throughout that cell. The cell is algebraically isomorphic to its unipotent group, hence to affine space in the coordinates x .

The coordinate ring on this principal open is the degree-zero part of the homogeneous coordinate ring localized at h . Consequently every polynomial in x is the restriction of q/h^n for some homogeneous coordinate polynomial q of degree n . Given a basis of the finite-dimensional space $\mathcal{R}_d$ of weighted degree d polynomials, choose one common n for all these expressions by multiplying numerators and denominators by powers of h . The same expressions are then available at every larger n .

For integral n , the line tensors of order n span the irreducible Cartan component of highest weight $n\lambda$. The dual highest-line tensors span its dual. They still span when their parameters are restricted to an arbitrary real open dual cell: a linear functional vanishing there is a polynomial vanishing on a real open subset of the real coordinate space, and thus on its complexification. After dehomogenization, it follows that the functions $F(y, \cdot)^n$, with y in that real open cell, span all restrictions of homogeneous coordinate polynomials of degree n . Taking the weighted degree d part gives

$$\mathcal{R}_d = \text{span}\{c_\alpha(n, \cdot) : |\alpha|_{\text{gr}} = d\} \quad \text{for all sufficiently large integers } n.$$

Here the equality follows because matching graded degrees makes the degree d part of $F(y, x)^n$ exactly $\sum_{|\alpha|_{\text{gr}}=d} y^\alpha c_\alpha(n, x)$.

In a fixed monomial basis of $\mathcal{R}_d$, choose a maximal minor of the coefficient matrix of the $c_\alpha(s, \cdot)$ which is nonzero at one such integer n . It is a nonzero polynomial in s , so it vanishes at only finitely many complex parameters. The union of these finite exceptional sets over all d is countable. For $s = i\theta$ outside this union, every polynomial in x is a linear combination of Taylor coefficients $c_\alpha(s, \cdot)$.

Taylor differentiation at $y = 0$ is the uniform limit on K_* of ordinary real difference quotients, because F stays positive on a neighborhood of $\{0\} \times K_*$. Each coefficient therefore belongs to the uniformly closed span of $F(y, \cdot)^s$ for arbitrarily small real y . Real-coordinate polynomials, with complex coefficients, contain constants, separate points of K_* , and are closed under conjugation. The Stone–Weierstrass Theorem makes their uniform closure $C(K_*)$. Finally $a(y)^s$ in Equation (12.1) is a nonzero scalar for each y , and multiplication by the fixed continuous nonvanishing function $b(x)^s$ is an automorphism of $C(K_*)$. These norm factors do not change density. Intersecting the exceptional complex parameters with the imaginary axis gives $\mathcal{E}$. $\square$

12.2. Reduction of a real panel link.

Lemma 12.2 (Separated projectivity blocks). *Consider a one-leg link omitting a single non-isolated archimedean type. Fix compact endpoint frames and two sufficiently small compact output windows U, L about distinct projectivities with the same endpoint panels. Let E_U, E_L be their output spectral projections. There is an ordinary dual-vertex neighborhood O such that, for generic source chamber labels Z ,*

$$(12.2) \quad E_U J(1 \otimes P_Z(O) X P_Z(O)) J^* E_L = 0 \quad \text{for every bounded } X.$$

For each fixed completion z , one can shrink O and find a compact set K_ of possible start values, transverse to all of O , such that*

$$(12.3) \quad E_U J(1 \otimes k_T(v, P_z)^{i\theta} b_{F,i}^{i\theta}) J^* E_L = 0 \quad (v \in O, \theta \in \mathbb{R}).$$

Here F is a tuple in the completed chamber containing the missing vertex. The block in Equation (12.3) is unchanged by restricting the P_z value to K_ , and its complement vanishes also with additional own- P_z multipliers.*

Proof. Two distinct projectivities have different inverse images at some point of the start pencil. Choose an ordinary dual vertex v_0 whose transverse projection onto that pencil is such a point. Shrink a neighborhood O of v_0 and the windows U, L , including their endpoint base windows, so that the two inverse-image sets remain separated with positive slack. Distinct base values already give zero blocks by sharpness and their commutation with the operators under consideration, so only a common base chart needs treatment.

For a generic source chamber Z , project its pertinent dual vertex onto the source start panel and denote the resulting completion by z_Z . On the ordered start cut $P_Z(O)$ the value at P_{z_Z} is the corresponding transverse projection. To see the operator statement, insert a gallery shortened from the longest word. The angular nonopposition rule of Section 5 forces its endpoint to be the unique completion of the panel nonopposite to that ordinary transverse dual vertex. The multiplied gallery identity gives this substitution together with the sharp base cuts.

Apply Equation (5.11) after that substitution. On the left output window U , the independent end value Q_{z_Z} is forced into the corresponding inverse-image cut; on the right window L , the adjoint argument forces it into the disjoint inverse-image cut. For precision, partition the compact base windows and the projected completion windows into sufficiently small patches, insert their sharp base and P_{z_Z} projections next to J on the appropriate side of $P_Z(O)$, and insert the independent Q_{z_Z} projection there. All these moves involve either a completion and its own base or separate tensor factors. The two final end cuts are orthogonal and act in the tensor factor untouched by X . Their product is zero for every X ; summing the finite partitions proves Equation (12.2). Compact exhaustion removes the preliminary frame restrictions.

Now compress Equation (10.12) with the isometry V of Equation (10.10), testing ordinary chamber vectors whose scalar supports are in O . On the source side the integrands are flanked by $P_Z(O)$: this follows directly from the formula $V(f \otimes \xi)(Z) = c_0^{-1/2} D_Z^{-1/2} P_Z(f) \xi$ and the own-coordinate commutations established in Section 10. Initially cut $D_Z^{-1/2}$ by bounded spectral truncations. Equation (12.2) then annihilates every integrand. The truncated images converge in their direct-integral Hilbert norm to the images under V . Pairing two such images with the bounded multiplier in Equation (10.12) passes to the limit by Cauchy–Schwarz. Thus the compressed operators converge ultraweakly, which suffices after multiplication by the bounded operators $E_U J$ and $J^* E_L$. Arbitrary scalar tests and Hilbert vectors supported in O now give Equation (12.3) for almost every ordinary dual vertex $v \in O$.

The multiplier between J, J^* commutes with the sharp starting panel, its own P_z , and the independent Q_z . Hence only common data for these slots can contribute to the block. By Equation (5.11), a possible start value must have a common inverse image under projectivities in U and L . The separated-inverse-image choice excludes a neighborhood of the completion projected from v_0 . Within the start panel this is exactly the locus which can fail to be transverse to v_0 . After shrinking the base and projectivity windows, all possible common start values therefore lie in a compact set K_* transverse to v_0 . Compact partitions in the commuting own slots show that every complementary contribution vanishes, also after inserting an additional own- P_z multiplier. This uses no arbitrary restriction inside K_* .

Shrink O so that $K_* \times O$ is transverse. On this compact set the function $v \mapsto k_T(v, \cdot)^{i\theta}$ is continuous in the uniform norm. The almost-everywhere zero blocks consequently extend to every v in the smaller O , proving the claim. $\square$

Proposition 12.3 (One-copy commutation in a real pencil). *The output projectivity PVM reduces $\text{ran } J$. On the input space its reduced calculus commutes with every P_z and Q_z , and Equation (5.11) becomes the joint spectral identity $P_z = gQ_z$. All P_z in the pencil commute mutually, and all Q_z in the pencil commute mutually.*

Proof. Fix the separated windows of Lemma 12.2. For $\theta \notin \mathcal{E}$, Lemma 12.1 and the compact restriction to K_* replace the pairing power in Equation (12.3) by any continuous function $f(P_z)$. Uniform approximation is legitimate because $b_{F,i}^{i\theta}$ is unitary and J is an isometry. Choose nonexceptional $\theta_n \rightarrow 0$. Strong continuity of this unitary group, with all angular factors bounded, gives

$$E_U J J^* E_L = 0, \quad E_U J (1 \otimes P_z(f)) J^* E_L = 0.$$

The compact restriction in the preceding lemma has removed all other contributions, so these equations hold for the full blocks. The argument with the start and end slots interchanged gives the same assertion for end multipliers.

Every two distinct projectivity values, and every two distinct base values, have separating windows of this kind. Exhaust by countably many compact charts. The vanishing of all these off-diagonal blocks says precisely that $J J^*$ and the displayed compressed operators commute with the output PVM. One can verify this criterion first for disjoint compact spectral sets, covering their product by finitely many separating rectangles, and then use spectral regularity and increasing exhaustion for arbitrary Borel cuts. Thus $\text{ran } J$ reduces the output PVM.

Pull its restriction back by J . All own input angular operators commute with this reduced projectivity calculus. Consequently the graph substitution of Equation (5.11), first on compact windows and then by exhaustion, reads $P_z = gQ_z$ in joint calculus. For any w , the operator $P_w(f)$ commutes with both g and Q_z , the latter because it acts on the other input factor. It therefore commutes with $P_z(f')$. This proves mutual commutation of the start family. Interchanging the two inputs proves the end assertion. $\square$

12.3. The homogeneous form of a real pencil map.

Proposition 12.4 (Projectivity-orbit classification). *On almost every sharp endpoint fibre, the one-copy real pencil measurements are evaluations of random homogeneous diffeomorphisms*

from the source pencil onto the target pencil. The maps and their inverses have jointly measurable versions as the endpoint fibres vary. Their evaluations recover the original PVMs at every fixed source label.

Proof. Fix a good pair of sharp endpoint fibres in Equation (5.11). The disintegration in Section 5, with a genuine continuous source lift when necessary, supplies strongly continuous source Levi representations on these fibres, transporting the labels while fixing the bases. Write $\mathcal{B}_S, \mathcal{B}_T$ for the source and target pencil boundaries, and S, T for their effective panel action groups. The identity action groups have noncompact simple real rank-one Lie algebras $\mathfrak{s}, \mathfrak{t}$. Compact normal factors and split centers are ineffective; finite components of the target projectivity group can be retained.

Choose faithful normal states on the separable abelian angular algebras of the two fibres. Their scalar spectral laws realize the evaluations as random measurable maps $\phi, \psi : \mathcal{B}_S \rightarrow \mathcal{B}_T$, initially modulo ordinary source null sets. Here is a useful measurability construction. Strong source continuity gives, for a countable family of continuous target functions, continuity of their evaluations into scalar convergence in measure. Finite measurable approximations of the source label, chosen with summable errors in measure on compact pieces, give jointly measurable limits. The countably many target relations identify a target point almost everywhere. Thus the map classes are random variables in the Polish space $L^0(\mathcal{B}_S, \mathcal{B}_T)$.

Use the product of the two faithful states. The start and end map variables are then independent. The reduced projectivity calculus of Proposition 12.3 witnesses, by Fubini, the relation $\phi = g\psi$ as map classes for product-almost every pair. The projectivity-orbit relation is Borel. Indeed, for each compact subset $K \subset T$, the set

$$\{(\phi, \psi) : \phi = t\psi \text{ for some } t \in K\}$$

is closed: take a convergent subsequence of the compact parameters and use continuity of postcomposition in measure. A countable compact exhaustion of T gives the claim. Fubini now supplies a representative ϕ_0 such that both independent map laws are concentrated on the one orbit $T\phi_0$.

This representative is not essentially constant. Otherwise all maps in the orbit would be constant, and source continuity would make every P_z the same PVM on the fibre. Choose a nonconstant polynomial coordinate test f on the ordinary incident-line embedding, a nonzero vector ξ , and put $\eta = P_{z_0}(f)\xi$. We would have $P_z(f)\xi - \eta = 0$ for all z . Fibre polynomial fullness from Section 5, after positive homogeneous normalization, would give $f(u)\xi - \eta = 0$ for every ordinary target residue value u . Two values of f contradict $\xi \neq 0$. Equivalently, a constant measured line cannot contain the tensor product of the fibre with the full ordinary incident-line span, which has dimension greater than one.

For each $s \in S$, source covariance is implemented unitarily on the fibre. It preserves the faithful spectral measure class, whether or not it preserves the chosen state. Thus the law after precomposition by s is still concentrated on $T\phi_0$. Precomposition commutes with the target action; intersecting the two full-measure orbit sets shows that $\phi_0 \circ s \in T\phi_0$. Consider

$$(12.4) \quad L = \{(s, t) \in S \times T : t\phi_0 = \phi_0 \circ s \text{ modulo ordinary null sets}\}.$$

It is a closed subgroup projecting onto S . For closedness, target composition is continuous by compactness of $\mathcal{B}_T$, and source composition is continuous in measure by approximation with continuous functions and locally uniform bounds on the smooth change-of-variables densities.

The Closed Subgroup Theorem makes L a Lie group. Its projection onto S is onto also on Lie algebras: otherwise the Constant Rank Theorem on countably many Lie charts would make its image a Haar-null subset of S . The radical of $\mathfrak{l}$ has zero image in the simple algebra $\mathfrak{s}$. A Levi decomposition [44, Appendix B, Theorem B.2] therefore has a simple summand mapping isomorphically onto $\mathfrak{s}$; distinct simple summands have commuting ideal images, so at most one can have a nonzero image. In $\mathfrak{s} \oplus \mathfrak{t}$, this summand is the graph

$$X \mapsto (X, \alpha(X))$$

of a Lie homomorphism $\alpha : \mathfrak{s} \rightarrow \mathfrak{t}$. If $\alpha = 0$, Equation (12.4) makes ϕ_0 invariant modulo null sets under the connected transitive source group, hence essentially constant. Thus $\alpha \neq 0$, and simplicity gives injectivity. For a panel of type i this proves $\dim \mathfrak{s}_i \leq \dim \mathfrak{s}_{p(i)}$. The archimedean types form a union of cycles of p by Section 6; going around each finite cycle forces equality at every step. Hence α is an isomorphism.

Let $\tilde{S}$ be the simply connected source cover and $a : \tilde{S} \rightarrow T^\circ$ the homomorphism integrating α . It is onto because its differential is onto. Let P_S be the full preimage of the source minimal parabolic, so $\mathcal{B}_S = \tilde{S}/P_S$. Equation (12.4) gives $\phi_0(sx) = a(s)\phi_0(x)$ for every s and almost every x . On $\tilde{S}$, the measurable function

$$r \mapsto a(r)^{-1}\phi_0(rP_S)$$

is therefore invariant modulo Haar-null sets under each left translation. Fubini, or convolution of its bounded scalar coordinate tests with a compactly supported approximate identity, makes it essentially constant, say equal to u_0 . Its right P_S transformation shows that $a(P_S)$ fixes u_0 . The recovered map is consequently

$$rP_S \mapsto a(r)u_0.$$

An isomorphism of real semisimple Lie algebras carries minimal parabolic Lie algebras to minimal parabolic Lie algebras [7]. Thus the inclusion of $\alpha(\text{Lie } P_S)$ in the stabilizer algebra of u_0 is equality. Boundary stabilizers are the full normalizers of these parabolic algebras. If $a(r)$ fixes u_0 , then r normalizes $\text{Lie } P_S$, and hence $r \in P_S$. This proves $a^{-1}(\text{Stab}_{T^\circ}(u_0)) = P_S$. The displayed map is a homogeneous diffeomorphism, rather than a map between nontrivial oriented covers. Central kernels and ineffective compact factors lie in the stabilizers; finite target components also act by diffeomorphisms.

Every almost-sure map class now has a unique continuous homeomorphism version. These versions are measurable without a measurable choice of ϕ_0 on different fibres. The space of homeomorphisms between compact metrizable pencils, with uniform convergence of maps and inverses, is Polish. Its injection into $L^0(\mathcal{B}_S, \mathcal{B}_T)$ is Borel and injective, since ordinary source measure has full support. The Lusin–Souslin Theorem [42, Corollary 15.2] gives a Borel inverse on its image. Apply that inverse in countably many compact endpoint charts. Finally, the original evaluations and these continuous evaluations agree for almost every label. Both are continuous in the label in scalar convergence in measure, the former by source transport and the latter by bounded convergence. They therefore agree for every fixed label on all continuous target tests, and hence as PVMs. $\square$

12.4. Commutation of the two normal copies.

Lemma 12.5 (Two-copy pencil commutation). *At every non-isolated position, the two normal copies of the one-copy pencil calculus commute with one another. In particular, adjacent chamber measurements commute, also when each chamber carries both copies.*

Proof. Disintegrate the common two-leg link over its joint sharp endpoint faces. These faces commute with their respective completion observations. Proposition 12.4 and the finite-place classification in Section 6 therefore give, in each copy $j = 1, 2$, a start homeomorphism B_j and an end homeomorphism A_j from the source pencil to the corresponding target pencil. Their continuous versions and inverses persist under normal copying and disintegration.

For each j separately, B_j and A_j act in independent input factors, so their joint spectral calculus defines the target-to-target map

$$C_j = B_j A_j^{-1}.$$

Write E_j for the output projectivity PVM in that coordinate of the two-leg link. For this fixed j , all evaluations of A_j and B_j commute. We may therefore multiply the residue graph identities in Equation (5.11) by further cuts from this same input calculus, imposing finitely many determining-label constraints together. Choose the labels in a countable dense set. Continuity makes their simultaneous graph equations $g_j A_j(z) = B_j(z)$ determine $g_j = B_j A_j^{-1}$.

Separation by compact projectivity windows, followed by spectral exhaustion, consequently gives

$$E_j(H)J = J1_H(C_j)$$

for every Borel projectivity cut H . This identity also puts the law of C_j on projectivities. The two output coordinate PVMs commute in the common link. Their intertwining identities therefore imply commutation of the two input composite calculi. Source transport in the start factor gives the same conclusion for

$$B_1 s A_1^{-1} \quad \text{and} \quad B_2 s A_2^{-1}, \quad s \in S.$$

This argument uses the joint calculus of (B_j, A_j) separately for each j ; commutation of B_1 with B_2 is still to be proved.

Let K_S be a compact subgroup transitive on the source pencil boundary; at a finite place use a special maximal compact. Its normalized Haar measure induces the ordinary pencil probability m_S . Fix disjoint compact sets $\mathcal{U}, \mathcal{L}$ in the start B_1 homeomorphism space, with spectral projections $R_{\mathcal{U}}, R_{\mathcal{L}}$. Fix a source transport s_n and a support point a_0 of the end A_1 law. For each common $k \in K_S$, the two compact systems

$$\{bs_n k a_0^{-1} : b \in \mathcal{U}\}, \quad \{bs_n k a_0^{-1} : b \in \mathcal{L}\}$$

are disjoint, since right composition is injective. Their separation in the topology of maps and inverses is uniform for this common k , by compactness of $\mathcal{U} \times \mathcal{L} \times K_S$. Hence a sufficiently small end neighborhood N_n of a_0 preserves the separation for all k . Its spectral projection is nonzero; choose a unit vector η_n in its range. The neighborhood and vector may depend on n .

Let f be a continuous function on the target pencil and v a target label for an evaluation of the composite map. The commuting-composite assertion gives zero between the two separated $B_1 s_n k A_1^{-1}$ cuts. Compressing the end factor to η_n yields

$$R_{\mathcal{U}}(\text{id} \otimes \langle \cdot, \eta_n, \eta_n \rangle)[f(B_2 s_n k A_2^{-1} v)] R_{\mathcal{L}} = 0.$$

Average over k . In the independent start and end calculus of copy 2, Haar averaging sends every $A_2^{-1} v$ to the same probability m_S , irrespective of its end value or of η_n . Thus

$$(12.5) \quad R_{\mathcal{U}} \left[\int f(B_2 s_n z) dm_S(z) \right] R_{\mathcal{L}} = 0.$$

For any prescribed source point z_0 , choose split transports, conjugated in the root-generated source group, with $(s_n)_* m_S \rightarrow \delta_{z_0}$. The usual rank-one contraction holds away from a proper repelling set of ordinary measure zero, both for smooth and for local-field root coordinates. For every realized homeomorphism B_2 , the integral in Equation (12.5) converges to $f(B_2 z_0)$. Its absolute value is bounded by $\|f\|_{\infty}$, so the convergence is strong in the single-copy spectral calculus. We obtain $R_{\mathcal{U}} f(B_2 z_0) R_{\mathcal{L}} = 0$. Compact exhaustion and the separated-block criterion show that every B_2 evaluation commutes with the whole B_1 calculus. Interchanging start and end proves the corresponding assertion for A_1, A_2 .

The fibre identities can first be imposed on countable dense tests and labels and then extended by continuous source transport with the bases fixed. The genuine source amplification of Section 5, when needed, allows exactly this fibrewise transport and descends all the resulting operator identities. Within a shared panel, the remaining chamber vertices already commute with the completion slots. Together with Section 11 at isolated positions, this proves the final assertion. $\square$

12.5. Reflection substitutions and global commutation.

Lemma 12.6 (Exact simultaneous reflection). *Let C, D be opposite source chambers. Reflect them simultaneously across their apartment panels of types i, i^* , obtaining C', D' . Let $\mathcal{F}_i$ be the corresponding simultaneous target reflection at types $p(i), p(i^*)$. On the open set of opposite target pairs, the ordered angular kernel $P_C P_D$ is transported exactly by $\mathcal{F}_i$ to $P_{C'} P_{D'}$.*

The assertion holds also for the joint two-copy measurements, with opposition required in each copy.

Proof. The type permutation preserves the Coxeter system, and opposition on types is respected. Thus $\mathcal{F}_i$ is the well-defined involution which reflects an opposite pair in its common apartment. On the open Bruhat orbit it is smooth at real places and continuous analytic at finite places. Equivalently, each reflected chamber is the unique completion of the prescribed panel which is not opposite to the other original chamber.

We prove the assertion first for smooth compact tests inside opposition at the real places, and finite clopen tests with slack in compact opposition patches at the finite places. Ordered integrals of the former are defined by summable separated expansions; the latter are finite sums. Insert the two new slots in the order

$$P_C P_{C'} P_{D'} P_D.$$

The first adjacent pair and the last adjacent pair commute by Lemma 12.5 and the isolated conclusion of Section 11. To constrain the C' value, interchange D, D' and insert the intermediate slots of a gallery from C' to D whose length is one less than the longest length in the pertinent component. Every consecutive pair shares its specified sharp panel, so the difference of any coordinate of that panel in the two slots annihilates the ordered kernel with arbitrary separated multipliers. The same is then true for the allowed joint test functions by their convergent expansions.

At a real place these shared-panel equalities define a regular submanifold: introduce the chamber variables successively, each time pulling back a panel equality along a submersion onto the panel base. In local coordinates their differences are independent normal coordinates. The integral form of the Taylor formula in these normal coordinates writes a smooth function vanishing on the submanifold as a sum of those differences times smooth functions. A compact partition of unity proves annihilation by the entire smooth vanishing ideal on the testing region. At a finite place, take finite clopen partitions fine enough in the compact region. Shared-panel equalities then force exact equality of the corresponding partition values. Refining the partitions imposes the same gallery constraint; continuity of the projection rule on opposition gives its exact substitution for every clopen test with slack.

On this gallery locus the C' value shares the designated panel of C and is not opposite to the D value: the shortened gallery, even with stutters, cannot have longest Weyl distance. On original opposition the panel projection rule therefore identifies it with the unique reflected completion. The preceding vanishing-ideal calculation proves this as a multiplied graph substitution, not just as a support assertion. The analogous constraint on D' follows after initially interchanging C, C' . The two individual constraints generate the simultaneous graph on the open region: in smooth graph coordinates the two new variables are determined functions of the old pair. The finite partition statement gives the same conclusion at finite places. It applies separately to both copy coordinates and therefore to their joint tests.

On opposition of the new values, the identical argument, interchanging within the adjacent pairs, gives the inverse graph substitution. For a test supported on a compact subset of original opposition, insert a compact cutoff equal to one on its reflected image. Apply the forward substitution in the four slots and then the inverse substitution to integrate out the old slots. The resulting ordered integral equals the integral against $P_{C'} P_{D'}$ of the transported test. All cutoffs stay inside their respective opposition open sets. At finite places choose the cutoffs clopen. This proves the asserted identity without crossing the singular boundary of opposition. A single isolated rank-one component already has global commutation and needs no shortened gallery argument. $\square$

Proposition 12.7 (Global joint angular commutation). *All chamber measurements, in both normal copies, commute. For opposite source chambers the joint target law is supported on opposition in each copy.*

Proof. Apply Lemma 12.6 successively along a reduced longest word, moving the opposite chamber simultaneously. In the source apartment the final pair is (D, C) . The permuted word is also reduced and longest, so the target pair is likewise interchanged. Thus the ordered kernels $P_C P_D$ and $P_D P_C$, in the same target variables, have equal integrals for all smooth or clopen compact tests in opposition. Approximation by such tests shows that

$$P_C(U)P_D(L) = P_D(L)P_C(U) \quad \text{when} \quad \bar{U} \times \bar{L} \subset \text{Opp}(B_T \times B_T).$$

The same holds for every Borel subcut of these patches, by spectral regularity. We first use this conclusion in one copy.

Let $R_{C,D}$ be the range join of these transverse products. Each product $R = P_C(U)P_D(L)$ is a projection. It reduces both full PVMs: for an arbitrary Borel A , the subrectangle $(A \cap U) \times L$ is still transverse, so $P_C(A)R = RP_C(A)$, and the same argument applies to P_D . The two restrictions to $\text{ran } R$ commute, again by the subrectangle assertion. These properties pass to the range join.

The projection $R_{C,D}$ is target invariant, since opposition is target invariant, and belongs to the finite algebra $\mathcal{N}$. It is fixed by a source split group fixing C, D and escaping in every noncompact factor. The mixing statement of Section 3 applied to $R_{C,D} - \tau(R_{C,D})1$ makes it zero. Thus $R_{C,D}$ is scalar. It is nonzero: if it vanished for one opposite source pair, source covariance and transitivity on the opposite-pair orbit would make it vanish for every such pair. Fix nonempty target patches U, L with compact transverse product. For fixed source C , almost every source D is opposite to C , so $P_C(U)P_D(L) = 0$ for a conull set of D . The conull-label angular fullness of Section 5 says that these $P_D(L)$ ranges span, giving $P_C(U) = 0$. Fullness for U is a contradiction. Therefore $R_{C,D} = 1$. In particular the one-copy pair law, now commuting, gives full mass to opposition.

For clarity, the same scalar-join argument for the joint copies has a separate nonzeroness step. Choose a nonzero transverse product $P_C^1(U)P_D^1(L)$ in copy 1. In copy 2 the preceding paragraph gives a commuting joint law supported on opposition. Exhaust this opposition set by finite disjoint unions of transverse rectangles, using a countable rectangle cover and finite refinements. The associated sums of products $P_C^2(U_a)P_D^2(L_a)$ increase strongly to 1. Insert these sums between the two copy-1 cuts. Their sandwiched sums converge strongly to the chosen nonzero product. Each summand is

$$(P_C^1(U)P_C^2(U_a))(P_D^2(L_a)P_D^1(L)),$$

because same-chamber copies commute. This is exactly a joint transverse product in both coordinates. At least one such product is nonzero. The joint reflection identity and scalar-join argument now show that their join is 1, so both joint chamber PVMs commute globally and their law is supported on joint opposition. Only separated products have been inserted; no discontinuous joint cutoff has been applied to noncommuting slots.

Finally approximate any source pair (C, D) by opposite pairs. Local source sections and strong source transport give strong convergence of each angular operator tested by a continuous target function. Uniform boundedness passes commutation to the limit. Continuous target tests generate the spectral algebras, so the assertion extends to all Borel cuts and to all source chamber pairs. $\square$

12.6. From panel bijections to chamber bijections.

Proposition 12.8 (Both directions of measure-class preservation). *In the classical spectral law of one copy, almost every chamber-map class has a measurable representative $\phi : B_S \rightarrow B_T$ which is a bijection between conull sets and satisfies*

$$\phi_* m_S \sim m_T,$$

where m_S, m_T are the ordinary compact chamber probabilities. The same assertion holds for each normal copy in their joint law.

Proof. Proposition 12.7, a faithful scalar spectral state, and continuity in measure of the chamber evaluations give jointly measurable random maps exactly as in the proof of Proposition 12.4. On almost every panel, for almost every realization, the restriction is a nonsingular homeomorphism with a measurable inverse. At real non-isolated positions this is Proposition 12.4; at finite positions it is Section 6; at isolated positions it is Section 11. The continuous versions, their inverse maps, and their evaluations are Borel in the endpoint charts. The measure statement follows in all cases from the respective smooth or analytic homogeneous identification. Panel incidence is preserved with the permutation p .

Choose a reduced word $(i_1, \dots, i_m)$ for the longest Weyl element of the full product. Sample C_0 with ordinary compact probability, and successively sample C_j in the i_j -panel of C_{j-1} with its compact rotation probability. These transition probabilities preserve the ordinary chamber probability: this is disintegration of the compact-invariant chamber-panel incidence measure. Thus each encountered chamber and panel has its ordinary measure class. Fubini over the spectral law and the finite sequence of choices shows that, for almost every realization and almost every starting chamber, every needed panel map is good for almost every path prefix. Its continuous version agrees with ϕ at the current chamber and at almost every next choice. The panel probabilities have no atoms, so nonstuttering has full measure, and the injective panel maps preserve nonstuttering.

Fix such a realization and starting chamber C_0 . The resulting map on gallery choices starts at $\phi(C_0)$ and has a triangular form: the map on the next panel is determined by the already mapped prefix. We verify that this map preserves measure classes in both directions. Suppose this has been proved through step j , with prefix map Φ_j and measurable inverse. Let μ_x and $\nu_{\Phi_j(x)}$ be the ordinary conditional probabilities on the next source and target pencils. Outside a null set of source prefixes there is a measurable bijection h_x satisfying

$$(h_x)_* \mu_x \sim \nu_{\Phi_j(x)}.$$

By the induction hypothesis, the exceptional prefix set has null image on the target side as well. For a measurable set A of extended target prefixes, conditional integration gives

$$\mu_{j+1}(\Phi_{j+1}^{-1}A) = \int \mu_x(h_x^{-1}(A_{\Phi_j(x)})) d\mu_j(x).$$

If $\nu_{j+1}(A) = 0$, its conditional sections are $\nu_{\Phi_j(x)}$ -null at almost every prefix. Fibre absolute continuity and $\Phi_{j*}\mu_j \ll \nu_j$ make the displayed integral zero. Conversely, if the integral is zero, nonnegativity makes its integrand zero almost everywhere. The reverse fibre absolute continuity and $\nu_j \ll \Phi_{j*}\mu_j$ imply $\nu_{j+1}(A) = 0$. This proves equivalence of the extended measures. The inverse first recovers the prefix through Φ_j^{-1} and then uses h_x^{-1} ; it is measurable and is inverse almost everywhere in both directions. The empty prefix begins the induction. We have therefore proved a measure-class isomorphism of the full gallery spaces.

The endpoint map identifies the generic galleries of this fixed reduced longest type from C_0 with its open opposition cell. The intermediate chambers are unique: they are recovered backwards by successive projections along the descents of the reduced word. In root coordinates this is the ordered unipotent Bruhat factorization [7, 64]. At a real place it is a smooth coordinate isomorphism with nonvanishing Jacobian; at a finite place it is an analytic coordinate isomorphism in the relative-root coordinates, hence preserves the local Haar class in both directions [13, 14]. The iterated compact panel probabilities have exactly these local classes. Products of the local identifications give the assertion for the full group. The complementary Bruhat cells have ordinary measure zero. The same statements hold for the target word $(p(i_1), \dots, p(i_m))$, since p preserves the Coxeter system.

Passing the gallery isomorphism through these two endpoint identifications gives a measurable bijection between conull subsets of B_S and B_T preserving both null directions. On almost every source gallery it agrees at the endpoint with ϕ , so it is a version of the original chamber-map class. This proves both essential bijectivity and $\phi_* m_S \sim m_T$. Normal copying

preserves all one-copy almost-sure statements, giving the assertion for both entries of the joint law. $\square$

13. ARITHMETIC RECOVERY AND FINITE MULTIPLICITIES

We first work in the weakly mixing setting of Section 3: $\Gamma < G$ is an actual ICC lattice, the commuting projective representations π, ρ on K are normal and finite over both endpoint factors, and $\text{Ad } \rho$ on $L^2(\pi(\Gamma)') \ominus \mathbb{C}1$ has no finite-dimensional subrepresentation. We then restore the compact sectors and pass to groups in $\mathcal{K}$.

Write B_S, B_T for the two copies of the full source flag space and m_S, m_T for their compact probability measures. The conclusion of Section 12 is used in its simultaneous form: the chamber evaluations are commuting spectral evaluations of nonsingular essentially invertible maps $B_S \rightarrow B_T$; the evaluations in the two comparisons sharing slot zero also commute. The induced spectral measure is equivariant for source transport and target postcomposition and is fixed by diagonal ρ . These simultaneous properties are needed below, not merely the existence of separate measurable boundary maps.

13.1. The measurable map space and uninduction.

Lemma 13.1 (The space of boundary isomorphisms). *The space $\mathcal{E}$ of nonsingular essentially invertible measurable maps $B_S \rightarrow B_T$, modulo equality almost everywhere, is a standard Borel space. Composition with the boundary actions is Borel, and target postcomposition is continuous for convergence in measure. The spectral maps supplied by Section 12 therefore define a PVM on $\mathcal{E}$.*

Proof. Equip the compact flag spaces with compatible bounded metrics. The space $L^0(B_S, B_T)$, with distance $\int d_T(f(x), g(x)) dm_S(x)$, is separable and completely metrizable. There are jointly Borel choices of representatives: approximate a map by the first member of a countable dense family of simple maps within a prescribed summable error, take an almost everywhere convergent subsequence with summable pointwise error bounds, and assign a fixed value where its pointwise limit does not exist. Consequently integration of a bounded Borel function of a map and a source point is Borel in the map.

Choose countable generating algebras $\mathcal{A}_S, \mathcal{A}_T$. For a map f , write $m_f = f_* m_S$. The condition $m_f \ll m_T$ is the countable uniform absolute-continuity test

$$\forall j \exists k \forall E \in \mathcal{A}_T : m_T(E) < 1/k \implies m_f(E) < 1/j.$$

It is Borel. The same test with the two measures interchanged recognizes $m_T \ll m_f$. The restriction to a generating algebra suffices, since that algebra approximates measurable sets in the finite measure $m_T + m_f$.

On this equivalence locus, essential invertibility is equivalent to

$$\forall E \in \mathcal{A}_S \forall j \exists F \in \mathcal{A}_T : m_S(E \triangle f^{-1}(F)) < 1/j.$$

This is again Borel. To verify the equivalence, the displayed condition says that pullbacks generate the completed source sigma algebra. Recover each member of a countable separating family of source indicators as a pullback of a target measurable indicator, using a summable subsequence of the displayed approximations. The resulting target indicator sequence codes a source point almost everywhere for m_f , and hence for m_T . The inverse of the Borel coding of the source gives a measurable map $g : B_T \rightarrow B_S$ with $gf = \text{id}$ almost everywhere. It follows that $fg = \text{id}$ almost everywhere for m_f , hence also for m_T . Conversely an essential inverse gives the displayed approximations. Thus $\mathcal{E}$ is a Borel subset of $L^0(B_S, B_T)$.

The jointly Borel representative construction also proves the composition assertion; nonsingularity makes composition independent of the chosen representatives. Target postcomposition is continuous by uniform continuity on compact flag spaces, also when the acting group element varies locally. Source precomposition is Borel by the same integral test.

Here is an explicit countable coding for the spectral pushforward. Choose a countable real continuous point-separating family (f_j) on B_T and use the source generating algebra $\mathcal{A}_S$ above:

$$I_{A,j}(\phi) = \int_A f_j(\phi(x)) dm_S(x), \quad A \in \mathcal{A}_S.$$

These coordinates are continuous for convergence in measure; equality of all of them gives equality of each $f_j \circ \phi$ almost everywhere by the monotone class theorem, and hence equality of the maps almost everywhere. The Lusin–Souslin Theorem [42, Corollary 15.2] therefore gives a Borel inverse to this coding on its image. In the common abelian spectral realization of Section 12, the weak integrals $\int_A P_x(f_j) dm_S(x)$ are precisely multiplication by $I_{A,j}(\phi_\zeta)$ for the jointly measurable random map ϕ_ζ . The chamber-bijection conclusion there puts $\phi_\zeta \in \mathcal{E}$ almost surely; faithfulness of the scalar spectral state therefore puts the joint operator PVM on the coding image of $\mathcal{E}$. Its pushforward under the Borel inverse is the asserted PVM, with covariance and common-slot commutation inherited from the same integrated operators. To recover a fixed boundary label x , take neighborhoods U_n shrinking to x . Source continuity gives

$$P_x(f_j) = \varprojlim_{n \rightarrow \infty} \frac{1}{m_S(U_n)} \int_{U_n} P_y(f_j) dm_S(y).$$

Each integral belongs to the coded operator algebra by L^1 approximation of source indicators from $\mathcal{A}_S$. Thus every fixed-label PVM is recovered, without evaluating an arbitrary representative of an L^0 class. $\square$

The coded PVM acts on $L^2(Y; K_0 \otimes K_1)$, where $Y = G/\Gamma$. The integrated angular operators used in its coding commute with the base multipliers $L^\infty(Y)$, so their spectral projections, and hence the coded PVM, are base-decomposable. Disintegration over Y gives PVM kernels T_y on $K_0 \otimes K_1$. Using the same base disintegration for both normal copies and countable determining families retains their common-slot commutation outside one null set. Use the section and cocycle of Section 3, so $c(a, y) = d_0(ay)^{-1}ad_0(y)$. Its covariance reads

$$(13.1) \quad \text{Ad } \pi_0(c(a, y))T_y(f) = T_{ay}(\phi \mapsto f(\phi \circ a)).$$

The identity holds for each fixed a outside a null set independent of f : first use a countable determining algebra and then normality.

Lemma 13.2 (Uninduction). *There is a PVM Θ on $\mathcal{E}$, acting on $K_0 \otimes K_1$, such that, for every $g, h \in \Gamma$,*

$$(13.2) \quad \text{Ad } \pi_0(g)\Theta(f) = \Theta(\phi \mapsto f(\phi \circ g)),$$

$$(13.3) \quad \text{Ad } \pi_1(h)\Theta(f) = \Theta(\phi \mapsto f(h^{-1} \circ \phi)).$$

It is fixed by $\text{Ad}(\rho_0(\lambda) \otimes \rho_1(\lambda))$ for every $\lambda \in \Lambda$. The two copies of Θ sharing slot zero commute.

Proof. For $g = d_0(y)h$ define the lifted kernel

$$F_g(f) = \text{Ad } \pi_0(h)^{-1}T_y(\phi \mapsto f(\phi \circ g)).$$

The tests depending on g are legitimate Borel kernel tests by Lemma 13.1; approximate them by simple Borel tests when applying the kernel identities. Since $ag = d_0(ay)c(a, y)h$, Equation (13.1) gives $F_{ag}(f) = F_g(f)$. All scalar matrix coefficients of a countable determining family are measurable and invariant under each left translation. Haar invariance and Fubini make them essentially constant. Choose one g_0 in the common conull set on which these coefficients have their constant values and the lifted kernel is a PVM, and define $\Theta = F_{g_0}$. For almost every g , equality on the generating algebra and uniqueness of the finite scalar matrix-coefficient measures give $F_g = \Theta$ on all Borel sets. Choose the same good g_0 for both normal copies and their countable determining commutators. This retains their common-slot commutation. Thus Θ is a PVM by construction; its countable additivity is inherited from that fibre.

Replacing g by $g\gamma$ in the lift gives

$$F_{g\gamma}(f) = \text{Ad } \pi_0(\gamma)^{-1} F_g(\phi \mapsto f(\phi \circ \gamma)),$$

which proves Equation (13.2). Target covariance gives Equation (13.3). The diagonal ρ -invariance and common-slot commutation hold for the induced kernels. Lifting the two comparisons simultaneously and taking the same countable determining families preserves those identities. $\square$

Lemma 13.3 (A target transversal). *Target postcomposition of Γ on $\mathcal{E}$ is free and admits a Borel transversal.*

Proof. If $h \circ \phi = \phi$ almost everywhere, equivalence of $\phi_* m_S$ and m_T implies that h fixes almost every target flag. Continuity and full support make it fix every flag, and faithfulness of the effective flag action gives $h = e$.

We prove the local slice property. Suppose $\phi_n \rightarrow \phi$ and $h_n \circ \phi_n \rightarrow \psi$ in measure in $\mathcal{E}$. An escaping sequence h_n has, after a subsequence, convergent compact Cartan frames and at least one diverging simple gap. In the corresponding proximal vertex module its normalized action contracts compact sets outside a limiting proper algebraic repeller to a single vertex. These are the local-field contraction properties of Section 3, applied to one factor if necessary. Pass to almost everywhere convergence of both map sequences. Since $\phi_* m_S \sim m_T$, its values avoid the repeller almost everywhere. At each such point, ϕ_n is eventually in a compact neighborhood away from the repeller, and the chosen vertex coordinate of $h_n \phi_n$ tends to the attracting vertex. The same coordinate of ψ is therefore constant almost everywhere, contradicting $\psi_* m_S \sim m_T$. Thus the translating sequence cannot escape.

For each ϕ there is consequently a neighborhood U with $hU \cap U = \emptyset$ for every $h \neq e$. Otherwise select pairs in progressively smaller neighborhoods with nonidentity translators. Escaping translators have just been excluded, and a constant subsequence would contradict freeness by continuity. Choose a countable cover by such neighborhoods, using separability. Each member meets an orbit at most once. Retain from the first member all its points, and from each successive member remove the countable saturation of the preceding members. The resulting Borel set meets each orbit exactly once. $\square$

13.2. From a simultaneous boundary law to a fixed partition. We now use the PVM Θ obtained by uninduction. Its two comparison copies have a common first module slot. Their commutation will let us regard the law in the second slot as a measurable family of ordinary PVMs. Each member of that family gives a finite-rank partition of K . The point of the argument is to show that these partitions are the same, even though their boundary-map labels can vary.

We first record the consequence of normality used to compare partitions.

Lemma 13.4 (Normal coefficient decay). *If a projective group representation extends normally to its twisted group factor, its matrix coefficients vanish at infinity. In particular, for finite-rank projections p, q ,*

$$\text{Tr}(p\pi(g)q\pi(g)^*) \longrightarrow 0 \quad (g \longrightarrow \infty).$$

Proof. A normal coefficient is given by an L^1 -density on the finite factor. Approximate that density in L^1 by a finite Fourier polynomial. The polynomial coefficient at u_g vanishes outside a finite set, and the approximation error is uniformly bounded by the L^1 -error. Expanding p, q in finite orthonormal bases gives the displayed conclusion. $\square$

Proposition 13.5 (A deterministic atom partition). *Retain the weakly mixing correspondence K from the start of this section. Let Θ be a PVM of total mass I on the boundary-isomorphism space $\mathcal{E}$, acting on $K_0 \otimes K_1$, with the precomposition and postcomposition covariances of Equations (13.2) and (13.3). Suppose that diagonal ρ fixes Θ and that its two copies on $K_0 \otimes K_1 \otimes K_2$ commute. Then K has an orthogonal partition into nonzero finite-dimensional subspaces permuted by $\pi(\Gamma)$ and $\rho(\Lambda)$. The Γ -action on the partition is free and has finitely many orbits.*

Proof. Disintegration and finite multiplicity. Take all normal slot-one slices of $\Theta(f)$ and let $\mathcal{A}_0$ be the von Neumann algebra they generate on K_0 . Apply two normal functionals, one in each branch slot, to the common-slot commutation identity. The resulting slices commute, also after taking adjoints; thus $\mathcal{A}_0$ is abelian. The covariance identities normalize it by π_0 and ρ_0 . The slice criterion gives

$$\Theta(f) \in \mathcal{A}_0 \bar{\otimes} B(K_1).$$

Use a faithful normal state to represent $\mathcal{A}_0$ over a standard probability space Z . The displayed inclusion disintegrates Θ into PVMs Θ_z on the fixed Hilbert space K_1 . This can be done first for one bounded real Borel coordinate generating $\mathcal{E}$ and then for its spectral calculus. Countably many generators and the countable group Γ make target covariance hold simultaneously outside one null set.

Let $T \subset \mathcal{E}$ be the Borel target transversal from Lemma 13.3, and set $q_z = \Theta_z(T)$. The projections $\pi_1(g)q_z\pi_1(g)^*$, $g \in \Gamma$, are orthogonal and sum to one. Hence

$$\ell^2(\Gamma) \otimes q_z K_1 \longrightarrow K_1, \quad \delta_g \otimes \eta \longmapsto \pi_1(g)\eta$$

is a unitary intertwining the first-factor twisted regular action with π_1 . In particular,

$$\dim_{L_\mu(\Gamma)} K = \dim_{\mathbb{C}} q_z K_1 < \infty.$$

The PVM on $q_z K_1$ therefore has finitely many nonzero atoms. Their Γ -translates give all of Θ_z , so its support consists of finitely many free target orbits of finite-rank atoms.

A measurable encoding of the projection partition. Choose a strictly positive trace-class operator D on K_1 . The measures

$$E \longmapsto \text{Tr}(D\Theta_z(E \cap T))$$

form a measurable kernel. Their nonzero atoms are exactly the nonzero atoms of the transversal PVM. After a Borel injection of $\mathcal{E}$ into $[0, 1]$, enumerate these atoms in increasing order: the first atom is the infimum of the rational cuts with positive mass; remove that mass and repeat. This is measurable and stops after at most $\dim q_z K_1$ steps. Kernel integration over the resulting Borel graphs makes the singleton projection weights measurable. Their matrix coefficients and finite ranks make them measurable also as elements of the separable Hilbert–Schmidt space.

Let $\mathcal{P}$ be the space of nonzero finite-rank orthogonal projections on K_1 . On its $\text{Ad } \pi_1(\Gamma)$ -orbit space put

$$d_\Gamma([p], [q]) = \inf_{g \in \Gamma} \|p - \pi_1(g)q\pi_1(g)^*\|_{\text{HS}}.$$

The triangle inequality follows from the isometric action. To see that the distance separates orbits, note that

$$\|p - \pi_1(g)q\pi_1(g)^*\|_{\text{HS}}^2 = \text{rank } p + \text{rank } q - 2 \text{Tr}(p\pi_1(g)q\pi_1(g)^*).$$

Distance zero requires equal ranks. Lemma 13.4 excludes an escaping minimizing sequence, while a nonescaping subsequence gives equality after one translation. The orbit metric is therefore separating and separable. Its nonempty finite subsets carry the separating, separable Hausdorff metric. These distances are measurable in the preceding enumeration, since every infimum is over the countable group Γ .

For each z , let $P(z)$ be the finite set of orbit classes of the transversal atom projections. This encoding forgets the boundary-map labels and the choice of a representative projection in each target orbit. It retains the full projection partition: take the union of the Γ -orbits of the projections represented by $P(z)$. Source precomposition relabels the boundary maps and commutes with target postcomposition. Returning a moved map to the chosen transversal T only conjugates its projection weight by $\pi_1(\Gamma)$. Thus source precomposition leaves $P(z)$ unchanged, and every bounded measurable function of $P(z)$ commutes with π_0 . Let $\mathcal{C}_0$ be the resulting abelian algebra. Then

$$\mathcal{C}_0 \subset \pi_0(\Gamma)'.$$

Weak mixing removes the remaining variation. The normalized trace on the finite factor $\pi_0(\Gamma)'$ gives the distribution of P an invariant probability measure. It has the same null projections on $\mathcal{C}_0$ as the faithful normal state used to disintegrate $\mathcal{A}_0$. A deterministic conclusion for this trace therefore also holds in the original spectral realization. Diagonal ρ -invariance of Θ gives, for bounded functions F ,

$$\text{Ad } \rho_0(h)F(P) = F(\text{Ad } \rho_1(h)^{-1}P).$$

Because ρ_1 commutes with π_1 , it acts isometrically on the projection-orbit space and on its finite-subset space.

Weak mixing of $\text{Ad } \rho$ on $L^2(\pi_0(\Gamma)') \ominus \mathbb{C}1$ makes the diagonal action on $\mathcal{C}_0 \bar{\otimes} \mathcal{C}_0$ ergodic. Indeed, a nonconstant invariant tensor with a mean-zero factor gives a nonzero Hilbert–Schmidt intertwiner. The associated positive compact operator would have a nonzero finite-dimensional invariant spectral subspace, contradicting weak mixing. The bounded truncation of the distance between two independent values of P is thus constant almost everywhere. That constant is zero: separability gives a countable cover by balls of arbitrarily small radius, and a ball of positive probability gives a positive probability of arbitrarily small pair-distance. Fubini now makes P itself constant almost everywhere.

The full projection partition recovered from this constant value is fixed on $K_1 = K$. Covariance makes it invariant as a partition under both endpoint groups. The free Γ -action and its finite orbit count are those already obtained from the transversal decomposition. $\square$

13.3. From a finite-rank partition to arithmetic models. The remaining group-theoretic step no longer uses the boundary law or weak mixing. We state its actual partition hypothesis so that the same result can later be applied after restoring the compact sectors.

Proposition 13.6 (Arithmetic models from a partition). *Let Γ, Λ be countably infinite ICC groups, with normalized scalar cocycles μ, ω , and write $M = L_\mu(\Gamma)$ and $N = L_\omega(\Lambda)$. Let K be a nonzero normal bifinite M – N correspondence. Suppose that K has an orthogonal partition into nonzero finite-dimensional subspaces permuted by $\pi(\Gamma)$ and $\rho(\Lambda)$, and that Γ acts freely on its labels with finitely many orbits. Then K is a finite orthogonal sum of the models $E(A, B, \delta, \sigma)$ of Theorem 1.2.*

Proof. Let X be the label set and fix $x \in X$. Its stabilizer in Λ is finite: an infinite stabilizer would preserve the corresponding finite-rank projection along an escaping sequence, contradicting Lemma 13.4. Since there are finitely many Γ -orbits, a finite-index subgroup of Λ preserves Γx . The two permutation actions commute, so moving x inside Γx does not change its Λ -stabilizer. This finite-index subgroup therefore normalizes that finite stabilizer. Each element of the finite stabilizer has a finite conjugacy class in Λ , and ICC makes the stabilizer trivial.

Thus Λ also acts freely. On each of its orbits, a basis of one atom gives that many copies of the twisted regular right module. Finite right module dimension makes the number of these orbits finite, so there are finitely many joint orbits as well.

Fix a joint orbit and its base atom V_0 . Let $A \leq \Gamma$ consist of the elements taking its label into its Λ -orbit, and define $B \leq \Lambda$ symmetrically. Their indices are finite by the two individual orbit counts. Separate freeness gives, for each $a \in A$, a unique $\delta(a) \in B$ such that $\pi(a)\rho(\delta(a))$ preserves V_0 . The joint stabilizer is a subgroup of $\Gamma \times \Lambda$ whose coordinate projections are injective. It is consequently the graph of an actual isomorphism $\delta : A \rightarrow B$. Its action on V_0 is

$$\sigma(a) = \pi(a)\rho(\delta(a))|_{V_0}, \quad \sigma(a)\sigma(b) = \frac{\mu(a, b)}{\omega(\delta(a), \delta(b))}\sigma(ab).$$

To identify the joint-orbit span with its arithmetic model, first define

$$F_0 : V_0 \otimes L^2(N) \longrightarrow K, \quad F_0(\xi \otimes v_h) = \xi v_h.$$

Orthogonality of the distinct Λ -translates makes F_0 a right-module isometry onto their closed span. The graph stabilizer gives

$$u_a \xi = \sigma(a) \xi v_{\delta(a)},$$

so F_0 intertwines the stipulated left subgroup action. This subgroup action is normal by projective regular absorption (Lemma 3.1) on each B -orbit in Λ .

Choose representatives $r \in \Gamma/A$. As a right subgroup module,

$$L^2(M) = \bigoplus_r u_r L^2(L_{\mu|A}(A)).$$

The relative tensor product defining $E(A, B, \delta, \sigma)$ is therefore the orthogonal sum of the spaces $u_r \otimes (V_0 \otimes L^2(N))$. On them set

$$u_r \otimes \eta \mapsto \pi(u_r) F_0(\eta).$$

Distinct cosets give distinct Λ -orbits of atoms, making this map isometric with range the full chosen joint-orbit span. Balancing and the subgroup identity give both endpoint intertwining identities on finite Fourier tensors, and hence on the completion. This is the required unitary with $E(A, B, \delta, \sigma)$. Summing over the finitely many joint orbits proves the proposition. $\square$

Proposition 13.7 (Arithmetic exhaustion). *For every $\Gamma \in \mathcal{K}$, every countably infinite ICC Λ , and arbitrary scalar cocycles μ, ω , every nonzero separable bifinite $L_\mu(\Gamma)$ – $L_\omega(\Lambda)$ correspondence is a closed summand of a finite direct sum of the arithmetic models in Theorem 1.2.*

Proof. Restrict the source to a finite-index subgroup actually isomorphic to a finite-index subgroup of a lattice, with its exactly pulled-back cocycle. The latter subgroup is itself an ICC lattice. By Lemma 3.2, a finite-index restriction of Λ splits the correspondence into finitely many blocks

$$K_z = V_c \otimes K', \quad \pi = 1 \otimes \pi', \quad \rho_h = v_h \otimes \rho'_h.$$

If v has multiplier ν , then ρ' has right-adjoint multiplier $\bar{\omega}/\nu$, and its right factor has cocycle $\omega\nu$. The reduced correspondence is normal, bifinite, and satisfies the weak mixing hypothesis used in the geometric construction and in Proposition 13.5.

Apply that proposition to K' and tensor its fixed atom partition with V_c . Both original actions permute the resulting finite-dimensional atoms. Their label actions are unchanged, so the Γ -action is still free with finitely many orbits. The hypotheses of Proposition 13.6 now hold for the original restricted correspondence, without a weak mixing assumption on its restored actions. It gives the arithmetic models with the original endpoint phases. Explicitly, on a recovered graph the reduced fibre multiplier is $\mu/(\delta^* \omega \delta^* \nu)$; tensoring with $a \mapsto v_{\delta(a)}$ multiplies it by $\delta^* \nu$ and restores $\mu/\delta^* \omega$. No finite-image assumption on v is used.

Finally apply Lemma 3.3 on both sides. The original correspondence embeds as a closed summand of the two-sided induction of its restriction. The normalized embeddings are the coset sums with terms

$$u_r \otimes \pi(u_r)^* \xi, \quad (\xi v_s^*) \otimes v_s,$$

respectively. Associativity of relative tensor products and the standard trace-module identifications show that inducing each restricted model gives

$$L^2(M) \otimes_{L_{\mu|A}(A)} (V_\sigma \otimes L^2(N)),$$

with the same graph subgroup, viewed now in the ambient groups. All indices and sums are finite, proving exhaustion. $\square$

Completion of Theorem 1.2. Proposition 13.7 gives the forward assertion for every nonzero correspondence; the zero correspondence is the zero summand. Corollary 2.3 gives normality and both dimensions of each elementary model. Normality and finite module dimension pass to finite direct sums and closed bimodule summands, as proved at the end of Section 2. This supplies the converse and completes the theorem. $\square$

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